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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP RESEARCH REPORT 214

BIM Beyond Design Guidebook

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2020

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AIRPORT COOPERATIVE RESEARCH PROGRAM

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FOREWORD

By Marci A. Greenberger
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ACRP Research Report 214: BIM Beyond Design Guidebook provides guidelines for how airport practitioners can build the business case to utilize building information modeling (BIM) for asset management once a construction project is completed. By using these guidelines, airport staff will be able to develop their BIM strategy, identify the stakeholders, and determine their BIM governance structure and the appropriate implementation metrics. Airport staff will also be able to identify how to scale the use of BIM for their organization considering the level of implementation sought with a current asset management system.

BIM is a well-known tool, first used by architects to design facilities. The cost benefit of using BIM has long been established within the construction industry, and therefore will continue to be used. Digital representation of a facility's physical and functional characteristics allows BIM to be leveraged as a comprehensive facility management system.

When the information in BIM is updated during construction, it generates as-built data that can be used in a computerized maintenance management system. BIM can also be scaled for construction projects of various sizes, renovations of existing facilities, and for more efficient management of planned utility shutdowns.

Under ACRP Project 09-15, CCI Engineering Services was tasked with developing guidance for airports on using BIM beyond design, that is, after construction, for facility management. The research included surveying and interviewing airports on their use of BIM, as well as organizations in other industries.

The guidance includes how to conduct the financial analysis to make the business case, data requirements, performance metrics, how to integrate BIM with existing systems, and case studies from domestic and international airports and other organizations. Airport staff in maintenance, facility management, and engineering and planning will benefit from the information provided in this report.

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SECTION 1

Introduction

Building information modeling (BIM) has transformed the design and construction industry over the last decade. It is now emerging as a key component to enhancing an asset life cycle management approach for many organizations. BIM supports a collaborative environment for documenting, analyzing, and sharing facility information across an entire organization. BIM not only reduces costs but also improves performance and operational readiness. For airport managers who oversee highly complex and mission-critical environments, implementing BIM can be a key strategic component that enables them to meet the needs of their airport's future growth in a cost-effective manner.

This Guidebook gives airport owners the basic knowledge required to implement successfully and maximize their benefits from BIM. It instructs airports and their consultants on best practices for implementing BIM in a life cycle management environment. The guidance focuses on BIM after construction, but the design and construction benefits are included in the discussion of the overall facility life cycle business case. Industry standards and best practices for life cycle BIM are still in the process of being developed. In areas where best practices have not yet been defined, this Guidebook presents options for leveraging approaches developed for the design and construction process and from available international standards. Where established best practices for life cycle BIM have not yet been established, this Guidebook provides academic references for life cycle BIM implementation, standards, and uses that airports can adopt to design their BIM strategies.

In the research conducted for this Guidebook, more than 60 airports responded to a survey that examined their current and planned use of BIM. Follow-up interviews were performed with some airports that indicated that they were utilizing or had immediate plans to utilize BIM as a life cycle tool beyond design and construction. These included Denver International Airport (DEN), San Francisco International Airport (SFO), the Port Authority of New York and New Jersey (PANYNJ), Dallas–Fort Worth International Airport (DFW), Los Angeles International Airport (LAX), and Massachusetts Port Authority (Massport). Although the specific implementation details of each airport differed, a common thread was the growing amounts and types of information required to manage airports and the siloed nature of this information across management stakeholder groups.

1.1 Why BIM?

Airports are implementing BIM to improve their asset information management and the reliability and life cycle maintenance of their infrastructure. They are also implementing BIM to enhance the ability of the organization to communicate and collaborate.

The complexity of airport management has grown dramatically in recent years, with increased security requirements, a focus on sustainability, increased competition, new technologies,

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and traffic growth. The continued growth of air traffic and service expectations of passengers, tenants, and airlines will demand increasing sophistication in airport information management. There are a number of innovations that will require airports to achieve a far greater level of information integration and interoperability than currently exists. These innovations include Internet of Things (IoT) sensor data that will collect and report performance information in real time across the entire airport, predictive maintenance systems that will all but eliminate unplanned outages, sensor-based baggage handling, smart parking systems, security system enhancements, and artificial intelligence–based decision support systems.

BIM can play an integral role in the evolution of airport information management by providing the three-dimensional (3D) virtual environment required for these progressive information systems to understand the location, context, and system connectivity of managed assets. Without BIM, assets are only entries in a database. With BIM, the operation of these assets can be simulated to augment airport planning, maintenance, and operations.

1.2 What Is BIM?

The National Institute of Building Sciences (NIBS) defines BIM as a "digital representation of physical and functional characteristics of a facility" that is a "shared knowledge resource" that provides a "reliable basis for decisions during its life cycle." The U.S. General Services Administration (GSA), which manages all federal public buildings, defines BIM as a "data rich, object-based, intelligent, and parametric digital representation of the facility." There are three primary characteristics of BIM: (1) visualization; (2) asset data; and (3) the intelligent, programmatic interface.

1.2.1 Visualization

BIM documents a facility in three dimensions rather than in two dimensions. Humans experience the world in three dimensions. While most facility owners can read traditional two-dimensional (2D) building plans, the conversion of 3D real-world data into 2D plans is imperfect. The 2D-to-3D translation adds time and cost to finding and extracting the as-built facility information managers require. It also greatly increases the possibility of errors in the plans or misinterpretation of the plans. Most airports have entire computer-aided design (CAD) manuals defining the standards for graphical representation of 3D facility data as 2D as-built drawings.

Figure 1-1 illustrates a model of a piece of mechanical equipment visualized in three dimensions. This 3D object model can be placed into a 3D facility model (Figure 1-2) to document its geospatial location (coordinates), its placement in the context of other surrounding equipment, and how it is connected as a system.

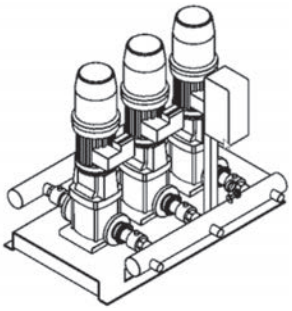


Figure 1-1. 3D object.

A 3D real-world view of as-built documentation can help facility operators find the data they require more quickly and with fewer errors.

1.2.2 Data Rich/Knowledge Resource

BIM provides more than just a 3D version of CAD; it also integrates the 3D model with asset data (see Figure 1-3) to provide a complete facility data environment. Unlike 3D CAD (where a 3D representation of a wall, door, or backup generator represents only the physical size, shape, and location of these objects), BIM draws on an intelligent facility database to represent assets more fully. Asset data include not only asset attributes such as the insulating value of

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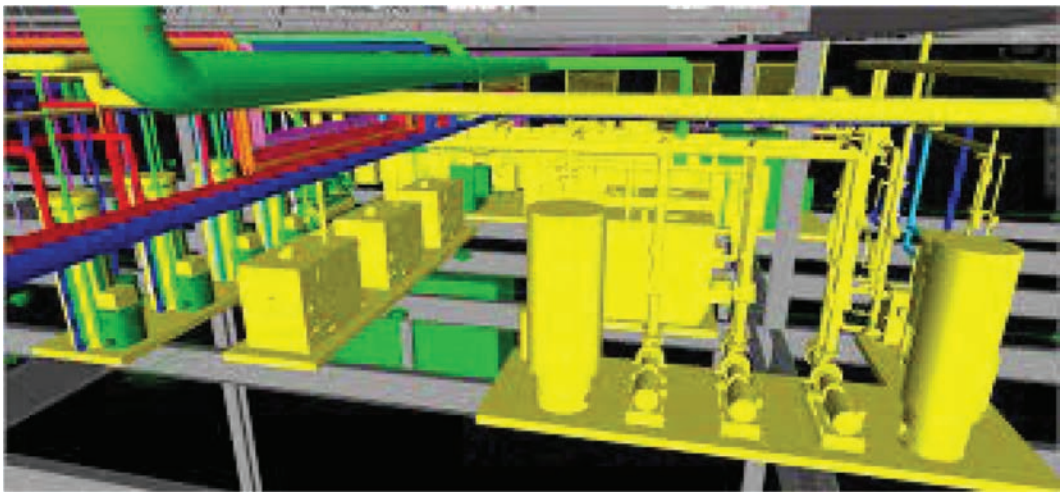


Figure 1-2. 3D facility model.

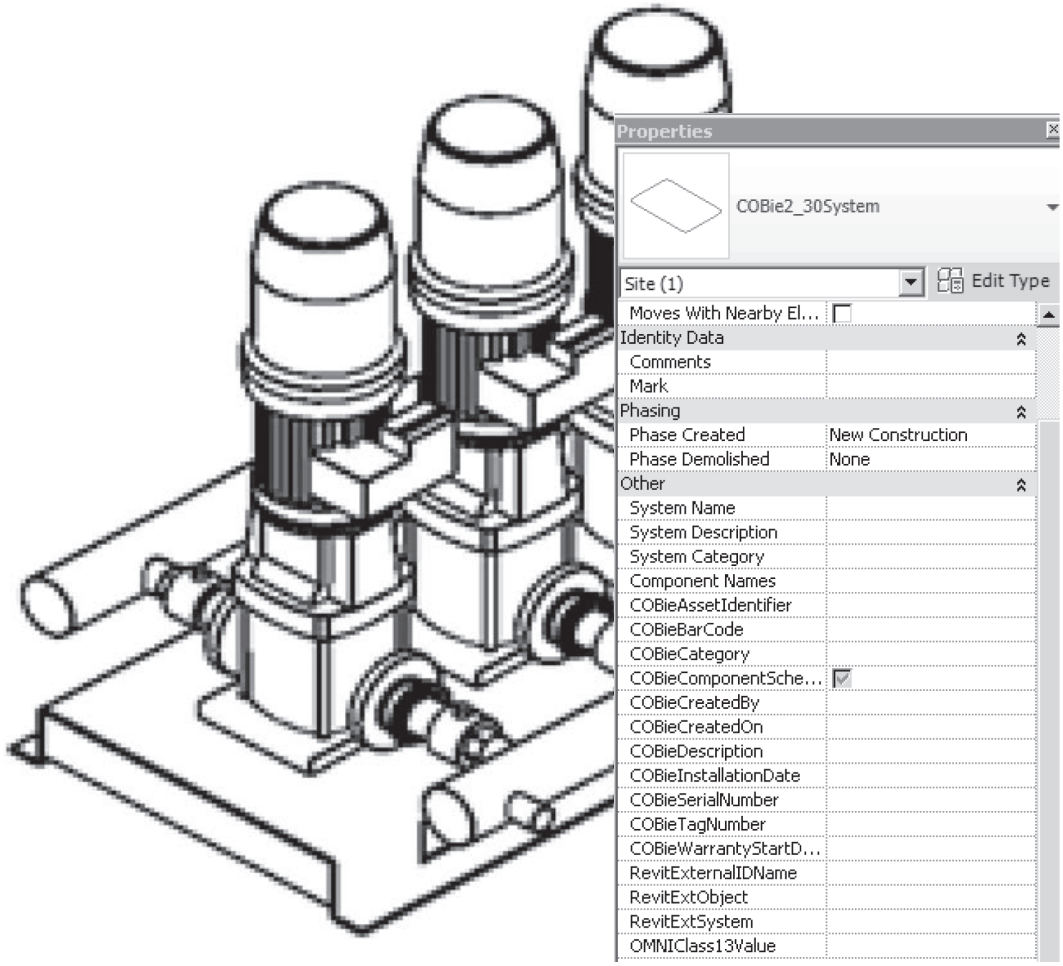


Figure 1-3. 3D object with data.

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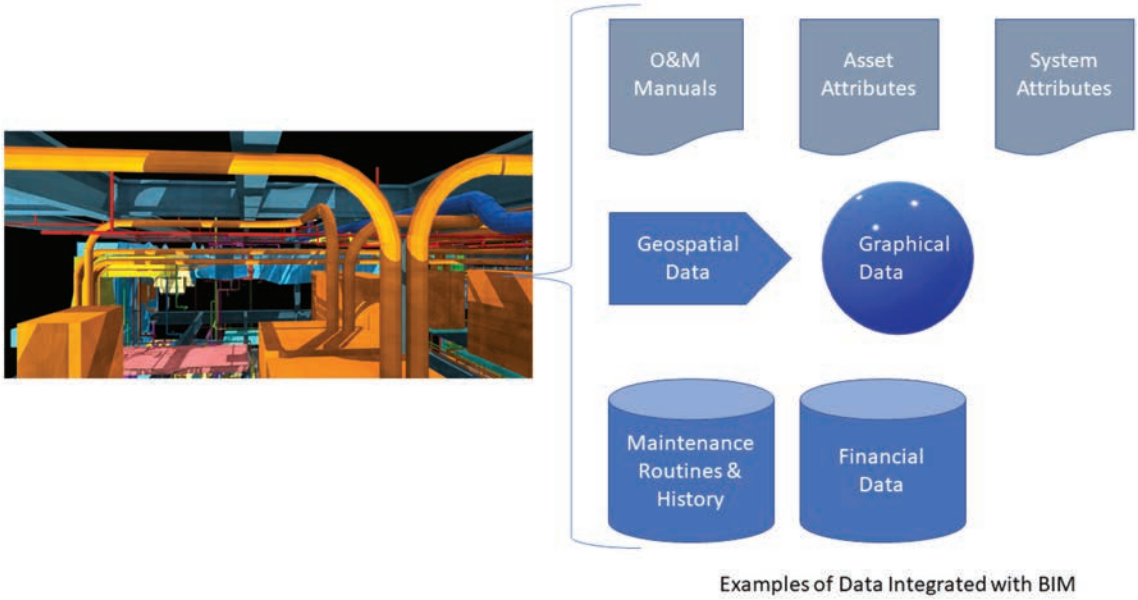


Figure 1-4. Data types integrated with BIM.

a window, but also the specific manufacturer, warranty data, installation date, maintenance schedules, and other data required for asset management.

These asset data combine to create a data-rich facility model that can be used by airport staff for planning and analysis on-site (via tablets and other mobile devices) or from desktops.

1.2.3 Intelligent/Programmatic

BIM provides a 3D, data-rich facility model in a digital environment, which extends access to the model's capabilities. External systems and programs can query and interact with BIM, which is not possible with a 2D building plan. This intelligent interface provides a platform that simulates the operations of the facility. Energy analysis tools can extract facility floor plans, volumes, and attributes related to walls, doors, and window insulation to predict energy utilization. Computerized maintenance management systems (CMMSs) can interact and share data with BIM to enhance maintenance planning and fault analysis.

Each asset in the BIM not only has 3D graphical elements for visualization, but also has attributes that can be customized to capture other important attributes. These include the asset manufacturer, model number, serial number, links to operations and maintenance (O&M) manuals and original manufacturer's specification sheets, installation date, warranty date, and work order history.

The ability to integrate BIM with other systems (see Figure 1-4) enables the sharing of BIM facility data in the format that each airport stakeholder group requires. It also provides a means for each group to share facility data that they are responsible for by updating the data in BIM.

1.3 The Value of BIM to Airport Operators

The current standard for facility data handover after construction is a set of 2D record plans, equipment specifications, O&M manuals, and possibly commissioning reports. Many airports have experienced, or will experience, BIM during design and construction because architects,

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engineers, and contractors have been adopting BIM over the last decade (even when it is not required by the contract).

BIM is at the forefront of a transformation of the construction industry. It enables the development of advanced construction logistics and workflow sequencing, cost estimating, value engineering, and automated construction. There has already been much written about the use of BIM for the design and construction portions of the facility life cycle, and there are many resources available to airports that wish to create standards for development of their capital projects. *ACRP Synthesis 70: Building Information Modeling for Airports* provides a broad overview of BIM practices at airports that is primarily focused on the design and construction phases (McCuen and Pittenger, 2016).

This Guidebook will primarily focus on how to utilize BIM after construction to enhance life cycle asset management. The use of BIM in this manner is fundamentally different from the use of BIM solely for design and construction. Life cycle BIM is about developing an ongoing process of asset creation, maintenance, and renewal. As such, BIM affects not only an airport's facility information infrastructure, but also the way various airport stakeholder groups interact, communicate, and collaborate. Life cycle BIM not only requires technology changes; it also requires cultural changes.

Airports face different challenges based on their size and location. Some are focused on how to best manage the projected growth of airline traffic over the next few decades. These airports are in the process of expanding facilities, adding runways, or completely rebuilding aging terminals and infrastructure. Other airports are focused on how to reduce their costs to make their rates and charges more competitive for airlines.

In some cases, airports may be strategically improving facilities and services as a means of economic development, serving as a gateway to promote regional growth. Regardless of the challenges faced, airports must find a way to fund improvements. BIM can be used to deliver benefits that reduce both capital and operational spending:

- **Capital expenses**. BIM can reduce the time and cost involved in designing and constructing new facilities. BIM can also enhance capital planning by providing more accurate and complete facility asset data.
- **Operational expenses**. BIM can lower the operational and maintenance costs of airport facilities through improved maintenance planning, asset management, and collaboration between the airport and its key stakeholder groups (airlines, tenants, and others). BIM can also promote the design of sustainable facilities.

1.4 Strategic Asset Management

BIM can be a key component in providing a facility information platform that is accessible across an organization. If an airport is considering using BIM for something besides design and construction, the airport is likely doing so as part of an overall effort to improve asset management. While detailed implementation of strategic asset management is beyond the scope of this Guidebook, the relationship of BIM with industry initiatives such as the American National Standards Institute (ANSI) Total Cost of Ownership (TCO) and the International Standards Organization (ISO)-55000 Standards for Asset Management will be discussed. An airport can benefit from the guidance herein without being committed to implementing these standards. The discussions of these standards will, however, enhance understanding of how BIM could be implemented as part of an airport's longer-term strategic asset management initiatives.

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1.5 Leveraging Future Innovations

The data-rich facility model that BIM provides will enable airports to benefit from other existing and emerging innovations that will redefine the facility management industry over the next decade. These innovations include IoT real-time data acquisition that supports building automation systems (BASs) and predictive maintenance systems, root cause analysis and decision-making support based on artificial intelligence (AI), and automated construction that will dramatically reduce the cost of new facilities.

These leading-edge technologies require accurate and complete facility data, which can be provided by BIM. BIM will reduce the future cost of building and managing airport facilities. It will also position airports to leverage facility management innovations that will emerge over the next few decades. The ability to rapidly adopt these innovations will enable airports to meet the challenges of an increasingly complex service environment.

The opportunity cost of not utilizing BIM over the next decade will mean forgoing the potential facility life cycle cost savings that could enhance an airport's competitive profile. Areas of enhancement include lower landing fees, lower cost per enplaned passenger, and improved service quality for airport passengers.

1.6 Using This Guidebook

This Guidebook will provide airports with an overview of BIM fundamentals and a process for implementing BIM and optimizing benefits received from BIM. The guidance is focused on BIM processes and standards and will not provide instruction on using specific commercial BIM tools.

1.6.1 Guidebook Content and Organization

The Guidebook is structured according to BIM development stages, as shown in Table 1-1.

A checklist is provided at the end of Sections 2 through 12 to provide guidance for the key activities suggested in each section.

1.6.2 Pre-BIM Activities

Sections 2 through 4 cover the activities involved in evaluating BIM for use at an airport. This evaluation includes performing an assessment to identify the needs and challenges related

Table 1-1. BIM development stages and Guidebook structure.

Pre-BIM Activities	BIM Organizational Assessment (Section 2) Preparing the Organization and Stakeholders for Implementation (Section 3) Financial Analysis (Section 4)
BIM Implementation	BIM Process (Section 5) Scaling BIM Implementation (Section 6) Technical Architecture (Section 7) Integration of BIM with Existing Systems (Section 8) Standards, Execution Plans, Required Data Elements (Section 9)
BIM Controls	Governance (Section 10) Progress Metrics (Section 11) Legal and Liability Issues (Section 12)

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to facility data management and the opportunities for BIM to provide benefit in that regard. Pre-BIM activities also include financial analysis to develop the business case for implementing a BIM program and measurement of the organization's capabilities and readiness to implement BIM and leverage its benefits.

1.6.3 BIM Implementation

Sections 5 through 9 describe the steps required to implement a BIM program. These include having the technical architecture required to implement and maintain BIM, outlining the BIM process, scaling BIM to meet existing and future needs, adhering to BIM standards to provide a basis for quality assurance/quality control (QA/QC), and preparing an organization to adopt BIM. Although BIM requires new technologies, it is primarily a process innovation that can require significant organizational and individual role changes.

1.6.4 BIM Controls

Sections 10 through 12 describe the requirements for measurement and creating a system of continuous improvement to ensure BIM delivers the expected benefits. These sections also include guidance on the establishment of progress metrics and key performance indicators (KPIs); BIM governance, including how BIM responsibilities are distributed across the organization; and the design and construction contractual framework to control legal and liability issues arising from BIM.

1.6.5 Guidebook Flowchart

The flowchart in Figure 1-5 represents the recommended approach to planning, implementing, and managing BIM at an airport. It is possible to take a more ad hoc approach to implementing BIM. However, the approach presented is recommended to ensure that the BIM program implemented fits an airport's organizational needs and capabilities and optimizes the potential benefits. Many airports interviewed as a part of this research did not perform a detailed evaluation of return on investment (ROI) of BIM as part of their business case. They implemented BIM with an a priori belief that it would be beneficial to their organization's performance. The options provided in this Guidebook serve as a menu of actions from which an individual airport can select based on its particular needs.

Figure 1-5 illustrates the process for developing and maintaining a facility BIM that meets the needs and requirements identified as part of an airport's formal, or informal, needs assessment.

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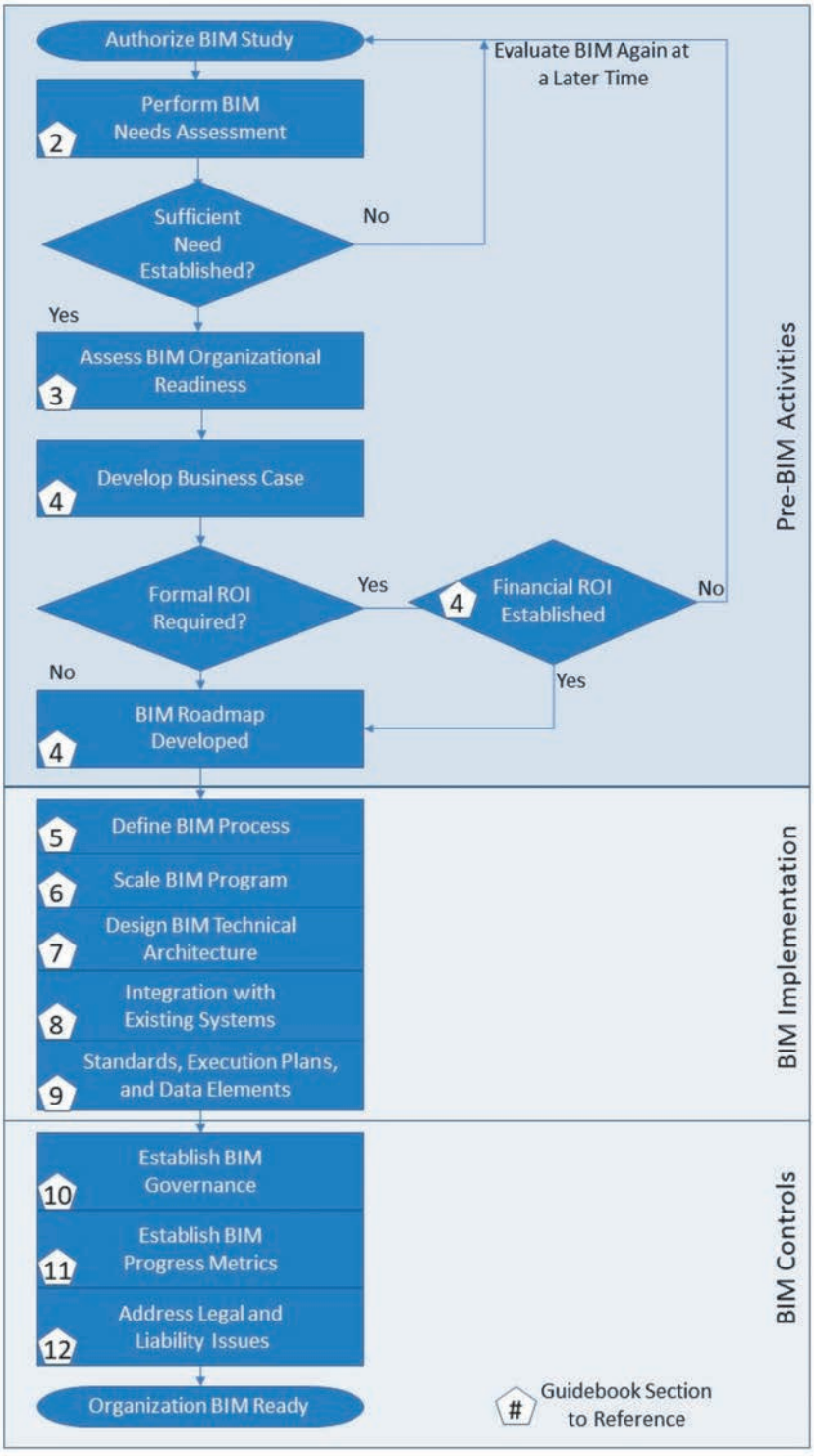


Figure 1-5. BIM development flowchart.

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SECTION 2

Pre-BIM Activities—BIM Organizational Assessment

BIM is a process that can have a wide-ranging impact on how an organization collects, manages, maintains, and shares facility information. Because of this, any implementation of BIM should begin with a BIM organizational assessment that includes two parts: (1) a BIM needs assessment and (2) a BIM capabilities assessment. The BIM needs assessment includes all airport stakeholder groups and engages them in a discussion of the opportunities and challenges related to facility information at their airport. The BIM capabilities assessment gauges an airport's readiness to implement BIM. With the data collected from these two efforts, a BIM program can be tailored to fit the unique characteristics of the airport and its operational and strategic goals. This section of the Guidebook will discuss how to perform the BIM organizational assessment.

2.1 Needs Assessment Process

The first step in customizing an airport's BIM program is to perform a needs assessment that identifies and prioritizes the facility information needs and challenges and aligns them with the features and benefits of BIM. A BIM needs assessment is designed to identify the following:

- What facility information, and in what format, is required by each airport stakeholder group to perform its job function?
- Why is this facility information required, and what are the alternatives?
- Who is responsible for creating and maintaining this facility information?
- Where is this facility information located, and how is it accessed?
- What are the current obstacles and challenges related to acquiring and maintaining this facility information? What are some specific examples of actual issues the stakeholders have faced or are currently facing? When possible, quantify the number of hours these obstacles may be adding to their workload.
- What are the opportunities to improve an airport stakeholder group's job function with improved facility information? Identify specific examples of the type of facility information that would benefit existing or upcoming work. When possible, quantify the number of hours that would be saved if this facility information were easily available.

The needs assessment process can be conducted with digital surveys, but a group interview process provides an opportunity to collect data from each stakeholder and to uncover valuable context at the same time. The group interview can also be done after analyzing digital survey results, but this will increase the amount of time spent completing the needs assessment effort.

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2.2 Data Flow Diagrams

After identification of the facility data needs and challenges, the next step in the needs assessment is to create a map of how facility data are currently collected, stored, maintained, and distributed at the airport. A useful tool for this purpose is a data flow diagram (DFD). A DFD is a map of airport facility data sources and types that also shows consumers of those data across an organization. A DFD provides a graphical representation of the flow of data and is an excellent tool for visualizing information. SFO utilized this tool when it analyzed its data environment (see Figure 2-1).

The diagram in Figure 2-1 shows how as-built documents are authored and distributed across stakeholder groups at SFO. The ellipse shapes in the diagram indicate internal stakeholders, while the central ellipse, labeled "D&C Library," identifies storage location of facility design and construction data. The connecting lines indicate where data are shared between (flow to and from) stakeholder groups. For example, the "Tenants" group on the lower right side of the diagram receives information from "Revenue Development and Management," which receives data from the aviation management (AVM) CAD Group. The design and construction library and networked server make the as-built facility plans available to the rest of the organization. "Project Management" acts as the gatekeeper to the external consultants working on projects at the airport and publishes updated as-built plans to the library.

DFDs can also include many forms of data in addition to as-built plans, such as maintenance data, asset management data, condition data, safety and regulatory reports, O&M manuals, and incident reports. Each data source can show the data types and available formats so that the gaps in data availability can be identified. (One example of a gap in data availability is a stakeholder group that requires facility data but either does not have access to those data or can't use the data in the form in which they are provided.) DFDs can be instrumental in getting each stakeholder group to think about airport facility data in a broader, more holistic manner. The process allows the groups to gain perspective on how the data they generate (whether through maintenance or financial reports) are used by other departments at the airport. Documenting how facility data are used at an airport is a vital element in building and maintaining the collaborative environment that BIM is designed to promote.

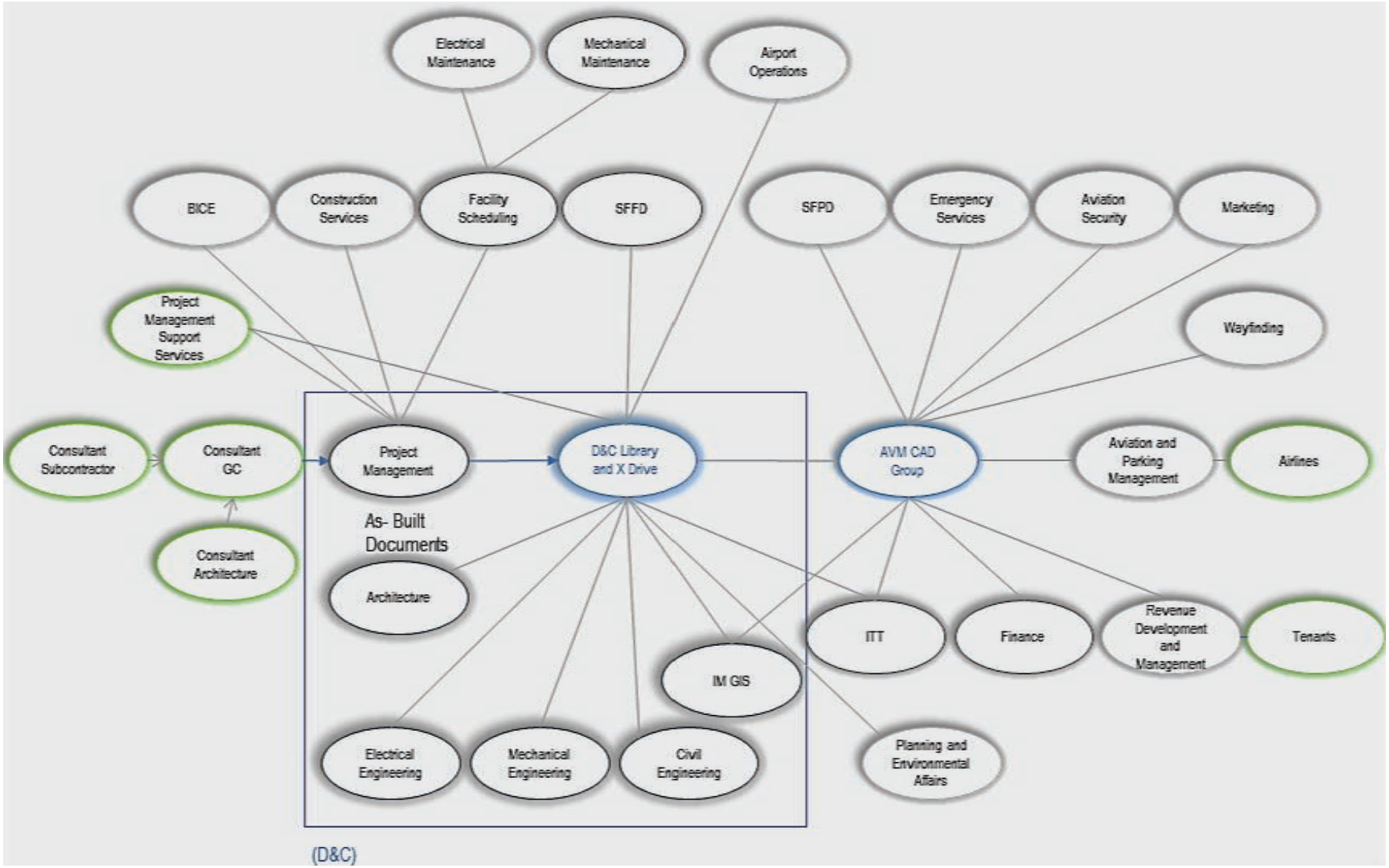
2.3 Prioritization

The final step in the needs assessment is to prioritize the results. Prioritizing facility data needs can be difficult. Even if it were possible to precisely measure the ROI resulting from improved facility data collection, management, maintenance, and sharing, there are always non-monetary priorities that need to be considered for an airport as well. These non-monetary priorities include safety and security, regulatory requirements, customer satisfaction, and broader concerns such as impacts on the local and regional economy.

As an alternative, the airport can take a consensus approach to determining the priority focus of improvements to facility data accuracy and accessibility. Measuring this consensus can be accomplished using a digital survey sent to a representative selection of members of each airport stakeholder group. If a more precise and structured approach is desired, an alternative would be to use a Delphi-driven methodology with the selected members of each airport stakeholder group as the panel of experts.

The DFD process collects the sources of facility data. The next step is prioritizing the facility data types that each stakeholder group uses in the performance of its work by determining

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Source: SFO
Note: AVM = aviation management; D&C = design and construction: GC = general contractor

Figure 2-1. SFO DFD.

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how important they are relative to other types of facility data the group requires. Examples of types of facility data utilized by airports include the following:

- As-built plans
- O&M manuals
- Maintenance history and warranty data
- Emergency procedures
- Asset inventory
- Asset condition
- Asset valuation
- Financial and capital planning data
- Facility floor layout plans
- Survey data
- Geotechnical data
- Underground utility locations
- Project management data

2.3.1 Digital Survey Method of Prioritization

Digital surveys are easy to distribute and take less time for respondents than attending a formal meeting. Industry studies have shown that survey length has the most influence in increasing response rates. A study published by the *Journal of MultiDisciplinary Evaluation* (Saleh and Bista, 2017) showed that 91% of respondents said that a survey duration of less than 15 minutes would increase their likelihood of responding. (For a survey internal to an organization, where there is the ability to follow up personally, a response rate approaching 100% should be achievable. Internal distribution makes the use of digital surveys more attractive than it might be otherwise.)

One accepted survey approach uses the Likert scale (named after its inventor, psychologist Rensis Likert) to rate the importance of each data type to the stakeholders. In this survey approach, respondents are asked to rank each type of facility data on a Likert scale of 1 to 5, with 1 indicating the highest mission criticality and 5 indicating little to no importance. Responses are collected and averaged across all airport stakeholder groups to calculate an average criticality across all groups. An example of the results of such a survey is provided in Figure 2-2. The numbers shown in Figure 2-2 are not meant to indicate what should be expected by any given airport. The departments listed in the top row are found at typical airports; for an airport using this survey approach, this list should be customized to reflect its organizational structure.

The rankings produced by a survey are not enough, by themselves, to set the overall priorities for using BIM to improve collection, management, maintenance, and sharing of facility data. However, the rankings will provide valuable information for decision making. For example, a consensus showing that accurate and accessible property layouts (i.e., facility floor plans) are critical to a wide range of airport stakeholder groups can be used to build the business case for creating the initial architectural BIM of the terminal and associated airport structures. As noted earlier, other, non-monetary strategic factors prioritized by the airport may influence the final rankings. Such factors include the need for improved disaster planning or sustainability analysis.

2.3.2 Delphi Method of Prioritization

The costs of collecting and maintaining different types of facility data also need to be considered. While these costs may not be tracked as a separate cost center, or be easily measurable, there may be a consensus that some types of data are costlier to collect than others. For example, an effort to collect the data to map all underground utility infrastructure would generally be more expensive than laser scanning and modeling airport terminal space. The Delphi method is better for this type of consensus-driven ranking.

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BIM Beyond Design Guidebook

Mission Criticality Matrix	Airfield Operations	Terminal Operations	Finance	Properties & Contracts	Maintenance	Planning & Construction	Engineering	Environmental	Information and Communications Technology	ARFF	Airlines	Concessions	Other Tenants	FAA	Average Criticality
Property Layout and Spatial Data	1	1	3	1	1	1	1	1	1	1	1	1	1	1	1.14
Emergency Procedures	1	1	4	1	1	1	1	3	3	1	1	1	1	1	1.50
Asset Inventory	1	1	1	1	1	1	1	1	1	1	3	3	3	3	1.57
Asset Location	1	1	1	1	1	1	1	1	1	1	3	3	3	3	1.57
Asset Condition	1	1	1	1	1	1	1	1	1	1	3	3	3	3	1.57
Asset Valuation	1	1	1	1	1	1	1	3	1	5	3	3	3	3	2.00
Project Management Data	2	1	4	1	2	1	1	1	4	2	2	2	2	4	2.07
As-Built Facility and Asset Information	3	1	4	1	1	1	1	1	1	1	4	4	4	4	2.21
Life, Safety and Health Compliance	5	1	4	1	4	1	4	4	4	1	1	1	1	1	2.36
Underground Utility Locations	4	1	3	3	1	3	1	1	4	1	4	4	4	1	2.50
Work Order Management/History	3	1	3	2	1	1	3	5	1	3	3	3	3	3	2.50
Capital Planning Data	3	3	3	2	3	2	1	3	3	5	2	2	2	2	2.57
Security/Incident Data	3	3	4	3	3	2	4	5	4	1	1	1	1	2	2.64
Financial Data	3	3	1	2	4	3	3	4	4	5	3	4	4	2	3.21
Survey Data	3	3	3	1	3	3	1	2	5	5	4	5	5	3	3.29
O&M Manuals	3	3	5	3	1	3	3	4	1	3	5	5	5	5	3.50
Geotechnical Data	5	5	5	3	5	5	1	1	5	3	4	5	5	4	4.00
Average Criticality	2.53	1.82	2.94	1.65	2.00	1.82	1.71	2.41	2.59	2.35	2.76	2.94	2.94	2.65	

Note: ARFF = Airport Rescue and Firefighting

Figure 2-2. Facility information prioritization example.

The Delphi method is a more structured approach than the digital survey approach just discussed. It uses a panel of experts to respond to questionnaires, similar to a digital survey. However, in the Delphi method, a facilitator provides feedback to each panel member on how the rest of the panel responded and why (without attribution to any particular panelist).

Panel members may revise their responses after receiving this information, and the process repeats until an adequate level of consensus has been achieved. Studies have shown this technique is highly effective at getting panelists' responses to converge if the process begins with a panel knowledgeable about the subject matter. This approach could be used to develop a more authoritative consensus on the cost to collect and maintain different types of facility data that, in turn, could be used to create a weighted cost average.

In the case of airport facility data, it is unlikely that there could be a more knowledgeable panel of experts than the airport stakeholders using the data daily. Because the Delphi method involves a facilitated discussion, the skills and knowledge of the facilitator are also an important factor. The ideal facilitator is someone who has a broad knowledge of airport operations and is well respected by the airport organization. An alternative is a professional outside facilitator who would be viewed as impartial.

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2.3.3 BIM Roadmap

Once an airport has prioritized its mission-critical facility data types, the next step is to organize the improvement of collection, management, maintenance, and sharing of the data as part of a BIM roadmap. The BIM roadmap is a phased implementation plan in which the most mission-critical and least costly facility data needs are prioritized, and the least important and most costly ones are deferred to later phases of the implementation plan. If an airport decides not to develop a comprehensive business case, a BIM roadmap can serve as an organization's guide to the long-term vision and anticipated benefits of BIM.

For example, developing a comprehensive, minimal "Level of Development" (LOD) dataset for an entire airport terminal can be done at a relatively low cost. It can be immediately shared to develop accurate property maps for tenant lease management, emergency response plans, incident reporting systems, and project management tools that show current and planned projects.

Detailed capital project as-built plans for CMMS integration and maintenance planning could be included in a later phase of a BIM roadmap, and creating an integrated facility and site-civil model might be scheduled to occur in subsequent phases based on cost and complexity. Examples of BIM roadmaps are included in Appendix D.

2.4 BIM and Asset Management Strategy

An additional aspect of the overall airport BIM needs assessment should be how BIM can support airport strategic asset management initiatives. If BIM is being evaluated as a tool for improving asset life cycle management, then implementing BIM as part of an overall restructuring of a standards-based strategic asset management methodology should be evaluated. The standards would be ANSI TCO and ISO 55000. BIM can play an important role in adopting one or both of these standards.

2.4.1 ISO 55000 Standard for Asset Management

The recognized standard for asset management is ISO 55000. ISO 55000 is a three-part standard composed of ISO 55000 (overview, principles, terminology), ISO 55001 (asset management system requirements), and ISO 55002 (additional guidance). This international standard addresses the management of a broad range of assets (not only physical assets) for the ultimate purpose of deriving the greatest value or ROI. The standard applies to all organizations and all industries. It embraces continuous improvement and prescribes methods for realizing maximum value from assets.

Preceding ISO 55000 was Publicly Available Specification (PAS) 55, published by the British Standards Institution (BSI). Still in use today, especially by firms that have achieved certification in the standard, PAS 55 focuses more narrowly on physical assets and not on the more transformative management systems that are the basis of ISO 55000. PAS 55 is the basis for British BIM Levels 1, 2, and 3, which define the "Digital Built Britain" mandates for the implementation of BIM (to reduce the cost of construction and facility management in Britain). As discussed previously, ISO 55000 addresses the organization as a whole to establish management systems that deliver maximum value and touch all aspects of business operations. ISO 55000 is broken down into 6 groups and 39 subject areas, as depicted in Figure 2-3.

Another consideration for the relationship of BIM with ISO 55000 is life cycle value realization, which involves balancing of costs and benefits to optimize the life cycle

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Source: Institute of Asset Management, 2015

Figure 2-3. Organization of ISO 55000.

About ISO and ANSI

ISO is an independent, international, non-governmental body with a membership of 161 national standards bodies. Among the 161 member standards bodies are

- ANSI, which coordinates the development of voluntary consensus standards in the United States, and
- BSI, established by the United Kingdom (UK) to promote trade (by developing common industrial standards), reduce waste (by simplifying production and distribution), and protect the consumer (through the use of licensed marks to identify conformity to standards).

Through its members, ISO brings together experts to share knowledge and develop voluntary, consensus-based standards that support innovation and provide solutions to contemporary global challenges. Conformance to ISO standards ensures that products and services are safe, reliable, and of high quality and are considered strategic business tools that reduce cost by minimizing waste and errors and increasing productivity.

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costing (LCC) and the value obtained from assets. LCC is also referred to as TCO, which is the subject of ANSI Standard APPA 1000-1 Total Cost of Ownership for Facilities Asset Management (TCO)—Part 1: Key Principles. This standard identifies BIM, and specifically "openBIM," as a foundational tool for establishing TCO and achieving life cycle value realization. The goal of each is to facilitate the optimum mix of benefits and life cycle costs to maximize ROI.

2.4.2 ANSI Standard APPA 1000-1 TCO

The ANSI Standard APPA 1000-1 TCO framework includes 18 cost elements classified across 5 life cycle phases. Figure 2-4 shows how the 18 cost elements map to the airport functions defined in *ACRP Report 19A* (Hazel et al., 2011). The table shown in Figure 2-4 is intended only as an example; airport organizations may define these responsibilities differently.

2.4.3 BIM for Life Cycle Value Realization or TCO

A key element needed for life cycle value realization, or TCO, is the ability to share accurate information about assets digitally across stakeholder groups without manual

TCO Framework Element	Airfield Operations	Air Service	ARFF	Cargo	Concessions	Energy Management	Environmental	Financial	Fuel	General Aviation	Grants	Human Resources	Information Technology	Legal	Maintenance	Parking	Planning/Construction	Police/Security	Properties/Contracts	Public Affairs	Safety/Risk Management	Service Quality	Terminal Operations
Initial Cost																							
Planning & Programing	✓	✓	✓	✓	✓		✓	✓		✓	✓		✓	✓	✓	✓	✓	✓		✓	✓	✓	✓
Acquisition		✓					✓	✓		✓	✓		✓	✓			✓		✓				
Design		✓								✓			✓				✓						
Construction/Site Development		✓								✓			✓				✓				✓		
Commissioning		✓								✓			✓				✓		✓		✓		
Operations and Maintenance																							
Lease or Rental		✓	✓	✓	✓			✓	✓	✓	✓		✓	✓		✓			✓				
Maintenance	✓	✓	✓	✓	✓				✓	✓	✓		✓		✓	✓					✓		✓
Operations	✓	✓	✓	✓	✓				✓	✓			✓		✓	✓					✓		✓
Overhead and Administration	✓	✓	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓	✓							✓
Utilities																							
Utilities	✓	✓	✓	✓	✓	✓	✓	✓		✓			✓	✓	✓	✓					✓		✓
Renewal																							
Replacement	✓		✓	✓	✓			✓		✓	✓		✓	✓		✓	✓		✓	✓	✓		✓
Programmatic Upgrades	✓		✓	✓	✓					✓	✓		✓			✓	✓						
Improvement/Enhancement	✓		✓	✓	✓					✓	✓		✓			✓	✓						
End of Useful/Functional Life																							
Sale/Adaptive Reuse			✓	✓	✓		✓	✓		✓			✓	✓		✓	✓		✓	✓	✓		
Resale Value/Salvage Value			✓	✓	✓		✓	✓		✓			✓	✓		✓	✓		✓	✓	✓		
Removal			✓	✓	✓		✓			✓			✓	✓		✓	✓				✓		
Site Restoration/Remediation			✓	✓	✓		✓			✓				✓		✓	✓				✓		
Deconstruction/Recycling			✓	✓	✓		✓			✓				✓		✓	✓				✓		

Source: CCI; ANSI TCO; and Hazel et al., 2011

Figure 2-4. Airport organizational unit and ANSI TCO element focus.

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data re-entry. BIM supports the development of this and other asset management building blocks by offering

- Accurate coordinate-based asset data.
- Integration and accessibility capabilities via digital interfaces with a wide variety of CMMSs, accounting systems, and project management systems.
- Complete system or portfolio views of how assets are connected/related, along with interdependencies.

Many parallels exist among ISO 55000, TCO, and BIM:

- Their effectiveness is dependent on well-defined standards for use.
- They require enterprise-wide attention to achieve the greatest value.
- They are most effectively implemented when viewed as management systems.
- They facilitate continuous improvement through the development of processes that connect assets to business value propositions.
- They are being embraced by both private and public entities outside of the United States (including the UK, Australia, and Canada).
- Requirements for ISO 55000 practices and BIM are increasingly included in public infrastructure requests for qualifications (RFQs)/requests for proposals (RFPs) within the United States, including those involving airports and airport infrastructure.

To ensure the success of ISO 55000, TCO, and BIM programs, adoption of these programs should be approached as an organizational change initiative that will significantly impact the culture, leadership, and communications of an organization. Table 2-1 compares TCO requirements with BIM-enabling functions.

Table 2-1. Comparison of TCO requirements with BIM-enabling functions.

TCO Requirements	BIM-Enabling Functions
Managed Assets shall be identified for inclusion in TCO analysis. Such assets apply to land parcels, facilities, structures, infrastructure, and equipment.	While BIM provides the most direct support for facilities, integration with 3D civil and geographic information system (GIS) data can be performed. Emerging tools are being developed to make this easier.
A detailed **Asset Inventory** shall be the basis for tracking costs and performance across the asset life cycle and upon which decisions are made.	A BIM coordinate-based asset inventory enables an accurate and verifiable asset inventory.
A fixed asset shall use a **Global Location Hierarchy** using Global Unique Identifiers. Location is a key element to differentiate similar assets from other assets.	BIM provides a coordinate-based asset location that fully supports geospatial coordinate systems and Global Unique Identifiers.
All assets shall be organized using an **Asset Classification** standard, which enables the use and ease of access to asset information. It also provides consistency for comparison and benchmarking of assets across the portfolio.	BIM based on the Industry Foundation Class (IFC) schema can be adapted to formal asset classification schemas using model view definitions (MVDs) such as COBie.
Asset Costing requires that all costs associated with facility assets are captured at regular intervals to support life cycle decision making.	5D BIM provides costing support for TCO.
A continuous process of **Asset Inspection** aligns asset condition with the allocation of resources.	BIM coordinate-based asset location reduces the amount of time required for routine asset inventory and condition assessment.
Asset Performance provides the metrics to improve resource allocation and strategic decision making.	BIM can be integrated with building automation systems (BASs) to collect real-time asset performance data and support root cause analyses.

(continued on next page)

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Table 2-1. (Continued).

TCO Requirements	BIM-Enabling Functions
Asset Decisions are enabled by TCO analysis by supporting the efficient and effective use of assets to achieve goals and objectives. This requires clearly defined decision-making objectives and methodologies to manage risk and resources.	BIM provides the complete facility view that can be used to simulate operational activities to analyze how well the facility layout and assets align with the facility function (strategic asset management plan).
Asset Annual Funding identifies an annual asset expense budget and actual asset expenses for the life of an asset.	BIM can support the analysis of asset performance and condition for input into spending decisions.
An **Asset Comprehensive Plan** provides a total view of assets by identifying and tracking current conditions, future growth, maintenance, and space needs. It is a key output for resource allocation decision making.	BIM provides location and context (how the asset is integrated with other systems), which are critical data to developing these plans.
Asset Information Sharing results in (a) the greatest efficiencies and effectiveness in the use of resources and (b) accuracy in data use. Sharing of asset data enhances transparency and improves trust in resource allocation and other decision-making processes. Sharing of core data, as needed, will also alleviate duplication of efforts and produce the greatest possible ROI.	BIM provides collaboration as a key feature.
Timely and meaningful **Asset Reporting** ensures that accurate information about assets is available to decision makers at strategic, tactical, and operating levels. This will satisfy organizational needs for effective planning, efficient resource management, risk management resiliency, continuous improvement, and the greatest ROI.	BIM drives collaboration and a single view of facility data.

Note: COBie = Construction Operations Building Information Exchange

2.5 Summary

Although airports share similar challenges, there is not a standard model for how airport organizations are structured or operated. A BIM organizational assessment process consisting of a needs assessment and a capabilities assessment will aid airports in identifying and prioritizing their needs for the improved accuracy and accessibility of facility data that BIM can provide.

BIM capability assessment tools can be used to measure the organizational readiness of the airport to successfully implement an effective BIM program. Organizational development strategies can be integrated into the BIM implementation plan to address any areas identified as needing further development.

Section 2 Checklist

1. Perform a BIM needs assessment.
2. Develop a DFD and prioritize facility data sources.
3. Perform a BIM capabilities assessment.
4. Develop a BIM roadmap.

Organizing priorities and capabilities requirements into a BIM roadmap (phasing plan) can communicate and guide the long-term vision for BIM throughout an airport's organization.

BIM is often implemented as part of a shift to an overall asset life cycle management (ALCM) strategy, such as those defined by the ANSI TCO or the ISO 55000 standards. For this reason, BIM standards should be developed with an ALCM perspective.

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SECTION 3

Pre-BIM Activities—Preparing the Organization and Stakeholders for Implementation

This Guidebook has established that BIM is a strategic tool for enabling continuous improvement of airport assets across the life cycle. As briefly mentioned in other sections, to achieve the greatest benefit from BIM, implementation should be treated as a program of organizational change and transformation. The effort will need to consider BIM's impact on all organizational functions and business processes and the contribution that each function and process can make to the success of the organization as a whole.

Organizational change is not easy. It is widely estimated that 70% or more of organizational change projects fail (McKinsey & Company, 2016). The primary causes of failure are employee resistance and inadequate management support. Organization-wide change requires a significant commitment from everyone in the organization, starting with leadership and including every stakeholder. Leaders need to be champions to support the initiative, and the organization needs to create an environment of trust, safety, and security so that the change is welcomed rather than resisted. Stakeholder roles should not be underestimated. Stakeholders should be given, and accept, ownership and responsibility for setting the direction for change to ensure buy-in.

Successful BIM programs begin by establishing the connection between the organization's mission and the BIM vision and building a roadmap of goals, initiatives, and activities to achieve the vision. This connection is graphically illustrated in Figure 3-1.

For every goal, initiative, and activity found in the BIM roadmap, the elements shown in Table 3-1 need to be addressed.

3.1 BIM Capabilities Maturity Tools

Several different approaches exist to develop a BIM capability maturity model (CMM) that airports can use to guide improvements in organizational BIM capabilities over time. CMM systems have been used for decades to benchmark capability maturity in industries such as engineering, software development, and product development. A CMM provides a structured methodology for measuring an organization's capability across some standardized scaling factors deemed critical for mastery of these capabilities. The Software Engineering Institute developed the standard CMM used by the software industry. The framework measures an organization on a scale of 1 to 5 based on the processes it has in place.

A CMM is not a standard that an organization can implement. Rather, a CMM measures whether an organization's processes are well defined, repeatable, measurable, regularly analyzed, effective, and managed through the process of continuous improvement. In some respects, a CMM can be seen as analogous to ISO 9000 in that it does not dictate work processes, but rather measures how well they are managed and aligned with the goals and objectives of the organization.

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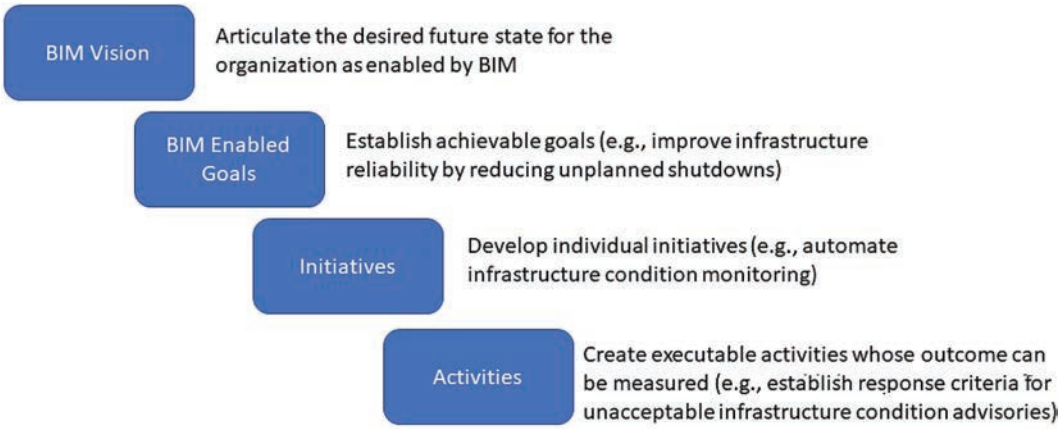


Figure 3-1. Preparing for BIM.

In the UK, there is a certification process to assess an organization's ability to conform to BIM Level 0, 1, 2, and 3 requirements:

- Level 0 indicates no BIM capability.
- Level 1 means a mix of 3D and 2D CAD capability.
- Level 2 describes a BIM using a collaborative environment.
- Level 3 indicates a set of BIM strategic initiatives for open standards design to encourage life cycle collaboration.

In the United States, there is no formal certification authority for measuring BIM CMM. There are a wide number of proposed BIM CMM frameworks that an airport can use to benchmark its existing BIM capabilities and progress toward higher levels of capability.

This section will introduce two BIM CMMs that airports can use to measure capabilities development. The first is included as part of the National BIM Standard—United States Version 3 (NBIMS-US V3), and the second is the owner's BIM Capabilities Assessment Tool (BIMCAT) developed by Raymond Issa and Brittany Giel. The NBIMS-US V3 CMM has a greater focus on BIM technology and processes, whereas the owner's BIMCAT is more focused on how an organization utilizes BIM from a life cycle view. If BIM is going to be primarily used for design and construction, then the NBIMS-US V3 CMM may be more straightforward to implement. Airports looking to utilize BIM as a life cycle tool would obtain greater benefit from using the owner's BIMCAT.

3.1.1 The NBIMS-US V3 CMM

The NBIMS-US V3 CMM provides a way for organizations to benchmark their BIM capabilities in 11 "areas of interest." Organizational capability in each of these areas is assessed against 10 benchmark levels, with "1" representing the minimum BIM skills and "10" representing the most advanced. The 11 areas of interest are shown in Table 3-2 with a brief description of each.

NBIMS defines a "Minimum BIM" that defines a level of achievement in each of 12 BIM capability criteria. If these levels are not achieved, NBIMS does not recognize that organization as sustaining the necessary activities to support the BIM process. Examples of these minimums include graphical interfaces that must include 3D intelligent graphics, a spatial capability that must include basic spatial locations, and delivery methods that are supported by limited web services. A given organization will likely score highly in some areas and not as highly in others, based on its internal priorities. The CMM areas of interest are designed to just provide a set of basic parameters to assess current capabilities and track growth.

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Table 3-1. Addressing the impact of implementing BIM.

Element	Information Required	Explanation
Leadership	What role will organizational leadership play to ensure success?	Leadership roles include not only executive staff but also leaders of both formal and informal groups, such as union leaders and recognized subject matter experts. (These are the program champions.)
Work Processes	Which processes will be impacted and how?	Create a BIM implementation roadmap and identify all of the touchpoints. Those touchpoints will illuminate the processes that will be impacted. Addressing this element should also include considering how organizational policies, both formal and informal, might connect with the BIM program, and whether they should also be reviewed for consistency with the BIM vision.
Organizational Structure	What structure will support the success and sustainability of the BIM program?	BIM governance and compliance functions should be established, and someone should be assigned to review the impact of external influences on the organization's use of BIM (e.g., regulatory and other organizational changes at a minimum). Roles and responsibilities for staff involved in the process touchpoints that become illuminated in the "work process" element should be reviewed and modified as necessary. The review may involve discussions about labor rules; third-party contracts; tenant agreements; and memoranda of understanding with property owners, granting bodies, government and regulatory agencies, law enforcement, and mutual aid partners, to name a few.
Staff Development	What skills and training do staff need?	These needs should be considered over the long term but be provided just in time for each step of the journey. Training needs should be considered for all points in the employee journey—recruitment (entry-level staff) and organizational-and self-directed advancement.
Technology	What technologies will support the end goal?	This element goes beyond BIM and addresses technologies to support all of the elements listed here. It includes such things as automation devices for data collection in the field, online training, and monitoring and reporting technology.
Communication	What information is required, and how will we communicate it?	Different stakeholders will require different messages and modes. This element builds on the element "Stakeholder Identification and Interrelationships" by suggesting that after all stakeholders and the role(s) they can play in achieving the vision are identified, then the types of messages and communications methods need to be identified to suit each stakeholder group.
Stakeholder Identification and Interrelationships (culture, buy-in, and support)	Identify all stakeholders, their interrelationships, and the roles they can play in achieving the vision.	This element offers the opportunity to build community and a shared purpose. The intent is to ensure inclusion so that all stakeholders see where they fit in and understand, own, and contribute to the outcome.
Incentives and Rewards	What compensation and rewards will be valued in return for the commitment expected of stakeholders?	Incentives and rewards include understanding how salaries and benefits packages for staff compare with best-in-class organizations and what motivates staff.

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Table 3-2. NBIMS-US V3 CMM areas of interest and descriptions.

Title	Description
Data Richness	Identifies the completeness of the building information model from initially very few pieces of unrelated data to the point of it becoming valuable information and ultimately corporate knowledge about a facility.
Life Cycle Views	Views refer to the phases of the project and identifying how many phases are to be covered by the BIM. One would start as individual stovepipes of information and then begin linking those together and taking advantage of information gathered by the authoritative source of information. This category has high-cost reduction, high-value implications based on the elimination of duplicative data gathering. The goal would be to support functions outside the traditional facility management roles, such as first responders.
Roles or Disciplines	Roles refer to the players involved in the business process and how the information flows. This is also critical to reducing the cost of data recollection. Disciplines are often involved in more than one view as either a provider or consumer of information. Our goal is to involve both internal and external roles as both providers and consumers of the same information so that data do not have to be recreated and that the authoritative source is the true provider of the information.
Change Management	Change management identifies a methodology used to change business processes that have been developed by an organization. If a business process is found to be flawed or in need of improvement, one institutes a "root cause analysis" of the problem and then adjusts the business process based on that analysis. Since this is related to the following item, business process, it should come after it.
Business Process	The business process defines how business is accomplished. If the data and information are gathered as part of the business process, then data gathering is a no-cost requirement. If data are gathered as a separate process, then the data will likely not be accurate. The goal is to have data both collected and maintained in a real-time environment, so as physical changes are made, they are reflected for others to access in their portion of the business process.
Timeliness/Response	While some information is more static than other information, it all changes, and up-to-the-minute accuracy may be critical in emergencies. The closer to accurate real-time information you can be the better quality the decisions that are made. Some of those decisions may be lifesaving in nature.
Delivery Method	Data delivery is also critical to success. If data are only available on one machine then sharing cannot occur other than by email or hard copy. In a structured networked environment, if the information is centrally stored or accessible, then some sharing will occur. If the model is a Systems-Oriented Architecture (SOA) in a web-enabled environment, the net-centricity will occur, and information will be available in a controlled environment to the appropriate players. Information assurance must be engineered into all phases.
Graphical Information	Often the starting point is a non-graphical environment. The advent of graphics helps paint a clearer picture for all involved. As standards are applied then information can begin to flow as the provider and receiver must have the same standards in place. As 3D images come into play, more consumers of the information will have a common view, and a higher level of understanding will occur. As time and cost are added, then the interfaces can be expanded significantly.
Spatial Capability	Understanding where something is in space is significant to many information interfaces and the richness of the information. Energy calculations must know where the heat gains will come from; first responders need to know where water supplies and utility cutoffs are located about the facility.
Information Accuracy	Having a way to ensure that information remains accurate is only possible through some mathematical ground truth capability. Having a mathematical product will also allow for better management by supporting difficult-to-measure metrics. These numbers can be used for occupancy, information collection completeness, and overall inventory calculations.
Interoperability/ IFC Support	Our ultimate goal is to ensure interoperability of information and getting accurate information to the party requiring the information. There are many ways to achieve this; however, the most effective way is to use a standards-based approach to ensure that information is in a form that can be shared and products are available that can read that standard form of information.

Source: Reproduced from data in NBIMS-US V3.
Note: IFC = Industry foundation class.

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NBIMS-US provides an interactive spreadsheet version of the CMM that can be downloaded freely from its website; this tool is called the Interactive BIM Capability Maturity Model v2.0. A tab on the spreadsheet provides category descriptions, which are listed in Table 3-2.

3.1.2 BIMCAT

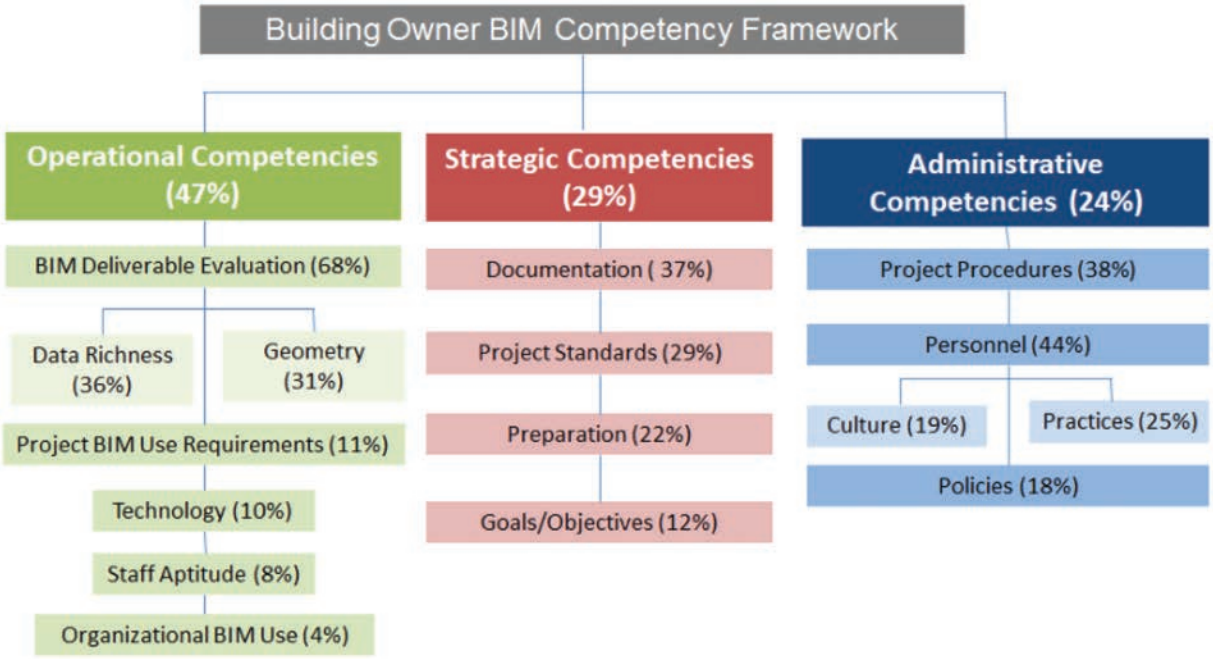
The BIMCAT BIM benchmark system proposed by Brittany Giel and Raymond Issa, from the University of Florida, provides benchmarking tools for an organization based on a BIM competency framework composed of operational, strategic, and administrative competencies.

Operational competencies are defined as factors that support the organization's ability to execute BIM at an organizational level and a project level. Strategic competencies are defined as those factors that support the organization's ability to plan and develop a course of action. Administrative competencies are defined as factors that support the organization's ability to manage resources and meet desired goals.

The percentages provided in Figure 3-2 show how much of the total BIM competency score is delivered by a particular kind of competency. For example, 47% of the total score comes from operational competencies, 29% comes from strategic competencies, and 24% from administrative competencies. The weighting of each BIM competency, as implemented in the Giel and Issa BIMCAT tool, shows the relative importance of the operational, administrative, and strategic elements in developing the BIM competency ranking.

The BIMCAT is a self-assessment tool developed to measure an organization's BIM capabilities and is composed of 124 questions. An organization could receive an overall score between 0 and 1,200, and a BIM competency ranking between Level 0 and Level 5 (see Table 3-3).

The subject areas within each core area of competence are shown in Figures 3-3 through 3-5 for strategic, operational, and administrative activities, respectively. Each diagram shows how



Source: Giel and Issa, 2016

Figure 3-2. Building owner BIM competency framework.

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Table 3-3. Building owner BIM competency levels.

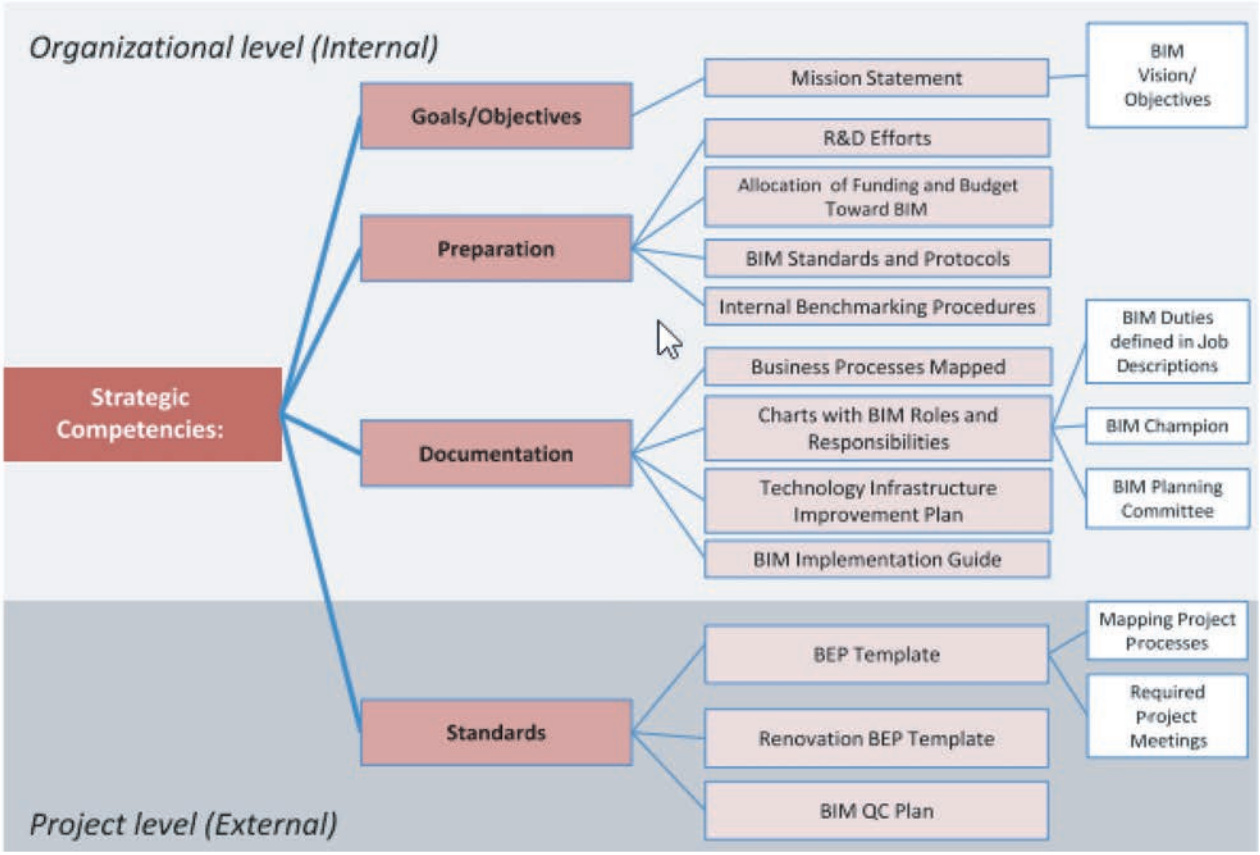
BIM Competency Level	Maturity	Score Range
Level 0	Non-existent	0–200
Level 1	Initialized	200–400
Level 2	Managed	400–600
Level 3	Defined	600–800
Level 4	Quantitatively Managed	800–1,000
Level 5	Optimizing	1,000–1,200

Source: Giel and Issa, 2016

BIM elements, defined on the right-hand side, support the operational, strategic, and administrative BIM competencies. For example, in Figure 3-3, the BIM Vision/Objectives need to be defined as part of the Mission Statement, which in turn needs to be reflected in the Goals and Objectives. The BIM job descriptions, the establishment of a BIM planning committee, and the identification of a BIM champion need to be formalized in the organizational chart.

Figure 3-4 illustrates the requirement to develop geometric and data richness evaluations of BIM deliverables across a wide variety of potential BIM-generated uses. It also covers software, hardware, networking infrastructure, staff training, and the development of BIM uses.

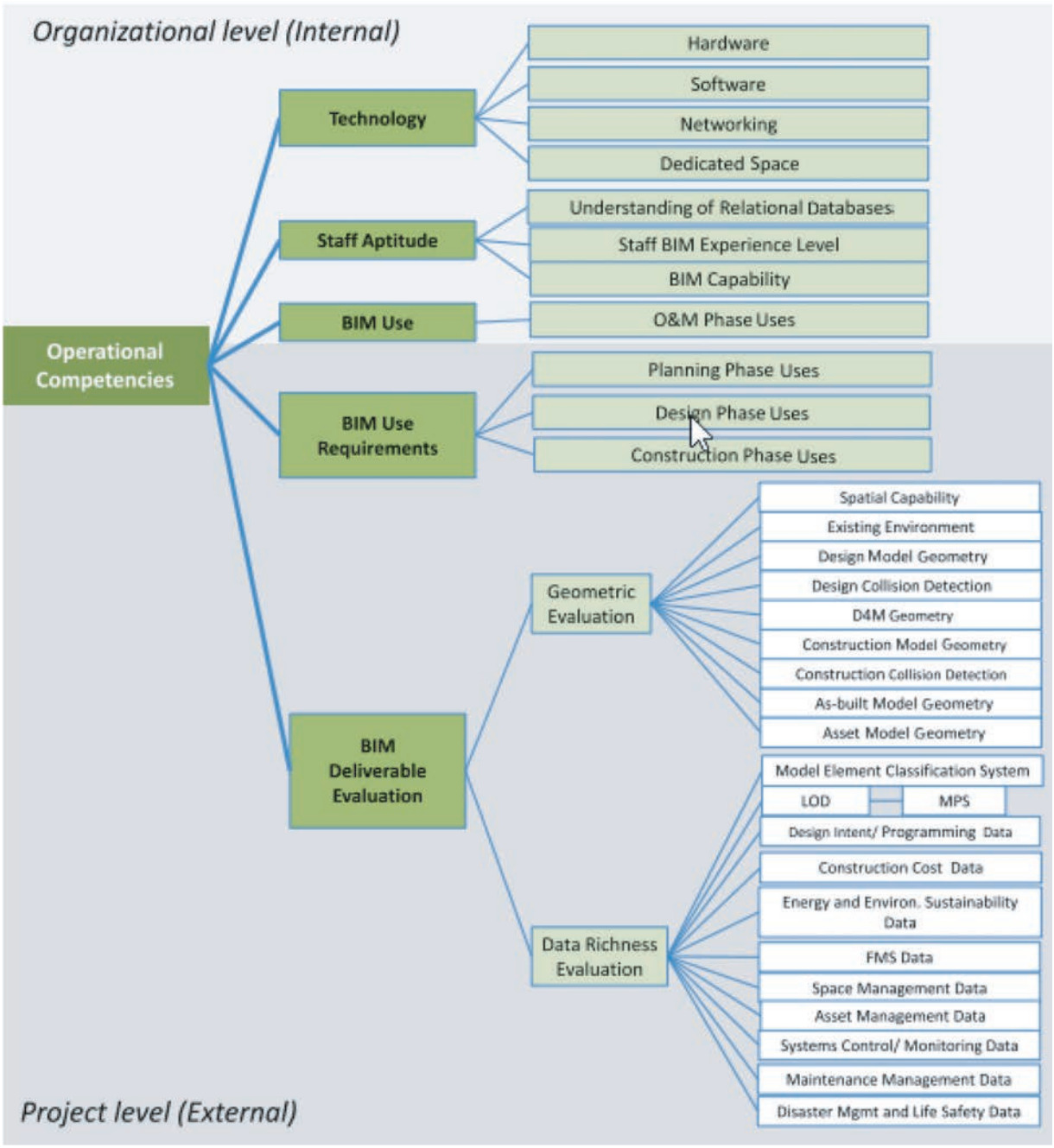
Figure 3-5 illustrates the BIM controls that need to be in place to properly manage the implementation and adoption of BIM processes across the airport. The focus is on the policies, procedures, and personnel (practices and culture) required to manage the use of BIM.



Source: Giel and Issa, 2016
Note: BEP = BIM execution plan

Figure 3-3. BIMCAT strategic competencies.

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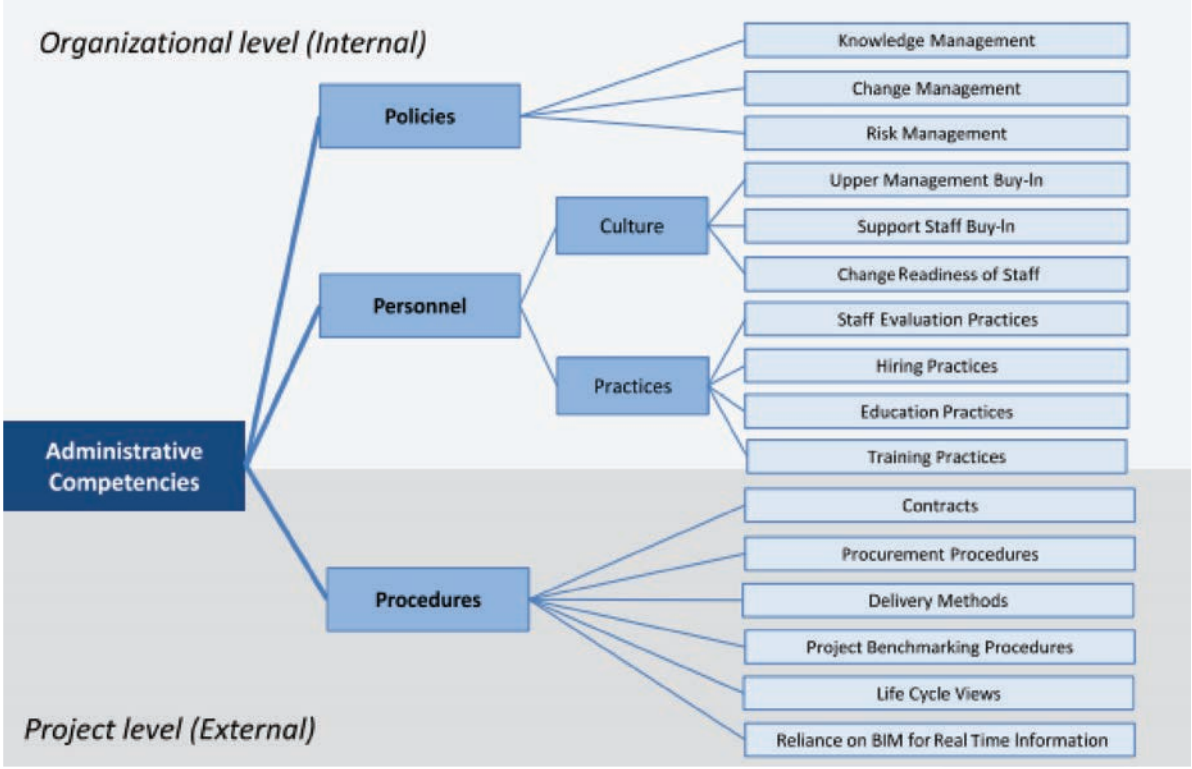
Source: Giel and Issa, 2016
Note: D4M = design for maintenance; FMS = facility management system; MPS = model progression specification

Figure 3-4. BIMCAT operational competencies.

The team that developed the BIMCAT also conducted a Delphi-based study of BIM experts that included owners, contractors, architects, consultants, and researchers who had significant BIM experience. This study identified the top 10 consensus factors contributing to an organization's capability to effectively implement BIM:

1. Upper management buy-in
2. BIM vision and objectives
3. BIM champion
4. BIM training practices

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Source: Giel and Issa, 2016

Figure 3-5. BIMCAT administrative competencies.

5. Construction clash detection evaluation
6. BIM education practices
7. Organizational mission statement
8. As-built model geometry evaluation
9. BIM job duties
10. BIM hiring practices

3.2 Other BIM CMM Options

The NBIMS-US V3 and the BIMCAT CMMs are just two options airports could evaluate for their use. There is not yet an established standard for measuring BIM capabilities. A BIM CMM feature comparison is shown in Figure 3-6.

Feature comparisons are provided in six key BIM planning elements: strategy, uses, process, information, infrastructure, and personnel. The evaluation criteria are measured using six different levels of achievement: non-existent, initial, managed, defined, quantitatively managed, and optimizing. "Quantitatively managed" refers to having specific measures in place, and "optimizing" refers to having a continuous process of reviewing and improving goals and vision. Fifteen elements (not shown in Figure 3-6) are used to assess the level of achievement. These elements are individually measured and ranked and include organizational mission and goals, BIM vision and objectives, management support, BIM champion, BIM planning committee, project uses, operational uses, project processes, organizational processes, model element breakdown, LOD, facility data, software, hardware, and physical spaces.

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Key Evaluation Criteria	NBIMS-US™ I-CMM NIBS (2007)	BIM Competency Index (BCI) Succar et al. (2013)	BIM Maturity Matrix (BIMMM) Succar (2010)	BIM Proficiency Matrix IU (2009)	BIM Quick Scan Van Berlo et al. (2012)	VDC Scorecard/ bimSCORE Kam et al. (2013)	Owner Maturity Matrix CIC (2012)	Owner's BIMCAT Giel and Issa (2013 a and b)
BIM Planning and Strategizing Efforts		•	•		•	•	•	•
Use of Technology and Tools		•	•		•	•	•	•
BIM Personnel: Mentality, Culture and Individual Competency		•	•		•	•	•	•
BIM Management Practices, Infrastructure and Administrative Policies	•	•	•	•	•	•	•	•
Processes and Operational Uses of BIM	•		•	•		•	•	•
Information Requirements	•		•	•	•	•	•	•
Geometric Requirements	•			•				•

Evaluation Context Used

Models — Organizations / Project Teams

Individuals — Owner Organizations

Source: National Building Information Modeling Standard, NBIM-US V3

Figure 3-6. BIM CMM feature comparison.

Also of note is the Civil Integrated Management (CIM) Capabilities Maturity Model that appears in Chapter 3 of *NCHRP Research Report 831: Civil Integrated Management (CIM) for Departments of Transportation, Volume 1* (O'Brien et al., 2016). CIM is the transportation corollary to BIM for vertical facilities. The CIM Capabilities Maturity Model can be used by agencies to analyze the current maturity of CIM across their divisions, and the model identifies the characteristics of a fully mature CIM/BIM.

3.3 Cultural Change Requirements

As a technology and as a process, BIM can not only change individual activities and workflows but also fundamentally change how individuals, groups, and departments within an airport's organization communicate and collaborate. BIM promotes transparency and sharing of data to eliminate redundant facility data collection and management activities and moves the organization toward a central facility information store.

3.3.1 BIM Champion

It can be difficult to overcome the tendency of departments within an organization to develop and maintain their own data stores to support their work activities. The presence of a strong BIM champion to demonstrate an airport's commitment to growing its BIM capabilities is essential to overcoming such difficulties in adopting and implementing BIM.

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Airports, like other large organizations, are likely to have experienced change initiatives in the past that were unsuccessful. Human tendency will be to stick with the status quo until sufficient momentum to change has been achieved. This takes time, and a BIM roadmap, or detailed business plan that is supported by the BIM champion, can help provide a long-term vision to the organization and reinforce the airport's commitment to implementing BIM.

Ideally, there is not just one BIM champion but a network of BIM champions in each department who can serve as the "local" trusted sources of how and why to use BIM. The Port of Seattle establishes "Centers of Excellence" when instituting port-wide organizational changes (such as BIM and sustainability efforts). These centers develop leaders within each department who can act as the internal resource and advocate.

3.3.2 BIM Manager

The existence of an internal BIM manager is critical to the future success of BIM within an airport organization. While the BIM champion works toward cultural change and organizational adoption of BIM, the BIM manager is responsible for ensuring the BIM implementation and standards fit the airport's needs and the requisite organizational structure is in place to maintain BIM. A McGraw-Hill survey of BIM users shows a strong correlation between BIM ROI and the level of engagement that the organization has with BIM (McGraw-Hill Construction, 2009). The research team's survey results also show that the existence of a BIM manager greatly increases the anticipated ROI, the number of fully completed BIM projects, and the O&M uses for BIM.

3.3.3 Internal Staff Training and Development

Although some airports (such as SFO and LAX) have had success implementing BIM programs (primarily in design and construction) by relying on long-term consultants, airports should consider developing the BIM skills of internal staff to further establish BIM as an organizational capability and not a temporary activity. DEN made a conscious decision to invest in internal BIM capabilities rather than relying on consultants, and this has enabled it to explore and expand the use of BIM beyond planning, design, and construction. Investing in internal staff training and development, or the hiring of new full-time BIM staff, will also demonstrate an airport's commitment to implementing BIM as a long-term solution.

Where BIM is supported only by external consultants, it (and most other technology initiatives) generally suffers from a lack of maintenance, cuts in funding due to the higher cost of maintenance versus using in-house personnel, and a lack of internal knowledge of how to sustain the program if the consultant's contract is not renewed. Consultant contracts should include assistance in establishing the BIM program and providing training for internal staff so that they are prepared to manage and maintain the BIM upon project delivery.

3.3.4 Incremental versus Big Bang

Although choosing a "big bang"' approach to implementing BIM (i.e., trying to implement BIM for all capital projects and for the entire airport organization all at once) may seem like it would maximize the airport's commitment to BIM, experience has shown this type of approach has limited long-term success. Developing BIM as a capability should be approached as 5- or 10-year effort. The BIM roadmap can be helpful in communicating this long-term commitment. Incremental BIM implementation that enables the airport organization to absorb change over time, and that is supported by demonstrations of short-term benefits, will have the highest

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chance of success. Airports such as SFO and DEN have more than a decade of BIM experience and their growth has been incremental, growing from a design and construction implementation to exploring post-construction activities.

3.4 Summary

Preparing an organization to support a life cycle BIM approach is a complex activity that will fundamentally affect the manner in which the airport operates. A comprehensive CMM should be performed to assess the readiness of the organization to utilize BIM and to identify where investments will be required to improve readiness. The CMM should be used to identify organizational capabilities after the BIM needs assessment. It should also be used as an ongoing tool to assess the growth of organizational BIM competencies and to identify where future investments in BIM training, education, tools, infrastructure, policies, and processes should occur.

The collaborative environment supported by BIM may not be achieved quickly. There will need to be investments in education and training and methods implemented to overcome the cultural resistance to change. The use of a BIM CMM can aid in the identification of short- and long-term investments needed to achieve the most effective BIM program for an airport.

Another key to successful implementation of a BIM program is for an airport to identify a BIM champion to navigate the challenges accompanying this type of change to organizational processes. Larger airports have successfully implemented BIM by hiring and developing in-house BIM staff or using semi-permanent extensions of staff. Without some reliable ability to maintain the BIM in house, it is unlikely that airport staff will trust BIM as their primary source of facility data.

The development of a BIM roadmap can communicate the long-term vision for BIM and further reinforce an airport's commitment to BIM. Organizational development is critical to the success of BIM. While consultants can play a key role in implementing BIM, investments in internal staff BIM skills will accelerate BIM adoption and identify areas where BIM is beneficial. An incremental approach to BIM should be followed to overcome cultural resistance and build widespread support.

Section 3 Checklist

1. Select and develop a BIM CMM.
2. Identify the airport's BIM champion.
3. Identify those who will maintain the airport's BIM.
4. Provide necessary BIM training to those maintaining and using the BIM.

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SECTION 4

Pre-BIM Activities—Financial Analysis

This section provides an overview of options for building a business case for BIM and methods for measuring ROI. Although several research reports and academic studies have been published on this topic, no formal methodology for developing a business case or measuring ROI has been standardized that airport owners can adopt directly. No airport spoken to as part of the research for this Guidebook reported formally measuring the ROI from BIM either as part of developing its business case or following BIM implementation to measure its effectiveness. There are aviation and non-aviation case studies where ROI was measured for elements of the BIM process. Data from these case studies will be presented in this section.

The airports that adopted BIM early shared the common elements of having a strong BIM champion and a vision of how BIM could contribute to meeting strategic and operational challenges. Larger airports have been leading the way on BIM implementation, expecting that it would improve their ability to rapidly and efficiently add new infrastructure and would enable them to meet projected growth demands. Large airports also saw BIM as a program that would improve communications and collaboration among their diverse group of airport stakeholders by breaking down the siloed data stores each group maintained to support its work processes.

Although ROI is difficult to measure precisely, the areas where airports can expect a positive ROI from BIM (see Section 4.5.4) include

- Maintenance costs avoided through improved maintenance planning.
- Uninterrupted operations through improved asset management.
- Labor productivity through improved planning, collaboration, and communication.
- Greater capital efficiency with shorter construction durations and improved management.

U.S. airport organizations that were early adopters of BIM are DEN, Massport, PANYNJ, and SFO. LAX and Seattle-Tacoma International Airport (SEA) are in the early stages of implementing BIM as a life cycle process and will begin the implementation phases in 2019–2020. LAX and SEA developed their preliminary business cases primarily on their experience with contractors on their capital projects who were using BIM. As it became clear that the architecture-engineering-construction (AEC) industry was rapidly adopting BIM, these airports thought they needed to understand BIM better to manage future projects. Also, they needed to understand BIM post-construction benefits.

This section will explore building a business case for the use of BIM at an airport and will present options for measuring the effects of BIM on specific operations. This section will not provide a precise way of measuring the total organizational ROI from BIM, but will identify mechanisms for receiving feedback on how BIM is performing. An airport should not introduce

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BIM with the expectation that it will produce immediate returns. The up-front investment and the time it will take an airport to adopt and integrate BIM processes into its day-to-day operations will likely take several years to recover.

Another option for airports building a business case for BIM is to include an investment in BIM as part of a larger business case for improving strategic facility asset management. This section will include a discussion of how BIM can support the adoption of a strategic facility ALCM approach such as in *ACRP Research Report 172: Guidebook for Considering Life-Cycle Costs in Airport Asset Procurement* (Fortin et al., 2017), the ANSI APPA 1000-1 – Total Cost of Ownership for Facilities Asset Management (TCO) – Part 1: Key Principles, and ISO 55000.

When evaluating the value of any business investment (whether in construction, capital investment, or an improvement to facility or business operations), the focus needs to be on the stakeholders and the intended outcome for those stakeholders. This focus means that the scope of the business case needs to include the impact on both current and future business processes. A common problem with many investments in technology-enabled improvements, such as BIM, is the failure to fully understand the business processes that will be affected. To avoid that misunderstanding, BIM's primary benefits include the following:

- Improving access to and use of facility data through one single source of truth (a database)
- Providing visualization of how resources are being utilized and how investments might be prioritized to address demonstrated deficiencies by establishing relationships between financial and physical assets
- The additional depth of analysis offered by adding geographic attributes to facility data

Any financial analysis of the benefits of BIM needs to incorporate end-to-end business processes throughout a facility or organization to capture all potential returns.

4.1 Developing a BIM Business Case

Business cases may take different forms and be developed on different levels depending on an organization's objectives. Because of this, there is no generally accepted standard for how a business case for BIM implementation should be developed. In the case of large airports, early adopters of BIM, the business case was informal and primarily driven by the vision of a strong internal BIM champion. Also, there was the recognition that BIM was being produced as part of large capital projects regardless of the airport's contractual requirements for plan production and delivery. In short, BIM was becoming a standard operating procedure in the AEC industry for major projects because it saves time and money for the AEC consultants and contractors. Airports recognized that there is a significant cost to not adopting BIM as a standard and to requiring AEC entities to convert their BIM to meet existing 2D plan delivery standards.

An alternative to developing a formal business case is to create a BIM roadmap that communicates the long-term vision and benefits of BIM in defined stages. A BIM roadmap is easier to develop and can be very useful in the early stages of BIM adoption to illustrate how BIM can benefit each airport stakeholder group. An example of a BIM roadmap for a Korean Rail project is included in Appendix D.

While a BIM roadmap or a strong BIM vision led by an internal champion may serve as a starting point for beginning BIM implementation at an airport, a more detailed business case may be required for approval of the budgetary resources needed for the BIM program. Regardless of scope, all business cases share one common goal: to provide a structured process for evaluating decisions and measuring their outcomes. While defining financial metrics

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plays a key role in the process, business cases can include both tangible (measurable) and intangible (difficult to measure but otherwise valuable) outcomes as part of the evaluation process. Business cases will include the following:

- **Reasons.** What improvements is an organization trying to make, or what problems is the organization trying to solve with this new program? What decision(s) is an organization building the business case to evaluate? Why is this new program needed?
- **Options.** What are all the possible solutions, and what happens if nothing is done? Why was the selected approach chosen?
- **Benefits/Disadvantages.** What are the expected tangible and intangible benefits? What are the specific levels of benefits, the time range for achieving those benefits, and the methods for measuring them? What are the potential costs, disadvantages, or trade-offs that may result from implementing BIM?
- **Timescale and Costs.** What are the expected time investment and costs required for the effort to achieve its goals in a phased approach?
- **Risks and Opportunities.** What are the major downside risks to the effort and upside opportunities that could enhance the expected benefits?

In the case of BIM, the "reason" the business case is being developed needs to be well understood. Is BIM being evaluated as a core, enterprise-wide organizational capability that supports strategic operational and asset management activities? Or is the business case being developed to assess the use of BIM more tactically, focused on one or more BIM uses (such as maintenance scheduling or energy analysis)?

As part of a BIM roadmap, the business case may include a mix of both enterprise-wide and tactical phases staged out over a few years. The CIC Research Program at Penn State has published a template for developing a BIM business case with its *BIM Planning Guide for Facility Owners—Version 2.0* (2013). This business case template, "Business Case for Organization BIM Integration—Version 1.02," outlines the sections of a BIM business case as the following:

- Executive summary
- Introduction and background
- Problem definition, goals, and objectives
- Proposed BIM uses
- Cost-benefit analysis
- Implementation timeline
- Final recommendations

The *BIM Planning Guide for Facility Owners—Version 2.0* and "Business Case for Organization BIM Integration—Version 1.02" are freely downloadable under a Creative Commons license ("Business Case for Organization BIM Integration—Version 1.02" is available as Appendix B of this report). The examples that follow of the development of an enterprise-wide and tactical BIM business case are based on the templates provided by the CIC Research Program. The standard sections and approach in the business case template should map well onto any standard format used at an airport. The following structure for a BIM business case is based on the CIC Research Program's "Business Case for Organization BIM Integration—Version 1.02."

4.2 Example 1: BIM Enterprise Business Case

A BIM enterprise business case would be used when an organization is evaluating an enterprise-wide adoption of BIM as a core tool/process for the development, management, and sharing of facility data. BIM may be supporting a shift to the adoption of a more comprehensive

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LCC approach to ALCM such as ISO-55000 or TCO. When airports were asked about their BIM uses for O&M as part of a survey for this Guidebook, the top three uses were asset management, maintenance management, and space management/planning. Other uses included work order management, condition assessment, internal QA/QC, capital planning, sustainability planning, operational management, and disaster planning.

4.2.1 Data Needs Assessment

It is important that the development of an enterprise-wide business case begin with an overall strategic planning effort that includes an organizational data needs assessment. This assessment will identify how facility data are being utilized, where there is a need for more accurate or accessible facility data, and where opportunities exist for BIM to improve overall organizational performance. The needs assessment provides information that is vital to the development of the business case problem statement and sets the stage for collaboration of all the organization's stakeholder groups in developing solutions. There is an organizational strategic plan template included with the *BIM Planning Guide for Facility Owners—Version 2.0* (CIC Research Program, 2013) that includes tools that can be used as a starting point if an airport does not already have a preferred strategic planning tool/methodology.

4.2.2 Problem Definition, Goals, and Objectives

Definition of a problem forms the basis of a BIM roadmap. The problem definition or statement should include a discussion of any impacts of the BIM initiative on organizational structure and culture. A core principle of BIM is the sharing of data to improve workflows, coordination, and communication. Siloed data and isolated workflows are common occurrences (not limited to airports); the potential change to a more collaborative culture may be challenging for some and so must be addressed in terms of the potential benefits received from implementing BIM and the potential risks of organizational resistance to new BIM processes.

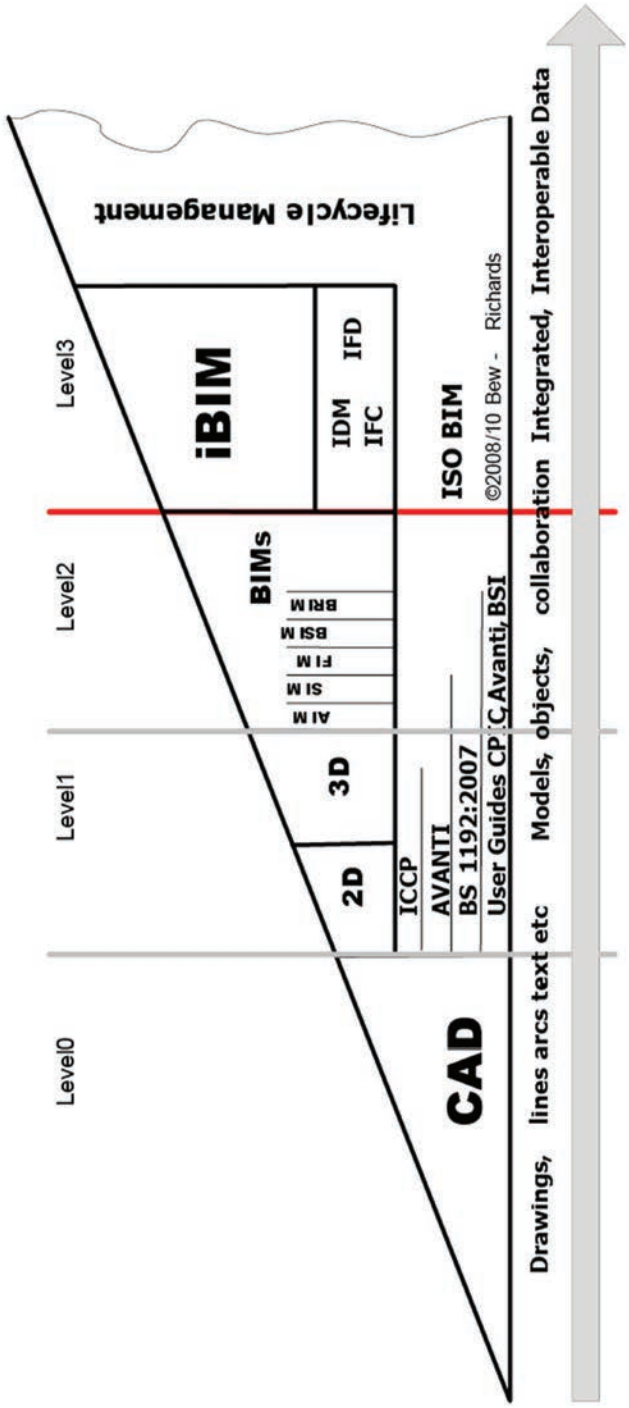
The Bew-Richards wedge was developed to illustrate the UK BIM Level 0 through 3 capabilities progression and is relevant to any strategic/enterprise BIM discussion. Reading the Figure 4-1 diagram from left to right shows how BIM supports a move from 2D CAD to a comprehensive life cycle management process.

A BIM CMM Matrix (see Section 3) is a tool that can be used as part of the needs assessment to define the organization's BIM capabilities and goals. These capabilities and goals would be measured against a variety of different elements such as the existence of a BIM champion, executive management support, BIM uses and project management processes, BIM LOD, facility data needs, software and hardware infrastructure, and the personnel with the required education and training. Current levels of each would be evaluated, and the target levels would be defined to support the goals defined in the enterprise BIM strategy.

4.2.3 BIM Uses and Cost-Benefit Analysis

The airport's overall strategy and goals for BIM will define which BIM uses will be required to support the achievement of those goals. Cost-benefit analyses should be performed for each of these BIM uses. The overall cost-benefit analysis for airports will include the contributions from each BIM use. It may also include cost savings due to resource sharing among BIM uses. For example, the spatial modeling done for energy analysis may substantially reduce the

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Source: Scottish Trust Organization

Figure 4-1. Bew-Richards BIM capabilities wedge.

amount of modeling needed to support space/property management and thereby reduce the cost of space/property management modeling. Also, there may be difficult-to-measure, or intangible, benefits that need to be evaluated on a qualitative basis (such as improved communication, culture, and knowledge management).

Each BIM use analysis should also include an assessment of associated risks or negative outcomes and the mitigation strategies needed to address those negative outcomes if they occur. These need to be as fully explored as potential benefits before approving a BIM use.

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4.2.4 Key Performance Indicators

To effectively manage organizational change, measures must be put in place to provide feedback on how well those changes are performing against the BIM implementation plan. Key performance indicators (KPIs) should accompany each BIM use business case. For example, the BIM use of "BIM for Energy Analysis" should establish a process by which to review and compare energy utilization against pre-BIM baseline figures between similar types of facilities. Another example might be the average time per asset to enter assets in the CMMS at project handover/commissioning. This feedback can be used to fine-tune BIM processes to improve the ROI or, in some cases, to discontinue a BIM use based on poor results. Likewise, tangible, measured, positive results can be used to justify the expansion of some BIM uses.

Enterprise BIM business cases need to be living documents that are updated and refined based not only on the internal business metrics collected but also on changes in the AEC and facilities management industries. BIM and other innovations in enterprise information management are still rapidly evolving. This will deliver additional opportunities for increasing efficiency over time and will expand currently evolving opportunities for the use of BIM beyond construction.

4.3 Example 2: Tactical Business Case for Individual BIM Uses

Rather than being evaluated as part of an overall organizational/enterprise strategy, BIM can be evaluated from the perspective of adopting the use of BIM for a specific purpose (i.e., for energy analysis or maintenance planning). This process would follow the same steps as those for the BIM enterprise business case, but the process would focus on a single BIM use. This use of tactical business case development may be the best approach for airports wanting to begin pilot studies of BIM before investing time and resources in evaluating a larger, strategic use of BIM. "Business Case for Organization BIM Integration—Version 1.02," the business case template developed by Penn State's CIC Research Program, could be used to develop a business case for a specific BIM use. The example that follows begins with the "Problem Definition." The "Business Case for Organization BIM Integration—Version 1.02" template also includes an "Introduction and Background" section, where an airport could include some specific details about its organization and any experience it has had with BIM.

4.3.1 Problem Definition, Goals, and Objectives for a Hypothetical Example

The sections that follow describe a hypothetical example of a business case that could be developed for the singular use of BIM to improve a backlog of facility maintenance work orders.

Problem Definition

In the hypothetical example, the backlog of preventive maintenance is typically exceeding the airport's benchmark goal of a maximum of 8 weeks. A time study has shown that the average planning time for each work order includes 2 hours of time to collect data about the site and to perform preliminary site investigations. The existing as-built data for the facility are organized by project, not by location, requiring maintenance staff to review multiple plan documents when detailed site data are required. A pilot using BIM for maintenance planning will test whether BIM can significantly reduce the hours spent in facility data collection as part of work order planning.

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Organizational Mission and BIM Vision

The mission of the Airport Maintenance Department in the hypothetical example is to manage, maintain, repair, and remodel the airport's infrastructure and facilities; provide a safe and comfortable experience for passengers, airlines, concessionaires, contractors, and airport staff; and to manage energy utilization and optimize sustainability of airport facilities.

Organizational Goals

Goals and objectives of the hypothetical pilot program to use BIM for maintenance planning are listed in Table 4-1, along with their priority.

Planning Team Members

For the development of a business case for any specific BIM use, a list of contributors must be developed identifying who will develop the business case, and who can answer related questions from the rest of the organization. For the hypothetical example of using BIM for maintenance planning, contributors who will develop the business case might be the following:

- Vice president of operations
- Vice president of engineering
- Vice president of finance
- Maintenance supervisor
- Director of asset management
- Director of information, communications, and technology

4.3.2 Proposed BIM Uses

Following the problem definition in the hypothetical example, the planning team develops proposed BIM uses and assesses their current and desired maturity levels (see Table 4-2). (Descriptions of Maturity Levels 0–5 are provided in Table 4-3.)

4.3.3 Cost-Benefit Analysis, Benefits, and Metrics

The following describes performing a cost-benefit analysis of the proposed BIM uses (work order planning, emergency response, and asset management) and metrics defined for the hypothetical example. The actual values identified are for purposes of illustration only and are not meant to benchmark.

Table 4-1. Goals and objectives of the hypothetical pilot program to use BIM for maintenance planning.

Priority	Goal Description	BIM Objective
High	Reduce unplanned outages affecting passengers and airlines	Use BIM to reduce preventive maintenance backlog, implement continuous improvement methodologies, and support improved reporting and response
High	Minimize downtime due to unplanned outages	Use BIM to reduce planning and response time to outages
High	Provide rapid response to emergencies	Share BIM with emergency responders to improve their knowledge of a site in preparation for a response as well as their readiness on arrival
Medium	Provide more accurate location data for asset inventory	Use BIM to supplement CMMS asset data with asset coordinates

Source: "Business Case for Organization BIM Integration—Version 1.02" template

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Table 4-2. Current and proposed maturity levels for proposed BIM uses in the hypothetical example.

X	Current Process	BIM Use	Description	Current Maturity	Desired Maturity
1	**Work order planning**	Maintenance planning	As-built facility data reference	0	3
2	**Emergency response**	Site visualization and analysis	Share 3D model for response scenario planning and hazard identification	0	4
3	**Asset management**	As-built facility data	Integrate BIM asset location data with CMMS facility data	0	4

Source: "Business Case for Organization BIM Integration—Version 1.02" template

BIM Use #1: Work Order Planning

In the hypothetical example, the first proposed BIM use is in work order planning. There are three goals for this use of BIM:

- Goal 1: To increase the percentage of work orders that can be completed within the initial site visit. BIM can provide an accurate location and list of necessary equipment for purposes of tool selection, work performance, and safety planning.
- Goal 2: To reduce the time spent per work order in facility data collection. A single, combined, complete, facility BIM can provide the required site analysis and context required for work order planning. BIM can also easily show how systems are connected to augment troubleshooting.
- Goal 3: To reduce the response time for unplanned utility outages. The same reduction in facility data collection for planned work orders will also benefit unplanned work orders.

Table 4-4 provides an example of how individual metrics/goals could be set and measured by an organization in order to evaluate the benefits derived from BIM. The values provided in

Table 4-3. Description of BIM maturity levels from "Business Case for Organization BIM Integration—Version 1.02," developed by Penn State's CIC Research Program.

Maturity Level	Description
(0) Non-Existent	A process has not yet been incorporated into current business processes and does not yet have established goals and objectives.
(1) Initial	A process produces results in which the specific goals are satisfied; however, they are usually ad hoc and chaotic. There is not a stable environment to support processes with the inability to repeat such and possible abandonment in time of crisis.
(2) Managed	A process is planned and executed in accordance with policy; employs skilled people having adequate resources to produce controlled outputs; involves relevant stakeholders; is monitored, controlled, and reviewed; and is evaluated for adherence to its process description.
(3) Defined	A process is tailored to the organization's standard processes according to the organization's guidelines, has a maintained process description, and contributes process related experiences to the organizational process assets.
(4) Quantitatively Managed	A process is managed using statistical and other quantitative techniques to build an understanding of the performance or predicted performance of processes in comparison to the project's or work group's quality and process performance objectives, and identifying the corrective action that may need to be taken.
(5) Optimizing	A process is continually improved through incremental and innovative process and technological improvements based on a quantitative understanding of its business objectives and performance needs and is tied to the overall organizational performance.

Source: "Business Case for Organization BIM Integration—Version 1.02" template

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Table 4-4. Example metrics/goals for using BIM to enhance work order planning.

Benefit	Goal/Objective	Benefit Calculation	Estimated Value
Average work completion time	Reduction of 2 hours per overall work order	30,000 average work orders/year resulting in 60,000 total maintenance work hours	$3 million
Average work order planning time	Reduction of 1 hour per work order	30,000 average work orders/year resulting in 30,000 saved work hours in planning	$1.5 million (included in average work order time savings)
Average # of site visits required to complete work order	Reduction of 50%	Reduced staff and vehicle traffic at the airport related to maintenance activities	Included in average work order time savings
% wrench time	Wrench time increased 15%	Additional 15% of total maintenance technician time available to reduce preventive maintenance backlog	Included in average work order time savings
Reduced preventive maintenance backlog	Average backlog reduced to 4 weeks, All peaks above 8 weeks eliminated	Timely preventive maintenance of assets will result in longer average asset life	10%–15% of total asset value
Total safety incidents related to work orders	Safety incidents reduced by 30%	Fewer trips required per work order reduce at-risk time spent on-site	30% claims reduction and lost hours

Table 4-4 are provided purely for purposes of demonstrating the process, not as examples of what an airport organization should be adopting as goals.

BIM Use #2: Emergency Response

In the hypothetical example, the second proposed BIM use is in emergency response. The main goal in this use is to improve effectiveness of emergency responders and improve understanding of potential hazards. BIM can be used by emergency responders (police, fire, and others) to review the airport's facilities virtually, examine hazards, and model different scenarios of response. Although it is difficult to assess a direct economic value of this BIM use, it strongly supports the airport's mission to provide a safe and secure travel environment.

Table 4-5 provides an example of how individual metrics/goals could be set and measured by an organization in order to evaluate the benefits derived from BIM. The content provided in the table is provided purely for purposes of demonstrating the process, not as examples of what an airport organization should be adopting as goals.

Table 4-5. Example metrics/goals for using BIM to enhance emergency response capabilities.

Benefit	Goal/Objective	Benefit Calculation	Estimated Value
Response time to emergencies	Accurate facility data available to all airport emergency responders	Reduction in response time	n/a
Emergency preparedness	25% reduction in time to develop and maintain a disaster plan	Improved preparedness of responders	n/a
Disaster scenario response simulation	Provision of input into the design/renewal of facilities	Improved security/safety design of new facilities	n/a

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Table 4-6. Example metrics/goals for using BIM to enhance asset management.

Benefit	Goal/Objective	Benefit Calculation	Estimated Value
Asset inventory accuracy and facility condition assessment	Establish coordinate-based asset locations, and reduce asset inventory time by 20%	Asset inventory time and facility condition assessment time reduced by 4,000 hours/year	$200K

BIM Use #3: Asset Management

In the hypothetical example, the third proposed BIM use is in asset management. The main goals in this use are to improve the quality and accuracy of asset location data within the CMMS and reduce the time to perform asset inventory and facility condition assessments.

Table 4-6 provides an example of how individual metrics/goals could be set and measured by an organization in order to evaluate the benefits derived from BIM. The values provided in Table 4-6 are provided purely for purposes of demonstrating the process, not as examples of what an airport organization should be adopting as goals.

4.4 Managing Cost and Risks

"Business Case for Organization BIM Integration—Version 1.02," the business case template developed by Penn State's CIC Research Program, presents a sample table for tracking costs that has been adapted here as Table 4-7. In an actual cost-benefit analysis, the table would be populated with actual and average salaries and estimates of the amount of time individuals would commit to implementing BIM.

Table 4-7. BIM business case cost.

Category	Item	Salary	Time	Factors	Cost
Planning Cost: Process and Standards, Business Case	BIM champion	Annual salary	Percentage of time allocated	Max and min of % of the time	% of time multiplied by salary
	Planning team members	Average salary	Average time all	Max and min number of staff	% of time multiplied by salary
Education and Training	Classes and on-the-job training	Average hourly rate	Hours of training	Max and min time required and # of staff	Average rate times total staff hours of training
Software	• Authoring • Viewer • Integration	n/a	n/a	Max and min # of licenses	# of licenses times the cost of licenses
Hardware	• Workstations • Servers • Tablets • Scanners	n/a	n/a	Max and min # of workstations, tablets, and scanners	# of devices times the cost of each device
BIM Development	Single facility BIM	n/a	n/a	Size of the facility and selected scope of BIM	BIM-authoring costs per square foot (sf)
Learning Curve	Additional work hours	Average hourly rate	Average hours until proficient	Max and min ranges for proficiency	Average hourly rate multiplied by average hours until proficient

Source: Adapted from the "Business Case for Organization BIM Integration—Version 1.02" template

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4.4.1 Risk Assessment

Organizational and cultural change can be difficult to manage and the use of BIM can fundamentally change the way teams communicate and collaborate. Organizational resistance is just one of the risks that BIM implementation can face. Table 4-8, adapted from "Business Case for Organization BIM Integration—Version 1.02," shows some risks organizations commonly face, but it is likely that each organization will also face risks unique to its own culture and work environment.

4.4.2 Implementation Timeline

Although there are significant financial and operational benefits to implementing BIM for some uses, the implementation timeline should be phased to enable the airport to measure whether cost-benefit returns are matching expectations. A phased implementation plan also allows staff the time to develop the skill sets required to maintain BIM for those uses. Additionally, enough time needs to be invested in organizational engagement and change management.

The example phased timelines that follow reflect an enterprise use of BIM. The length of each phase will vary based on organizational size and complexity. This phasing could be reused as part of a BIM roadmap to illustrate the long-term vision of the airport organization.

Phase 1: 0–12 Months	Needs assessment, baseline measurements, process development
Phase 2: 12–18 Months	Pilot BIM facility development and testing, training
Phase 3: 18–30 Months	Initial pilot, data collection and measurements, process refinement
Phase 4: 30–36 Months	Test refinement, assess results
Phase 5: 3–5 Years	Expand program to other facilities and BIM uses

4.4.3 Recommendations

The final sections of the business case would include final recommendations and any associated data used to support those recommendations (e.g., preliminary time studies, PARETO breakdowns of maintenance activities). For example, maintenance PARETO (see Figure 4-2) analysis predicts that 80% of maintenance resources are consumed and driven by 20% of the assets. Focusing on modeling these assets and associated systems can yield the most immediate benefits.

The two business case examples previously discussed are provided only for guidance in the process and are not meant to emphasize these BIM uses as priorities. The figures used are not meant to reflect any industry benchmarks. The next section will explore methods for calculating a financial ROI from BIM uses. While the three example BIM uses and prioritization discussed in Sections 4.3 and 4.4 provide a method for analyzing the impact of BIM on an

Table 4-8. BIM business case risk assessment table.

Risk	Risk Likelihood	Risk Impact	Risk Mitigation Measures	Recommendation
Organizational resistance	Medium	High	Maximize stakeholder engagement before BIM initiatives	Acceptable
Cost too high	Medium	Medium	Deploy in phases to carefully assess results before larger commitments	Acceptable
Unable to find qualified staff to maintain BIM	High	High	Invest in early training and developing of new staff. BIM phasing needs to allow enough training time	Acceptable

Source: Adapted from the "Business Case for Organization BIM Integration—Version 1.02" template

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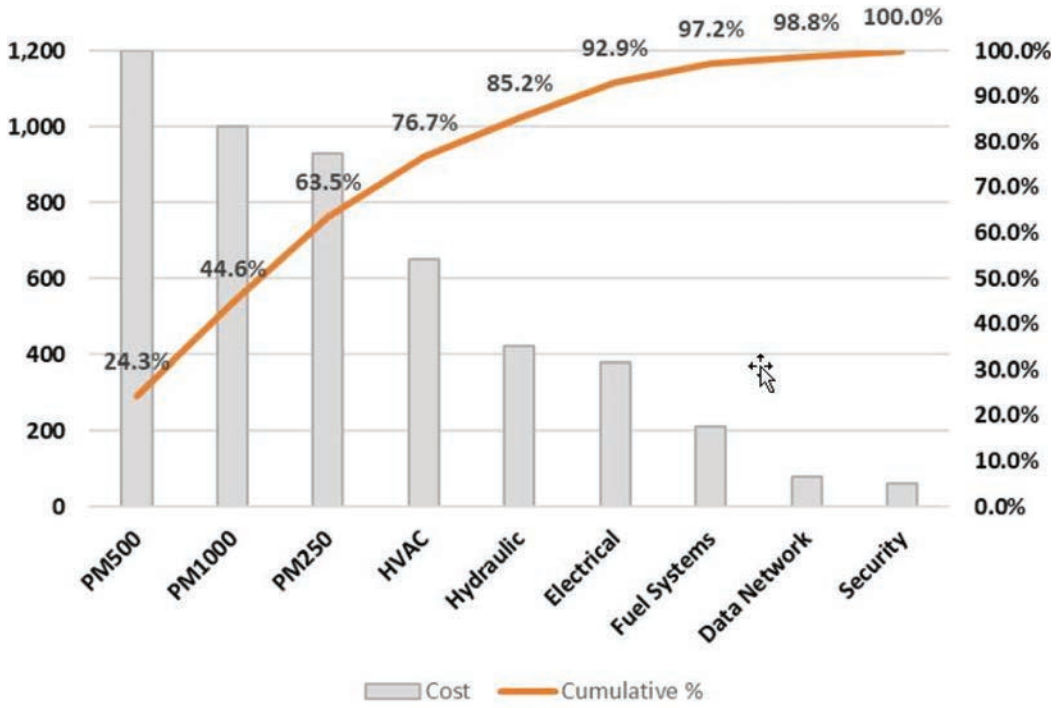


Figure 4-2. Facilities maintenance PARETO chart example.

airport's ability to support its vision and mission, an estimation of ROI provides a more detailed, financial view of the investment in BIM (see Section 4.5).

4.5 Measuring BIM ROI

ROI from BIM is difficult to measure comprehensively. The difficulty in measurement is because of the complex interactions BIM can have with the development of facility data and on the difficulty in measuring the benefits of improved communications and collaboration on an organization's performance. A few case studies exist where ROI was measured with some precision. One study involves two very similar buildings, one that was designed using BIM and one that was not; the similarity of the two buildings meant that performance metrics could be directly compared with one another (Giel and Issa, 2013a). This type of situation is a rare circumstance. For complex organizations like airports, making this kind of comparison precisely is difficult, and it is thus hard to predict ROI. Consequently, many studies have chosen to survey "ROI expectation levels" in the absence of methodologies to capture more precise figures.

Measuring ROI from the technology and process improvements delivered utilizing BIM can be a complex task. While industry studies consistently identify high BIM ROI expectations from owners, contractors, and architects during design and construction, these studies are often qualitative. The facility management industry has not produced many quantitative BIM ROI studies. The quantitative studies that exist have been focused on a narrow aspect of BIM, such as using BIM for asset creation (Eastman et al., 2008) or reduction of change orders and request for information (RFI) during construction. The industry experience with the use of BIM during design and construction has created the expectation that there will be a corresponding positive ROI from the use of BIM during facility O&M as well.

ROI can be used in various situations and for various purposes. In the financial industry, where investment and value are precisely measured and monitored, ROI is calculated and

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distributed routinely and expeditiously. It is not the same case for a technology investment impacting assets and processes, such as BIM. This is especially true in the operation and maintenance period, which often gets lower priority than the design and construction phases, and where there are typically fewer data available. Despite this, there are approaches to consider in determining ROI (or value derived) from investing in BIM to support the operation and maintenance of airport assets and facilities in the post-construction period.

4.5.1 ROI Methodologies

A distinction needs to be made between an ROI methodology and an ROI calculation. The methodology is the overall approach to making the calculation. Inherent to all the calculations is the comparison of return (value/benefit/gain) to the amount of investment made. Some calculations involve the time value of money and discounted cash flow analysis, while others use only nominal amounts. ROI can also be realized by risks mitigated through the use of BIM-enhanced preventive/predictive maintenance abilities to anticipate future failures and prevent them.

The selected ROI approach can depend on a variety of factors, including the following:

- The complexity of the project/asset being analyzed
- The useful life of the project/asset, which serves as the basis for the analysis period of useful life
- The required assumptions, level, and quality of necessary data
- The purpose of the analysis being performed (to justify a decision or to select an option/alternative)

In selecting methodologies and calculations, it is helpful to consider an overall ROI framework and its separate elements. *NCHRP Research Report 866: Return on Investment in Transportation Asset Management Systems and Practices* (Spy Pond Partners, LLC, et al., 2018) utilized the following:

- Definition of investment and base cases
- Identification of benefit and cost categories
- Methods of calculating/estimating benefits
- Relationship of benefits to performance measurement
- Determining the ROI and reporting
- Consideration of uncertainty

4.5.2 Baseline and Investment Cases

Establishing baseline and investment cases for BIM ROI purposes is often problematic. The comparison can be retrospective (where the investment has been made, and data are available from both before and after the investment was made) or prospective (where the baseline is the current state, and the investment case is based on the most likely future estimated outcomes). There is another approach that has been used by the Oregon Department of Transportation (DOT). Its ROI methodology was a current state based on 5-year-old integrated geographic information system (GIS) technology. Through a survey of staff and available current data, the Oregon DOT calculated the ROI based on the estimated efficiency and cost savings applied over the entire prior 5-year period. A pre-BIM state served as the baseline.

These cases can be based on a facility using pre- and post-implementation data (if available) or comparing similar projects (one using BIM and one without BIM). The latter approach was used in some of the early BIM ROI analyses, focusing only on the design and construction phases (Giel and Issa, 2013a). Another approach for BIM ROI analysis is to find a similar BIM-based project in a comparable industry and adapt it to the base case to determine the impact of the investment.

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When there are not suitable BIM investment cases available, established industry benchmarks can be used to create models to simulate the most likely future results by applying them to the base case (IFMA et al., 2013). No matter which approach is used, it is best to have sufficiently detailed baseline data to make the comparison meaningful.

4.5.3 Investment/Cost

The BIM investment should include the full extent of implementation costs. These investments can generally be categorized as follows (CIC Research Program, 2013):

- Software purchase and maintenance costs
- The labor cost of new/relocated staff
- Staff training costs, including travel time and lost productivity
- The labor cost of a BIM planning team, its leader, and any BIM champion
- Workstations and maintenance costs
- Network infrastructure improvements and maintenance
- Process change costs—organizational adjustments, process documentation, and BIM learning curve
- Maintenance updates of BIM
- Design authoring of BIM
- Integration costs with other organizational data systems

This last area of integration with other data systems is important in that it allows BIM benefits to be maximized over the entire airport. BIM integration with both CMMS and financial reporting systems enables financial and asset management data to be better coordinated and utilized in decision making, not only for improved management of assets but also for analysis of alternatives, rate setting, and cost recovery. When it comes to BIM benefits, however, this integration of systems can make it difficult to differentiate between those benefits derived within one system and those derived in another and to decide how to best allocate benefits between the two.

4.5.4 BIM Benefits

BIM provides benefits over the entire life cycle of a project: from design and construction, through O&M, and ultimately through asset decommissioning. This reduces the TCO of an asset. Within the operating and maintenance component of the life cycle, the items listed below were cited as BIM ROI areas to focus on at a 2013 National Facilities Management and Technologies (NFMT) conference (Starkov et al., 2013):

- Labor utilization savings (shorter work order time)
- Utility cost reduction (energy savings/efficiencies)
- Fuel and material savings (less travel and waste)
- Risk management (quicker response times)
- Regulation compliance (auto checking codes)
- Space optimization (smart algorithms)
- Improved inventory management (spare parts)
- Data accuracy/quality (asset location and condition)

A more comprehensive list of BIM benefits for facility management can be found in the "GSA Building Information Modeling Guide" (2011).

Within an ROI framework, benefits can also be categorized as directly impacting the asset/facility or indirectly providing value. For example, energy cost savings will directly show up as a facility cost reduction. Other benefits will not result in direct cost reductions in a specific facility,

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but will result in cost savings or other efficiencies across the overall airport (i.e., time savings through regulation compliance). There are also other BIM benefits that can be termed "strategic" or "enterprise level." These relate to improving the long-term condition of assets and enhancing value through avoiding maintenance costs, as well as extending an asset's life. These enterprise-level benefits are measured through both financial and non-financial strategic performance measures. Improved airport customer satisfaction is an important non-financial example.

This stratification of benefits can be viewed as a three-tiered approach to returns from BIM investment that can be applied to assets/projects across an entire organization:

1. Direct ROI/value—where the return can be measured at the asset/project level
2. Indirect ROI/value—where the return is shared across the asset/facility
3. Strategic value—where the benefits accrue to the entire organization and are measured or evaluated against overall strategic objectives

While most BIM benefit areas are based on cost or time savings, there can also be situations were BIM can positively impact airport revenues. For example, space optimization can increase rentable square feet within a terminal or cargo facility. Accelerating the delivery of a revenue-producing capital project can lead to quicker receipt of revenues and an increased value of the revenue on a net present value (NPV) basis.

4.5.5 Determining Benefits—Measurement, Calculation, and Estimation

The most significant, and typically most complicated, component of an ROI framework can be the determination of benefit value: how to quantify, calculate, or estimate it. The nature and level of available data will drive this process. In situations where there are integrated financial and asset management systems, direct financial data are likely more available. Depending on the manner in which labor is tracked and the level of detail tracked and accounted for, time and efficiency savings can often be measured and monetized using appropriate available hourly rates. When data are less available, other options can include applying savings rates from other, similar projects/assets or industry benchmarks to the base costs or surveying staff involved in the process to develop estimated savings rates that will be applied to respective functions/activities.

4.5.6 BIM Benefits/ROI and Asset Performance Measurement

Many organizations maintain financial and asset management information in separate silos and include coordination and reconciliation of related data only when necessary as part of the planning and reporting processes. However, there is widespread recognition of the critical role that financial and asset management information systems provide in sustaining any ongoing operation or organization. Without the assets, there is no basis for the generation of revenues and income (key components of any financial return).

BIM as a tool for improved asset management and asset maintenance can provide a substantial benefit to the financial management of the airport. BIM can also improve the quality/reliability of the airport's infrastructure in providing services to passengers, airlines, vendors, and the local community it serves. Performance measurements for BIM can be organized as shown in Table 4-9.

The direct financial impact from BIM, in terms of measurable revenue enhancement, is difficult to determine except in a few emerging applications (such as using BIM to identify the best locations for ad placement within an airport terminal). Determining indirect financial benefits, in terms of

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Table 4-9. BIM asset performance benefits.

Value Area	Value Measure
Financial—Direct	Directly Measurable Revenues/Savings
Financial—Indirect	Measures Convertible to Cost in Dollars (Time/Labor)
Asset Condition	Facility Condition Index/Avoided Costs/TCO
Risk & Management Safety	The Probability of Avoided Costs
Level of Service	Operating Statistics
Customer Satisfaction	Survey Results
Other Strategic Elements	Tailored to Strategic Objectives

time saved, is a better focus for BIM ROI at airports. The value of time can be looked at in a variety of ways. The value of time can be determined by looking at time saved

- Directly through operation and maintenance efficiencies,
- Indirectly through organization and process improvements, and
- Value realized through cost savings/productivity improvements.

The value of time can also be determined by looking at time gained

- Through extending asset life—higher residual value—balance sheet impact and
- Through more useful years of revenue production—impact on long-term financing capability/capital capacity.

Each of these areas must be examined when measuring the potential indirect impact of BIM on an airport's overall performance. In addition, the benefits delivered through improved asset protection (inventory, condition assessment, and risk assessment) have potential value in building strategic customer relationships with airlines, cargo carriers, concessionaires, and other tenants that play a key role in strategic airport growth. While difficult to measure, these factors must also be considered as part of the intangible contribution to ROI.

4.5.7 Performing the ROI Calculation

Several critical aspects of measuring ROI have been discussed in this section, but how does an airport select the best approach to meet its needs? While some methodologies have been discussed, the ROI calculation should adopt the following basic steps.

Establish Objectives

Starting with a business case will define the airport's objectives, or desired goals and expected benefits, from BIM. More than likely, BIM is an enabling element of a larger strategic objective at the airport and needs to be understood in that context. The strategic objectives may be attracting new airlines or route traffic, improving asset management, expanding sustainability programs, or managing existing growth trends.

Define Metrics/Benefits

Select metrics that will provide measures of the airport's objectives. These might include service-level improvements important to airlines, such as reducing unplanned outages/gate failures; improved response times to unplanned outages; and improved facility condition indexes. Below this strategic level, the BIM-specific metrics that support these higher-level objectives should be defined. For example, asset locations based on BIM coordinates enable

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more rapid response time to unplanned outages and may also enable improved preventive maintenance programs that reduce unplanned outages and improve facility condition.

As noted previously, the benefits can be direct and/or indirect and tangible and/or intangible. Performing a survey of airport staff may be a useful tool for measuring intangible benefits. In measuring ROI, it is important to include all types of benefits in the calculation, as these will be balanced against the costs. Failing to identify key benefits will make the ROI fall short of actual results.

Define the Costs

Initial start-up costs and recurring maintenance costs must be defined. Also, achieving the forecast relation between costs and benefits for a BIM program will take time and will not progress in a linear pattern; there will always be a learning curve for new technology and processes. In most cases, the adoption process for new technology will follow the "J-curve" (see Figure 4-3), which predicts that a period of learning (in which performance decreases) precedes the period of expected performance improvements. This initial lag in predicted gains should be accounted for in the ROI. The benefits of implementing BIM across an organization may not be immediately apparent and may require several years of program refinement.

Costs of implementing BIM include the following:

- Software—authoring, viewing, analysis, collaboration, document control
- Integration—custom programming required for integrating BIM into CMMS, enterprise asset management (EAM), space management, and other legacy data systems
- Network—network and security upgrades for airport access to BIM
- Hardware—upgraded workstations, tablets for field uses, scanners for as-built capture
- Training—formal classroom education
- On-the-job—field experience to achieve competency
- Consultants—if required for pilot or to support migration to BIM
- Process and standards—developing standards and processes for internal airport organizational use and external use by contractors/consultants/vendors
- Facility BIM development—cost to establish initial BIM for an airport (or selected portions)
- External—cost for contractors, vendors, and other airport suppliers/stakeholders to adjust to new standards and requirements

Recurring costs also need to be identified:

- Facility BIM—cost to maintain BIM to accurate existing conditions
- Training—cost to maintain trained staff
- Staff—additional permanent staff required to maintain BIM
- Consultants—if required to support BIM

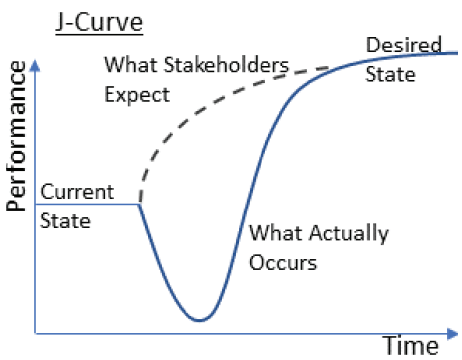


Figure 4-3. J-curve for new technology adoption.

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Measure the Baseline

If the metrics selected are well established and baseline data already exist, then the data need to be collected and analyzed to understand how these metrics have performed. If no historical data exist to establish a baseline for the metrics, then the airport should begin a program to measure these data to create a baseline. The maxim that "if you can't measure it, you can't manage it" applies here; without a baseline, an airport cannot measure the benefits or costs of BIM.

Measure the Pilot Program

An airport that wants to establish a BIM program should establish BIM processes and standards and perform a pilot program. It is vital to establish standards rather than implement BIM in an ad hoc manner, to be sure of what the airport should be measuring. Deploying BIM is more than purchasing software or hiring consultants. To accurately measure ROI, the outcomes (measured metrics) must be measured against the inputs (the process refinement contributed by BIM). Failing to control either side of this equation will provide inaccurate data.

If the pilot program is built around one project, it is important to fully understand how the selected project is similar to or deviates from historical projects that are being used as a baseline.

If an enterprise-wide pilot is being implemented, try to maintain similar team structures within the organization for both the baseline measurements and the measurement metrics taken during the pilot program. The goal is to collect the metrics in as similar an environment as possible, with the single variable being the use of BIM.

Make certain an effort is made to collect data and input from external stakeholders. Although the costs of BIM borne by contractors and vendors may not be measured, the effects of these costs may be that some vendors no longer bid on projects, increase their bids, or include larger contingencies. Although directly measuring the cost to external stakeholders is not feasible, surveys can provide qualitative input for the cost side of the ROI equation.

Interim ROI Calculation

After a pilot project has been completed, perform ROI calculations. The formula can be represented as

$$\text{ROI} = [(\text{Value of BIM Benefits} - \text{Value of BIM Costs})/(\text{Value of BIM Costs})]$$

This formula is simple, but the components are composed of a variety of complex elements. For example, the Value of BIM Benefits and Value of BIM Costs can be reviewed over 1 year or over 10- to 20-year periods (which may be more appropriate if factors related to extended asset life are included). In these cases, the benefits/costs needed to be examined from an NPV accounting perspective so that future benefits/costs are evaluated regarding the current value of those investments and can be compared equally with other possible investments.

How are intangible benefits used in ROI calculations? Intangible benefits are those that cannot be accurately measured (or are difficult to measure). In the first part of this process, the BIM metrics were tied to specific strategic objectives. The intangible benefits can be used to show how potential ROI from BIM (or other possible investments) connects with the airport's strategic objectives. If an airport is choosing between potential investments, these intangible benefits can be used to show which investments most strongly support the airport's strategy.

4.6 ROI Industry Standards

A variety of methodologies that can be used to measure BIM ROI within an organization or on a project basis have been discussed. Although data are sparse with regard to measured ROI outside of the areas of design and construction, there have been some published case studies

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addressing specific BIM uses and applications. These studies may provide some guidance in the early stages of building the business case for BIM before having any directly measurable ROI. The BIM Benefits for Owners (Institute for BIM Canada, 2013) provides some guidelines for life cycle BIM ROI metrics and what levels of return facility owners can expect to receive from those metrics.

The BIM Canada guide separates the facility life cycle into planning, construction, handover, and operations. Planning is further subdivided into three elements: pre-design, design, and integration. During the planning of the project, the guide projects that there will be a 17–35% savings in time to complete the planning process with the use of BIM. Construction is divided into two elements: change orders and construction schedule. The guide projects that there will be a reduction in change orders from the expected 8–10% of the total construction cost to 0–2% with the use of BIM. In addition, the guide states that the overall construction schedule duration will be reduced by 20–28% with the use of BIM. The handover phase is split into two elements: commissioning and capital cost. The guide says that commissioning cost will be reduced by 50–98% with the use of BIM. The guidance on capital cost is that owners should anticipate a reduction in cost overruns, but the guide does not provide a specific level of savings that should be expected. The operations phase is divided into two elements: (1) operations and (2) maintenance. The guide states that owners can anticipate a 17% overall savings across the total life cycle cost of maintenance with use of BIM. For the operational element, the guide states that there is "anecdotal cost recovery" reported by owners, but a specific level of cost reduction is not provided.

A summary of additional anecdotal cost savings related to non-design BIM uses is provided in Table 4-10.

4.7 Airport ROI Expectations

A survey of commercial aviation airports conducted in this research asked some questions related to BIM ROI. Airports were not required to answer all questions, so the results shown herein vary in the number of total answers received.

4.7.1 What Expectations Does the Airport Have for BIM ROI?

An expectation was expressed by 19 of 29 responding airports (65% of respondents) that BIM would have a positive ROI (see Table 4-11). A break-even or better result was expected by 22 of 29 respondents (75%). Only three respondents indicated they did not think BIM would be worth the cost involved.

4.7.2 What Does Your Airport Consider to Be the Primary Contributors to ROI?

Responses to the survey indicated that the strongest perceived ROI was in design and construction, where BIM has been most widely used (as shown in Table 4-12).

Using BIM beyond design and construction, however, is still of substantial interest to airports. Table 4-9 shows that there was interest in using BIM to support preventive maintenance (57%) and to enable predictive maintenance (27%). BIM's ability to provide early collaboration among airport stakeholders in design optimized for maintenance (57%) and operations (47%) is also seen as a strong potential ROI source. Enhanced disaster planning and emergency response times (20%) and enhanced safety and security (17%) appear to be BIM uses that still require further development.

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Table 4-10. BIM ROI sources.

BIM ROI Source	ROI Source	Reference
Maintainability of Facility	40% through enabling lean maintenance design 60% of latent defects avoidable through better design	"Maintainability Approach for Lean Maintenance" (De Silva et al., 2012)
	28% reduction in maintenance cost	Construction Industry Institute's RT-142 Design for Maintainability, http://construction-institute.org/resources/knowledgebase/knowledge-areas/project-planning/topics/rt-142
Reduced Cost of Maintenance Planning	½ hr to 2 hrs per work order	*Cost Analysis of Inadequate Interoperability in the U.S. Capital Facilities Industry* (Gallaher et al., 2004)
Reduced Maintenance Labor and Overtime Cost	½ hr per day, per staff member, and eliminated overtime	Case study of Denver International Airport (see Section 14 of this report)
Operability of Facility	5%–35% increase in operational efficiency	"Excellence in Facility Management, Five Federal Case Studies" (NIBS, 1998)
Accurate Condition Asset Inventory	Greatly reduced time spent locating assets for annual compliance report: 3%–6% savings	"GSA Building Information Modeling Guide Series: 08 – GSA BIM Guide for Facility Management Version 1" (GSA, 2011)
Sustainability-Driven Cost Reductions	3% energy cost savings due to increased preventive maintenance measures and lower energy cost, improved lighting, LEED compliance, and reporting	"GSA Building Information Modeling Guide Series: 08 – GSA BIM Guide for Facility Management Version 1" (GSA, 2011)

Note: LEED = Leadership in Energy and Environmental Design

4.8 Strategic ALCM and BIM

Building the business case for BIM may be simplified if it is included as part of building a business case for improving the overall strategic ALCM at the airport. Much study has been invested in the long-term benefits of improved ALCM, and many airports already have programs to evaluate the emerging new systems and standards such as ISO 55000 and

Table 4-11. Airport ROI expectations.

ROI Expectation	% of Respondents	
More than 100%	0.00%	0
50%–99%	13.79%	4
25%-49%	17.24%	5
10%–25%	17.24%	5
Less than 10%	17.24%	5
Break even	10.34%	3
Net cost but worth the investment	13.79%	4
Not worth the cost	10.34%	3
	Answered	**29**

Source: ACRP Project 09-15 Digital Survey

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Table 4-12. Airport BIM primary ROI sources.

BIM Uses Contributing to ROI	Responses	
Improved planning	60.00%	18
Reduced design/construction cost	70.00%	21
Improved communication/collaboration	56.67%	17
Improved capital planning	30.00%	9
Enhanced preventive maintenance	56.67%	17
Enhanced predictive maintenance	26.67%	8
Operational efficiencies	33.33%	10
Reduced downtime	30.00%	9
Design optimized for maintenance	56.67%	17
Design optimized for operations	46.67%	14
Enhanced disaster planning and emergency response times	20.00%	6
Enhanced safety and security	16.67%	5
Other (please specify)	6.67%	2
	Answered	**30**

Source: ACRP Project 09-15 Digital Survey

ANSI TCO. The following will provide some basic background in ALCM fundamentals and standards for those airports that wish to consider implementing BIM as part of one of these programs. More detail can be found in the ANSI 1000-1 TCO Standard and the ISO 55000 Standard.

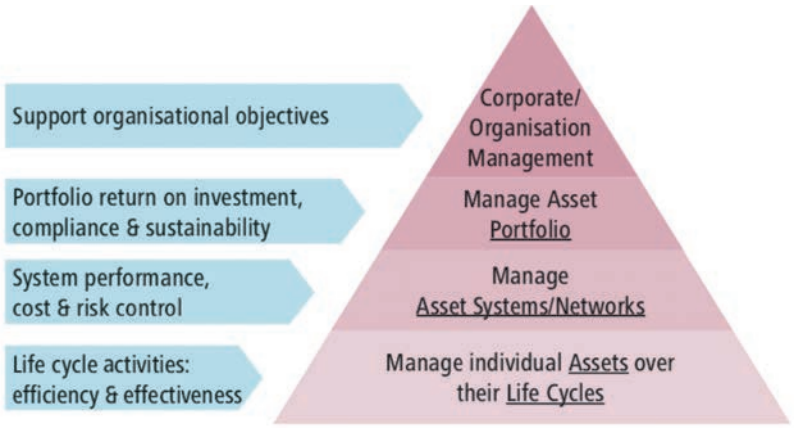
Airports are heavily invested in infrastructure and other physical assets. The care and functionality of airport facilities are not only highly regulated but also often subject to stiff competition from airports around the country. Investment in facilities to keep up with current trends in passenger needs is continuous, and ensuring capacity to meet the demands of all airport stakeholders is a top priority. The demand for growth, the need to provide and maintain a secure and safe environment for air travel, and meeting new regulatory requirements are all drivers for an improved approach to strategic facility asset management and operations. BIM, with its ability to provide a collaborative and transparent platform for sharing facility asset data across airport stakeholder groups, can play a key role in supporting these larger initiatives. Since these larger strategic needs may drive the motivation to evaluate BIM, it is important that these initiatives be reviewed herein. The following is a brief overview of the developing Strategic Asset Life Cycle initiatives and a discussion of how BIM fits in with them.

Airports are businesses, and businesses exist to serve stakeholders. Among the factors that weigh heavily in the success of an airport are the variety of airlines and destinations served and traveler satisfaction with airport facilities, as measured through indicators of the effectiveness of infrastructure and airport operations. The 2017 *North America Airport Satisfaction Study* by J.D. Power measured overall traveler satisfaction by examining six factors (in order of importance): (1) terminal facilities, (2) airport accessibility, (3) security check, (4) baggage claim, (5) check-in/baggage check, and (6) food/beverage/retail. Clearly, airport assets and their utility are core determinants of airport success. The recognition that airport assets are key determinants of business success points to the need to address management of assets strategically, by aligning them with corporate goals and objectives.

Alignment of asset management to corporate goals and objectives is a key principle for achieving optimum value from assets (see Figure 4-4). Alignment, also called "line of sight," ensures that everyone in an organization, or in this case, an airport, knows how they contribute to its success.

"Line of sight" involves translating corporate objectives into asset management policy, strategy, and objectives, which then become more detailed asset management plans and activities. Alignment ensures that the actions of employees, whether they are decisions from

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Source: Institute of Asset Management, 2015

Figure 4-4. Asset management hierarchy.

the executive level or they are O&M tasks carried out at the facility, are based on their impact to assets and the achievement of organizational goals.

4.8.1 Asset Value

According to ISO 55000, "Assets exist to provide value to the organization and its stakeholders" (ISO-55000: 2014, 2.4.2, Fundamentals). It is easy to see how this quote from ISO 55000 (the internationally recognized standard for asset management) applies to airports when the criticality of such assets as pavement, lighting, signage, access control systems, airport rescue and firefighting (also known as ARFF) vehicles, and aircraft fueling systems are taken into consideration. Stringent regulatory requirements dictate the functionality, inspection, and in-service or uptime minimums for these airport-maintained assets. Non-compliance with these regulatory requirements places an airport at risk of decertification. Airport assets also include terminal buildings, cargo hangars, and baggage delivery systems. Although regulatory requirements also guide the performance of these assets for issues of public safety, it is clear from this and the former list that an airport would not be open for business if these assets were not available and performing to meet customer needs.

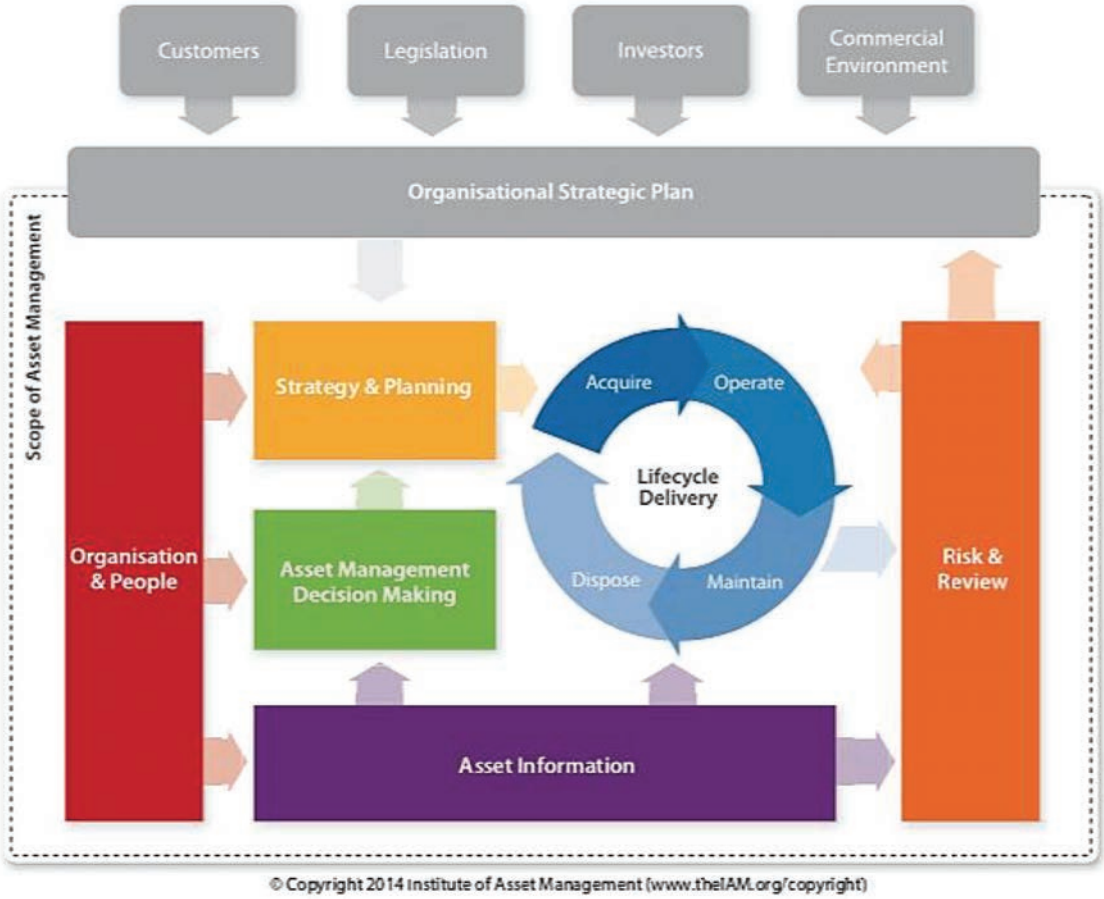
A comprehensive approach to asset management focuses on achieving the best value for the money through careful consideration, or analysis, of the trade-offs among performance, cost, and risk throughout the entire life cycle of an asset. There are many methods for performing this type of analysis. The first and most important thing required is reasonably accurate information about asset cost, asset performance, and business risk. When there is alignment between corporate goals and asset management, all this information will be found in applications that provide a single source of truth about assets.

Optimum value is achieved when organizations address six primary aspects of asset management (see Figure 4-5).

The six primary areas are the following:

- **Strategy and Planning**. An asset management policy is in place that is consistent with the organizational strategic plan and that incorporates external requirements. The policy establishes a framework for the development of asset management strategies, plans, and objectives.
- **Asset Management Decision Making**. Processes exist to evaluate and analyze capital investments and operation and maintenance requirements. Activities balance the costs and

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Source: Institute of Asset Management, 2015 (reproduced with permission)

Figure 4-5. Strategic asset management framework.

benefits of asset renewal/maintenance/overhaul/disposition events. Methods ensure the best total value of asset systems or portfolios in consideration of life cycle activities and the optimal combination of costs, risks, performance, and sustainability. Resourcing of people, infrastructure, tools, and materials and strategies for shutdowns and outages are cost-effective, efficient, and safe.

- **Life Cycle Delivery**. Processes are in place to implement asset management plans in compliance with technical standards and legislation. Processes include gates for asset acquisition, creation, installation, and commissioning, as well as approval and release of funding, handover to operations, and monitoring and capture of actual costs and benefits. Standards exist and are in use for maintenance delivery and reliability engineering, asset operations, resource management, shutdown and outage management, fault and incident response, asset decommissioning, and disposal.
- **Asset Information**. A strategic approach is taken to the definition, collection, management, reporting, and overall governance of asset information to support the implementation of asset management strategy and objectives. Standards exist for structure and format for the collection and storage of information and reporting and include support for management activities and decision making.
- **Organization and People**. Processes ensure alignment of both insourced and outsourced asset management activities. Organizational leadership, structure, culture, and processes support the delivery of asset management objectives. Processes systematically develop and maintain the needed supply of competent and motivated people to fulfill asset management objectives at all levels of the organization.

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- **Risk and Review**. Policies and processes identify, quantify, and mitigate risk while exploiting opportunities. Processes and systems support business continuity. Processes are interdisciplinary and collaborate to ensure a balanced, sustainable, progressive approach to economic, environmental, and social issues. Performance indicators assess asset health and current or historical performance. Business processes exist to review and audit asset management processes and asset management systems. Processes define and capture as-built, maintenance, and renewal costs for valuation and depreciation of assets. Methods are prescribed to engage with stakeholders.

4.8.2 Measuring and Driving Asset Value

Value and how it is measured will be unique to the objectives of each organization. Value will likely include both tangible and intangible components. For airports, measurement is usually driven by a combination of statutory and industry conventions or requirements, which, at least from the financial perspective, are probably well established in an airport's management structure. Several widely accepted measures are offered here for illustration. So far in this Guidebook, ROI has been the single term used for a financial measurement in support of a project's or business's performance. For enterprise-wide measurements, ROI is modified by ROA (return on assets) and ROE (return on equity). In the simplest terms, equity is what business owners have invested in a firm, whereas assets usually include owners' equity plus borrowed capital invested in the firm. ROA and ROE are two of the basic ratios that financial managers use in assessing a business's balance sheet (assets and liabilities).

ROA is an indicator of how well assets are being managed to generate income, while ROE gauges how well the owner's investment is generating income (investment as distinct from debt). In the discussion of ROI in previous sections of the Guidebook, no distinction was made between the business owner's money and borrowed money in business investment (investment equals debt plus equity). At the EAM level, however, analysts must drill deeper. When everything is accounted for and entered into a balance sheet, ROA equals net income divided by total assets, and ROE equals net income divided by equity. ROA and ROE are expressed as percentages, and both are tricky to interpret and only as good as the component assignments of asset value. That said, in the hands of qualified financial analysts (perhaps working for bond rating agencies, lenders, or regulators), ROA and ROE are tools for comparing business performance and assessing financial health.

As a simple example, let's say an airport management team decides to buy a small general aviation airport for $1 million using $500,000 of its own money and borrowing $500,000 at 3% interest. If the small general aviation airport generates $100,000 in net income the following year, ROA would be 9% ($100,000 income minus $15,000 interest divided by $1 million asset) and ROE would be 17% ($100,000 minus $15,000 interest, divided by $500,000 equity). Of course, this is too simplistic to help management evaluate all the different kinds of assets that make up an airport or the different funding sources and obligations they entail (even at a small airport). For one thing, a meaningful analysis should be looking at performance over several years. On another level, the simple example doesn't account for the fact that the borrowed principal has to be paid back, not just the interest.

However, with the introduction of a few more ratios, a widely used method developed in the DuPont Company can provide a better sense of how ROA and ROE can help managers measure and drive asset value to serve their strategic objectives.

The DuPont model of financial analysis created by E. Donaldson Brown (aka "the Father of ROI") is the basis for most ROI models. Several variations exist with endorsements from various

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associations. While the variations are often couched in terms of private enterprise, it is nevertheless useful to look at a couple of them to get a better sense of the components that cause shifts in "equity," which is the business owner's essential capital available for use or reinvestment in future periods. The DuPont analysis uses three metrics to break down ROE:

- Operating efficiency (net income/equity)
- Asset use efficiency (revenue/assets)
- Financial leverage, as couched in terms of the "equity multiplier" (assets/equity)

The International Facilities Management Association (IFMA) identifies one metric for value as ROA. IFMA follows the DuPont model in defining this metric. According to IFMA, ROA is income before debt service divided by fixed assets. ROA can be used to measure operating performance, or an organization's ability to generate net sales from fixed asset investments, net of depreciation:

ROA = Net Income/Total Assets

This is consistent with previous discussion. To look at the contributions of assets in deriving economic benefit, the DuPont components can be applied to ROE (which to this point has been discussed as just net income divided by equity):

ROE = Operating Efficiency × Asset Use Efficiency × Financial Leverage

= [Net Income/Revenue] × [Revenue/Total Assets] × [Total Assets/Equity]

= ROA × Total Assets/Equity

To look at fixed asset investments in terms of identifying and analyzing specific asset contributions, the ROE model is reduced to

ROA = Operating Efficiency × Asset Use Efficiency

= Net Income/Revenue × Revenue/Total Assets

Using this model, ROA can be improved by increasing either operating efficiency or asset use efficiency or both.

If the newly acquired general aviation airport in the simple example previously discussed is treated as a single asset, and it is assumed to have operating expenses of $200,000:

Operating Efficiency = $85,000/$300,000 = 28%

Asset Use Efficiency = $300,000/$1,000,000 = 30%

And, as said before:

ROA = 9% = [28% × 30%]

A different scenario would be an airport that has a mature BIM program, integrated with an EAM program, all of which is implementable at its new general aviation facility (which had been a sleepy, paper-based operation). One aspect of the BIM and EAM programs is that maintenance staff use tablet computers in the field from which they have access and input capabilities to the BIM and EAM systems. As discussed in previous sections as well as in the case studies, one benefit of this system is savings on overtime cost. It is further assumed in

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this scenario that the transfer of the BIM and EAM systems to the new airport results in a $50,000 reduction in operating expenses, so they drop from $200,000 to $150,000 and stay at that level forever. As a result, operating efficiency improves to 45%, asset use efficiency remains at 30%, and ROA increases to 14%.

On the other hand, if management imposes some EAM measures at the general aviation airport at a cost of $50,000 per year, the expense combined with the reduced overtime costs could result in overall expenses remaining the same.

However, the EAM improvements cause customer satisfaction to increase so much that the fixed-based operator (FBO) handling aircraft operations can charge more for its services and airport managers are able to renegotiate the FBO's lease, resulting in a $50,000 increase in rent. Under this scenario, operating efficiency increases to 39% and asset use efficiency also increases to 35%, resulting in ROA of 14%, as well.

Moving away from quantifiable measurements, two additional approaches offered by the Institute of Asset Management are worth noting:

- **The value stream**—a "lean" operations concept, focusing on customer satisfaction and business process mapping to analyze and contrast the current state with a future state to reduce waste.
- **The value chain**—a strategic concept focusing on competitive advantage, where value to the customer is measured in profits (in for-profit organizations) and cost of service (in not-for-profit organizations). Values are assigned to assets that support a customer value proposition.

These two approaches are especially significant for airports because they are removed to some degree from the financial accounting for net income and other strictly quantifiable measures of value. Unlike profit-driven private enterprises, airports can hurt their long-term viability by over-reliance on ROI in driving their strategic plan. This is not to say that ROI, ROE, and ROA are not important measures of an airport's economic self-sufficiency for bonding authorities, fiduciary overseers, and other entities looking for its financial viability and value.

Going back to the simplified scenario, improving both the newly acquired airport's operating efficiency (via reduced overtime) and its asset use efficiency (via increased revenue) resulted in the same ROA (14%) as if only the measures that improved operating efficiency had been implemented and the savings were used to pay for something unrelated.

What the quantitative financial analysis has missed, however, is the qualitative improvement in customer satisfaction. The Institute of Asset Management's value stream and value chain concepts are a way to address these oversights, at least in strategic planning and management.

4.8.3 Asset Management Maturity

Early studies in total quality management by Ledet and Paich (1994) looked at how various maintenance methods in manufacturing could improve equipment uptime through

- Planning optimization,
- Scheduling optimization, and
- Preventive and predictive maintenance.

The results indicated that only minimal improvement could be realized through any one strategy (<1%). However, when the strategies were used in combination, the improvement rose to 5%. When techniques for defect elimination were added, the improvements rose

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to 19%. The exponential improvements were attributed to the motivation, or passion, of staff when tasked with the problem.

The results of Ledet and Paich's study (1994) illustrate how applying a comprehensive management system and achieving alignment, or line of sight, can improve asset management synergistically and advance asset management maturity:

> Asset Management maturity is the extent to which the capabilities, performance and ongoing assurance of an organization are fit for purpose to meet the current and future needs of its stakeholders, including the ability of an organization to foresee and respond to its operating context. Organizations that demonstrate Asset Management maturity should be able to foresee and respond to both the changing business environment and changing stakeholder needs in a manner that retains alignment of the various activities within the organization. (GFMAM 2015)

Figure 4-6 demonstrates how asset management maturity results from addressing the six primary areas of ISO 55000.

Asset management maturity means that an organization is not only taking a life cycle view of its assets, but that it also understands its risk, integrates asset data and financial data, and is committed to continual improvement.

To mature in asset management means that the organization is learning and improving from some baseline or benchmark, preferably on a continuous basis. This means that an organization has reliable data for its baseline and is applying a proven model for change such as the Deming Plan-Do-Check-Act Cycle (see Figure 4-7).



Source: Institute of Asset Management, 2015

Figure 4-6. Strategic asset management maturity.

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Figure 4-7. Deming Plan-Do-Check-Act Cycle.

4.8.4 How BIM Supports Strategic Asset Management

> The biggest challenge in asset management has not been finding technology solutions, invoking meaningful change, or applying management standards, but rather to justify that asset management is worthy of strategic consideration and investment. (Fogel and Swanepoel, 2014)

A historical bent toward not identifying the life cycle costs or total costs of ownership for assets leaves a void for the identification of historical costs and, therefore, meaningful cost-benefit analyses to support technology solutions. CMMSs have been established to be fundamental tools for the management of assets, regulatory compliance, and legal defense. However, systems that offer functionality beyond work order management, such as those included in EAM and BIM, still face skepticism. From all indications, this has resulted in two distinct positions among BIM consumers throughout the United States:

1. **Asset owners/operators:** Reluctant to invest early for fear of implementation failure, asset owners/operators wait for others to identify clear benefits to justify the expense.
2. **Contractors and developers:** Contractors and developers see BIM as a differentiator. Contractors include those in design-build, design-build-operate-maintain, and design-build-finance-operate-maintain business. For firms using BIM only for design and construction, it is a specialty that can result in longer-term business for future design changes and additional service offerings. For firms that are in the operation and maintenance end of the business, BIM can provide cost efficiencies that lead to more competitive pricing and more desirable management services, including contemporary asset management methods, metrics, and reports.

Strategic asset management implies that best practices such as LCC or TCO are in use. LCC is the accounting of costs for managing an asset throughout its life. TCO is the alignment of an organization's mission with its investment strategy for an asset's life cycle, including all related infrastructure and business process costs. Best practices have also come to include the use of an EAM system (formerly a CMMS) to automate planning and scheduling of preventive, corrective, and inspection work and to ensure a single source of truth about assets over their life cycle.

TCO standards readily acknowledge the use of BIM as an enabler for asset management. The ANSI standard for TCO endorses BIM and suggests that the minimum location data required to support a "transparent, holistic, and efficient approach to financial management, asset management, and resource allocation" would incorporate asset attributes for geospatial locations. Core BIM capabilities include not only the collection and storage of location data

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but also the capability to share that data throughout the organization for collaboration among staff with different asset management roles and responsibilities to optimize ROI.

Overall, for its collaborative features, its ability to place assets in a geographic context, and its ability to integrate with EAM systems to model concepts of operations and life cycle events (asset modification), BIM appears to be the best investment to achieve long-term effectiveness and maturity in a comprehensive asset management program. Its acceptance in the United States has been slow to date, but proof of its use for advancing asset management is expected to come from countries that have adopted it as public policy.

4.9 Summary

When evaluating the use of BIM at an airport, it is recommended that the process begin with the development of a business case that evaluates the strategic objectives/outcomes that would be supported by the adoption of BIM. To fully assess the business case for BIM, it is necessary to understand how BIM and its specific uses will support those strategic objectives. The airport financial environment has some unique aspects that must be considered in developing the initial strategic objectives properly. Depending on the specific airport's financial environment, the relevance of the BIM business case to airlines and other tenants may differ. Larger airports may look to aggregate long-term operating costs as part of capital budgets by negotiating long-term warranties for new facilities. Including the cost of developing BIM and collecting asset data as part of a capital program may be an attractive solution to funding BIM. For smaller airports that may need to fund BIM as an operational expense, the scope of BIM requires a more selective approach focusing on where BIM would deliver the most value.

BIM as a supporting element of gathering more accurate and complete facility asset data can be strongly aligned with strategic asset management systems such as ISO 55000, TCO, and other ALCM methodologies. If an airport is currently evaluating asset management improvements, the cost-benefit analysis of BIM should be included.

While the business case defines these high-level strategic objectives for utilizing BIM, ROI is the tool for defining financial objectives and the framework for measuring the success of BIM in achieving these objectives. The industry does not yet have the foundation on which to base an accurate benchmark for an ROI that airports should expect from BIM. Nonetheless, airports can design a framework to measure future ROI if they define BIM metrics and develop a baseline for those metrics before implementing BIM.

Section 4 Checklist

1. Determine a BIM pilot program and parameters (BIM uses).
2. Determine the benefits and costs of implementing BIM and establish current baseline cost.
3. Develop a business case for implementing BIM.
4. Establish BIM ROI goals.

It is recommended that in 5 years an ACRP project should reassess the topic of BIM ROI, as more entities around the world start to use BIM beyond construction. ACRP-managed research could review BIM-affected cost metrics from the airports now investing and implementing BIM, including DEN, LAX, SEA, SFO, DFW, PANYNJ, and some of the newly constructed international airports that plan on using BIM-developed facility plans for O&M. These include the Grand Istanbul International Airport (Turkey), Changi International Airport (Singapore), and Beijing Daxing International Airport (China). Also, the future research should review the many non-aviation entities worldwide that are on the cusp of implementing BIM beyond construction. This will provide airports with more accurate expectations of ROI and the best framework for measurement.

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SECTION 5

BIM Implementation—BIM Process

5.1 Developing the BIM Life Cycle Process

Unlike BIM for design and construction, life cycle BIM is concerned with a process of developing *and maintaining* the data for an entire facility, not just for one project. This Guidebook refers to the BIM life cycle phases as planning, design, construction, operation, and renewal. The primary use of BIM is to maintain accurate facility data and make these data accessible to the organization. Construction projects deliver a set of project as-builts that may only provide documentation on a portion of a facility. As renovations occur, there may not be an as-built set of plans that represents the entire facility. Facility managers may have to sort through multiple sets of plans to get all the data they need. A life cycle BIM contains a representation of an entire facility and is maintained as a complete model over its useful life. As a result, life cycle BIM is more process oriented than project oriented.

BIM is primarily a process for collecting, documenting, and maintaining facility data in a format that enables the simulation of the facility life cycle. BIM is often highlighted as a collaboration tool for its ability to provide a shared view of the facility. BIM enables team members to work cooperatively to analyze, plan, test, and perform facility life cycle activities. This section will focus on providing guidance on the BIM process after construction handover and for the O&M portions of the facility life cycle. *ACRP Synthesis of Practice 70: Building Information Modeling for Airports* provided an overview of BIM for design and construction at airports; additional references for the use of BIM in design and construction are provided in the bibliography of *ACRP Synthesis of Practice 70* (McCuen and Pittenger, 2016).

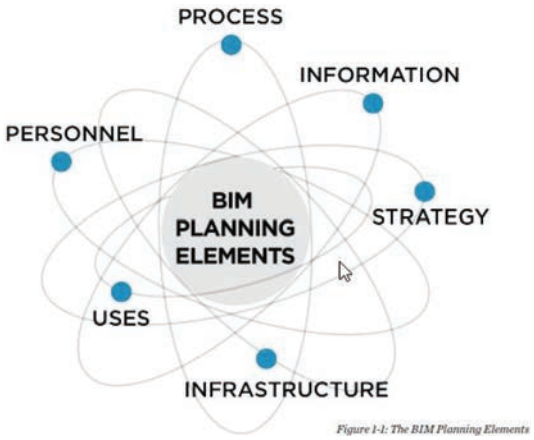
Developing the BIM life cycle process in an airport involves establishing BIM within the organization, known as "BIM implementation planning." The *BIM Planning Guide for Facility Owners—Version 2.0* (CIC Research Program, 2013) provides a good reference for the BIM development process and defines six key planning elements (Figure 5-1):

1. Strategy—goals and objectives
2. BIM Uses—applications supported by BIM
3. Process—workflow descriptions
4. Information—facility data requirements for BIM
5. Infrastructure—technology requirements
6. Personnel—roles, responsibilities, and training requirements

5.2 Strategy

The first step in implementing BIM is to identify the goals and objectives of the BIM program. The data collected during a needs assessment, business case, and capabilities assessment need

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Source: CIC Research Program, 2013

Figure 5-1. BIM planning elements.

to be developed into an implementation strategy. A phased plan or roadmap that outlines the objectives can communicate the activities required to achieve those objectives.

At a minimum, the strategy needs to include the following elements:

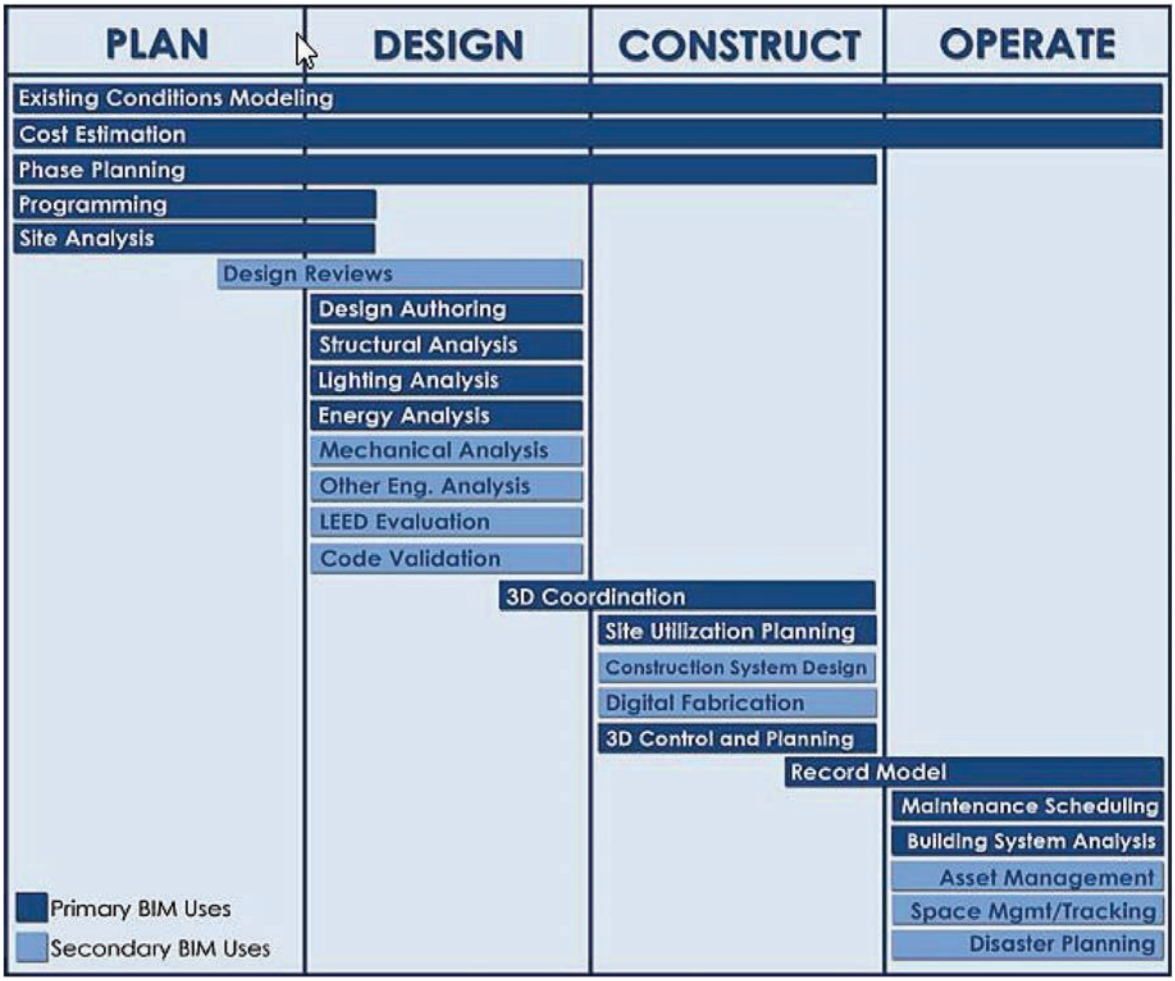
- **BIM goals.** What are the specific financial and operational goals for BIM? Extend asset life of existing facilities? Accelerate development and handover of new facilities? Improve communications and coordination with airport tenants and airlines?
- **Readiness.** What is required to properly prepare the airport organization to adopt the BIM process and leverage the potential benefits? Requirements include training and new technology infrastructure, as well as addressing cultural changes to develop increased collaboration.
- **Timeline.** Fully implementing a BIM process takes time. Near-term, high-priority objectives can be the focus and provide early benefits to encourage organizational BIM adoption, but developing airport-wide, accurate, and complete facility BIM may take many years. Fully integrating BIM with all airport management systems may take even longer. It is not unusual for large airports to take a decade to complete a BIM development program. Creating a roadmap that defines the BIM objectives over time gives the organization near-term goals and a long-term vision.
- **Resources.** A committed budget needs to be defined that anticipates that benefits may lag investment costs by some years. Not only does there need to be an investment in infrastructure to support BIM, but there will also naturally be organizational resistance to changing work processes and transparency. Also, there must be dedicated resources regarding personnel and upper management support.

5.3 BIM Uses

BIM uses describe the specific life cycle activities that utilize BIM. Examples of these include maintenance scheduling/planning, space planning, asset inventory, energy analysis, Leadership in Energy and Environmental Design (LEED) compliance analysis, lighting photometric measurements, and others.

It is important to understand the intended uses of BIM and their priority before developing the phased implementation plan for BIM at an airport. If the BIM uses are not well defined and prioritized, the airport may overinvest by developing a more detailed model than is needed or not invest sufficiently to develop a model that fits the needs of the airport stakeholders.

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Source: Messner et al., 2019

Figure 5-2. Examples of BIM uses by life cycle phase.

For example, considerable cost and effort can be expended in modeling all the mechanical, electrical, and plumbing (MEP) infrastructure—including piping, conduits, ductwork, process mechanical, cable trays, and hangars. If the initial uses of BIM are to be space planning and energy analysis, the effort to model MEP infrastructure could be postponed to a later phase, and only the facility spaces and critical mechanical equipment would be initially included in the BIM.

Uses of BIM may be prioritized as part of the strategy and integrated as goals throughout the BIM roadmap. Some uses of BIM, such as energy analysis, may only happen occasionally and, thus, the airport will only occasionally incur the cost associated with them. Other uses of BIM, such as maintenance scheduling, will occur continuously and will require a separate commitment of time and resources to ensure the BIM is maintained.

A partial list of uses of BIM is shown in Figure 5-2. However, these applications are evolving, and new uses for BIM continue to emerge.

5.4 Process Mapping

The BIM process can be complex and, by the very nature of its objective to increase collaboration, may include input from many different stakeholders. To design and manage these workflows, BIM process maps have been developed.

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BIM process maps are the standard mechanism for documenting life cycle activities utilizing BIM. They are flowcharts that identify specific actions, required facility data inputs, facility data created, and who is responsible for each activity. BIM is designed to provide a framework for managing and utilizing complex facility information. A process map ensures that the airport organization clearly understands who is responsible for collecting, maintaining, and using a particular set of data. Without a well-defined and documented process, BIM can quickly become unusable due to poor or inconsistently maintained data.

Process maps enable each stakeholder group to clearly understand what data they can expect to receive and in what format, and what data they are expected to maintain and deliver. The maps should be customized to fit the organization and operations of an airport, and it should be anticipated that it may take several iterations of design to produce process maps that fully support the needs of all stakeholder groups.

Process maps are flowcharts and have several distinct elements (see an example in Figure 5-3):

- **Process step.** The element indicates the BIM use and who is responsible for performing that activity. This is represented by a segmented rectangle with the use identified in the top section and the owner of that use in the bottom section.
- **Reference information.** This element indicates that data from outside systems are required as input into a BIM process. Data could include warranty data, collected sensor data, maintenance history, or other CMMS/EAM data. This is represented by a document page icon with the lower right-hand portion folded up. The data source is identified with text above the icon. These will be shown as inputs into the process flow.
- **Information exchange.** Facility information produced or required as input into a BIM process. This is also represented with the document page icon but is shown as outputs from or inputs to the process flow.

In Figure 5-3, an example process map is shown from the Penn State CIC BIM Process Templates for Maintenance Scheduling. In this example, productivity information is collected by the facility manager to adjust the maintenance schedule. Maintenance is performed, and equipment performance is validated. The process is repeated until the maintenance routine has been optimized. The last step is updating the maintenance data in the BIM and updating the maintenance schedule data.

5.5 Asset Information

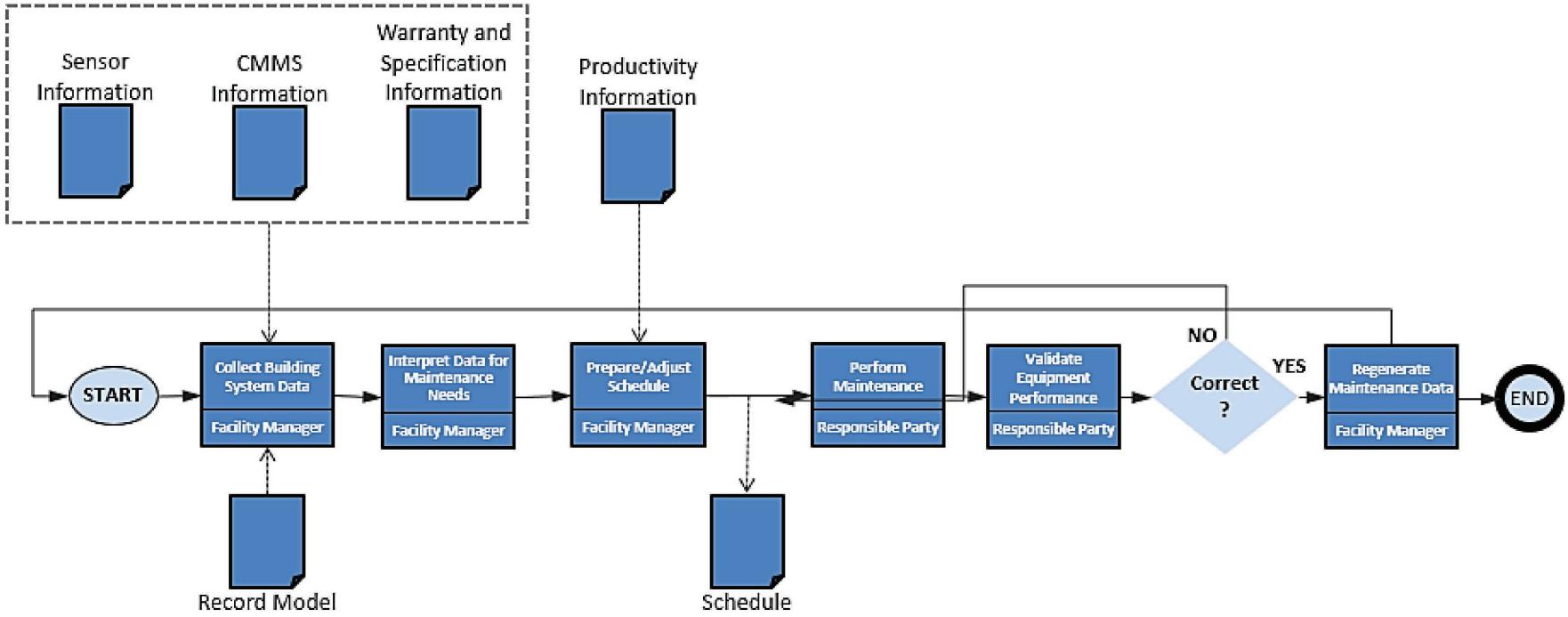
While a 3D model is useful for coordination during design and construction, it is less useful during O&M without the integration of the underlying asset information. In practice, asset information at airports can become highly localized, with each airport stakeholder group maintaining the facility asset data it requires to perform its work functions effectively.

Integrating these data with BIM can provide a means of sharing these data with and distributing these data to airport stakeholder groups that could benefit from the facility information. Without BIM, maintenance staff may develop facility floor plans for maintenance scheduling, property management staff may develop separate floor plan layouts for tenant leasing and space management, and planning staff may separately develop layouts to scope options for future growth. Using BIM, they could all share the same layout plans and eliminate redundant effort.

Also, without a consolidated as-built model of the complete facility, it is quite possible in the example just described that none of the three layout plans is accurate or complete.

When developing facility information requirements, there are three primary considerations: what assets will be in the BIM, to what LOD should the BIM representation of assets be created, and what are the asset attributes that must be collected?

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Source: Penn State CIC

Figure 5-3. BIM process map example.

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5.5.1 Asset Selection

When developing BIM for design and construction, a high LOD is required to coordinate the construction activities. Even minor assets that are not modeled can create unforeseen conflicts when the trades are installing heating, ventilation, and air conditioning (HVAC); plumbing; electrical; fire protection; and technology systems.

In creating and maintaining BIM for O&M purposes, the cost of including all these types of elements should be weighed against the benefits. For maintenance purposes, it may be more cost-effective to review work order histories and focus on modeling the architectural facility layout, high-maintenance or high-criticality assets, and their associated systems. Prioritizing which assets are initially included in the BIM will deliver benefits in maintenance planning and scheduling with minimal up-front costs. Additional assets and infrastructure can be added over time as part of the overall BIM roadmap.

5.5.2 LOD

While there is a natural tendency to create BIM to the highest LOD possible from the beginning, there is limited benefit to making the model photorealistic. Although such an effort may aid in the identification of assets and provide some increased ability to perform very detailed clash detection and spatial analysis, these potential benefits need to be weighed against the cost of developing and maintaining BIM for O&M at that level.

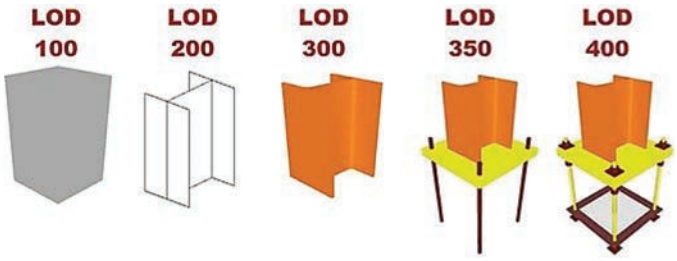
Over-specifying the LOD for BIM elements can greatly increase the cost of development and the size of the data files, potentially making them difficult to use for the rest of the organization.

As seen in Figure 5-4, LOD 100 or 200 would often be adequate for O&M purposes. Specific assets or spaces within the BIM that require a higher LOD can be identified on a case-by-case basis to minimize cost. BIM facility LOD can be divided into two categories—graphical detail and asset attribute data developed.

5.5.3 Asset Attributes

Asset information requirements need to be well defined and consistently maintained in BIM. The type and format of asset information requirements should be documented before BIM is developed.

BIM uses will require certain classes of information to be available and accurate to provide analysis of value to owners. For example, BIM energy analysis requires insulation factors for windows and doors and an accurate layout of heating and cooling zones, HVAC capacity, and other data elements. If the BIM implementation strategy included integrating BIM with an existing CMMS, then these data may already reside in the CMMS. However, ensuring that the asset identifications match in the CMMS and BIM will then become critical.



Source: Level of Development (LOD) Specification 2018 Part II, created by BIMForum. © 2018 Project Modeling LLC (reproduced with permission)

Figure 5-4. LOD examples.

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BIM Beyond Design Guidebook

While there is no standard reference for asset attributes, there are classification standards that can be referenced if there is not an established internal airport standard in place. The standard asset classification systems include OmniClass, UniFormat, Construction Operations Building Information Exchange (COBie) or Industry Foundation Class (IFC) (see Section 8 for standards).

If an existing non-standard airport asset attribute classification is currently in use, the BIM implementation strategy may include evaluating migration to a standards-based system as part of the BIM roadmap. As the industry continues to develop BIM uses, tools, and technologies, they will be developed to utilize standards-based asset data. Although data conversion can always be performed, it will add time and cost to utilizing future innovations.

5.6 Infrastructure

Strong technology infrastructure needs to be in place if the benefits of a BIM program are to be maximized. For those involved in maintaining BIM, high-end workstations and software will be required. BIM files can be very large. They require large server storage capacity and a high-speed network in order to be shared across an organization. The size of the files can be managed to some extent by controlling the level of detail on the assets included in the BIM, but the files may still be larger than most files that are typically shared across an airport. Information technology (IT) infrastructure needs to be accounted for in the business case and coordinated with airport IT staff. Typical BIM infrastructure elements include the following:

- Upgraded workstations—high-end for those maintaining/updating BIM
- Software licensing—BIM authoring, reviewing, and analysis tools
- Network—backbone for transferring and sharing large files
- Storage servers—capacity for storing potentially very large files (>1 terabyte)
- Mobile access—field access to BIM with Wi-Fi and laptop, tablet-based applications
- Cloud-based—field access to BIM, external sharing with consultants/vendors
- Security—secure access technology to protect facility data

Infrastructure technology is a rapidly evolving environment. It is likely that the airport will need to anticipate a technology upgrade cycle every 2 to 3 years.

5.7 Personnel

Personnel will require training not only on new tools and technologies, but also on new work processes. A capabilities maturity assessment will provide some visibility to the training that will be required. A change management strategy will need to be developed that addresses how to overcome the natural organizational resistance to change. This BIM CMM process is discussed in more detail in Section 2.

For BIM implementation to be successful, it needs to have strong executive support and one or more BIM champions across the organization who can provide day-to-day support for addressing the challenges that will naturally arise as part of any major technology and process change.

Roles and responsibilities need to be very clearly defined so that individuals in the organization know who to go to for specific assistance, and those individuals need to have the training and experience to perform these roles. Training needs to be developed, and time should be given to staff to receive this training. There is a significant investment in time involved in preparing an organization to use BIM. However, without this investment, it is likely BIM will not return the anticipated benefits.

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BIM roles and responsibilities include the following:

- BIM champion—a member of the executive team who promotes the BIM vision
- BIM steering committee—the interdepartmental team, created to define strategy, develop the business case, and work with executive staff to get approval and funding for the BIM strategy and implementation
- BIM manager—has overall responsibility for executing the BIM strategy
- BIM designers—responsible for creating and maintaining BIM
- BIM asset managers—responsible for defining and maintaining the accuracy and completeness of BIM
- BIM procurement/legal—responsible for identifying and maintaining BIM contractual and legal requirements
- BIM IT—responsible for maintaining the network infrastructure to support secure BIM
- BIM leads—BIM leaders within each department who are knowledgeable and can promote the use of BIM to improve department processes and provide feedback on potential opportunities for improvement

A decision will need to be made on where the role of BIM manager falls. The survey of airports conducted for this research showed that most airports placed BIM management within an engineering or asset management department. Aligning BIM with asset management will align it with the overall life cycle focus of each group.

5.8 Phasing

Once the strategic short-term and long-term goals are established and other key elements of BIM implementation are defined, it is important to structure a phasing plan. The "big bang" approach to implementing BIM, where an organization implements a complete BIM program for all its capital projects and facilities, is difficult. This difficulty is even more pronounced when introducing a complete BIM program into an organization as complex as an airport, with its diverse stakeholders and types of facilities.

A phasing plan that identifies specific goals, along with the BIM implementation activities required to deliver on those goals, is critical to the success of the implementation. The plan may be developed as a formal project schedule, a strategic narrative, or a BIM roadmap.

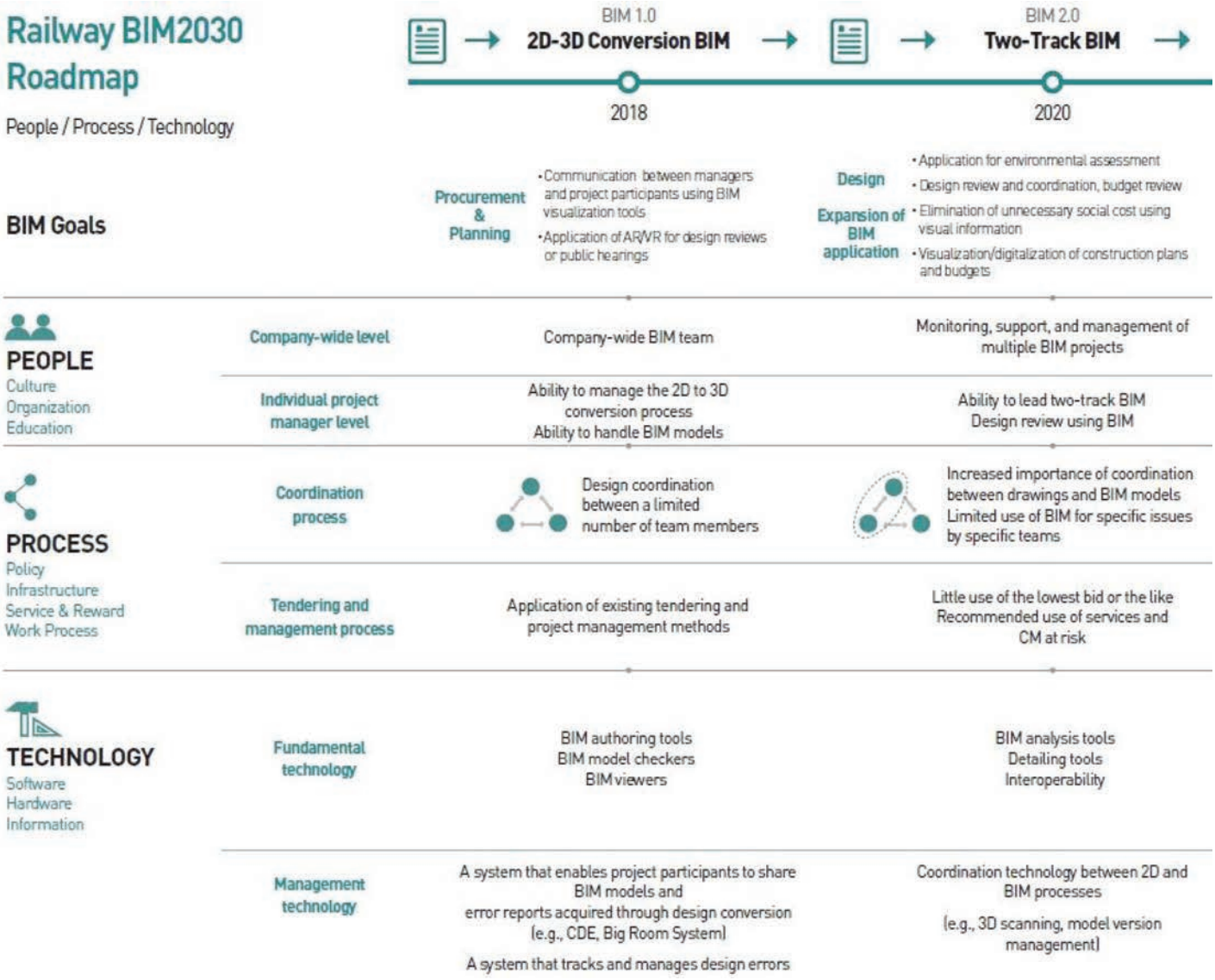
At a minimum, the phasing plan should define the BIM goals, associated tasks, and timeline for the required changes related to people, process and technology. A roadmap example, from the Korean Rail BIM 2030 program, is provided in Appendix D. A portion of this roadmap is shown in Figure 5-5. The full roadmap defines five phases:

- 2018 BIM 1.0 2D-3D Conversion BIM—develop 3D facility data
- 2020 BIM 2.0 Two-Track BIM—migration phase, a mix of BIM and non-BIM
- 2022 BIM 3.0 Integrated BIM—BIM-driven project and facility management
- 2024 BIM 4.0 Lean BIM—BIM to develop operational efficiencies
- 2030 BIM 5.0 AI BIM—BIM to leverage IoT, AI decision making, and real-time data

The roadmap tracks BIM goals in three categories—people, process, and technology—which each have two subcategories:

- **People** (culture, organization, education)
  - Company-wide
  - Individual project manager level
- **Process** (policy, infrastructure, service & reward, work process)
  - Coordination process
  - Tendering and management process

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Source: Professor Ghang Lee, Yonsei University, Korea

Figure 5-5. A portion of the Korean Rail BIM 2030 Roadmap.

- **Technology** (software, hardware, information)
  - Fundamental technology
  - Management technology

The value of the roadmap is that it communicates the overall direction of the BIM strategy over time documenting the key tactical elements needed as well as sharing the long-term vision for the effort. Examples from Phases 4.0 and 5.0 of the Korean Rail BIM 2030 Roadmap show the details provided.

In Phase 4.0, under "Process," the coordination process is defined as including a "Lean-based management process, final scheduling system, and another pull-planning/scheduling system, manufacture-to-order (MTO) or engineer-to-order (ETO)."

In Phase 5.0, under "Process," the coordination process is defined as including "Data-driven decision-making processes, Data-driven design, construction, and FM/AM, Automated data processing and exchange, Automated production for on-site and offsite construction."

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5.9 BIM Development and Maintenance

Unlike BIM developed to support design and construction, a life cycle BIM must be maintained over the life of the entire facility to provide any long-term benefit. Developing BIM handover requirements on capital projects can be a means of creating a BIM foundation for an airport. This section will outline the steps involved in developing and maintaining an asset information model (AIM) suitable for post-construction life cycle O&M activities.

5.9.1 Existing Conditions Capture

Before BIM development can begin, the existing conditions of the facilities must be captured. For recently developed facilities there may be an existing construction-level BIM to use as a starting point. For projects that have no existing BIM to use as a starting point, existing conditions must be developed from as-built plans, from manual measurements, or by using laser-scanning technologies to generate point clouds from which a BIM can be created.

BIM Delivered by Capital Project Development Projects

Construction BIM may be too detailed for the airport to easily maintain. If construction-level BIM is available, then the airport should evaluate the intended BIM uses during O&M to determine the desired LOD for the AIM. If BIM is delivered as part of a construction record model, BIM will likely be at an LOD of 350/400/500. This may not be required for a BIM use like maintenance scheduling and structural construction details such as rebar and equipment hangers. Manufacturer's BIM families may make the file size of the BIM too large to be easily accessed by field staff. Strategically reducing the LOD can make the BIM easier to use and more accessible to the airport stakeholders.

Existing Conditions Capture

Even at large airports where new capital construction is ongoing for large portions of the airport infrastructure, there will still be significant areas where BIM will need to be developed outside of a design and construction workflow. In these cases, a workflow to capture the existing conditions of these facilities will be required to create accurate BIM.

As-built drawings, depending on their age, may not accurately reflect existing conditions. Techniques such as laser data scanning and BIM authoring from point cloud data produced from laser scans can be a means of efficiently developing BIM with a high degree of confidence in accuracy. Small unmanned aerial survey (sUAS) technology can be used to rapidly and accurately capture existing facility exterior conditions and site-civil topography.

Laser data scanning will capture line-of-sight spatial data, but will not see between walls, above ceilings, infrastructure embedded in concrete, or underground. If there is value in capturing these data with high precision, then owners can deploy more intrusive and more costly methods such as lifting ceiling tiles, opening walls, excavating, or using ground-penetrating radar.

Manual validation of existing as-built drawings with walk-throughs and spot measurements is another standard technique for collecting the required input to begin BIM authoring. The manual technique may not capture all deviations from existing plans, but may be used if there is a high degree of confidence in the existing as-built plans.

5.9.2 Consolidated Asset Information Model

The consolidated AIM will be a single, or federated, BIM that represents the complete airport facility. A federated model is a collection of facility BIMs or portions of a facility BIM, such as the HVAC or electrical systems. Construction BIM, if it is not for a completely new facility, may

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only update a portion of the overall facility. The airport must create and maintain a consolidated BIM that, at a minimum, covers an entire facility. If the airport's BIM is maintained as separate BIM systems that are organized by project, it will not be able to serve as the single trusted source of most up-to-date facility data. This means airport stakeholders will still need to search through multiple models to assess which is the most accurate for their purposes. Having multiple BIM systems with potentially overlapping data will also limit other BIM uses (such as maintenance scheduling and planning, energy usage, an asset inventory, and condition assessments).

Maintaining a core consolidated BIM will provide the most flexibility and benefits for integration with other asset management systems and with airport stakeholders needing accurate and complete facility data. As new projects are bid, portions of the consolidated BIM can be supplied to the contractors to update, maintain through construction, and return as handover requirements for reintegration into the consolidated BIM.

5.9.3 BIM Development and Maintenance Workflow

Figure 5-6 outlines a starting point for how airports can manage the development of BIM that is being delivered via new capital projects or for legacy facilities, using technologies that

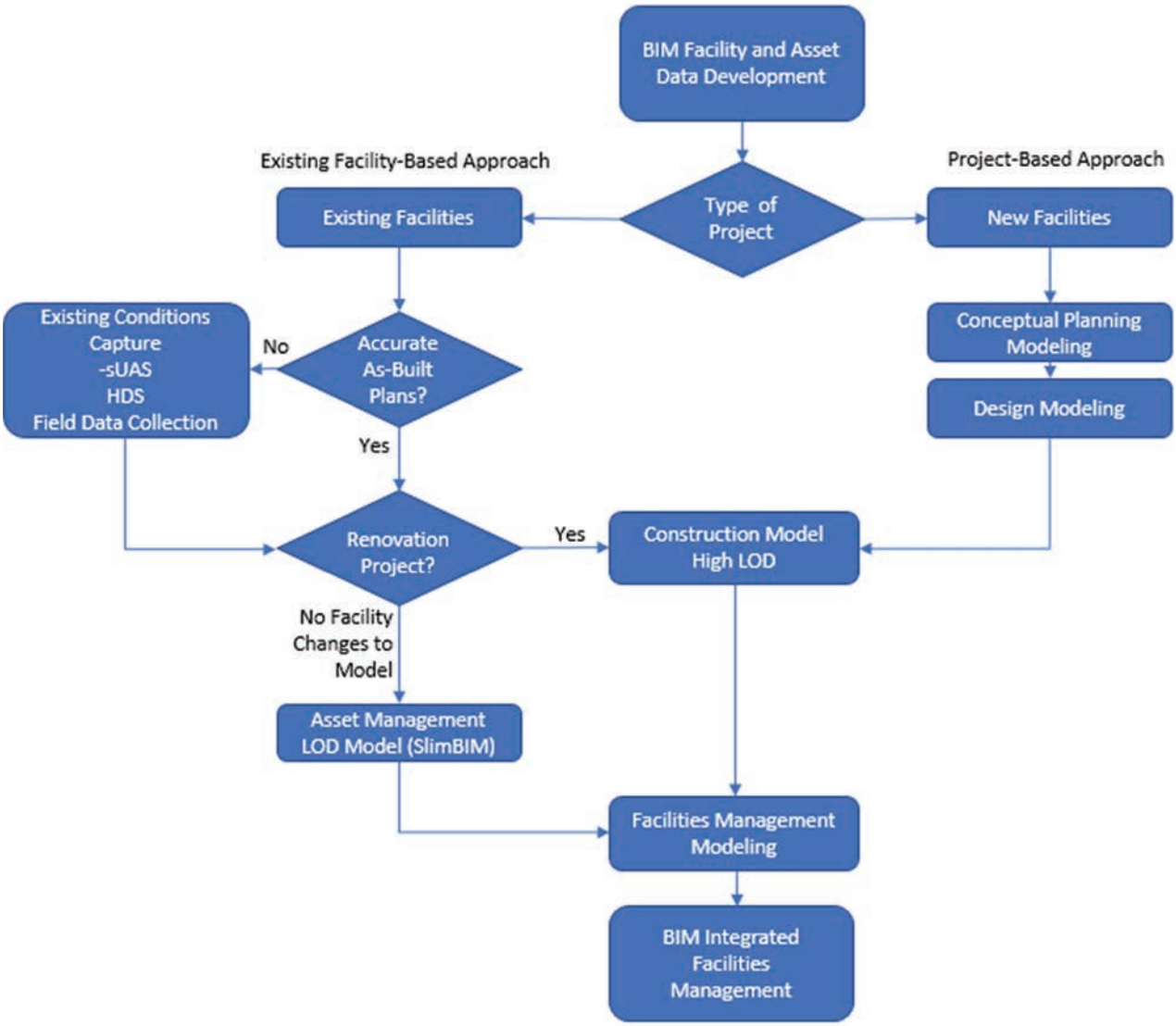


Figure 5-6. BIM flowchart—existing buildings versus new construction.

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can capture existing conditions or manual validation of existing as-built drawings. While these two approaches seem different, they follow a similar workflow, defined in the following:

- **Define asset information requirements.** Define the data required to support BIM uses across the facility asset life cycle.
- **Define LOD requirements.** Define the level of completeness and the level of graphical detail required to support the facility asset life cycle.
- **Acquire existing spatial conditions.** For new construction, this is delivered as a record model that documents the as-built handover conditions of the facility. Otherwise, the BIM may be created using 2D record plans or manual measurement techniques or laser scanning the facility and creating BIM from the point cloud data.
- **Acquire asset data**. For new construction, this will be defined in the asset information requirements (AIRs) and delivered as part of the BIM. Otherwise, BIM asset data must be collected from product specification sheets, O&M manuals, CMMSs and/or EAM systems, or another asset/facility database.
- **Develop BIM.** For new construction, this will be delivered, and the owner needs to perform QA/QC to ensure it meets the defined requirements. Otherwise, the BIM will need to be developed by the owner or outside consultants.
- **Integrate BIM data.** BIM should be viewed as a system of spatial data and asset data. Not all these data need to reside within the native file formats of the BIM-authoring software; they may exist in a CMMS and/or EAM system or other facility databases if there is a clear mapping of the assets within the BIM and asset data in other data systems. It is easier to maintain asset data outside of the files generated by BIM-authoring software.
- **Create BIM applications.** Create work processes and integrations to support the desired BIM uses to support the job functions of the varied airport stakeholder groups.
- **Maintain BIM.** Create QA/QC processes and automated scripts to review updates to the BIM and the criteria for making BIM updates. The value of BIM is in its accessibility, accuracy, and completeness. A decision needs to be made on each project as to whether it justifies a BIM update. If it changes the location, orientation, or type of asset, the BIM must be updated. If it is not updated, the value of the BIM will greatly diminish over time.

Section 5 Checklist

1. Establish the airport's BIM strategy.
2. Identify the airport's BIM uses.
3. Create BIM process maps.
4. Determine asset information (what assets will be in the BIM, to what LOD, and what asset attributes must be collected).
5. Identify the required technology infrastructure to support BIM (hardware and software).
6. Determine the required BIM personnel and organizational structure.
7. Develop a roadmap or phasing plan for the implementation of BIM.
8. Identify the steps required to develop and maintain an AIM for O&M.

5.10 Summary

The process for the development of a life cycle BIM can have many different requirements. While BIM for design and construction is project focused and is oriented toward a specific endpoint of handing over the project to the owner, a life cycle BIM is focused on a continuous process of maintaining the facility.

The scope of the life cycle BIM requirements must be carefully measured against the benefits to maximize the returns and minimize the cost of maintaining the BIM. Unlike construction BIM, a consolidated BIM must be built and maintained to ensure that facility asset management uses (such as maintenance scheduling, planning, and asset inventory) represent a complete view of the facility. Multiple BIM systems in disconnected models will make maintenance difficult and limit future usefulness.

Airport owners must develop a strategy for developing BIM through future capital project development and capturing existing conditions of legacy facilities. A roadmap should be created that outlines short-term and long-term strategies, and that identifies the vision, commitment, and resources for BIM implementation.

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SECTION 6

BIM Implementation—Scaling BIM Implementation

This section will discuss the scalable aspects of BIM that need to be evaluated in the BIM roadmap and how these aspects might fit with the available resources and needs of smaller airports. While larger airports may have an existing CMMS and/or EAM system in place to fully leverage BIM data, smaller airports with several non-integrated and separate management systems may want to focus on a subset of high-value, lower-cost BIM uses. Every airport should take a phased approach to BIM implementation, to take advantage of lessons learned as the implementation progresses.

For most airport organizations, the initial entry point into BIM will be through deliverables received from major new, or renovated, facility capital projects. Even in the absence of contractual owner requirements, many architects, engineers, and general contractors (GCs) have adopted BIM as a standard process because it saves them time and money in the project production process.

The various scaling factors presented in Sections 6.1 through 6.5 will assist airports in determining how to develop a BIM implementation plan that is appropriate for their specific needs. Level 0 indicates the most basic integration, and Level 3 indicates the most advanced. Section 6.6 provides an example of how the following scaling factors can be used to identify BIM roadmap parameters for a small airport.

6.1 Life Cycle Scaling

Level	Life Cycle Phase
0	**Design and Construction:** While design and construction BIM is quite complex, it is likely the first life cycle phase where airports will have the opportunity to receive BIM deliverables.
1	**Planning:** Reusing design and construction BIM to develop reduced LOD AIMs for financial planning, space planning, and property management requires the least effort. It also has immediate benefits, such as more accurate facility data available for planning purposes. Further, it provides a greatly enhanced platform for the initial conceptual planning of future facility renovations.
2	**Maintenance:** BIM for facility maintenance requires more integration effort but maximizes benefits. Such benefits are dependent upon developing AIRs and upon the greater involvement of maintenance staff in the early design and construction process to maximize design for maintainability.
3	**Full Asset Life Cycle:** This level represents developing BIM as part of a comprehensive, strategic facility asset management focus by established standards such as ANSI TCO or ISO 55000.

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6.2 Facilities Included

Level	Facility Scope
0	**Project Driven:** BIM is managed on a project-by-project basis with the focus on reducing the cost and time required for new facility development. Project- driven BIM enhances collaboration with airport O&M staff early in the design process to maximize facility maintainability.
1	**High-Priority Facilities:** All major facilities are developed as a single AIM, not as separate and isolated BIM deliverables organized by capital project.
2	**All Facilities:** All facilities are integrated into one airport AIM. This includes many legacy facilities that will not have BIM deliverables from recent design and construction. Existing condition capture methods (such as laser data scanning and point cloud modeling) will be needed if accurate as-built plans are not available to create the BIM.
3	**All Infrastructure:** This level represents a single, comprehensive facility and site-civil infrastructure BIM. This is difficult to achieve with the software tools available today, but industry efforts currently in development are designed to improve BIM, GIS, and 3D site-civil design tools, which will make this level less costly to achieve in the future.

6.3 LOD

Note: The LOD definitions used below are the same as those discussed in Section 5.5.2.

Level	Facility Model LOD
0	**Spatial Model:** This LOD, known as LOD 200, accurately captures floor layout plans, areas, volumes, and features (such as doors, windows, and ceilings). It is suitable for property/lease management, concepts of operations (e.g., disaster planning), and very basic energy analysis. It may include high-priority managed assets as part of a minimum level of BIM, or SlimBIM, approach to supporting maintenance scheduling enhancements, asset inventory, and condition assessment.
1	**Asset Model:** LOD 300 includes all spatial layouts and features, all managed assets, and asset data to support a CMMS/EAM/IWMS (integrated workplace management system) integration.
2	**Design and Construction Model:** LOD 350/400/500 provides the level of detail required for the coordination of construction activities. For new construction and renovation projects, airports will typically receive a BIM of LOD of 350 or 400 as a final project deliverable. LOD 500 is defined as the default standard for representing the "field-verified" BIM deliverable, but the exact meaning of this has not been well demonstrated and is difficult for owners to verify. The LOD 350/400 BIM can be maintained and developed into AIM-level BIM for future maintenance.

6.4 Asset Management

Level	Degree of Asset Integration
0	**Stand Alone:** The BIM is not integrated with any external CMMS/EAM or IWMS. The BIM can still provide an independent platform for detailed, coordinate-based, asset inventory to enhance the CMMS asset data. It can also be used as a planning tool for maintenance scheduling, sustainability, and asset condition.
1	**CMMS/EAM Manual Integration:** Information exchange between BIM and a CMMS and/or EAM system is a manual export and import process. Integration provides BIM detailed facilities data across all the CMMS/EAM applications. BIM also provides a process for rapid asset data creation after construction project delivery.
2	**CMMS/EAM Integrated:** Full integration provides real-time information exchange between BIM and CMMS/EAM platforms. Direct real-time integration eliminates the possibility that either platform has inaccurate data. Achieving direct integration may require a middleware software solution to synchronize the data in an automated fashion. Full integration may also allow users of the CMMS and/or EAM system to view all or portions of the graphical model, in addition to the facility asset data.

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6.5 Data Accessibility

Level	The Degree of Data Accessibility
0	**Archive Server:** BIM is only available as an archived file that can be viewed on a limited number of workstations that have access to the native BIM-authoring software tools. 2D plans derived from the BIM are part of the engineering archive and are managed independently, just like any other project documentation.
1	**Network Server:** BIM is available from a networked server to all airport stakeholders. Each group maintains local copies used for their purposes.
2	**Enterprise Server:** A BIM server platform makes a centrally maintained BIM available to all airport stakeholder groups. Modifications and updates to the BIM are shared by all BIM users.
3	**Mobile/Cloud-Based Server:** BIM is available via mobile applications to staff working in the field, greatly enhancing the ability of field staff to access current and accurate facility data.

6.6 Small Airport Scalability

Smaller airports can benefit from the use of BIM by using the scaling factors noted above to design a BIM program that meets their requirements. While large airports may be implementing BIM with a focus on enabling growth demands and the corresponding capital development, smaller airports may be more focused on using BIM to reduce operational costs. Large airports will have BIM delivered for significant portions of their facilities as part of the capital cost of new development, while smaller airports will need to fund BIM for their existing facilities from their operational budgets.

Following is an example of what BIM implementation might look like at a small airport, based on the scaling factors previously discussed.

Level	Life Cycle Scaling	Small Airport Features
1	Life Cycle—Planning	Small airports can benefit from the planning level AIM for space planning, maintenance scheduling to enhance preventive maintenance, and reduced cost of asset inventory. If they do not have large facility capital projects, they will need to build their AIM from scratch, unlike larger airports that can scale down capital project construction LOD BIM.
1	Facilities Included—High-Priority Facilities	Small airports may not have projects being delivered as BIM. If they develop BIM directly from as-built conditions, they would not have existing BIM files that required merging.
0	LOD—Spatial Model	An LOD 200 AIM would support basic space, maintenance, and asset management BIM uses that would have the greatest potential impact on reducing operational expenses.
0	Asset Management—Stand Alone	BIM will be used as a stand-alone system, with the primary user interface being the native BIM-authoring software that was used to develop BIM. Some manual exports to other systems may be developed to reduce data migration costs.
1	Data Accessibility—Network Server	A small airport will not develop BIM unless it has a specific use in mind, so BIM will never just be archived to a storage server. It is possible that BIM would be stored on a single workstation that could be accessed remotely by other users when needed.

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6.7 Other Small Airport Implementation Considerations

6.7.1 Outsourcing BIM

While the research team has recommended investing in building internal capabilities for airports to lower costs and to strengthen the organizational adoption of BIM, for smaller airports outsourcing BIM may be a better adoption. Large and medium airports will likely have sufficient work to justify one or more full-time dedicated BIM support staff. Small airports may not be able to justify a full-time staff member to support BIM, making it more economical to hire trained and experienced consultants to use on an as-needed basis.

6.7.2 Open-Source BIM Tools

Small airports may want to evaluate the use of open-source BIM tools to lower the cost of airport stakeholder groups accessing the BIM. Open-source BIM tools that utilize the BIM IFC data format are rapidly making advancements in their ability to modify, maintain, and analyze BIM. While these tools still lack many of the high-level capabilities of commercial BIM-authoring platforms, they can provide a great deal of functionality at little or no cost.

6.8 Summary

There are several BIM scaling factors to consider when designing a BIM roadmap or implementation plan. These are high-level factors that define the overall scope and desired functionality of an airport's BIM program and that will dictate the framework for the detailed standards and processes that must be developed. These factors include the overall life cycle support focus for the BIM, the facilities that will be covered, their LOD, the level of BIM integration, and the level of accessibility required. These scaling factors will also facilitate the design of a long-term BIM roadmap that manages the evolution of BIM at an airport through the BIM scaling factors. Such a roadmap will guide BIM development at an airport over several years and enable BIM capabilities to develop over a managed timeline.

While large airports that are managing many large capital projects have primarily been the early adopters of BIM, small airports can also benefit from BIM by scaling their programs to meet their operational needs.

Section 6 Checklist

1. Determine life cycle phase scaling factor.
2. Determine facilities to include in the BIM.
3. Determine facility model LOD.
4. Determine degree of asset integration.
5. Determine degree of data accessibility.
6. For small airports, determine life cycle scaling benefits for use of BIM.

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SECTION 7

BIM Implementation—Technical Architecture

Prior sections have reviewed the basic framework of the BIM process and how airports can evaluate using BIM to improve their O&M. This section will address the underlying design requirements for a BIM system: the asset data schema and the interfaces necessary to fully support its collaborative benefits.

The technical architecture underlying BIM is primarily focused on the concept of information exchanges. An information exchange comprises the standards and methods by which BIM data are shared between stakeholder groups and other facility management systems such as CMMS, EAM, IWMS, and property/lease management systems.

The concept of standardized information exchanges is the primary enabler of the collaborative benefits of BIM. Data visualization is a form of information exchange, establishing graphical standards that all users can quickly view, analyze, and integrate into their decision making. Underlying the graphical information, however, is a data-rich environment of asset attribute information. The ability to easily exchange information among stakeholder groups throughout the facility or asset life cycle is critical to realizing BIM's full benefits.

A traditional facility life cycle proceeds in a compartmentalized fashion through planning, design, construction, operations, and maintenance. Each phase of the life cycle has its priorities and set of defined deliverables that are passed on to the next phase. Planning evaluates different options and requirements and produces a conceptual facility plan and scope. The design team converts this scope into a detailed facility design and bid package. Construction uses the design to construct the facility and produce shop drawings, product schedules, record drawings, O&M manuals, and other handover data. O&M uses the record drawings and O&M manuals to develop maintenance programs and manage the facility assets to the renewal and replacement phase.

Figure 7-1 illustrates the flow of facility data across the facility life cycle. At each stage, there is a formalized exchange of facility data, but much of the knowledge acquired during one phase is not passed on to the next phase. The owner develops scope documents for the architects, but there may be internal needs assessment information that the owner collected from staff and tenants that is lost when the scope is formalized. The architect and trades engineers create a formal set of building plans that communicate how the facility is built, but do not communicate the design intent needed to guide future decision making when conflicts arise. The contractor creates plans and shop drawings necessary to construct the facility, but only to the degree the contractor needs to coordinate and build that facility. As-built documents are often left incomplete.

Figure 7-2 shows how BIM creates an information exchange that enables a collaborative life cycle approach, in order to eliminate the separate data silos and enhance the sharing of vital facility data. This data sharing not only reduces the data loss at handover but also improves

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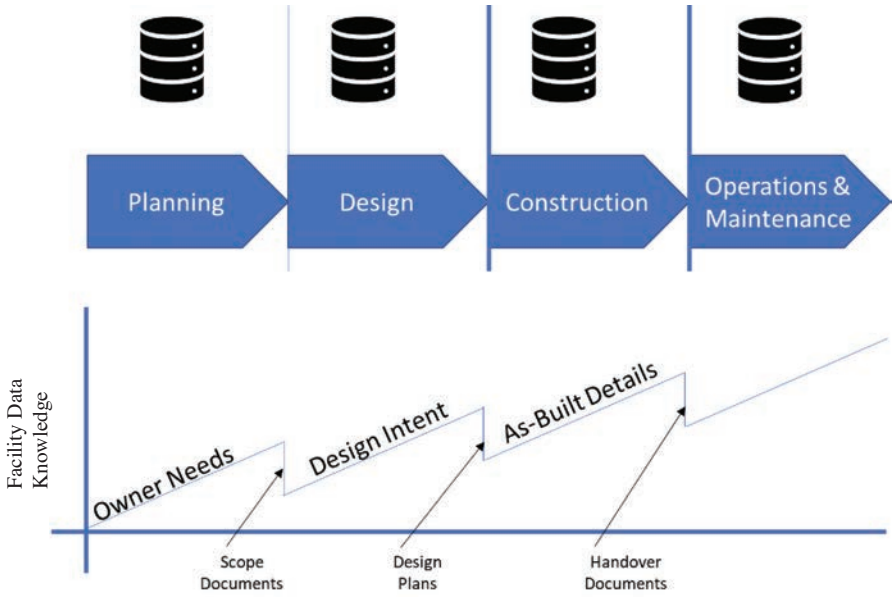


Figure 7-1. Facility data loss during facility life cycle development.

the collection of required data within each life cycle phase. Designers can more fully integrate constructability and maintainability requirements early in the process, where such integration can have the largest effect on reducing the overall life cycle cost of the facility. Contractors have a greater understanding of the owner's and designer's intent, reducing downstream change orders and producing a facility that fully meets the owner's needs.

7.1 System Architecture

What are the required elements and interfaces to support this collaborative information process? System architecture describes the existing software systems, hardware, network infrastructure, and the required information exchanges to make these systems work together.

Traditionally, paper facility record plans were stored in document rooms along with other handover documentation. Digital versions of these documents were stored on archive data servers that could be shared across an organization but did little to encourage collaboration. The BIM-enabled environment allows an organization not only to share facility information digitally but also to maintain and update these data in real time across the organization and with enterprise applications relying upon accurate facility data. Examples of current facility management applications are CMMS, EAM, CAFM (computer-aided facility management), BAS, and IWMS.

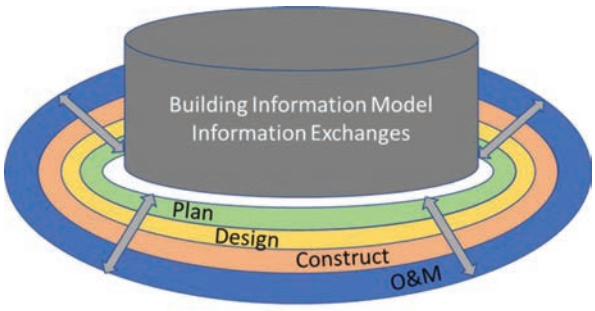


Figure 7-2. BIM information exchange.

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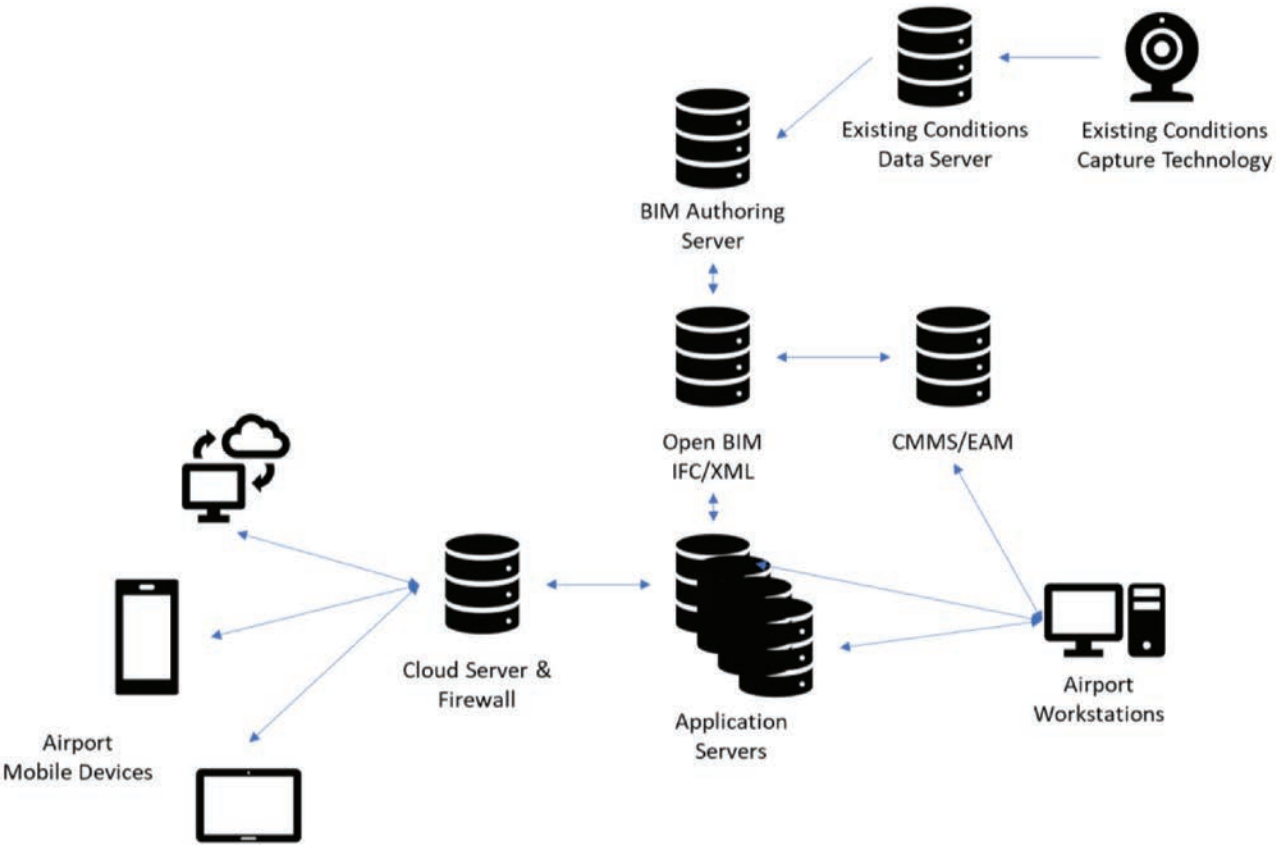


Figure 7-3. Sample BIM network architecture.

While these applications are designed to manage large amounts of facility data, the complexity of facility management will only increase over time. The information revolution has produced an exponential growth in the facility data collected. As next-generation technologies such as IoT sensors and real-time data collection emerge, data collection is likely to grow by several orders of magnitude. BIM will enable this growth by putting the data that are collected into a spatial context to understand how the data are related and distributing the data in a format that can be consumed by the wide range of applications performing the analysis. Big data solutions and artificial-intelligence-supported decision-making tools that will be available over the next decade will require a robust information infrastructure.

Figure 7-3 illustrates a typical network architecture designed to be scalable to meet the rigorous data-processing and storage demands of BIM. A brief description of each component follows.

7.2 Existing Conditions Data-Processing Server

While capital construction projects may deliver a complete facility BIM, in most cases there will be a significant portion of airport facilities that have incomplete or dated as-built data from which to create the BIM. Laser data scanning (also called "high definition survey") can expedite the capture of existing conditions as input into the BIM-authoring process. A laser scanner is positioned on a tripod and collects distance data by scanning all the surfaces within the line of sight of the scanner. Accurately mapping a facility with enough detail may require hundreds or even thousands of separate scans that are registered together as a "point cloud." It is called a "point cloud" because it represents the facility as a highly dense collection of points;

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in most cases, this can include billions of separate points, each with unique coordinates. Although laser scanning sounds time consuming, it is a relatively rapid process (it is possible to scan a 2,000 sf room within 5 to10 minutes, depending on the complexity of the room layout).

BIM-authoring software can use point clouds to build the 3D solid models utilized by most user applications. Separate server(s) need to be designated for processing these data, as even small facilities can be highly computationally intensive, and file sizes can easily exceed 1 terabyte or more before processing begins.

Lidar data and sUAS photogrammetry can also benefit from having a designated data-processing server to register their data and create site-civil surface contours and facility 3D models. Photogrammetry can also be used to generate internal 3D geometry as the input for BIM, although not with the precision of laser scanning. However, the level of precision achievable with photogrammetry may be enough for many BIM applications, and photogrammetry can be accomplished more quickly and at a lower cost than laser scanning with current technology.

7.3 BIM-Authoring Servers

The BIM-authoring environment has become increasingly collaborative. Those designing and updating BIM may be working from workstations, updating or annotating BIM from tablets, or even updating BIM asset data from their mobile phones. The native file format of the central BIM is typically on a shared server or, increasingly, on a cloud-based server that maximizes its accessibility both across local area networks and wide area networks. The authoring platform gives users the most robust set of tools for managing BIM, but comes with per-user license fees that do not make this financially attractive as the primary stakeholder interface for BIM for most airports.

The BIM-authoring server will provide an application programming interface (API) that is defined by a vendor-provided software development kit that provides application development for individual users to develop custom applications tailored to their requirements. Identifying off-the-shelf software solutions will always be more economical, but the API may provide a specialized solution where no commercial solution currently exists. In most cases, data will be converted for use by external applications using exporting data as IFC, extensible markup language (XML), or COBie (see Section 8).

7.4 Open BIM Server

An open BIM server enables sharing of BIM data using IFC-based standards. ISO 16379: Industry Foundation Class (IFC) defines the data standard that provides the greatest interoperability between all data systems and BIM applications. It is a structured plain text data definition that translates the asset data, known as attributes, and asset geometry into a standardized format. IFC, although text-based, is difficult to read and maintain. Model view definitions (MVDs) have been created to map the IFC data into formats that are more readable by humans and focused on specific BIM applications. The most popular of these is COBie, which has been designed to standardize the handover of critical facility data to facility owners after construction. While COBie is not the "BIM data standard," it does provide a simple and standard approach to collecting, sharing, and maintaining facility data. As such, many facility management software vendors provide COBie-compliant data interfaces. Facility management software applications include CMMS, EAM, CAFM, and BAS software providers.

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Regardless of the types of data formats an organization settles on as its standard, a stand-alone open BIM server will provide the organization with the greatest flexibility in supporting existing and future data requirements.

7.5 CMMS/EAM

Previous sections have explained the wide variety of BIM uses or applications available during the facility asset life cycle. It is these applications that provide the direct benefit to the stakeholder groups. Many stakeholder groups will continue to interface with facility data through the existing CMMS and the supported functions utilized by the airport. Supported functions could include work order management, asset inventory, maintenance planning, capital planning, and other management systems. However, while the interface would be largely unchanged, the facility information, accuracy, and completeness would be greatly enhanced. A CMMS integrated with BIM can provide coordinate-based location, visual mapping, and system data for managed assets. This allows the CMMS to provide users with the complete spatial and system context of assets for purposes of inventory, maintenance planning, space management, and other CMMS applications.

Most large CMMS platforms now support COBie to simplify integration. Some platforms may support real-time, bi-directional synchronizations that allow changes to the BIM to propagate to all CMMS applications automatically. Likewise, changes to the CMMS asset data are automatically propagated to the BIM, where they are available to field staff. In other cases, "middleware," a separate software that manages the synchronization and translation of BIM data, may be necessary. Alternatively, the exchange may be performed manually with exports and imports of data from each system.

7.6 Application Servers

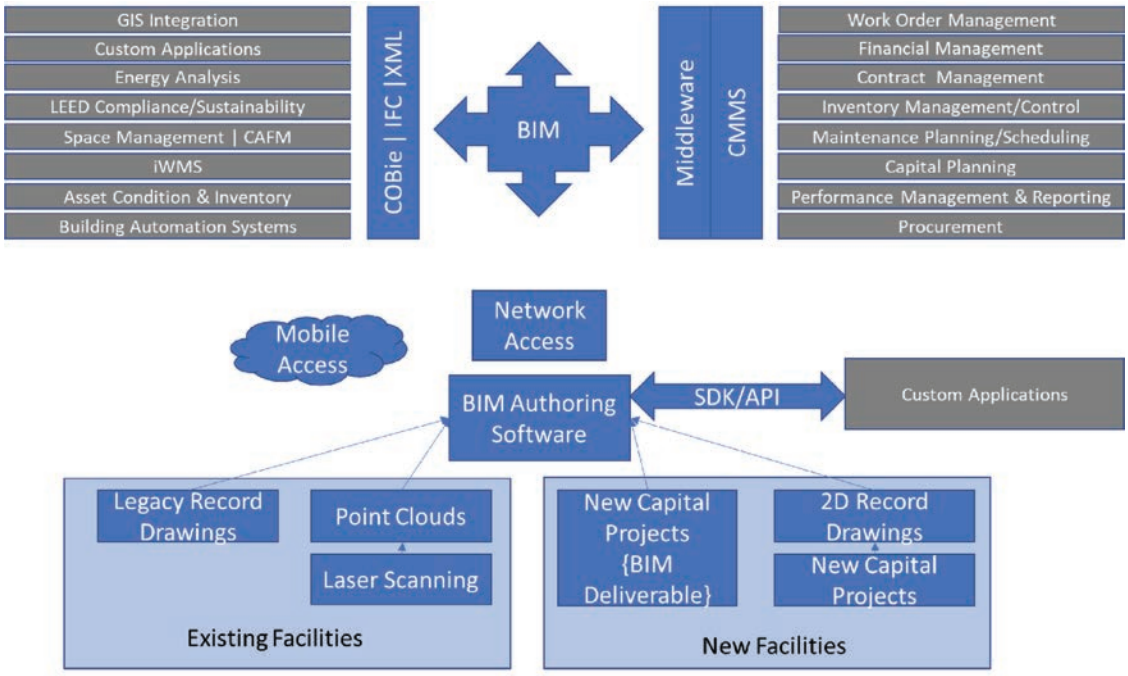
While a CMMS provides a convenient, central clearinghouse for managing and analyzing facility data collected by BIM, many airports will have separate systems maintained for distinct management functions and maintained by different departments. The open BIM server will provide the robust interface for translating and delivering the BIM-generated facility with these data systems. Potential data systems could include IWMS, CAFM, property management systems, asset inventory systems, and others.

There will also be a class of facility management applications that the CMMS does not support or where a non-CMMS integrated solution may have additional desired features or be available at a better price point. Integrated solutions might include energy analysis and usage applications, LEED compliance analysis, and BAS. Having one or more application servers provides the most flexibility in deploying, maintaining, and upgrading these applications with the least disruption to the other applications. It is possible to bundle multiple applications on a single server if care is taken to understand the data-processing, storage, and networking requirements of each application.

7.7 Cloud Server and Firewall

While BIM provides a powerful tool for collecting and sharing facility data, this benefit can also introduce increased risk and exposure to the airport. The sharing of these data across Wi-Fi or cloud-based networks opens the risk of network intrusion. A strong local firewall, or GSA-approved, cloud-based service provider, can minimize these risks.

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Note: SDK – software development kit

Figure 7-4. BIM architecture.

7.8 BIM Application Map

An application architecture map is shown in Figure 7-4 that illustrates how data flow from the BIM to the various application types that may exist within the airport. As noted previously, the BIM interfaces with external applications through the following mechanisms:

- Export of IFC Extensible Markup Language (IFCXML) asset attributes and graphical data
- Export of COBie asset and facility data (does not include graphical data)
- Use of middleware to provide this translation and support bi-directional synchronization
- Direct interface between native BIM-authoring tools and an application with native APIs specific to vendor software packages

Section 7 Checklist

1. Determine the system architecture required to facilitate data exchange among the airport's various facility data management software applications.
2. Document the required facility data exchange in a BIM application architecture map.

7.9 Summary

While BIM-supported user applications can provide significant benefits to productivity across an airport organization, the infrastructure required to support BIM can be extensive. Technology to capture existing conditions can require very high-end processing power and data storage requirements. The translation of native BIM spatial and asset data requires careful planning and a robust data network to support the information exchange requirements between BIM and CMMS/EAM/CAFM and other BIM applications. While one of BIM's most significant benefits is enhancing organizational collaboration across the facility life cycle, this requires a network architecture that meets the facility data access needs of all airport stakeholders, regardless of their physical location.

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SECTION 8

BIM Implementation—Integration of BIM with Existing Systems

Prior sections have discussed how BIM can be a process for sharing facility data and building a collaborative life cycle approach to managing facilities. This section will describe the primary data types, structures, and formats that are generally associated with BIM, and the most common methods of information exchange between BIM and other facility management applications.

BIM can provide accurate and complete facility data, but if these data are not accessible to the organization, little benefit will be realized. To facilitate BIM accessibility, it is critical that the facility data within BIM be integrated into the other existing information management systems so that these data provide direct productivity and quality improvements to O&M activities across the airport.

While major BIM-authoring software applications allow an ad hoc approach to asset data definitions, there are two primary considerations in ensuring the smooth integration of BIM with external systems. The first consideration is the format of the data. The second consideration, particularly important with the integration of BIM and a CMMS and/or EAM system, is the notion of asset classification and mapping in different classification systems. Both considerations will be reviewed in this section, followed by an example of how they are tied together to provide an integrated data management process.

8.1 Data Formats

8.1.1 IFC

IFC, defined by ISO 16739, is the core BIM data standard that is used to provide interoperability with BIM and other external applications. It is a structured, plain text format that can be opened and viewed with any text editor. It also comes in a format (an IFCXML format) that is structured for the IFC text to be easily viewed using web browsers. IFC includes both a definition of asset data types and graphical primitives (commands on how to draw an item in 3D) that can reproduce the size and shape of the assets developed within the BIM. IFC also includes data on how assets are tied together into systems within BIM.

IFC is one of the primary methods used to share facility models developed within one vendor's BIM-authoring system with another. IFC as an integration method works well to some degree, but there may be productivity features within one BIM-authoring tool that do not carry over in the translation to another tool. This is because IFC has no corresponding support for those features. Unsupported features include the parametric design aspect of objects within a BIM that would enable how it is inserted and sized within a BIM. Using IFC as an exchange mechanism between one BIM-authoring platform and another would result in the loss of this parametric object functionality and bring over a static object.

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One area IFC does excel in is supporting open-source, inexpensive viewing platforms that can be used to publish BIM data across an organization. To the extent that access to BIM should be maximized, having open-source options for sharing BIM is valuable.

Other benefits of IFC are apparent in BIM coordination during construction, when various subcontractors are using BIM-authoring platforms from different vendors, and in analysis applications that provide support for performing energy analysis, lighting analysis, LEED compliance analysis, or structural analysis.

Although a detailed analysis of the IFC format is beyond the scope of this research, a snippet of IFC code related to a door shows the complexity that would be involved in using this as a primary human interface:

```
#548 = IFCRELASSOCIATESCLASSIFICATION('3BZU0$SsX19PepvRtakq3K',#41,
       'Uniformat Classification','',(#533),#547);

#552 = IFCCARTESIANPOINT((1.38777878078145E-17,0.));

#554 = IFCAXIS2PLACEMENT2D(#552,#23);

#555 = IFCRECTANGLEPROFILEDEF(.AREA.,'36" x 48"', #554,
       0.0416666666666667, 2.58333333333333);
```

This sample of code illustrates that while IFC can be read by humans, in practice, it is more relevant as a means of data transfer between software applications.

8.1.2 COBie—MVD

The COBie MVD of IFC was designed to make working with IFC data easier. An MVD of IFC defines a subset of the IFC focused on one or more information exchange requirements needed to support end-user applications. While this sounds complex, it is designed to get the IFC mapped into a manageable format.

COBie is defined in Chapter 4.2 of the NBIMS-US V3 National BIM Standards. Additional guidance is provided in "The COBie Guide: A Commentary to the NBIMS-US COBie Standard," (East and Carrasquillo-Mangual, 2013), which provides additional details about implementing COBie.

COBie was developed as a standard to manage the handover of critical O&M data to owners after construction to support facility management applications. Because of this, COBie is the most common means of transferring data from BIM into CMMS and EAM systems.

COBie is not the only means of integrating data between BIM and a CMMS. The data within a BIM can be formatted and classified using any ad hoc approach the owner wishes. However, if the owner decides to use a non-COBie approach, the owner will be responsible for creating the approach's unique mapping of BIM data attributes to the asset classifications used in the owner's CMMS. Using COBie provides a standard consensus approach that is supported by many CMMSs and other management application vendors. A standards-based approach enables the owner to more freely migrate to new systems or replace systems without the added complexity of having to invest in an additional programming effort.

Unlike IFC, COBie is primarily focused on facility and asset data and not on representing the detailed size and shape of each asset within the facility. Other MVDs focus on the geometric data and include the reference view and coordination view. These examples demonstrate how valuable the use of MVD structure is in limiting the size and complexity of the IFC data to fit the need of the application.

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COBie data are very easy to read and manage and have value outside of BIM. Data are often developed, shared, and managed within spreadsheets, and can be easily exported directly from BIM with most major BIM-authoring software. COBie can be viewed and managed as spreadsheets, which are defined by the SpreadsheetML standard; this is generally the format that owners will be using.

COBie data define a facility as a set of zones, spaces, and floors, with assets having types composed of specific components that may be connected as systems. Within the COBie spreadsheet, there will be separate worksheets for contacts, facility, floor, space, zone, asset type, component, and systems. Spatial assets, such as floors and rooms, are identified by space and floor worksheets. Spaces can also be defined using a zone worksheet, where zones might include data on HVAC circulation zones, security zones, fire protection, or other facility space management structures. It is important to note that while spaces must be unique, spaces can be defined as existing in multiple zones.

Equipment is identified in the type and component worksheets, and specific equipment asset information is in the attribute worksheet. Optionally, equipment can be identified as belonging to a system. Figure 8-1 shows COBie's typical data structure.

COBie does not define an asset-naming convention; it only requires that asset names are unique. The COBie commentary from the NIBS suggests a method for ensuring unique asset names by structuring the names based on the COBie space and asset type names. The formula recommended is the following:

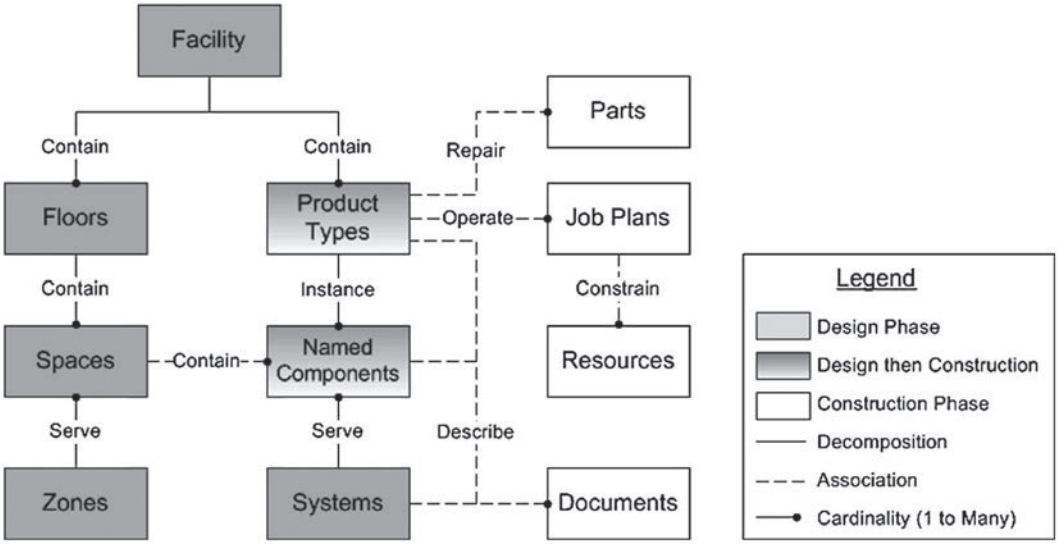
COBie.Type.Name – COBie.Space.Name – Item Count in Space

For example, light fixtures in a lobby might be defined as the following:

Lobby.overhead light – 001, Lobby.overhead light – 002, and so forth

If the COBie data are going to be combined with several other facilities, then the asset names can be prefixed with the COBie.Facility.Name:

TerminalA.Lobby.overhead light – 001



Source: East, 2013

Figure 8-1. COBie data structure.

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Other asset-naming schemas are possible and will work with COBie if the naming conventions ensure uniqueness.

The NBIMS-US V3 COBie standard does not define specific equipment attributes that each component must provide. The COBie commentary (East and Carrasquillo-Mangual, 2013) does provide some guidance on the attributes that should be collected. An example from the commentary is shown as Figure 8-2. The shaded rows are optional entries while the non-shaded rows are required data.

Each owner may include additional attributes; COBie does not restrict the inclusion of additional asset data.

Example COBie Space Worksheet

Figure 8-3 is an example of a space worksheet. Each unique space should have a line and a unique name in the worksheet. In the example, the room tag is used as the space name, and the "Category" column identifiers are using OmniClass classifications (described later in this section).

Example Component Worksheet

Each unique component of the facility must first be defined as a type in a COBie type worksheet. For example, each type of door would have a line in the worksheet shown in Figure 8-4. Likewise, similar doors from different manufacturers would each have a separate line.

The COBie component worksheet would look as shown in Figure 8-5. In this worksheet, each line represents a unique asset. Every door has a specific location and name. Note that the type name for the component must match one of those defined in the type worksheet. Also, note that, in the case of a door, the space description includes the two COBie spaces and the door connects. These COBie spaces must be those defined in the COBie space worksheet.

Heading	Typical Unit
Name	Boiler-TypeXX-Space#-01
Type	Boiler-TypeXX
SpecificationSection	(as identified in client's contract)
Location	(Space Name)
Current	Amps
Voltage	Volts
Frequency	Hz
Input Power	kW
Output Power	kW
Water Flow	m^3/hr
Entering Water Temp	C
Leaving Water Temp	C
Vent Diameter	mm
Passes	-
Pressure Drop	kPa
Energy Source	-
Fuel Type	-
Output Media	-

Source: East and Carrasquillo-Mangual, 2013

Figure 8-2. COBie recommended asset attributes example.

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Category	FloorName	Description	RoomTag	UsableHeight	GrossArea
13-65 13 13 : Toilet Space	Level 1	Toilet	102	2440	4.1929
13-65 13 : Bathroom	Level 1	Bath	106	2440	4.0406
13-57 17 13 : Break Room	Level 1	Break Room	108	2440	15.3995
13-57 13 13 : Kitchen Space	Level 1	Kitchen	112	2750	37.132
13-63 13 11 : Storage Room	Level 1	Outdoor Storage	115	3600	5.2405
13-23 11 13 : Stairway	Level 1	Stair 2	116	3800	11.862
13-23 19 : Utility Equipment Room	Level 1	Mechanical Room	105	3600	36.7902
13-25 : Circulation Spaces	Level 1	Hall	117	2440	36.893
13-57 13 15 : Dining and Drinking Spaces	Level 1	Recreation/Dining Room	118	3000	76.6595
13-23 11 13 : Stairway	Level 1	Stair 1	125	3800	9.7479
13-63 13 11 : Storage Room	Level 1	Food Storage	113	2750	13.3101
13-65 : Private Residential Spaces	Level 1	Living Room	121	2750	17.5909

Figure 8-3. Example COBie space worksheet.

Category	Description	AssetType	Manufacturer	ModelNumber
23-17 11 13 : Metal Doors	External Single Door	Fixed	n/a	n/a
23-17 11 15 : Wood Doors	Internal Single Door	Fixed	n/a	n/a
23-17 11 15 : Wood Doors	Internal Single Door	Fixed	n/a	n/a
23-17 11 15 : Wood Doors	Internal Single Door	Fixed	n/a	n/a
23-17 11 15 : Wood Doors	Internal Double Door	Fixed	n/a	n/a
23-17 11 13 : Metal Doors	External Double Door	Fixed	n/a	n/a
23-17 11 15 : Wood Doors	Internal Single Door With Glazing	Fixed	n/a	n/a
23-17 11 13 : Metal Doors	External Door Part of a Window Unit	Fixed	n/a	n/a
23-17 11 27 17 : Roof Hatches	Roof Access Hatch	Fixed	n/a	n/a

Figure 8-4. Example COBie type worksheet.

TypeName	Space	Description	SerialNumber	InstallationDate	WarrantyStartDate	TagNumber
DR:T1E	209,209	Internal Double Door	n/a	10/1/2017	2/1/2018	100118
DR:T1E	211,211	Internal Double Door	n/a	10/1/2017	2/1/2018	100119
DR:T1E	213,213	Internal Double Door	n/a	10/1/2017	2/1/2018	100120
DR:T1E	215,215	Internal Double Door	n/a	10/1/2017	2/1/2018	100121
DR:T2A	105,128	External Double Door	n/a	10/1/2017	2/1/2018	100122
DR:T2A	115,128	External Double Door	n/a	10/1/2017	2/1/2018	100123
DR:T4A	101,117	Internal Single Door With Glazing	n/a	10/3/2017	2/1/2018	100124
DR:T4A	121,123	Internal Single Door With Glazing	n/a	10/3/2017	2/1/2018	100125
DR:T6A	123,128	External Door Part of a Window unit	n/a	10/3/2017	2/1/2018	100126
DR:T7A	202,301	Roof Access Hatch	n/a	10/3/2017	2/1/2018	100127

Figure 8-5. Example COBie component space definitions.

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A detailed tutorial on COBie is beyond the scope of this Guidebook. Additional COBie resources are included in the bibliography. The critical aspects of COBie that airport owners should understand are the following:

- COBie can provide a structured system for asset data handover after construction.
- COBie defines asset data, not 3D geometry.
- Using COBie provides a consensus standard approach that is supported by many CMMSs and other management application vendors. A standards-based approach enables the owner to more freely migrate to new systems or replace systems without the added complexity of having to invest in an additional programming effort.
- While COBie is a good option for data integration with a CMMS, full IFC is a better option for integration that requires 3D visualization, floor plans, and other spatial geometry.

8.2 Classification Systems

BIM supports a standard for asset classification and, because OmniClass is the most prevalently used system, it is a good starting point for the discussion of how to classify airport assets.

8.2.1 OmniClass/UniFormat

OmniClass and UniFormat are classification systems developed for the construction industry. OmniClass includes 15 tables that classify construction environment information:

- Table 11—Construction Entities by Function
- Table 12—Construction Entities by Form
- Table 13—Spaces by Function
- Table 14—Spaces by Form
- Table 21—Elements
- Table 22—Work Results
- Table 23—Products
- Table 31—Phases
- Table 32—Services
- Table 33—Disciplines
- Table 34—Organizational Roles
- Table 35—Tools
- Table 36—Information
- Table 41—Materials
- Table 49—Properties

OmniClass, as its name suggests, encompasses several other classification systems. The Construction Specifications Institute (CSI), Master Format standard, is part of "Table 22—Work Results." This table classifies a facility regarding the specific components required to build it and to support cost estimating. The CSI UniFormat standard is part of "Table 21—Elements," which identifies specific facility elements. CSI Master Format "Table 23—Products" specifies standard product types currently used in the construction and operation of facilities.

While an owner's non-standardized asset classification system can be fully integrated with BIM, the use of a standardized system is a step toward creating a sustainable asset life cycle approach that can leverage industry standard tools, processes, and experience to lower overall facility management costs.

8.2.2 Custom Asset Data Schema

Migrating to a new asset classification system can be a time-consuming and costly process. Adopting a standards-based data schema will eliminate the cost and the potential introduction

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of errors of the data translation that will be required to utilize industry-developed productivity and collaboration tools that require standardized interfaces. The alternative is to invest in creating a custom data mapping standard that will show how the airport's custom asset classification maps to existing standards. The considerations for integration are the following:

- Defining the AIRs before BIM development
- Defining an AIR quality control process to validate asset data before integration
- Mapping BIM asset unique identifiers with assets in the CMMS
- Being able to classify assets by their BIM systems
- Being able to classify asset zones and spaces within BIM

8.3 Integration Process

Much of the preceding narrative has been focused on BIM data integration with a CMMS system. This narrative covered the general data formats and asset classification systems that airport owners will need to understand. The following sections provide a recommended process for creating a maintainable BIM–CMMS integration.

8.3.1 Create BIM AIRs

Airports are composed of many facility types. As BIM is developed for these facilities and delivered either through new construction or from as-built data, it is important that a consistent asset data schema, or AIRs, is used. Included in the AIRs are the requirements for the AIM that define what LOD is required to support the desired external integration with asset management, space management, sustainability analysis, disaster planning, and other operational analysis applications.

With the current level of technology, a record model BIM delivered as part of a construction handover process may be overly complex and cause performance issues with the integration. The construction LOD required for clash detection, workflow coordination, and sequencing is greater than that required for asset inventory or maintenance planning. Also, the BIM objects that many manufacturers supply are often overly complex and designed to look photorealistic rather than designed to provide the minimum graphical model requirements (size, shape, critical inputs/outputs, and access points). For space-planning purposes, an LOD of 200 may be enough to support the integration of floor layouts. LOD 300 is likely enough for most asset management applications. The reduced detail model is the AIM. Over time, as the BIM software and hardware continue to improve and reduce issues with handling large complex files, the need to optimize files or reduce complexity will be minimized.

8.3.2 Define Mapping of AIR to CMMS Classification and Component Elements

Integration of BIM with end-user applications requires mapping both component-level elements and space elements and addressing how they are organized into systems. Integration requires mapping the AIR elements onto the CMMS data dictionary. In the absence of standards, this mapping can be a time-consuming and costly task. The use of COBie and a CMMS that supports COBie can greatly accelerate this task.

8.3.3 Integrate BIM and the CMMS

The term "integration" can represent many things. Exporting data from BIM into a standard format (such as COBie or IFC) and exporting it into another application is a form of

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integration. It is not a tight integration, such as a bi-directional interface between two systems that ensures updates into one system are updated in the other. The range of automated integration capabilities varies across different CMMS or other facility management software platforms, and this should be well understood before beginning this process. If the CMMS is not capable of the desired level of integration, middleware software is available that can provide a bridge between the CMMS and BIM to provide these types of capabilities. Table 8-1 is a high-level mapping of COBie data to an IBM Maximo data system.

Some end-user systems will not only support the exchange of component and space data but will also support the exchange of graphical data. For example, selecting a component in the CMMS links the user to the BIM and to a view of the selected component that allows the user to review the location and surrounding environment.

8.3.4 Integrate BIM and BASs

Another area of intense interest is integrating BIM with BASs.

With the emergence and rapid adoption of IoT real-time sensors, the data available to BASs and other facility management applications will enable optimization and performance levels not previously achievable in buildings. BIM is important to BASs because it provides a coordinate-based spatial context to the real-time sensor data collected. BIM integration enables BASs and other building analytic software to correlate events and provide a level of root cause analysis.

The integration tools for BIM and BAS (see Figure 8-6) rely upon IFC. There is currently a Building Automation Modeling Information Exchange (BAMie) in development that is a subset of IFC and is focused on building automation (just as COBie was focused on facility data handover after construction). This standard has not yet been fully released, or adopted widely, so the IFC integration still relies upon customizing the BIM.

8.3.5 Integrate BIM and GIS

GIS is typically used for the site-civil asset documentation, while BIM provides a view of the airport's building facilities. While there is a likelihood that these data types will converge in the future to provide a complete digital model of airport facility infrastructure, today they are still documented in very different types of systems: GIS and BIM.

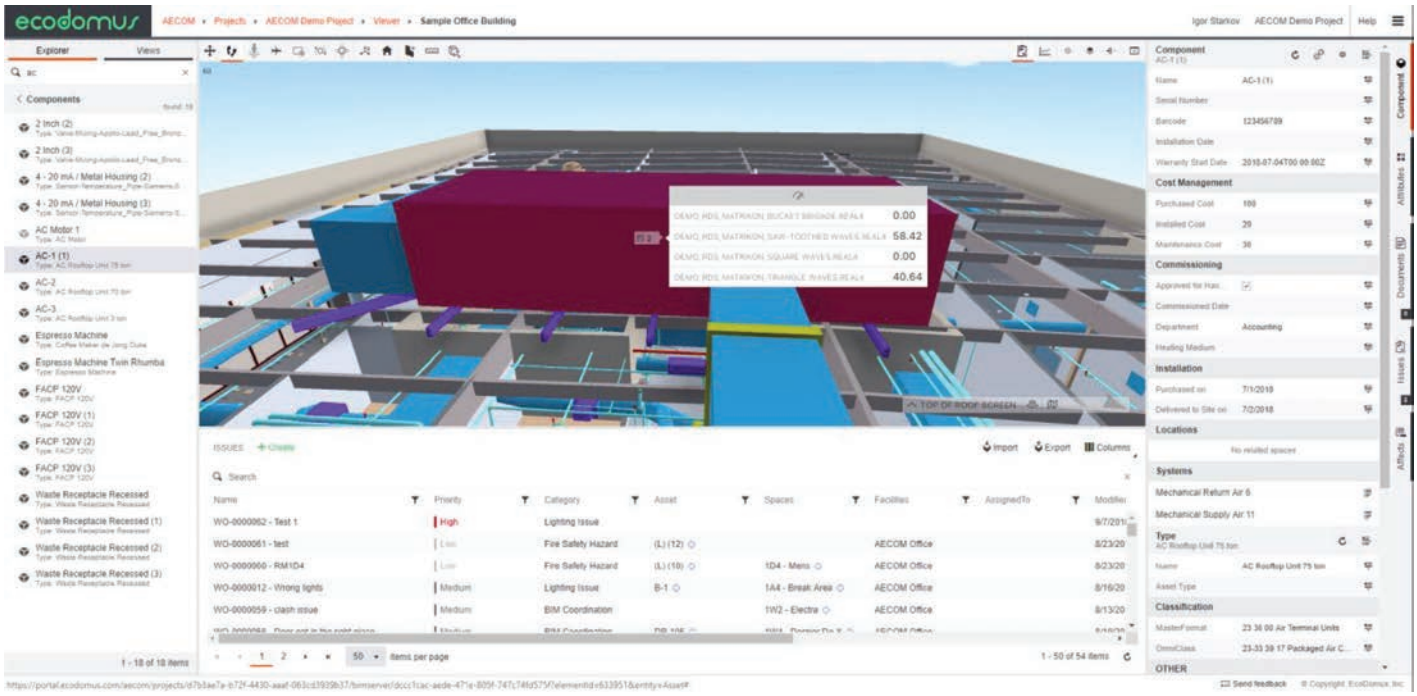
GIS and BIM can be seen as developing out of the 2D CAD environment. GIS added a rich database environment and programmatic interface onto 2D maps. BIM similarly added a

Table 8-1. High-level mapping of COBie data to an IBM Maximo data system.

COBie Data Type	Maximo Object Type
Contacts	PERSON, COMPANIES, COMPANY CONTACT
Facility	LOCATION
Floor	LOCATION
Space	LOCATION
Component	LOCATION AND ASSET RECORD
Zone	LOCSYSTEM
System	LOCSYSTEM
Attributes	Values in SPECIFICATION TABLE of ASSET TYPE

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Source: Ecodumus

Figure 8-6. BIM–BAS integration interface.

database of facility asset data and a programmatic interface for simulations to 2D CAD. While GIS was primarily 2D, it now supports 3D elements.

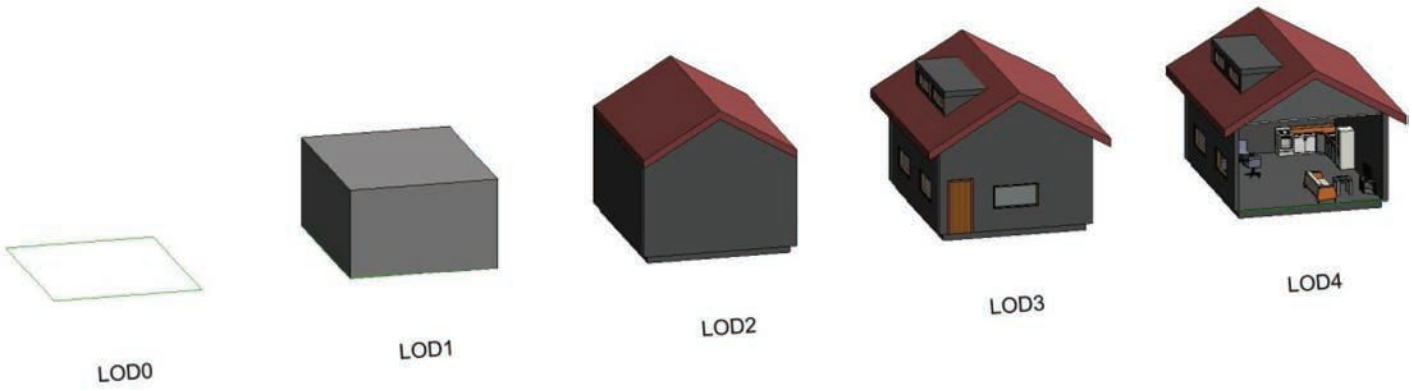
It would be desirable to be able to leverage the considerable amount of information on horizontal facility infrastructure at airports (such as underground utilities, roadways, and airside pavement) within BIM to provide a more comprehensive view of the entire airport for planning, design, construction, operations, and maintenance activities. Major GIS and BIM software vendors have begun to actively cooperate to simplify the information exchange between GIS and BIM. However, this coordination is still in the early stages, and more manually intensive methods are still required. Depending on the airport stakeholder group, it may be more beneficial either to integrate BIM data into the GIS or to integrate the GIS data into the BIM. Land survey staff may prefer the ability to see facility BIM data as part of their GIS base maps. Facility managers may prefer to see GIS underground utility asset locations and data in the context of their facility BIM. While there is currently no standard method for this exchange, two options are provided in the following.

8.3.6 BIM Data Integrated into GIS

One approach is to use the IFC export from BIM and map these data onto a City Geography Markup Language (CityGML) format that can be used with GIS. CityGML is an open information model standard developed by the Open Geospatial Consortium for the visualization and exchange of 3D city model data. It also has a concept of LOD similar to BIM, in which LOD 4 has not only accurate building shapes but also accurate internal floor plans.

Another approach is to use the WebGL (Web Graphics Library) to combine GIS and BIM data (see Figure 8-7) into a single view. This approach does not bring the BIM data into

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Source: CityGML

Figure 8-7. CityGML LOD example.

the GIS database, but it does provide a common interface for viewing and accessing the data within BIM and GIS.

One of the greatest challenges with BIM and GIS integration is BIM not being aligned with real-world geospatial coordinate systems. While BIM can utilize real-world spatial coordinate systems, in practice most architectural and construction models are built around internal reference coordinate systems. Converting internal reference coordinate systems to real-world geospatial coordinate systems after construction can be problematic and time consuming, so requiring the use of geospatial coordinates as part of the BIM-authoring standards is recommended.

8.3.7 GIS Data Integrated into BIM

Until the specific BIM-authoring tools to provide closer integration with tools available in the GIS world are more fully developed, bringing GIS data into native BIM-authoring tools will require some intermediate integration to convert the data into a format more usable by BIM. The required workflow would bring GIS data into a 3D site-civil design tool where contour surface maps could be created. GIS shapefiles would need to be used as a guide to developing 3D civil infrastructures such as roadways, taxiways, pavements, and underground utilities. The site-civil 3D infrastructure developed from the GIS, along with the GIS attribute data, can then be integrated into a BIM-authoring platform. The integration is somewhat straightforward if tools from the same software vendors are being used, but can also be done using IFC, if necessary.

8.4 Summary

Integration of BIM with other facility data management systems can be a complex activity. Although software and technology vendors are working toward more highly integrated solutions, it is still early in the process, and available solutions will still require some level of manual effort or customization. The value of BIM, however, increases dramatically as the information available in BIM is shared across the facility information management infrastructure. The more systems that share common data, the more value BIM delivers.

Standards-based BIM asset data schemas will maximize the benefits of interoperability while at the same time minimizing the cost of integration. Custom data translation is possible

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to support existing non-standard data classification systems; however, there will be a cost associated with integration, and it could introduce the possibility of future data translation errors.

The facility management industry is rapidly evolving to accommodate innovation in facility information management. These innovations promise to radically improve the information and real-time controls available to facility managers. These include IoT real-time sensor data, BASs, predictive maintenance systems, AI-based decision-making support, and automated construction and maintenance systems.

Developing a BIM with an open standards-based technical architecture will position the airport to leverage existing BIM tools, processes, and standards, and to benefit from future innovations.

Section 8 Checklist

1. Determine the BIM data standard to be used to provide interoperability with BIM and other external applications.
2. Create BIM AIRs.
3. Define mapping of AIR to CMMS classification and component elements.
4. Integrate BIM and the CMMS.
5. Integrate BIM and BASs.
6. Integrate BIM and GIS.

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SECTION 9

BIM Implementation—Standards, Execution Plans, Required Data Elements

Previous sections of the Guidebook have presented an overview of the BIM process, an evaluation of how BIM can address an airport's specific organizational needs, and direction on how to build a business case to implement BIM. This section will address how to begin the design of a BIM program. The following elements should be established by an airport that wants to create an effective BIM program:

- Standards—guidelines for the design of the BIM program that will maximize the interoperability of the BIM system with other, existing BIM systems and guidelines for the tools and processes that will provide the best path to reusability with future developments and innovations maximizing the longevity of the BIM system.
- Execution plans—templates for managing the BIM process that define the roles, responsibilities, BIM data and spatial requirements, process diagrams, and overall structure of the BIM program.
- Data elements—options for the design of the data structure of the facility, systems, and assets within the BIM.

The United States does not have a national BIM mandate that requires the use of BIM and does not require the adoption of a specific standard for BIM (such as the UK BIM Mandate). While each airport could completely customize its own BIM programs, adopting a standards-driven approach will greatly reduce the time and cost of developing those programs.

Most BIM programs have been developed to manage BIM on a project basis for the planning, design, and construction of new or renovated facilities. The use of BIM for the entire facility life cycle (and particularly for O&M) is still not widespread. However, the project-oriented BIM standards and processes developed for design and construction can be adapted for use in O&M as well.

9.1 BIM Standards

The United States does have a National BIM Standard, NBIMS-US V3, which was developed by NIBS. NIBS is a non-profit agency established in 1974 by the U.S. Congress with a mission to "unite the entire building community in advancing building science and technology." NBIMS-US V3 is a consensus document that has been developed by representatives of private industry and public agencies to identify and provide solutions to "facilitate the efficient life cycle management of the built environment supported by digital technology." The end goal is to create a reliable and accurate process for the creation, maintenance, and management of BIM data throughout the facility life cycle.

The stated goals of NBIMS-US V3 are the following:

- Reducing the TCO of the built environment and its impact on the natural environment via timely, accurate, reusable information for the management of a project through its life cycle.

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- Enabling collaboration and information sharing among all shareholders via established products, methods, and information formats.
- Prescribing information development and sharing via consensus documents that promote a consistent, common path forward when multiple divergent paths were once available.
- Creating a standard expectation of BIM processes and deliverables, thus creating predictability and consistency in costs and outcomes.
- Sharing information with software vendors, as well as other product and service providers, to build solutions that support the consensus agreements of practitioners.

NBIMS-US V3 standards encompass the following elements:

- Technology reference standards
- Practice standards

Although there are a variety of international standards, the core concepts described in this document will be the same. AAAE is in the process of developing a BIM standard for airports.

9.2 Technology Reference Standards

BIM is composed of model elements that represent the spatial characteristics of a facility and the data structures that represent the underlying asset information. While the details of each technology reference standard are not necessary for this Guidebook, they are included as reference points for use in building an airport's BIM requirements and capabilities.

9.2.1 ISO 16739 IFC 2x3

IFC is the underlying data schema that defines a neutral "exchange format" for BIM. An information exchange format is a standardized listing of what data need to be defined for each type of asset in a BIM. By creating a standard list with standardized names, an "information exchange" can facilitate the sharing of data between a BIM developed using one system and a BIM developed using different BIM-authoring and development tools. These include software applications for energy analysis, BASs, and code compliance. BIM-authoring tools use IFC exports as the most interoperable information exchange export option. The most widely used form of IFC data is in the plain text STEP format; however, there is also an IFCXML format that conforms to the standards for XML encoding defined by the World Wide Web Consortium (W3C). The IFCXML file sizes are larger and primarily used to exchange portions of the model. The benefit of XML is the wide range of existing web-based tools supporting XML data exchange.

9.2.2 Green Building Extensible Markup Language

The Green Building Extensible Markup Language (gbXML) format was developed specifically to support the exchange of BIM data with applications supporting architectural and engineering sustainability analysis such as energy utilization and HVAC load calculations.

9.2.3 OmniClass

OmniClass is a classification system that was generated for the construction industry to organize asset data for electronic databases. Although it was developed for the construction industry, this classification system works equally well for the O&M phase of a facility's life cycle. OmniClass has classifications at the facility level, system levels, and specific asset level. It also has classifications for other construction-related elements such as phasing, organizational

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management roles, tools, materials, and properties. Utilizing an OmniClass classification system in a CMMS will provide the best interoperability between the BIM and CMMS to support life cycle facility management services.

9.2.4 UniFormat

OmniClass has worked to provide interoperability with other classification systems such as UniFormat and MasterFormat. UniFormat provides a classification system that is focused on just the physical parts of the facility (systems and assemblies). It does not include classifications for the non-asset data such as organizational roles and responsibilities. UniFormat is often utilized in cost-estimating applications based on its streamlined form. Although UniFormat cannot classify elements of the overall organization and process, its streamlined focus on specific assets can make it a good choice for many organizations.

9.2.5 National CAD Standard

While BIM is a 3D modeling environment, the reality is that many airports will continue to utilize 2D plans as a primary source of facility data for their O&M workflows. NBIMS-US and the National CAD Standard (NCS) are standards that were developed and are maintained by NIBS. Although NBIMS-US supersedes some aspects of the NCS, most of the NCS requirements can be implemented with BIM processes and tools. Using NCS as a guide for 2D plan production can simplify the 3D-to-2D conversions, although this is not a requirement.

9.2.6 W3C XML

W3C XML is a standard developed to facilitate the exchange of BIM data in an Internet-based environment. This standard may take on greater importance as mobile-based BIM applications find greater adoption in the industry. Several of the major BIM/GIS software vendors are looking at W3C-compliant systems for sharing data between platforms.

9.3 Practice Standards

NBIMS-US V3 also provides guidance on planning for facility owners for three types of procedures: strategic planning, implementation planning, and procurement planning:

1. **Strategic planning** is focused on an organizational needs assessment and on creating a set of BIM goals that will address these strategic needs.
2. **Implementation planning** is focused on developing the details of how the organization will create, operate, and maintain BIM to meet its strategic goals.
3. **Procurement planning** is focused on how BIM can enhance the contracting process.

Six core NBIMS "BIM planning elements" are listed below:

1. **Strategy—**defines the BIM goals and objectives, assesses change readiness, and considers management and resource support.
2. **BIM uses—**identifies the methods through which BIM will be implemented for gathering, generating, processing, communicating, and realizing information about the owner's facilities.
3. **Process—**describes the means used to accomplish BIM goals and objectives through documenting the current methods, designing new processes to leverage BIM, and developing transition plans.
4. **Information—**documents the information needs of the organization, including the model element breakdown, LOD, and facility data.

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5. **Infrastructure—**determines the technology infrastructure to support BIM including computer software, hardware, networks, and physical workspaces.
6. **Personnel—**establishes the roles, responsibilities, education, and training of the active participants in the BIM processes established.

9.4 BIM Execution Plan Templates

The "BIM Project Execution Plan" template developed by the CIC Research Program at Penn State still serves as the primary tool to manage BIM projects throughout the United States. A link to Version 2.0 of this template is included in Appendix G of the *BIM Project Execution Planning Guide, Version 2.2*: https://psu.pb.unizin.org/bimprojectexecutionplanningv2x2/back-matter/appendix-g-bim-project-execution-plan-template/ (Messner et al., 2019). While a wide variety of other BIM execution plan templates exist (such as those developed by Stanford's Center for Integrated Facility Engineering and the Massachusetts Institute of Technology), the Penn State "BIM Project Execution Plan" template still serves as the starting point for most organizations in developing their systems. The "BIM Project Execution Plan" template is oriented toward managing construction projects, although the elements of the "BIM Project Execution Plan" can be adapted to execute BIM across an organization for O&M purposes. The elements of the "BIM Project Execution Plan" are described in the following.

9.4.1 Project Information

This information includes data about the owner, the project, and the project phasing. For an organizational BIM development effort, the "project" would include the BIM strategy and anticipated timeline.

9.4.2 Roles/Responsibilities

This element includes key leaders in the BIM development effort and their roles. This will enable stakeholders across the airport's organization to identify who to contact about issues/questions.

9.4.3 Project Goals/BIM Uses

This is a prioritized list of the strategic/operational goals for the BIM implementation including information on how particular BIM uses will support achieving those goals (see Figures 9-1 and 9-2). A mapping of BIM uses by life cycle is included. BIM uses shown in Figure 9-2 are for illustration purposes only.

9.4.4 Organizational Staffing

This element identifies new organizational roles required, existing roles that may require training, staffing levels, leaders, and location of staff. New roles might include BIM managers,

PRIORITY (HIGH/ MED/ LOW)	GOAL DESCRIPTION	POTENTIAL BIM USES

Source: "BIM Project Execution Plan—Version 2.0" template

Figure 9-1. BIM project execution plan—project goals.

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X	PLAN	X	DESIGN	X	CONSTRUCT	X	OPERATE
	PROGRAMMING		DESIGN AUTHORING		SITE UTILIZATION PLANNING		BUILDING MAINTENANCE SCHEDULING
	SITE ANALYSIS		DESIGN REVIEWS		CONSTRUCTION SYSTEM DESIGN		BUILDING SYSTEM ANALYSIS
			3D COORDINATION		3D COORDINATION		ASSET MANAGEMENT
			STRUCTURAL ANALYSIS		DIGITAL FABRICATION		SPACE MANAGEMENT / TRACKING
			LIGHTING ANALYSIS		3D CONTROL AND PLANNING		DISASTER PLANNING
			ENERGY ANALYSIS		RECORD MODELING		RECORD MODELING
			MECHANICAL ANALYSIS				
			OTHER ENG. ANALYSIS				
			SUSTAINABILITY (LEED) EVALUATION				
			CODE VALIDATION				
	PHASE PLANNING (4D MODELING)		PHASE PLANNING (4D MODELING)		PHASE PLANNING (4D MODELING)		PHASE PLANNING (4D MODELING)
	COST ESTIMATION		COST ESTIMATION		COST ESTIMATION		COST ESTIMATION
	EXISTING CONDITIONS MODELING		EXISTING CONDITIONS MODELING		EXISTING CONDITIONS MODELING		EXISTING CONDITIONS MODELING

Source: "BIM Project Execution Plan—Version 2.0" template

Figure 9-2. BIM project execution plan—BIM uses.

BIM designers, and BIM IT technicians. Roles requiring training could include project managers, maintenance technicians and planners, airport planners, CAD/GIS technicians, and asset management support. (See Figure 9-3.)

9.4.5 Process Map Design

The BIM process can be complex and will require input from a wide variety of stakeholders. The use of process maps (flowcharts) can be used to manage how BIM data are created, maintained, and used throughout the airport organization. Additional detail about building process maps can be found in Section 5. Figure 5-3 provides a BIM process map example.

9.4.6 Information Exchanges

There are two types of information exchange worksheets. The first identifies what model elements are being produced and who is responsible for them. The second shows the model

BIM Use	Organization	Number of Total Staff for BIM Use	Estimated Worker Hours	Location(s)	Lead Contact
Maintenance Scheduling	Aviation Maintenance	12	18,500	Airport	Maintenance Supervisors
Energy Analysis	Sustainability	3	300	Airport	Director of Asset Management
Asset Inventory	Finance	3	3,000	Airport-Admin Office	Director of Finance

Source: "BIM Project Execution Plan—Version 2.0" template

Figure 9-3. BIM project execution plan—staffing plan.

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progression by life cycle phase, from planning, design, construction, to O&M. If BIM is being developed for an existing facility that is not under construction, only the operations column would be needed to identify which facility elements would be modeled. Each item on the worksheets includes an information descriptor and responsible party, as shown in Figure 9-4.

While the list of contributors under "Responsible Party" in Figure 9-4 was primarily developed for construction BIM, this list can be amended to include other airport stakeholder groups if BIM is being developed internally for existing facilities. For example, airport engineering, survey, or maintenance management could be assigned tasks in developing the BIM model and asset data.

The information ranking of A, B, and C (shown under "Information" in Figure 9-4) indicates the LOD ranking (A, B, or C) of each portion of the planned BIM. A numeric ranking from 100 to 500 is more typically used, where 500 represents the most detailed and complete data.

The abbreviation "LOD" is used to represent both "level of development" and "level of detail," but it is important to note that these are different things. "Level of development" designates the overall level of completeness of BIM spatial and facility data, whereas "level of detail" designates the level of graphical accuracy in model elements (in this Guidebook, LOD stands only for "level of development"). In evaluating a BIM, it is important to understand that the LOD is being assessed and not just the graphical level of detail. The "BIMForum 2018 Level of Development" is an excellent guide to understanding and implementing level-of-development standards. LOD is covered in greater detail as the BIM required data elements.

9.4.7 BIM Data Requirements

This term includes all the facility data requirements. The Penn State CIC Research Program's "BIM Project Execution Plan—Version 2.0" template does not provide a format for how this information should be presented, but this section is critical to successfully utilizing BIM during O&M and to integrating BIM with a CMMS and other data systems.

Since there isn't a standard template, each airport organization can develop standard methodology for defining BIM data requirements that match its asset management methodology, asset classification structure, and asset data dictionary. Alternatively, a standard BIM MVD, such as COBie, can be used to define the required data attributes. One advantage to using COBie is that many software vendors have integrated COBie data integration modules into their systems to facilitate the transfer of data from BIM into their systems. COBie is sometimes perceived as the standard BIM data format, but BIM does not require COBie-formatted data, and COBie does not require BIM. COBie data can be, and often are, supplied as a simple spreadsheet of facility asset data.

Information	
A	Accurate Size & Location, include materials and object parameters
B	General Size & Location, include parameter data
C	Schematic Size & Location

Responsible Party	
ARCH	Architect
CON	Contractor
CE	Civil Engineer
FM	Facility Manager
MEP	MEP Engineer
SE	Structural Engineer
TC	Trade Contractors

Source: Messner et al., 2019

Figure 9-4. BIM project execution plan—information exchange.

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MVD refers to the mapping of an asset data model to the IFC data model. A variety of MVDs exist, not just COBie, including the Coordination Model View, the Electrical System Information Exchange (SPARKie), the HVAC information exchange (HVACie), and the Water System information exchange (WSie). These systems have not been widely used but are available for use as standards or as a starting point for developing custom model views.

Regardless of the asset model defined, it is important that the asset data model is defined early in the process to ensure that, when the BIM is authored, all the needed model attributes are included in the BIM. This effort can occur after the BIM is authored but performing this effort early on can prevent rework. The process of defining the required asset data often leads to developing a more precise definition of what needs to be included in the BIM.

9.4.8 Collaboration Strategy

For design and construction projects, this defines how the architect, contractor, and subcontractors use BIM to improve collaboration. This portion of a BIM project execution plan defines who is involved at each stage and how frequently they update and review the BIM.

The collaboration strategy for facility management–driven projects defines how the BIM is initially developed and who is involved in facility data acquisition, BIM authoring, and integration. It also defines which systems will be integrated and when. After BIM development, the collaboration strategy should include how BIM will be maintained and how frequently, who is responsible for maintaining the BIM, and how BIM accessibility is provided to the various airport stakeholders who can benefit from BIM. Performing a BIM needs assessment is an excellent opportunity to identify the collaboration framework for BIM at an airport. The BIM needs assessment process is described in more detail in Section 2.

9.4.9 Technology Needs

This element defines what the technology infrastructure requirements are for the BIM program. Technology needs include software, hardware, tablets/mobile devices, network infrastructure, laser scanners, drones, augmented reality/virtual reality equipment, and other visualization devices.

9.4.10 Quality Control

This defines the procedures for quality control of BIM development and maintenance, who is responsible, and with what frequency. Quality control is a critical element of the success of BIM as a collaborative operational tool. If shared data are seen to be inaccurate or dated, each airport stakeholder group may return to developing and maintaining its own facility data sources, eliminating the organizational benefits achieved through improved collaboration and communication. Figure 9-5 is a sample quality control table from the Penn State CIC Research Program's "BIM Project Execution Plan—Version 2.0" template.

The template also requires model accuracy and tolerance. It is important not only to define the tolerances, but also to have the capability to measure whether those tolerances are being met (through the use of laser scanning, photogrammetry, or other means of measurement).

9.4.11 Model Structure

It is important to define the structure of the BIM before it is developed, since it will be time consuming to make changes after the BIM has been developed. The structure of the BIM defines what subsystems make up the overall BIM. Typically, the model structure would be

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CHECKS	DEFINITION	RESPONSIBLE PARTY	SOFTWARE PROGRAM	FREQUENCY
VISUAL CHECK	Ensure there are no unintended model components and the design intent has been followed			
INTERFERENCE CHECK	Detect problems in the model where two building components are clashing (including soft and hard clashes)			
STANDARDS CHECK	Ensure that the BIM and AEC CAD standards have been followed (fonts, dimensions, line styles, levels/layers, etc.)			
MODEL INTEGRITY CHECKS	Describe the QC validation process used to ensure that the Project Facility Dataset has no undefined, incorrectly defined, or duplicated elements and that there is a reporting process for non-compliant elements and corrective action plans			

Source: "BIM Project Execution Plan—Version 2.0" template

Figure 9-5. BIM project execution plan—QC plan.

used to clearly distinguish between different facility systems such as the architectural, civil, structural, HVAC, plumbing, and electrical models.

Additional structure can be defined by building, floors, functional areas, and areas of responsibility (such as tenant, airlines, and airport spaces). In some cases, this type of structure may already exist in the CMMS/asset management system, and this breakdown could be reflected in the BIM structure as well. The naming convention of the models should be consistent to enable ease of combining facility BIMs for O&M workflows. Figure 9-6 shows the proposed standard for the naming of the separate model types that will make up the overall combined model, known as a federated model.

The model structure also includes the asset location coordinate system. While connecting BIM to a true coordinate system may not be necessary for a single facility BIM, it becomes increasingly important when developing a BIM composed of multiple facilities and linear civil infrastructure (such as utilities, runways, parking lots, and roadways). If the BIM is not developed using a common coordinate system, it will be difficult to combine the BIM into an overall BIM of the airport or to integrate that model with external data sources with coordinate

FILE NAMES FOR MODELS SHOULD BE FORMATTED AS:	
DISCIPLINE - PROJECT NUMBER – BUILDING NUMBER.XYZ	
ARCHITECTURAL MODEL	ARCH-
CIVIL MODEL	CIVIL-
MECHANICAL MODEL	MECH-
PLUMBING MODEL	PLUMB-
ELECTRICAL MODEL	ELEC-
STRUCTURAL MODEL	STRUCT-
ENERGY MODEL	ENERGY-
CONSTRUCTION MODEL	CONST-
COORDINATION MODEL	COORD-

Source: "BIM Project Execution Plan—Version 2.0" template

Figure 9-6. BIM project execution plan—model structure.

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translations. Ideally, the BIM should be tied to a world coordinate system for maximum interoperability with external databases and to optimize BIM and GIS integration; however, nothing in the BIM standards requires this. The "model structure" component also identifies the relevant BIM and CAD standards adopted by the airport.

9.4.12 Project Deliverables

Although this element is oriented toward the project deliverables associated with a construction project, it can also be used to define the phased deliverables in developing a BIM of existing facilities and the BIM infrastructure/integration efforts needed to support O&M BIM uses across an airport.

9.4.13 Delivery Strategy

As with previous elements, this one has a focus on the contract and team selection for new construction/renovation projects. However, it can be used to define the overall BIM implementation strategy by designating which activities are performed using internal resources, which activities require external consultants, and what the criteria are for identifying and selecting external consultants. For long-term BIM implementation development efforts, this element can also include the option of training and developing internal staff with the skills needed to perform future phases of the BIM program.

9.5 Required Data Elements

The BIMForum, which is the U.S. chapter of buildingSMART International, has a detailed LOD specification that includes examples of graphical level of detail for various facility asset types and a listing of recommended baseline asset attributes that should be included in a facility BIM. Defining the required data elements within an airport's BIM should be a collaborative effort that involves airport stakeholders who utilize facility data across the entire asset life cycle. In the absence of quality input from the potential users of BIM at the airport, it is very easy to over-specify the requirements and inflate the costs to a level where a business case is difficult to make. Without quality input from each airport stakeholder group, it would also be easy to under-specify the BIM requirements and end up with a system that is not usable by the end users. Using the BIMForum (or other open standard) as a starting point to involve airport stakeholders in a collaborative effort to define the BIM LOD requirements necessary to support their job functions will be the most cost-effective and thorough approach.

9.5.1 BIMForum 2018 LOD Attribute Definitions

The BIMForum standards include a LOD specification spreadsheet that identifies asset attributes for a wide variety of asset types. The attributes are identified as "baseline" attributes that all assets of classification should incorporate and "additional" attributes that an organization may incorporate. The following is an excerpt of the LOD definitions from the BIMForum (BIMForum, 2018):

LOD 100

The Model Element may be graphically represented in the Model with a symbol or other generic representation, but it does not satisfy the requirements for LOD 200. Information related to the Model Element (i.e., cost per square foot, the tonnage of HVAC, etc.) can be derived from other Model Elements.

BIMForum Interpretation: LOD 100 elements are not geometric representations. Examples are information attached to other model elements or symbols showing the existence of a component but not

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its shape, size, or precise location. Any information derived from LOD 100 elements must be considered approximate.

LOD 200

The Model Element is graphically represented within the Model as a generic system, object, or assembly with approximate quantities, size, shape, location, and orientation. Non-graphic information may also be attached to the Model Element.

BIMForum interpretation: At this LOD, elements are generic placeholders. They may be recognizable as the components they represent, or they may be volumes for space reservation. Any information derived from LOD 200 elements must be considered approximate.

LOD 300

The Model Element is graphically represented within the Model as a specific system, object, or assembly regarding quantity, size, shape, location, and orientation. Non-graphic information may also be attached to the Model Element.

BIMForum interpretation: The quantity, size, shape, location, and orientation of the element as designed can be measured directly from the model without referring to non-modeled information such as notes or dimension call-outs. The location of the element is accurately located within the defined project coordinate system.

LOD 350

The Model Element is graphically represented within the Model as a specific system, object, or assembly regarding quantity, size, shape, location, orientation, and interfaces with other building systems. Non-graphic information may also be attached to the Model Element.

BIMForum interpretation: Parts necessary for coordination of the element with nearby or attached elements are modeled. These parts will include such items as supports and connections. The quantity, size, shape, location, and orientation of the element as designed can be measured directly from the model without referring to non-modeled information such as notes or dimension call-outs.

LOD 400

The Model Element is graphically represented within the Model as a specific system, object, or assembly regarding size, shape, location, quantity, and orientation with detailing fabrication, assembly, and installation information. Non-graphic information may also be attached to the Model Element.

BIMForum interpretation: An LOD 400 element is modeled at sufficient detail and accuracy for fabrication of the represented component. The quantity, size, shape, location, and orientation of the element as designed can be measured directly from the model without referring to non-modeled information (such as notes or dimension call-outs).

LOD 500 [NOT USED]

The Model Element is a field-verified representation regarding size, shape, location, quantity, and orientation. Non-graphic information may also be attached to the Model Elements.

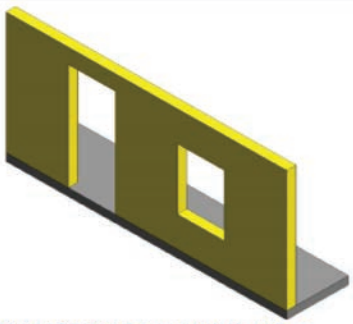
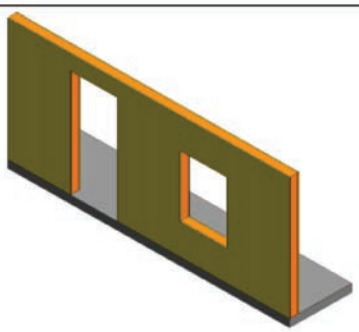
BIMForum interpretation: Since LOD 500 relates to field verification and is not an indication of progression to a higher level of model element geometry or non-graphic information, this Specification does not define or illustrate it.

An example of the LOD progression for an Exterior Wall (Cold-Form Metal Framing) is shown in Figure 9-7a and Figure 9-7b for purposes of illustration. This example is from *Level of Development (LOD) Specification Part 1 & Commentary: For Building Information Models and Data; Version 2018* (BIMForum, 2018).

9.5.2 Attribute Spreadsheet Example

An example of an attribute spreadsheet for the exterior wall is shown in Figure 9-8. Note that the first seven items are identified as the "baseline" attributes. These include construction, material skin/finish, material substrate/structure, insulation material, wall type, thermal resistance, and thermal transmittance. Additional attributes for this asset type include target LOD (the LOD desired) and current LOD. This is an example of an attribute that would

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200	Generic wall objects separated by type of material (e.g. brick wall vs. terracotta). Approximate thickness of layer represented by a single assembly. Layouts and locations still flexible.	 72 B2010.05-LOD-200 Exterior Wall (Cold-Form Metal Framing)
300	Specific wall modeled to actual dimensions. Penetrations are modeled to nominal dimensions for major wall openings such as windows, doors, and large mechanical elements. Shear panels	 73 B2010.05-LOD-300 Exterior Wall (Cold-Form Metal Framing)

Source: BIMForum, 2018. © 2018 Project Modeling LLC (graphic reproduced with permission)

Figure 9-7a. LOD 200 and 300 examples.

be updated during design and installation. Other attributes include wind load capacity, fire rating, impact resistance, UV resistance, sound transmission, and air infiltration. The last attributes, if included, could enable engineering analysis of a facility with regard to fire protection, wind resistance, noise studies, resistance to solar radiation, and security ratings. While these attributes are not required, they can increase the value of the BIM regarding operational analysis capabilities after the facility BIM has been created.

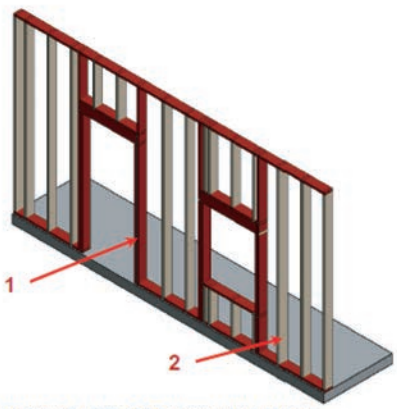
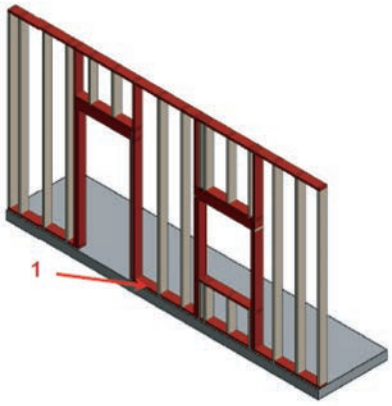
9.5.3 National Building Specification BIM Object Element Matrix

Another example of an asset data matrix is one established in Australia as part of its National Building Specification (NATSPEC) BIM standards. Looking at the NATSPEC object for the exterior wall from the last example shows a slightly different organization of data, which is more IFC-centric. The attributes are also organized by LOD, so it is easier to see how the attribute data evolve in parallel with the graphical attributes. Note that the entire NATSPEC asset data matrix has not been reproduced herein; selected portions from LOD 100–500 follow:

- LOD 100 Conceptual—the exterior wall basic dimensions and conceptual cost.
- LOD 200 Approximate geometry—adds position requirements, LEED items, and programming requirements for glazing, fire rating, room, and building type.
- LOD 300 Precise geometry—adds coordinate data to the position, manufacturer, wall specifications such as exterior and interior finishes, molding, and core type. Also adds assembly base costing, energy analysis values, sustainable materials types, and code compliance specifications. This phase also adds the facilities asset management system descriptors.

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350	Wood framing is developed with sufficient elements to support detailed interface coordination with other systems such as MEP. All penetrations are modeled at actual rough-opening dimensions. Openings modeled with support framing around openings *Image notes:* *1) Elements in red are critical wall support elements that cannot be easily cut for coordination of MEP opening through the walls.* *2) Infill wood framing modeling may be omitted at this LOD if stated in the BXP.* *3) Cladding and sheathing are not shown for clarity in this image.*	 *70 B2010.06-LOD-350 Exterior Wall (Wood)*
400	Wood framing is developed with sufficient elements that support the fabrication of the wood framing system. *Image notes:* *1) Connection content is development in the wall elements. This includes but is not limited to fasteners, anchor rods, and other related hardware.* *2) Cladding and sheathing are not shown for clarity in this image.*	 *71 B2010.06-LOD-400 Exterior Wall (Wood)*

Source: BIMForum, 2018. © 2018 Project Modeling LLC (graphic reproduced with permission)

Figure 9-7b. LOD 350 and 400 examples.

- LOD 400 Fabrication—this level adds the as-installed data for warranties, model and serial numbers, spare parts, expected life, certifications, and replacement costs. Typically, many of the required O&M data do not get populated during design and construction until this LOD phase.
- LOD 500 As-built—this LOD adds a field-measured GIS and GPS (global positioning system) tag/position, asset condition, defects, recorded actual costs, and LEED documentation.

9.5.4 COBie Data

The COBie format standardizes delivery of information about scheduled products and equipment. Before beginning operations of a new facility, maintenance, operations, and asset management system data in COBie format may be imported without rekeying. Over 30 commercial software programs, including software for designers and facility managers, have been tested for their ability to export and/or import COBie data. BIM software may simplify the initial creation of COBie data. However, the majority of COBie information, such as manufacturer, model number, and preventive maintenance schedules, is created in the traditional construction administration process.

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BIMForum LOD Specification 2018 Part II

B – Ext. Wall

Baseline
Additional

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Attribute	Data Type	Units - Imp.	Units - Metric	Part 1 - Attribute Description: Option Examples	
Construction	Text			framed, unit masonry, panelized, EIFS, etc.	
Material - Skin	Text			tiles, composite, sheet metal, etc.	
Material - Substrate	Text			corrugated metal, plywood, composite panels, etc.	
Material - Insulation	Text				
Wall Type	Text				
Thermal Resistance	Number	h·ft2·°F/Btu **(R)**	$m^{2o}C/W$ **(R)**		
Thermal Transmittance	Number	Btu/(h·ft2·°F/Btu **(U)**	W/($m^{2o}C$) **(U)**		
Target LOD	Text			100, 200, 300, 350, 400	
Current LOD	Text			100, 200, 300, 350, 400	
Wind Load Capacity (drag)	Number	psf	Pa		
Wind Load Capacity (pressure)	Number	psf	Pa		
Fire Rating	Text			options: [UL label - A,B,C,D,E,S]	
Impact resistance	Text			options:[T/F, class]	
UV Resistance	Text			options:[T/F, class]	
Air Infiltration	Text			options:[T/F, class]	
Sound Transmission					

Source: BIMForum, 2018. © 2018 Project Modeling LLC (graphic reproduced with permission)

Figure 9-8. Attribute worksheet example.

Although COBie information is already part of traditional construction handover documents, contractors may consider COBie to be something new. It is not. COBie is simply an internationally recognized way to organize existing handover information. To obtain a quality COBie deliverable, COBie requirements must be correctly specified and enforced. The COBie standard was published in 2015 in NBIMS-US V3 by the NIBS. COBie training is available from the University of Florida's COBie Academy. The United States Institute for Building Documentation has formed a subcommittee to develop a COBie certification process. A separate COBie Commentary published by Bill East (see Figure 9-9, for example) provides recommendations for COBie asset attributes and data types.

9.6 International Standards

Several efforts have been designed to establish and mandate the use of BIM standards in various countries to improve construction productivity and reduce costs. Following is a description of these BIM standards initiatives that are being undertaken at a national level. They are included here as examples that have been developed to address post-construction BIM life cycle requirements.

9.6.1 United Kingdom

The UK is currently the most progressive country in establishing nationwide BIM standards and goals and has developed some initial standards to support BIM in operations as well as construction. The UK Government Construction Strategy was developed to require a fully

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Heading	Typical Unit
Name	AHU-TypeXX-Space#-01
Type	AHU-TypeXX
Specification Section	(as identified in client's contract)
Location	(Space Name)
Current	Amps
Voltage	Volts
Frequency	Hz
Fan Flow – Maximum	L/s
Fan Flow – Nominal	L/s
Fan Outside Flow	L/s
Fan Ext Pressure Drop	kPa
Fan Motor Power	kW
Fan Speed	RPM
Fan Sound Level	dB
Coil Flow	L/s
Coil Velocity	m/min
Coil Capacity	W
EnteringAirTempDB	C
EnteringAirTempWB	C
LeavingAirTempDB	C
LeavingAirTempWB	C
Entering Water Temp	C
Leaving Water Temp	C
Chilled Water Rate	L/s
Runout Inlet Size	Mm
Runout Outlet Size	Mm
Coil Air Pressure Drop	Pa
Coil Water Pressure Drop	kPa

Source: Bill East, COBie Commentary

Figure 9-9. COBie asset specifications.

collaborative BIM on all centrally procured public projects. The goal was to mandate a minimum BIM "Level 2" by 2016, with the goal of achieving a 20% construction cost reduction from pre-mandate construction costs. The UK BIM levels are defined as the following:

- Level 0 BIM—2D CAD drafting
- Level 1 BIM—3D/2D CAD with electronic data sharing
- Level 2 BIM—(current target) collaborative BIM with IFC or COBie data
- Level 3 BIM—(future) BIM and open standards for data sharing, contracting, and management

The UK's BIM standards are developed first as PASs and then, after a period of review, are evaluated as to whether they should be formalized as British Standards. The BSI manages this process and currently has published the following standards:

- **PAS 1192-2: 2013** Specification for information management for the capital/delivery phase of construction projects using building information modeling.
- **PAS 1192-3: 2014** Specification for information management for the operational phase of assets using building information modeling (BIM).
- **PAS 1192-4: 2014** Collaborative production of information. This standard describes the employer's information exchange requirements using COBie.
- **PAS 1192-5: 2015** Specification for security-minded building information modeling, digital built environments, and smart asset management.

9.6.2 Australia

Australia, like the UK, has been very progressive in mandating the use of BIM to improve construction productivity and facility life cycle asset management. AUS-SPEC defines the

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Australian government system for life cycle asset management used for all public facilities. NATSPEC has developed the Australian BIM standards and specifications in widest use. The NATSPEC *National BIM Guide* is a collection of BIM standards and templates that includes the following:

- BIM Guide and Project BIM Brief—fulfills a role similar to the BIM project execution plans.
- BIM Reference Schedule—a list of suggested BIM standards and resources.
- BIM Object/Element Matrix—mapping of BIM objects/elements and their standardized properties organized by UniFormat/OmniClass classifications and expected LOD at each life cycle phase of BIM development.

9.6.3 Finland

BIM standards have been developed in Finland with the assistance of buildingSMART Finland. Finland's standards currently consist of the following:

- **Common BIM Requirements (COBIM) Yleiset Tietomalli Vaatimukset (YTV) 2012** that define life cycle BIM standards (including facility management uses). The BIM uses defined in the COBIM include support for facility management, space management, energy and environmental management, maintenance budgeting, long-term planning, and performance monitoring.
- **Common InfraBIM YIV 2015** that establish BIM standards for infrastructure projects.
- **Intramodel Data Exchange** that defines an open standard for exchange of infrastructure data based on Land Extensible Markup Language (LandXML) standards.

9.6.4 Singapore

The Singapore Building and Construction Authority is developing BIM standards and processes to improve productivity and the level of BIM integration across disciplines in the facility life cycle. The "Singapore BIM Guide Version 2.0" establishes the national BIM standards. The "BIM Essential Guides" provide a series of best practices for BIM focused on key roles, responsibilities, and activities. These include the following:

- BIM Essential Guide for Adoption in an Organization
- BIM Essential Guide for BIM Execution Plan
- BIM Essential Guide for Architectural Consultants
- BIM Essential Guide of Collaborative Virtual Design and Construction
- BIM for DfMA (Design for Manufacturing and Assembly) Essential Guide
- BIM Essential Guide for C&S Consultants
- BIM Essential Guide for MEP Consultants
- BIM Essential Guide for Contractors
- BIM Essential Guide for Building Performance Analysis
- BIM Essential Guide for Land Surveyors

The Building and Construction Authority also offers BIM training and certification programs to develop the local workforce's BIM capabilities and expertise.

9.7 Summary

While implementing BIM does not require compliance with national or international standards, complying with these standards will reduce the long-term cost and increase the long-term value of BIM. Establishing BIM standards enables the airport to leverage existing tools, resources, and workforce to the maximum possible extent. In the United States, NBIMS-US V3

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presents a comprehensive framework of standards, processes, and tools to support a BIM life cycle management approach that is focused on optimizing asset life. While many of these standards and resources were developed initially with a focus on design and construction processes, they are adaptable to supporting the full asset life cycle.

Creating a BIM execution plan will provide a mechanism to define and measure BIM deliverables from consultants, contractors, and internal BIM resources. There are a wide variety of BIM execution templates available that airports can use as a starting point for their own. One of the best known is the "BIM Project Execution Plan" developed by the Penn State CIC Research Program.

LOD is a key standard for designing and measuring the completeness of a BIM program and encompasses both the level of graphical detail and the level of completeness of asset data. The BIMForum publishes a specification that can act as a basic guide for what kind of asset data should be collected. Australia's NATSPEC standards are another example of a BIM asset data specification.

COBie provides a standard framework for facility equipment data that enables interoperability with other management systems that conform to COBie data requirements. COBie supports the exchange of facility equipment information, but not for fixed assets (such as walls and floors).

Before developing a BIM program, an airport should develop an asset data model as part of a collaborative BIM needs assessment. A standard data model (such as COBie) should be the starting point and should be customized as needed to preserve as much interoperability with standards-based systems as possible. This will allow the airport to achieve the maximum use and cost-effectiveness of its BIM program.

Section 9 Checklist

1. Establish BIM standards.
2. Create a BIM execution plan template.
3. Determine the data structure of the facility, systems, and assets within the BIM.

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SECTION 10

BIM Controls—Governance

Once an airport has developed the business case for implementing BIM, it must determine who will manage the BIM program and what policies, procedures, and infrastructure are required to deliver BIM across the organization effectively. This section will focus on options for where the accountability for BIM should reside, the roles and responsibilities that need to be established, and the policies and processes required to maintain BIM as a shared facility information resource.

10.1 Organizational Structure

There is no established best practice for where the primary accountability for BIM management should lie within an airport organization. As a tool that is designed to enhance collaboration, an argument could be made that all airport stakeholder groups should share in BIM management to maximize its benefits. However, this is rarely practical. An engineering department; asset management department; information and communications technology (ICT) department; or a new department that is focused solely on developing, maintaining, and distributing facility information such as BIM are good candidates for managing an airport's BIM program.

10.1.1 Engineering Department

An engineering department providing planning, design, and construction management services for the airport may be selected as the primary owner of the airport BIM, since it has the most experience with BIM as a design and construction technology solution. As a result, this department will also have the staff most skilled in the use of BIM-authoring tools and in the ability to manage changes to the BIM. Relying on an engineering department to own and manage the use of BIM will also require that it be closely engaged throughout the life cycle of an asset and, particularly, that the department work closely with the airport staff responsible for the operation and maintenance of the asset. This increased engagement will help in identifying and capturing the data needed to keep the BIM current and in analyzing those data for trends and areas needing attention.

10.1.2 Asset Management Department

For BIM uses beyond the project design phase, an asset management department (or the equivalent) will be in the best position to identify where BIM can provide benefits to an organization from the planning, design, construction, operations, maintenance, and renewal stages

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of its assets. The asset management department may stand alone in an organization or be part of the engineering department. In either case, the departments must work closely together to take full advantage of the expertise that resides in each one if they are to maximize the benefits of BIM for the airport. If BIM is being used as part of an airport's overall migration toward a TCO or ISO 55000 strategic facility asset management approach, then the asset management department would be in the best position to define how BIM will be used throughout the airport.

10.1.3 ICT Department

ICT is typically focused on technology infrastructure, but with a collaborative information model such as BIM, ICT may be in the best position to maintain BIM accessibility. ICT is typically not responsible for maintaining the accuracy and completeness of data, so broadening ICT's role to include complete management of a BIM would require changing the typical ICT responsibilities significantly.

10.1.4 A New Department

Another option is to develop a new organizational unit focused solely on developing, maintaining, and distributing facility information such as BIM. Massport has taken this approach with its Design Technology Integration Group. The manufacturing industry is approaching a fourth industrial revolution (4IR) that features emerging technologies such as robotics, AI, IoT, and 3D printing, all of which require the management of increasingly detailed facility information. Thus, there is a good reason for larger airports to establish a separate airport group focused on the collection, validation, maintenance, and distribution of airport facility information. This information may provide benefits to the airport that go beyond those of asset management, including detailed leasing information, cost recovery data, etc., as the new department works with all the airport's stakeholders and identifies their facility data information interests and needs. An independent BIM/facility information department also helps to enable the enforcement of the BIM program across the airport, since it will not be a sub-function of an existing department.

10.2 Roles and Responsibilities

Implementing BIM as a collaborative tool across an airport organization requires not only the training of existing staff but also adding new roles and functions. It is possible that outside consultants or contractors can perform these roles for some period. However, the airport will generally achieve greater benefits if these roles are filled by permanent staff who are better aligned with the vision of the airport.

10.2.1 BIM Manager

The BIM manager is responsible for the overall development, validation, and maintenance of the BIM program across the airport organization. The BIM manager

- Ensures that the BIM processes and standards are well defined in contracts.
- Confirms that proper BIM controls are in place to ensure that all BIM delivered through new capital development and facility renovation projects, whether by the airport or its tenants, conforms to the airport's BIM standards.

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- Manages the maintenance of the airport's BIM.
- Manages the integration of existing facilities into BIM.
- Works with ICT to ensure BIM accessibility and security.
- Provides BIM educational resources and training plans for the rest of the organization.
- Works with individual airport departments on integration requirements to support their internal systems and to identify and capture their BIM informational needs.

10.2.2 BIM Designer/Technician

The BIM designer is skilled with the use of BIM-authoring tools (such as Autodesk Revit and Navisworks, Bentley software, and Graphisoft ArchiCAD). The BIM designer provides internal support for the maintenance of BIM due to small capital projects without BIM requirements and for internally performed asset modifications or facility renovations.

10.2.3 BIM Information Analyst

The BIM information analyst is a skilled programmer who can support custom or off-the-shelf commercial integration solutions to airport stakeholder groups that want to utilize BIM data within their internal systems.

10.2.4 BIM Project Manager

The BIM project manager provides BIM management on the capital project development of new and renovated facilities. This person participates in the full life cycle of the facility's development, from planning through project delivery and operation, to ensure that airport BIM processes and standards are understood and followed. Also, the BIM project manager facilitates the involvement of other airport stakeholder groups (such as airport maintenance and operations) during design and construction to ensure that information they may need following the turnover of the asset is identified and captured during the design and construction of the asset. The BIM project manager role is separate from that of a capital project manager, whose responsibilities will typically end once the construction of an asset is completed.

10.2.5 BIM Trainer

The BIM trainer develops and provides training resources on the use of BIM to airport stakeholder groups and acts as a helpdesk resource. For a large airport, this may be a full-time role, or it may be a shared duty among other BIM staff. Without staff across the airport who are sufficiently trained in the use of BIM for their needs, the use of BIM within an airport may stall, and the potential benefits may be greatly diminished.

10.2.6 Other BIM Roles

While not full-time roles, there are other critical roles that must be filled to make the adoption of BIM go smoothly.

Executive-Level BIM Champion

BIM requires strong executive support to be successful. An executive-level BIM champion encourages and supports the airport's BIM program and provides continued support as implementation challenges and issues arise. This BIM champion needs to be the guiding

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hand that manages the airport's BIM vision through the inevitable rough patches any such program will experience.

Department-Level BIM Champion

For a large organization, each major stakeholder group may need to have its own BIM champion who is seen as a trusted member of that group. The process and cultural changes that a multi-faceted BIM program introduces require day-to-day reinforcement that an executive-level BIM champion will not be able to provide. Additionally, staff-level issues and questions are more effectively managed at the departmental level.

BIM Executive Steering Committee

As noted earlier, many public airports are managed as part of a larger organization with multiple areas of responsibility. The use of BIM is often part of a larger organizational policy driven by various factors including the recognition of BIM's benefits, the desire to stay current with technological advancements, etc. A BIM executive steering committee, chaired by the executive-level BIM champion and including all the department-level BIM champions, can provide a forum for the open discussion and resolution of BIM implementation concerns, how new developments at other organizations or in the underlying BIM technologies will be handled, etc.

BIM Outreach Education

One challenge in establishing BIM processes and standards will be finding local contractors and subcontractors who have the requisite skill levels to perform projects with BIM requirements. A need for BIM skills development may exist for small and disadvantaged business consultants and contractors. Providing training opportunities and resources to these entities can be a long-term investment in building the skilled labor force the airport needs to sustain future growth. It can also be a means for airports to provide local economic development to the communities they serve.

10.3 BIM Policies

The development of BIM policies through the establishment of BIM standards and contract language will provide governance of BIM facility data delivered as part of new capital construction and major renovation projects. Also, there must be a corresponding set of policies and processes to manage BIM data for internally designed, small-scale projects and operationally driven facility changes that require changes to the BIM. Policies and processes are also needed for determining how use of BIM on tenant projects (which may include facilities as large as new passenger terminals) will be managed. A few policy issues must be addressed outside of the design and construction life cycle.

10.3.1 Operational BIM Changes

How and when should the BIM be updated when small operational changes are made to the airport's facilities? For example, when a pump fails and is replaced with a different type that requires retooling of electrical and plumbing facilities, this work is often performed with in-house resources. If these changes are not captured in the BIM, the accumulation of many such small asset changes may degrade the trusted value of the BIM. Airport staff will then resort to relying on the project plans as the most accurate resource. Also, contractors may bid on inaccurate plans, resulting in change orders.

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There must be a policy in place that captures asset and facility modifications that require a change to the BIM. If the airport has internal BIM designers, these changes can roll into a task list for these individuals to address in the BIM. For contractors, an up-to-date log of any differences between existing conditions and the BIM will ensure that they have accurate and accessible data even if small changes to the BIM are not maintained in real time.

10.3.2 Security

The collaborative benefits of BIM can easily come into direct conflict with airport information security policies. BIM will contain critical infrastructure information necessary for airport staff to efficiently perform their work functions, but which could be misused in the wrong hands. The exposure of this information increases as the BIM is made more accessible. Limiting access decreases the benefits of BIM. Mobile access to BIM data maximizes those benefits.

Information security technology evolves rapidly. It is outside the scope of this Guidebook to detail the various BIM data security options. However, it is vital that the airport ICT staff, or qualified consultants, be involved in establishing and updating BIM security policies to ensure the proper use of and access to BIM.

10.3.3 Project Selection

While it is easy to make the business case for the use of BIM on large capital projects, the use of BIM on smaller projects may be viewed as adding cost and providing little value. Some organizations establish BIM use criteria based on the size of the expected construction cost (exceeding $1 million or $5 million, for example). Others may use a criterion based on how much of the construction budget is composed of MEP-related activities (complex MEP benefits more from BIM coordination).

If an airport is moving toward a life cycle BIM approach, it is important for BIM to be incorporated into all its capital projects and throughout its existing facilities in a prioritized fashion. This is true of airport tenants' capital projects and existing facilities as well; many of these become airport property at the conclusion of the lease term. Having part of the airport's facility plans available in BIM and part of them available as 2D plans can easily encourage the airport staff to rely on 2D plans only, rather than having to learn two different systems to access facility data.

Another approach would be to allow small projects to be delivered without BIM and use internal resources to modify the existing BIM from the 2D plans delivered. The airport will need to evaluate whether this is more efficient than simply requiring the use of BIM on all projects.

If there is a project selection criterion established that does not require BIM to be used on all projects, there will need to be a corresponding internal process to update the overall facility BIM from the as-built data delivered on project handover.

10.3.4 Legal Policies

There are a few contracts and legal policies that must be considered regarding the status of BIM as a contract requirement. These policies are covered in more detail in Section 12. If BIM is going to be used as a tool for facility management, it is critical that design and construction contracts require that the level of accuracy in the BIM deliverable matches the level of accuracy required in the record models. Often BIM will be used to coordinate

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activities, and the details that are not required for coordination are left out. For a smooth transition at the end of construction, the data within the project's record plans must also be reflected in the BIM.

10.4 Summary

The use of BIM for ALCM requires a different governance approach than the use of BIM only for design and construction. While design and construction BIM allow the governance of BIM to be primarily focused on the contractual requirements, life cycle BIM requires a more comprehensive governance structure that includes organizational changes, the definition of new roles and responsibilities, and the development of new policies that provide the controls necessary to maintain, distribute, and secure BIM data.

Section 10 Checklist

1. Determine where the primary accountability for BIM management should lie within an airport's organizational structure.
2. Determine the staffing, roles, and responsibilities required to maintain the BIM.
3. Identify and provide the training required for those maintaining and using the BIM.
4. Develop BIM policies and processes for BIM facility data delivered as part of capital projects, for the management of BIM data for internally designed projects and operationally driven facility changes, and for the use of BIM on tenant projects.

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SECTION 11

BIM Controls—Progress Metrics

Once an airport has established a BIM program, it is important for the airport to measure the effectiveness of the program to guide future corrective actions and improvements. This section identifies metrics that provide an airport with the data it needs to ensure that BIM is achieving established goals. As discussed in Section 4, the metrics are often KPIs, developed and validated in an airport's strategic planning and organization assessment exercises. The metrics that an airport employs may refine or expand on the ones described herein.

11.1 Maintenance Planning

As discussed in prior sections, BIM can deliver both tangible and intangible benefits across an organization. The effects of using BIM as an improved source of facility data for maintenance planning can be measured in terms of hours saved (tangible). It is more difficult to measure the overall effect of BIM as a source of improved communication across the organization (intangible). Below are some basic, tangible metrics that can provide a framework for managing BIM and measuring progress against an airport's BIM roadmap.

The tangible benefits of BIM fall into two primary categories: time saved (hours) and asset utilization (extended asset life). The metrics listed may be collected as part of CMMS/EAM functions or by other airport data systems. The following metrics are focused on measuring improvements related to maintenance activities.

11.1.1 Average Time to Complete Work Orders (hours)

This metric should measure the time to complete a work order starting from when it is assigned to a maintenance worker. This metric is meant to provide data on how much BIM improves a maintenance worker's access to facility data in the performance of the work order. Using BIM should reduce the amount of time needed to identify the location of the asset, assess the working conditions around the asset, identify access points and tools required, and identify available specifications and maintenance procedures. With BIM, many work orders that previously required preliminary site visits to create a work plan can be completed in one trip to the asset site. At a large airport, this can sometimes save hours per work order.

11.1.2 Wrench Time (percentage)

This metric is used by maintenance departments to measure the percentage of total hours expended by maintenance technicians that is devoted to actual, hands-on work on the asset/equipment versus other planning and overhead activities. Some organizations will include travel time as part of the wrench time, and some will include only the time on-site at the

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equipment's location. Regardless, BIM will increase the wrench time percentage; however, the measured effect will be less if travel time is not considered a part of wrench time.

11.1.3 Preventive Maintenance Backlog (days, weeks, months)

This metric tracks the average delay (in days, weeks, months) in performing asset preventive maintenance activities from an established baseline maintenance schedule. Preventive maintenance activities are those designed to keep an asset in optimal working condition (such as regular tightening, lubrication, and cleaning). Regular preventive maintenance not only extends the life of an asset, but also contributes to sustainability efforts: well-maintained equipment will operate more efficiently and consume less power. Longer asset life reduces the cost of disposing of end-of-life assets.

As a maintenance planning tool, BIM will help in identifying preventive maintenance–bundling opportunities to optimize preventive maintenance efficiency. Also, by improving wrench time, BIM indirectly frees up additional hours for the existing maintenance staff to perform preventive maintenance activities rather than addressing unplanned maintenance issues. As a result, the preventive maintenance backlog should see a reduction because of BIM.

11.1.4 Asset Uptime (hours)

In some cases, an asset uptime metric will be tracked that shows the time (in hours) the asset has been in continuous operation without any unplanned outages. Although BIM does not improve equipment efficiency, the asset uptime metric should improve as unplanned outages related to preventive maintenance delays decrease.

11.2 Asset Management

The following suggested BIM metrics are related to asset management activities as they relate to asset inventory, condition assessment, and property management. Tracking these metrics will optimize the ability to measure the cost and benefits of BIM in an organization.

11.2.1 Asset Data Integration (hours)

This metric tracks the amount of time (in hours) for new assets to be entered into the airport's CMMS and/or EAM system (primarily after new construction and renovation projects) and for accurate space plans to be incorporated into property management systems. By standardizing the asset data requirements that GCs must meet upon project handover, this process can be reduced from years or months down to weeks. This reduction in handover time is an example of a tangible benefit—reduced labor hours for manual (or semi-automated) data entry—that is directly measurable. An intangible benefit is also provided to the financial asset management of a new facility by having these data available more quickly.

11.2.2 Asset Inventory (hours)

This metric tracks the amount of time (in hours) required to perform periodic asset inventory that is required due to external regulatory requirements and airport operating procedures. The asset location within a CMMS and/or EAM system is often generalized to rooms or even entire facilities. The lack of precise asset location data can make the process of tracking down specific

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assets for inventory time consuming and can potentially enable mistakes due to a lack of precise asset locations. An example is when multiple assets of the same type near one another are being inventoried. By providing a precise coordinate-based location for each asset, BIM reduces the time to locate assets and eliminates the possible faulty identification of assets.

11.2.3 Asset Condition Assessment (hours)

In addition to physical inventory, the airport is responsible for the periodic condition assessment of its assets. The inclusion of condition assessment data within BIM allows the airport to use coordinate-based condition data to analyze whether condition deficiencies can be correlated with other facility condition issues or with the asset's work order history, to support a root cause analysis of unplanned failures.

11.3 Additional Metrics

The following metrics are suggested as tools that could be valuable in providing feedback on the economic benefits of BIM and on the extent to which BIM is being utilized.

11.3.1 Design for Maintainability (hours)

If the airport's maintenance department is included in early design reviews or as part of operational readiness and airport transfer (ORAT) activities, the corrective maintainability actions identified by maintenance should be tracked as a life cycle cost savings after project handover. For example, if maintenance staff identified in the design phase that the location of a proposed access panel would add an hour of labor during maintenance, then this should be tracked as a life cycle cost savings each time this maintenance activity is performed. This will provide some measurable ROI for the use of BIM as a collaborative tool.

11.3.2 BIM Utilization (number of accesses)

Since BIM is a centralized facility data repository, the access history can easily be tracked to provide the airport with a metric of how often BIM is being accessed to retrieve facility data. Early in the BIM adoption phase, it is likely that departments will continue to utilize their internal stores of facility data, and this would translate to a low volume of BIM data access. If BIM is gradually gaining acceptance, this metric will increase. These data could also be used to focus future investments in BIM infrastructure based on the types of BIM data that were most actively accessed. This metric also identifies facilities where an investment in additional BIM LOD would have a greater return on that investment.

11.4 Data Collection

If the airport has an established CMMS and/or EAM system, the data required to track the metrics discussed herein should be available. However, not all airports have such a system available to them. In these cases, the collection of BIM metric data should be approached similarly to other time-tracking processes currently in place. These time-tracking systems could be via a work order management system or just a focused staff-hour reporting process where activity codes for the metrics previously discussed could be added and tracked. Collecting these data will take additional effort and processing time but will give the airport the feedback on BIM usages required to guide future investments.

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11.5 Summary

BIM is a complex process that provides tangible and intangible benefits across an airport organization. To efficiently manage the implementation of BIM and to maximize the realized benefits to an airport, it is vital to establish a program of BIM metrics. While intangible benefits may be hard to quantify, there are also some tangible BIM metrics that can be collected to inform future BIM implementation and investment decisions. Access to a CMMS and/or EAM system makes these data more easily collectible, but even smaller airports can implement time-tracking efforts to collect these data. Also, data on BIM benefits are critical in building organizational support for the adoption of BIM, as these data can be shared with airport staff to show how the effort is adding value.

Section 11 Checklist

1. Establish BIM metrics to measure the effectiveness of the program and guide future corrective actions and improvements.
2. Establish a schedule and methods for communicating BIM metrics performance to the airport organization, consultants, and contractors.

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SECTION 12

BIM Controls—Legal and Liability Issues

BIM is designed to provide a collaborative data environment that breaks down the traditional silos among the various life cycle phases of a facility. While this approach provides benefits in improved communication and coordination of activities across those life cycle phases, it comes into direct conflict with the traditional manner in which risk and liability have been managed contractually. In most cases, architects, GCs, and owners all seek to control and limit their risk and liability. Each team member would prefer to tightly control its participation and communication within the facility development and management life cycle to minimize its risk. Development of a new contractual framework that incentivizes open communication and the sharing of facility information is necessary to fully benefit from BIM.

For design and construction, two primary BIM contract models have been developed. The American Institute of Architects (AIA) developed the E202 BIM Protocol Exhibit, and the Associated General Contractors (AGC) developed the Consensus Document 303 BIM Addendum. Much of the language in these contract templates focuses on the handover from design professionals (architects and engineers) to GCs. There are similar handover issues experienced by owners when BIM is delivered to them after construction using either template. This section will discuss the primary legal and liability issues owners need to consider when implementing BIM for their airport organizations. These include

- Liability
- Ownership of the BIM
- BIM legal status
- Digital data delivery

12.1 Liability

Traditional project delivery models encourage owners, architects, engineers, and GCs to work toward minimizing their risk and liability by shifting it to other members of the team. This shifting of risk does nothing to reduce the overall risks associated with the development of the facility but rather seeks to protect the interests of each contributor. A description of how this works follows.

12.1.1 Architects/Engineers

The doctrine of privity has traditionally protected architects and engineers from claims made by GCs utilizing their designs. The doctrine provides that a contract cannot confer rights or impose obligations upon any person who is not a party to the contract. Privity protects the architects and engineers from parties with whom they do not have contracts. In a

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design-bid-build contract, there is no direct contractual relationship between the architects and engineers and the contractors. The extent to which the privity doctrine is accepted varies among legal jurisdictions.

In collaborative contractual environments such as design-build, construction management at risk, and integrated project delivery, the architects and engineers *do* share a common contract with the general contractor. Evidence of collaboration reduces any protections privity might afford them. While this decreases the ability to shift risks to other parties, the improved communications among all the contributors to the project help to identify and resolve design issues much earlier in the life cycle process. BIM may reduce each entity's overall risk and liability exposure.

12.1.2 General Contractors

The Spearin doctrine is the primary means utilized by GCs to reduce their potential liability on projects. This doctrine states that there is an "implied warranty" that the design documents are "free from defect." This enables the GC to recover costs for any changes required during construction based on design errors and omissions. Spearin does not protect a GC from performance-based specifications that specify how to build a system and the operational performance level the system must achieve. For example, while construction documents include shop drawings for the layout of HVAC mechanical equipment and ductwork, a performance specification might state that a minimum of 360 cfm is delivered to each office. The burden is then on the GC to construct to that performance specification and test to the satisfaction of the architects and engineers that this has been achieved.

BIM diminishes the protections of the Spearin doctrine because, in a collaborative environment, the GC can contribute to the detailed design. This opens an exception to Spearin that removes the implied warranty. While BIM may diminish the protections of the Spearin doctrine, it also enables the GC to produce a more constructible facility early in the design process, reducing the GC's overall risk.

12.1.3 Owners

Airport owners must manage their risk and liability with the use of the contracts developed to manage the delivery process. If the airport wishes to utilize a full life cycle BIM approach, these contracts must also include the digital data delivery specifications that are required to utilize the BIM to support O&M activities after project handover. Unless CMMS integration is included as part of the project delivery requirements, or as part of a commissioning requirement, the liability for properly specifying the AIRs is the owner's. Thus, the owner should be actively involved early in the process of detailing the data that need to be collected and the delivery format. The owner must remain actively involved in the commissioning process to ensure that the owner's data needs have been met *before* acceptance of the asset for operation and use.

12.2 Ownership of BIM

Standard AIA-based contracts specifically state that "no ownership" right of the design documents is transferred to any other entity. These contracts only allow for a limited license to use or modify the design. The limited license can be an issue if the owner wants to reuse the design documents for future renovations and may limit the transfer of the design data to other consultants for facility management applications.

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The use of BIM potentially includes intellectual property developed by the architects and engineers, such as 3D model libraries, that would subject those libraries to being copied and illegally distributed if shared with other consultants working for the airport.

Architects and engineers are also concerned with liability that might arise from the reuse of their designs by parties who might modify the model without the input of the design architects and engineers. The following licensing requirements can address these issues:

- Owner maintains ownership and copyright of BIM after project handover.
- Designer's liability is minimized for any future uses or modifications.
- Designer's model libraries are contractually protected from reuse outside the scope of the project BIM (i.e., other consultants may not copy or reuse).

There may be architects and engineers unwilling to agree to such terms. The legal framework for BIM was still evolving at the time this Guidebook was being written. However, the licensing requirement terms listed previously will provide the owner with the broadest rights necessary to modify and reuse the BIM for the O&M life cycle. Such terms are already in use in the design and construction contracts at DEN, and their sample contract language is included as Appendix A.

12.3 BIM Legal Status

The legal status of BIM as a contractual document is still fluid. Signed and sealed 2D plans are still the primary basis for existing contracts. There is no settled determination on the legal status of a "record BIM" that is delivered at project handover; however, owners have a few options they can follow (see Sections 12.3.1 and 12.3.2).

12.3.1 BIM as Supporting Specification

Under this option, the 2D plans and specifications remain the legal standard, and BIM is supplied as "additional information" with no contractual legal status. It is supplied for the owner's use, but with no corresponding guarantee of completeness or accuracy.

12.3.2 BIM as Contract Document

Under this option, the BIM is delivered as a legal document, and the 2D plans are verified as being generated from the BIM. Even if the 2D plans are generated and linked to the BIM, the BIM and 2D plans may still contain additional information. The 2D plans may contain additional annotation. The BIM will contain not only additional asset information that is not included in the plans and specifications but also information on asset interdependencies that is only accessible in a programmatic manner.

If both BIM and 2D plans and specifications are contract documents, then an order of precedence will be required to identify which one has the primary legal authority if there is a conflict between them. Federal Acquisition Regulation 52.236-21 states that, if a conflict arises between specifications and drawings, the specifications will have precedence. The AIA suggests precedence of drawings over specifications. BIM encompasses the data in both the drawings and the specifications, so clear legal guidance is not yet established on the best practice (other than that there should be clear precedence specified to resolve potential conflicts when they arise). A further complication is that BIM also captures shop drawing information used to secure final architects' and engineers' approvals before fabrication or installation of an asset.

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12.4 Digital Data Delivery

BIM is a combination of data traditionally received from drawings and data received from specifications. However, the BIM contract requirements need to define how the digital specification data are going to be delivered to be interoperable with the airport's existing facility management infrastructure. An AIR should be developed that identifies what types of asset information need to be collected and in what format the information should be delivered. The UK's PAS 55-2:2008 Asset Management Guidelines provide some guidance on what should go into an AIR, including legal, commercial, financial, technical, and managerial asset data.

The AIR can be organization specific, based on the existing airport asset data dictionary and facility management structures, or it can be structured around a standards-based information exchange (such as COBie) to maximize interoperability with other standards-based facility management tools and systems. Regardless of the standards, the digital data delivery requirements must be detailed as part of the project delivery contracts to minimize the cost of using BIM for facility management and to minimize the time required for digital data integration from BIM.

12.5 Summary

The development of legal structures that will incorporate BIM is still fluid, and case law is not extensive enough to establish best practices. Owners need to structure their design and construction contracts for new facility construction and renovation to prioritize their interests and identify their requirements for acceptance of the asset, particularly if they intend to utilize BIM to assist in the maintenance of that asset. While the AIA and AGC have developed standard BIM contract exhibits, they seek to limit their own respective risks and liabilities while maximizing protection of their ownership of the BIM. Owners should develop contracts that protect their own interest. BIM for life cycle asset management dictates that additional contractual considerations be evaluated by owners to ensure their ownership rights and digital data delivery requirements are met to support their facility management needs.

Section 12 Checklist

1. Develop design and construction contract language for BIM that addresses the issues of liability, ownership of the BIM, the legal status of BIM as a contractual document, and digital data delivery.
2. Provide outreach to airport consultants and contractors on new contracting requirements related to BIM.

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SECTION 13

Conclusions

BIM for post-construction facility life cycle management is still in the early phase of adoption around the world. Standards for the processes, technology, and implementation have not yet been well defined. This Guidebook has presented airport owners with options and best practices for implementing BIM with a focus on post-construction facility management activities. Given the rapid adoption of BIM by the design and construction industry, there is considerable interest in exploring how BIM can benefit facility management (where 85% to 90% of the overall facility life cycle cost is realized). Developing a comprehensive ROI for BIM post construction is difficult, and there are few examples where BIM ROI has been measured in a rigorous manner. The airports contacted as part of the development of this Guidebook moved forward with adopting BIM based on informal business cases and on the strength of internal BIM champions. These BIM champions had a strong belief that the benefits of BIM could be extended throughout the facility life cycle and would enhance the ability of their airports to manage growth and reduce operational and maintenance costs.

As an alternative to formal, detailed, financial analysis, developing an informal business case, or BIM roadmap, may be sufficient to drive initial BIM implementation. Performing a BIM needs assessment can identify specific facility information enhancement opportunities that BIM can address. Prioritizing these opportunities can demonstrate the benefits of the BIM to the overall organization. Focused pilot studies to address these opportunities can limit the initial cost while building organizational support and experience.

Establishing core progress metrics for BIM will enable an airport to document the benefits of BIM more effectively. The most immediately accessible metrics include the post-construction CMMS asset creation time and reductions in work order completion time due to the use of BIM for improved maintenance planning (virtual site visits, maintenance bundling). Design for maintainability and operations can also deliver substantial long-term facility life cycle cost benefits, but these are more difficult to measure.

One or more BIM champions are needed to manage change and communicate the long-term vision. Consultants can be used strategically, but there needs to be a commitment to building internal staff capabilities if BIM is to be truly internalized as a core process.

Core BIM standards beyond the design and construction life cycle are still emerging; however, by following the U.S. NBIMS standards and leveraging best practices [such as the *BIM Planning Guide for Facility Owners—Version 2.0* (CIC Research Program, 2013)] being developed, industry leaders can lower the risk involved in airports' investment in BIM.

The greatest benefits from BIM will be realized from the integration of BIM with existing systems. BIM-authoring platforms were not designed to be facility management systems, so the ability to link the BIM graphical and asset data to external systems for analysis and

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enhanced decision making is critical to maximizing benefits. CMMSs and/or EAM systems can benefit from the precise asset location data provided by BIM. Asset inventory time can be significantly reduced.

The ability of BIM to enhance the development of accurate and accessible facility asset data is a key component for airports that are exploring the implementation of comprehensive strategic asset management systems such as ISO 55000 or ANSI TCO. These asset management systems can fundamentally transform the manner in which airport assets are managed and maintained. BIM can be included as part of the business case for these efforts.

Implementing BIM must be approached with long-term vision and the idea that it will evolve over years or decades. This is not only because of the wide-ranging effect it can have on the organization's communication and collaboration, but also because the technology and standards are still evolving. Implementing BIM today will enable airports to build the infrastructure and processes required to leverage not only the existing benefits of BIM, but the much greater benefits to be delivered through BIM in the future.

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SECTION 14

Case Studies

This section presents 12 case studies from the following organizations:

1. Denver International Airport
2. San Francisco International Airport
3. Massachusetts Port Authority
4. Heathrow International Airport
5. John F. Kennedy International Airport
6. The Ohio State University
7. Western Michigan University
8. New York City Public Housing Authority
9. New York Presbyterian Hospital
10. Perth Children's Hospital
11. New Royal Adelaide Hospital
12. Sydney Opera House

14.1 Denver International Airport: Hotel and Transit Center Post-Construction BIM

DEN has standardized BIM for all new capital development projects. BIM is used throughout the asset life cycle from planning, design, and construction through O&M. The recent $544M hotel and transit center project was the first major project managed under the new BIM standards.

IBM Maximo was selected as the asset management platform. DEN utilizes Maximo's BIM integration capabilities to enhance work order management and planning, field data collection, and bulk asset creation/updates. DEN has generated a BIM architectural-level model for 95% of the airport terminal. Also, all future projects will have BIM requirements, which will incrementally add BIM detail to the model.

BIM Uses

DEN used BIM for the following:

- Design and construction
- Asset management
- Asset creation/inventory
- Warranty inspection
- Maintenance planning
- Condition assessments

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Pilot Project

To leverage the airport's investment in BIM to improve O&M, DEN deployed BIM 360 Field to perform quality inspections and manage warranty data on newly installed equipment.

The initial goal was to have a common platform for all asset data collected in the field by contractors, but it quickly grew beyond that as a tool for enhancing internal O&M. All project data, including construction plans, specifications, and O&M manuals, were now available to all inspectors via BIM-enabled tablets.

Business Case—ROI

Costs included approximately $70,000/year in licensing fees for BIM mobile software applications and $50,000 in initial costs for training, tablets, and a central server for the BIM mobile software.

DEN reported a payback on this investment in 18 months. This ROI was derived primarily from two categories of BIM benefits: (1) time savings and (2) improvement of data quality.

Savings Details

Savings from BIM were the following:

- Virtual elimination of overtime hours for inspectors. Inspectors had been averaging 10 to 15 hours a week.
- Improved data quality from inspections due to there being no transcription errors.
- Reduced average time to close warranty issues.
- Reduced rework time, as maintenance staff had access to the most current and accurate plans and data.
- Fewer return trips, as staff could use BIM to assess the job-site and task requirements before visiting the site.
- Dramatically lower printing costs, as inspectors and maintenance staff no longer needed to carry bundles of prints with them.
- Savings in printing costs for plans of the hotel and transit center project that were enough to pay for the cost of DEN's initial BIM mobile software licensing cost.
- Accurate field data available to project managers.
- Elimination of additional charges from contractors for using the contractor's BIM mobile software.

Governance

DEN's BIM program has had strong support from an executive-level champion. There was uncertainty about which department should have ownership of the BIM program. The Sustainability Department was eventually selected, based on its long-term life cycle view of the airport's facilities and the role that BIM could play in supporting a collaborative approach to ALCM.

Key Challenges

Field inspectors were initially reluctant to adopt a new approach to performing their warranty and field inspections. Paper checklists were used to capture field data and notes. The data were

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then manually entered into the CMMS back in their offices. This duplication of effort was time consuming and introduced the possibility of errors in data entry.

Organizational resistance was quickly overcome by running a number of small pilot tests with selected inspectors using BIM-enabled tablets that allowed the inspectors to collect their data, notes, and photos in a format that could be directly uploaded to the CMMS.

The tablets could also provide access to all of the building plans, O&M manuals, equipment specifications, and other facility documents. The use of tablets eliminated the need to carry around large printed sets of plans and the need for return trips to the office to collect additional data.

The feedback from the initial BIM pilot project and the positive review of the field inspectors who participated soon had the rest of the inspectors requesting access to BIM via field tablets.

Lessons Learned

- Internal staff will adopt new technologies if the application enables them to perform their work more efficiently without adding unnecessary complexity.
- Introducing BIM through selected trials can build the business case and organizational acceptance of BIM.
- Initial capital investment in equipment and training had a rapid return in work order efficiency and accuracy.
- The use of BIM supports green office and sustainability initiatives via dramatically reduced printing of construction plans and O&M documents.

14.2 San Francisco International Airport: Asset Creation

SFO manages 13 million square feet of facility space and serves more than 56 million annual passengers who fly on 48 separate airlines. SFO has four terminals, more than 100 gates, and 604 separate structures. Total land use area is 5,200 acres.

SFO began implementation of BIM in 2008 as part of its migration toward a more collaborative and cost-effective contracting approach using a progressive design-build (PDB) methodology. An initial PDB pilot program for the secure connector between Terminal 3 and the International Terminal finished ahead of schedule and under budget. Additional PDB projects completed since then include Terminal 3 East, Terminal 2, and the new air traffic control tower.

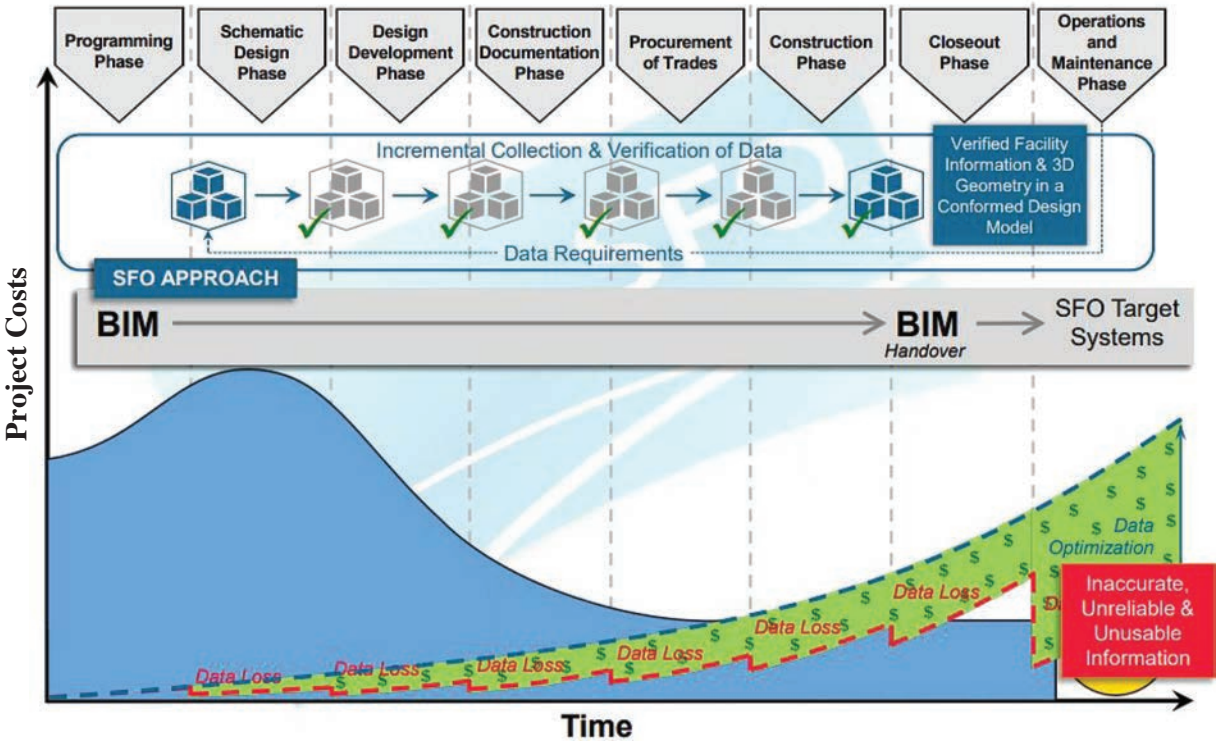
PDB creates a collaborative environment for the designer, contractor, subcontractors, and owner that can accelerate project delivery and lower costs. To achieve these benefits, effective communication among team members is critical. BIM contributes to this by providing a shared collaboration platform for the facility from planning through project delivery and greatly accelerating the handover of facility data to the airport for O&M. Figure 14-1 shows how the BIM-supported life cycle project costs (area shaded below the curve) compare with the traditional approach (area shaded with dollar signs) that experiences facility data loss between each phase. The facility data loss is primarily due to the lack of support for collaboration in traditional contract models.

BIM Uses

BIM uses are the following:

- Asset management
- Rapid asset creation

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Source: Neumayr, 2015

Figure 14-1. SFO PDB and BIM.

- Energy modeling and analysis
- BIM-GIS integration
- Design and construction

Business Case—ROI

BIM as part of a PDB facility development approach contributes ROI from direct cost savings and indirect cost savings from reduced construction schedules. A reduction in the construction schedule means the airport facilities are brought into service earlier.

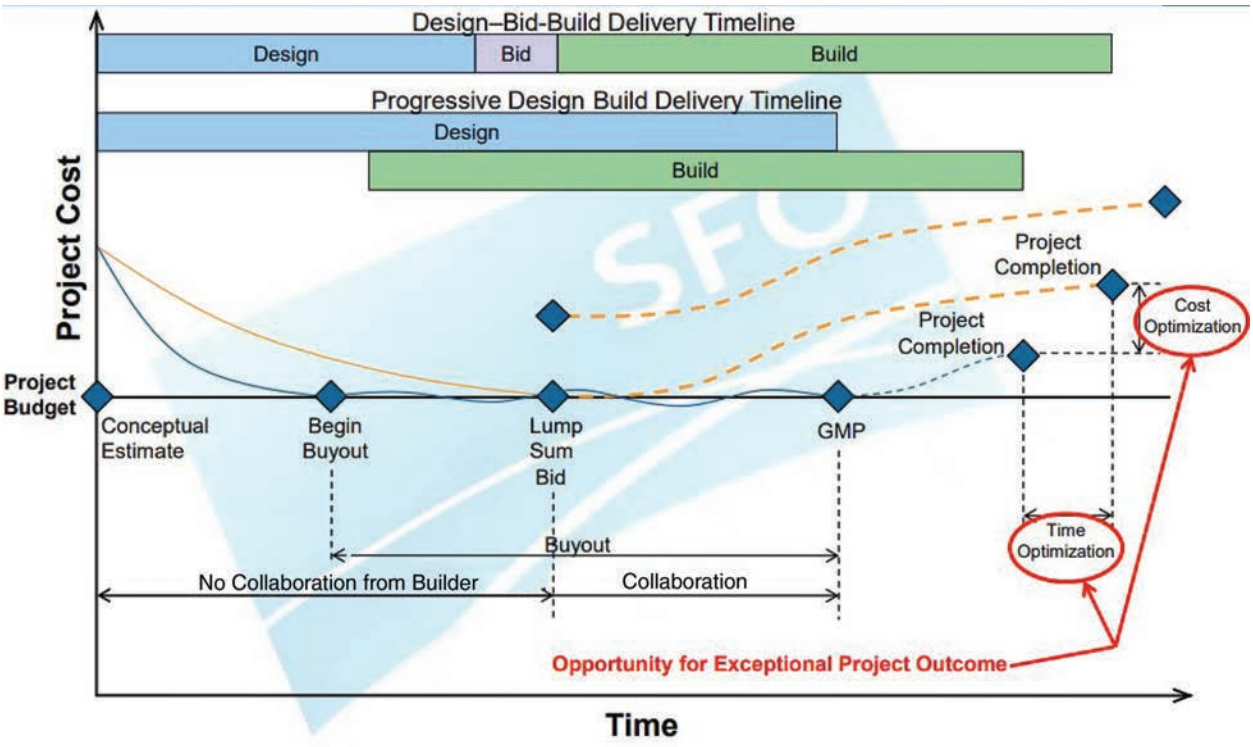
Post construction, the facilities must be handed over to SFO. The facility data handover process before BIM could take as long as a year because all asset data had to be manually entered into the CMMS and maintenance routines had to be established.

As BIM was established, this post-construction handover time was reduced from a baseline of nearly a year to only a few months on each major project. The goal is to be able to perform this activity in less than a week. This will be enabled by the direct integration of BIM data into the CMMS. Accelerated facility data transfer enables facility management to more quickly integrate the facility with its work order management system to ensure full readiness and rapid response to outages during the initial facility shakedown period.

Organizational Readiness

Developing collaborative team environments as part of the PDB contracting approach (see Figure 14-2) required overcoming the existing working relationships of architects, engineers, contractors, and owners. SFO not only had to develop its internal organizational capabilities but also had to work to identify and develop contractors' and local subcontractors' capabilities to adopt a PDB approach.

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Source: Neumayr, 2015

Figure 14-2. SFO PDB cost and time savings.

SFO maintained consultants as long-term extensions of staff to support BIM development, provide training, and identify BIM uses.

Key Challenges

Key challenges were

- Developing a pool of contractors and subcontractors who know how to use BIM.
- Internal organizational development to use BIM.

Lessons Learned

Lessons learned included the following:

- Establishing a trusted, collaborative environment promotes excellence in project outcomes. BIM as a collaboration tool can be a key.
- Early stakeholder engagement builds high-performance teams. BIM can provide a platform for sharing data and collecting stakeholder input.
- Metrics and measurements are key to improving the process. BIM data can be used to track collaboration effectiveness.

14.3 Massachusetts Port Authority: BIM Roadmap

Massachusetts Port Authority (Massport) owns and operates Boston Logan International Airport, which is located on 1,700 acres. The airport has four terminals that each have their ticketing, baggage claim, and ground transportation facilities. There are 94 gates with jet bridges and nine regional jet gates that serve more than 50 airlines. Massport also owns and operates L.G. Hanscom Field and the Worcester Regional Airport.

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The Massport BIM Vision states:

> Massport will utilize BIM and related technologies to empower its staff and service providers to design and build outstanding structures, and then manage and sustain these facilities to Massport's mission.
>
> As stewards of these digital assets, Massport will maintain BIM, GIS, and facility maintenance asset data to support strategic planning, sound decisions, sustainability, environmental responsiveness, and improved processes to add value to managing the building life cycle, and to aid Massport in better reinvestment of available funds based upon organizational missions and operational requirements.

(Dr. Lucian Burdi, International Association of the AIA, contributed to this section.)

Goals

The goals are the following:

- Lean BIM process development—reduces the cost of facility data management.
- Normalized facility asset data integrated with BIM.
- Integrated facility management platforms—life cycle planning and management.
- Collecting maintenance data to implement predictive maintenance systems.

BIM Process

The Massport BIM development process, as presented in the Massport BIM standards, is outlined in Figure 14-3. The Massport process is designed to support lean management processes. The integrated BIM facility data environment directly supports data measurements that the value-stream mapping and optimization processes require.

Massport's BIM process is built wherever possible on industry standards including NBIMS, NCS, OmniClass, and UniFormat asset classification standards.

BIM Uses

BIM uses include the following:

- Post-construction asset creation in the CMMS
- Maintenance planning
- Maintenance scheduling
- Sustainability planning
- Lean process optimization

Business Case—ROI

The business case was presented as a BIM roadmap that highlighted the short-term and long-term vision for BIM. A formal process for measuring a financial ROI for BIM was not developed.

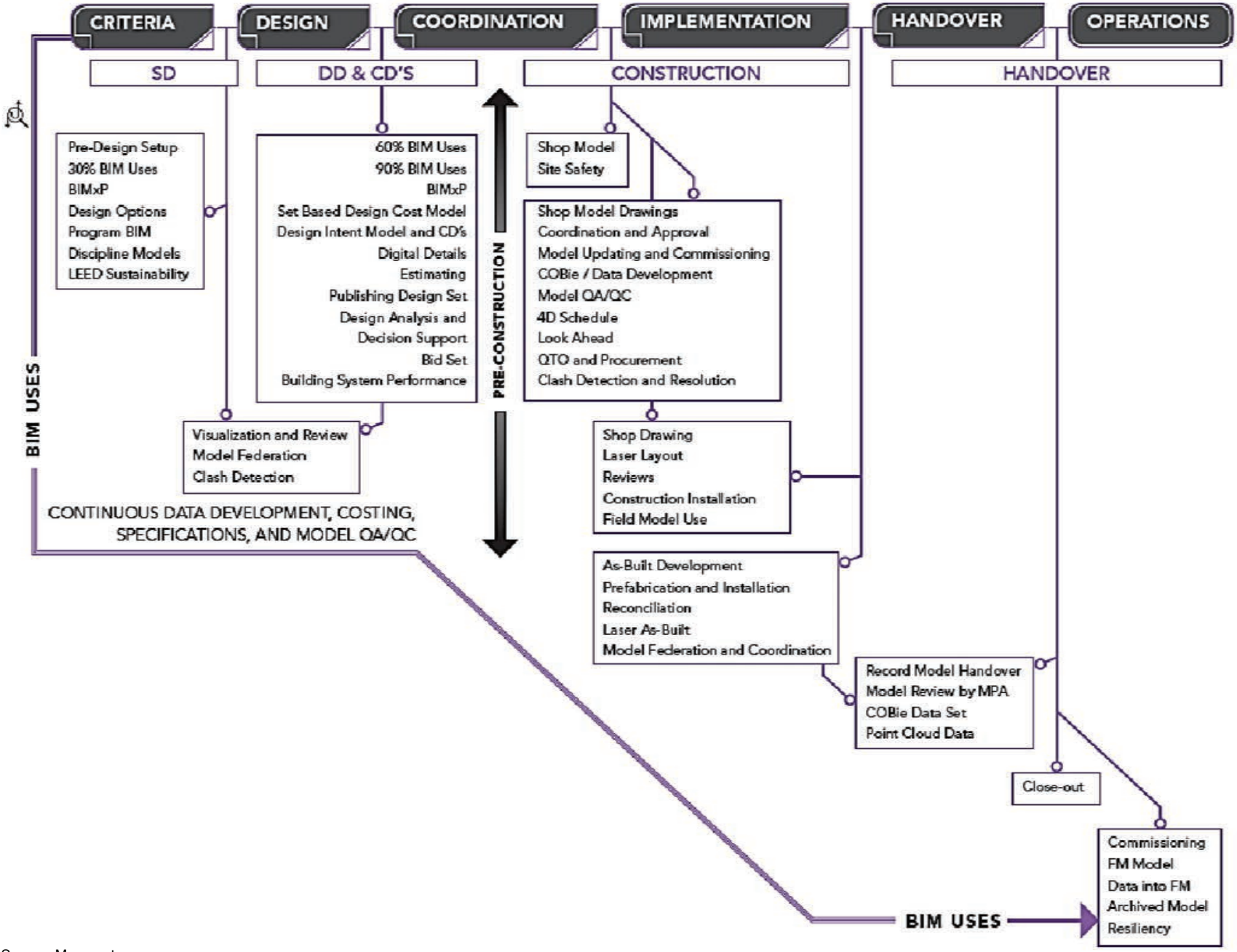
The BIM roadmap implementation has three phases: normalize, optimize, and institutionalize.

The first step (see Figure 14-4) is to build the foundation for Massport to use BIM.

The second step (see Figure 14-5) is optimizing the use of BIM to provide benefits to the airport stakeholders.

The third step is to utilize BIM as part of a full suite of EAM tools and processes as the primary source of trusted facility data (see Figure 14-6).

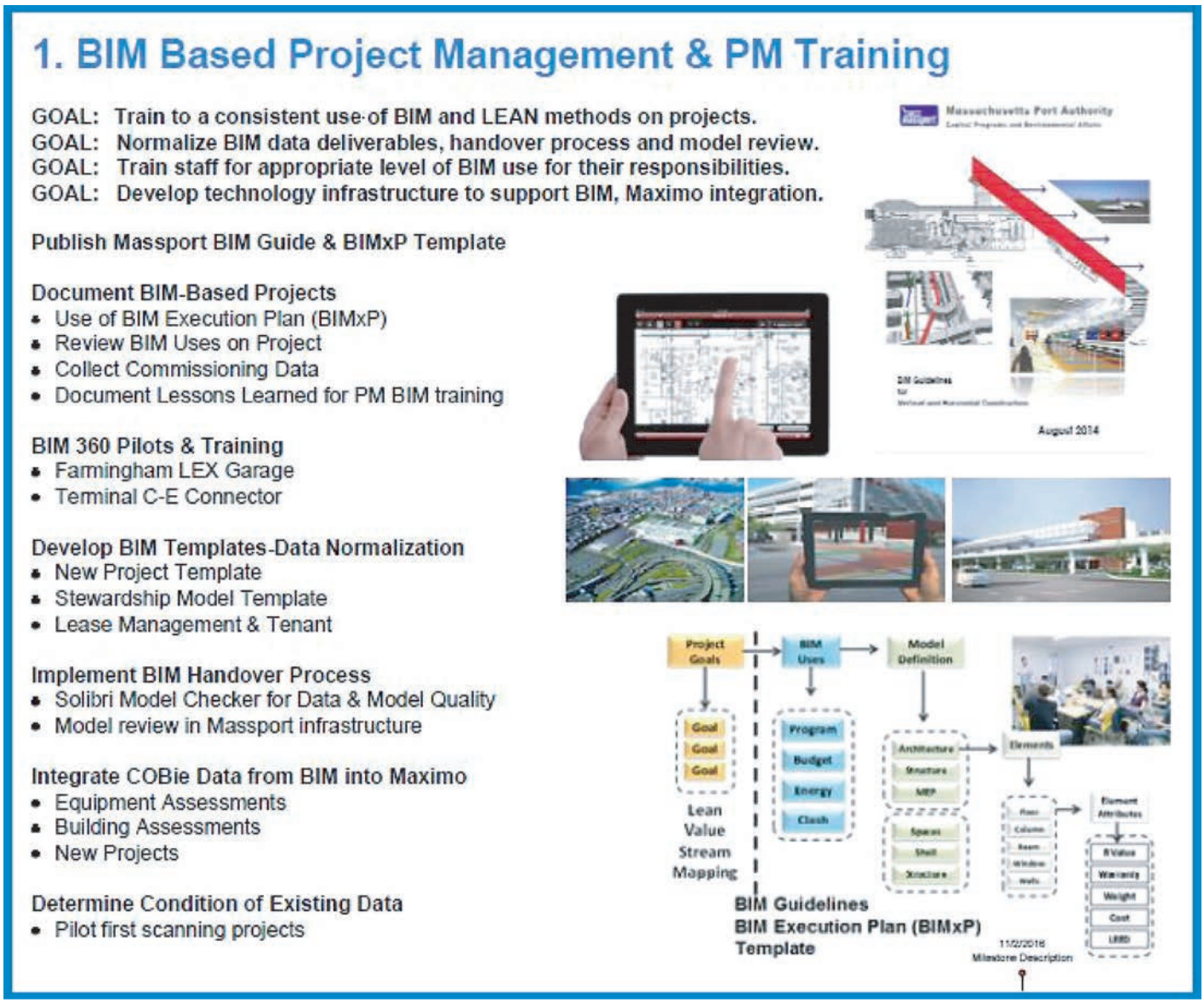
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Source: Massport
Note: QTO = Quantity Take Offs

Figure 14-3. BIM life cycle process map.

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Source: Massport

Figure 14-4. BIM roadmap phase 1.

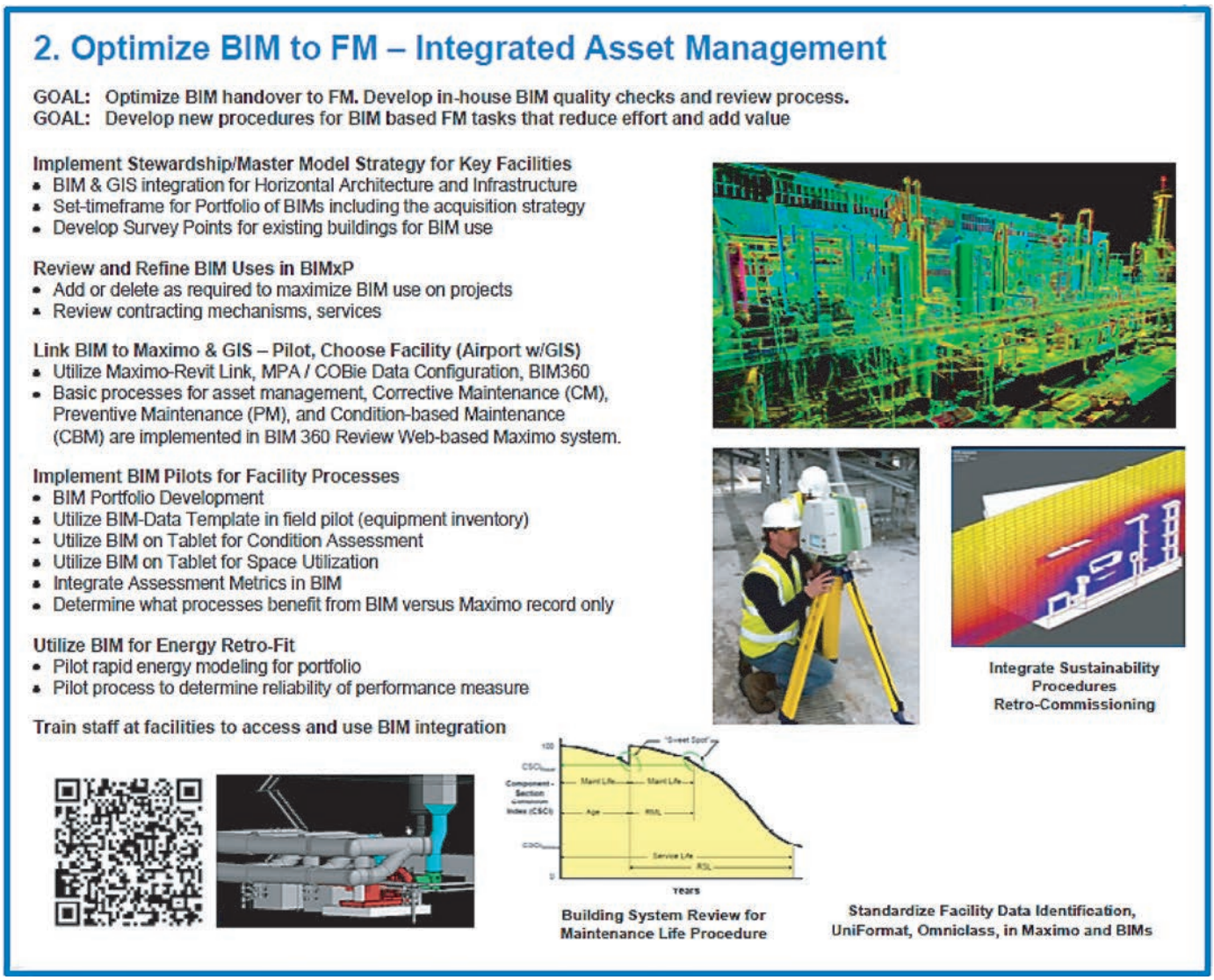
Applications include the following:

- IWMS
- CMMS
- BAS
- Energy usage simulation

Organizational Readiness

Massport created a Design Technology Integration Group to provide organizational support for developing BIM infrastructure, workflows, asset reporting, and data integration. Having a group solely focused on BIM and other emerging facility information innovations positions Massport to be able to maximize its use of BIM and its benefits.

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Source: Massport

Figure 14-5. BIM roadmap phase 2.

The Design Technology Integration Group BIM manager's responsibilities, as stated in the Massport BIM Guidelines, include the following:

- Help create specific BIM scopes of work on projects.
- Provide BIM project support for Massport programs.
- Support lean BIM.
- Help project managers with BIM facilitation on projects.
- Facilitate BIM project execution plan development on projects.
- Ensure compliance with the BIM project execution plan template.
- Facilitate model progression schedule showing BIM use and element LOD.
- Review models for quality and conformity to Massachusetts Port Authority BIM standards at the handover.
- Coordinate BIM, CMMS, and GIS integration.
- Manage updates to Massport technologies within the Massport information infrastructure.
- Review emerging technologies and standards for incorporation into Massport's BIM program.

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Source: Massport

Figure 14-6. BIM roadmap phase 3.

Key Challenges

Key challenges are the following:

- BIM is not internally developed, so facilities management–driven internal changes to the facility are not updated until the next project is bid that affects the area. Updating BIM to as-built conditions will be built into deliverables.
- Development of internal organizational BIM capabilities.
- Organizational acceptance for BIM-lean processes.

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Lessons Learned

Lessons learned are the following:

- Achieve early win at reasonable/low cost by revising contract and procurement requirements to meet best-of-class BIM standards.
- BIM implementation may take longer than expected; original 2018 goals were extended to 2020.
- Organizational readiness must be a focus if the program is to achieve its goals.
- Adopt standards whenever possible.
- Develop a group with BIM as a primary responsibility in order to maintain a focus on BIM.

14.4 Heathrow International Airport: Terminal 5

The Heathrow International Airport is the second busiest airport in the world and implemented BIM during the 1980s for the Terminal 3 project. The infrastructure at Heathrow includes 72 miles of high-pressure fire mains, 45,000 manholes, and 81 miles of aviation fuel piping. The Terminal 5 development was expected to serve over 30 million annual passengers with 42 gates. It included intermodal rail facilities. At the time, the £4.3 terminal expansion was one of the most complex projects ever developed in Britain, involving over 50,000 people and over 100 separate contractors.

Goals

The goals included

- Improved life cycle asset management.
- Improved communication and collaboration across large and diverse airport stakeholder groups involved in the facility life cycle.
- Common data environment to improve facility life cycle cost.

BIM Process

BIM has been a deliverable on all projects since 2003. A common language for delivering asset information in a defined structure was adopted in 2008 (Rhoades, 2015). "Heathrow Map Live" extended BIM visibility to the entire business in 2015. UK BIM Level 2, the United Kingdom's mandated BIM design requirement, was achieved in 2016.

British Airports Authority (now Heathrow Airports Holdings) created a framework agreement for the new Terminal 5 that aimed at creating a collaborative environment for the design and construction team and at establishing a common data environment that would support life cycle asset management. The British Airports Authority common data environment initiative was named AMA (Acquire and Maintain Assets) and streamlined the flow of information into its Maximo CMMS.

Passenger and Baggage-Handling Load Balancing

The Heathrow Airport Operations Center was designed to manage the flow of aircraft, vehicles, passengers, and baggage through the airport and provide real-time alerting when indicators were triggered due to performance issues. BIM is used as part of this model and simulates the flow of passengers through the terminals and the flow of baggage through the baggage-handling systems. Based on these simulations, Heathrow will stack aircraft landing sequences to load balance passengers from flights based on connections and number of bags. This information is shared between the airlines and the airport.

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Asset Management and Infrastructure Data Organization

Heathrow developed a comprehensive Common Language Mapping system that organized the airport facility infrastructure elements in a location-centric structure rather than using the historical-project-centric approach. The new common language standardized asset attribute data delivery and instituted a progressive handover of information throughout projects. This enabled the airport to better manage its equipment information requirements from contractors and reduced the time to commission new assets after project completion.

The asset common language framework organized asset data into the following infrastructure systems: moving planes systems, moving people systems, built environment (that impact passenger experience), security systems, safety systems, utility/environment systems, transportation links, exposed structures, ancillary systems, and other systems.

Root Cause Analysis

In the Heathrow Terminal 5 project, metrics showed that the terminal had the highest level of escalator accidents in the entire terminal network. As Terminal 5 was the busiest terminal, this did not immediately stand out as unusual. When safety incidents were mapped in BIM, it was evident that just one escalator in Terminal 5 was the primary source of incidents. A site visit and observation identified a single billboard near the top of the escalator that was diverting attention. Removing the advertisement reduced the level of incidents to a baseline level. While this type of analysis could have been performed without BIM, it was performed much more rapidly with BIM and illustrates the efficiency of data visualization in problem-solving and planning.

BIM Uses

BIM uses are the following:

- Design for operability
- Design for maintainability
- Maintenance planning and root cause analysis
- Asset management
- Space management
- Asset inventory
- Condition assessment

Business Case—ROI

No formal ROI analysis was performed.

Organizational Readiness

Heathrow developed two separate BIM groups: one that focused on the use of BIM for design and construction efficiency and another that focused on research and development to identify how BIM could be most effectively used throughout the facility life cycle.

Key Challenges

Key challenges include the following:

- Difficulty in fully capturing Heathrow's massive infrastructure in BIM.
- Meeting the funding requirements to fully model Heathrow's facilities.

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Lessons Learned

Lessons learned include the following:

- BIM and a common data environment can improve design and construction delivery on large, complex projects and can deliver life cycle asset management benefits.
- The use of BIM for design for operability can eliminate future O&M costs related to congestion and poorly distributed passenger resources.

14.5 John F. Kennedy International Airport: Terminal 5 Redevelopment

John F. Kennedy International Airport (JFK) is New York's busiest airport and covers a total area of 4,930 acres. JFK is also the 22nd largest airport in the world and serves as the single largest international gateway airport in North America. In 2017, JFK processed over 59 million passengers, and 20 million of those passengers went through Terminal 5. Terminal 5 was a $550 million, 635,000 sf terminal developed as a hub for Jet Blue.

Goals

Goals include the following:

- Improved life cycle asset management
- Improved space management
- Faster information exchanges and management
- Higher process automation
- Improved communication

BIM Process

BIM was utilized during the design of the terminal redevelopment to optimize passenger flows into and out of the terminal using passenger microsimulation. Before redevelopment of the space, consultants modeled design concepts in BIM and used those to model passenger behavior. The modeling tools leveraged the floor spaces, layouts, stairs, escalators, and security gates within the BIM to define the parameters of the simulation.

Passenger behavior could be modeled to simulate the movements of different types of passengers. Business travelers would move through express lanes and carry few bags, while tourists would move through slower lines and carry more bags. Different balances of originating passengers versus those on connecting flights could be modeled. The simulation allowed measurement of the congestion within the terminal.

The results of the effort greatly impacted the redesign of the terminal; the design evolved from a central terminal with spokes to a triangular design with a long and curved drop-off/pick-up space and ticketing in the center. Twenty-six gates align the other two sides of the triangle, organically funneling traffic to the airline gates. The simulation also drove the adoption of the Terminal 5 security checkpoint, which is 340 feet wide and has 20 checkpoint lanes. The furthest gate is only 5 minutes from the checkpoint.

As an additional operational design element, BIM was used to model the acoustics in the terminal using a sound simulation software package to ensure that the loudspeaker system could be heard over normal crowd noise.

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BIM Beyond Design Guidebook

BIM Uses

BIM was used to design for operations, which included passenger transit studies and acoustics modeling.

Business Case—ROI

There was no formal ROI analysis, but benefits of using the BIM process for redevelopment included

- Reduced congestion that eliminates excessive wear caused by crowding.
- Improved safety readiness in the event of emergencies.

Organizational Readiness

This BIM use did not require internal organization changes to realize ongoing benefits. The improved design of the terminal was a life cycle benefit. With little or no formal BIM training, O&M staff can participate in design for operations and sustainability through BIM's visualization capabilities.

Key Challenges

Custom programming needed to be developed and required ongoing consultant support to maintain.

Lessons Learned

BIM can deliver benefits to an airport through improved design for operations and sustainability without any changes to O&M procedures, BIM infrastructure, or internal training.

14.6 The Ohio State University: Buckeye BIM Initiative

The Ohio State University's (OSU's) main campus in Columbus, Ohio, is composed of 593 buildings. The capital investment budget for 2018 was $922 million and grew to $1.48 billion in 2019. The facilities include classrooms, research laboratories, hospitals and clinics, libraries, administrative offices, sports centers, and agricultural facilities.

The Buckeye BIM initiative was formally kicked off in 2015 by the Facilities Information & Technology Services Department with a goal of developing BIM for more than 34 million sf of facility space.

Goals

Goals include the following:

- Use BIM to reduce TCO.
- Improve the speed of decision making and the quality of those decisions.
- Develop a single source of truth for facility data.
- Develop a collaborative environment for developing and maintaining accurate facility data.
- Accelerate post-construction facility data handover and integration.
- Improve design for sustainability and maintainability.

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BIM Process

Existing buildings were developed in BIM from 2D, CAD, as-built plans that were field verified for accuracy. Each building took approximately a week to field verify and audit.

Newly constructed buildings, or major renovations, have BIM deliverable requirements that conform to the OSU building design standards.

OSU is using COBie-formatted BIM asset data for integration with its Asset Information Management (AiM) System. The COBie workflow is shown in Figure 14-7.

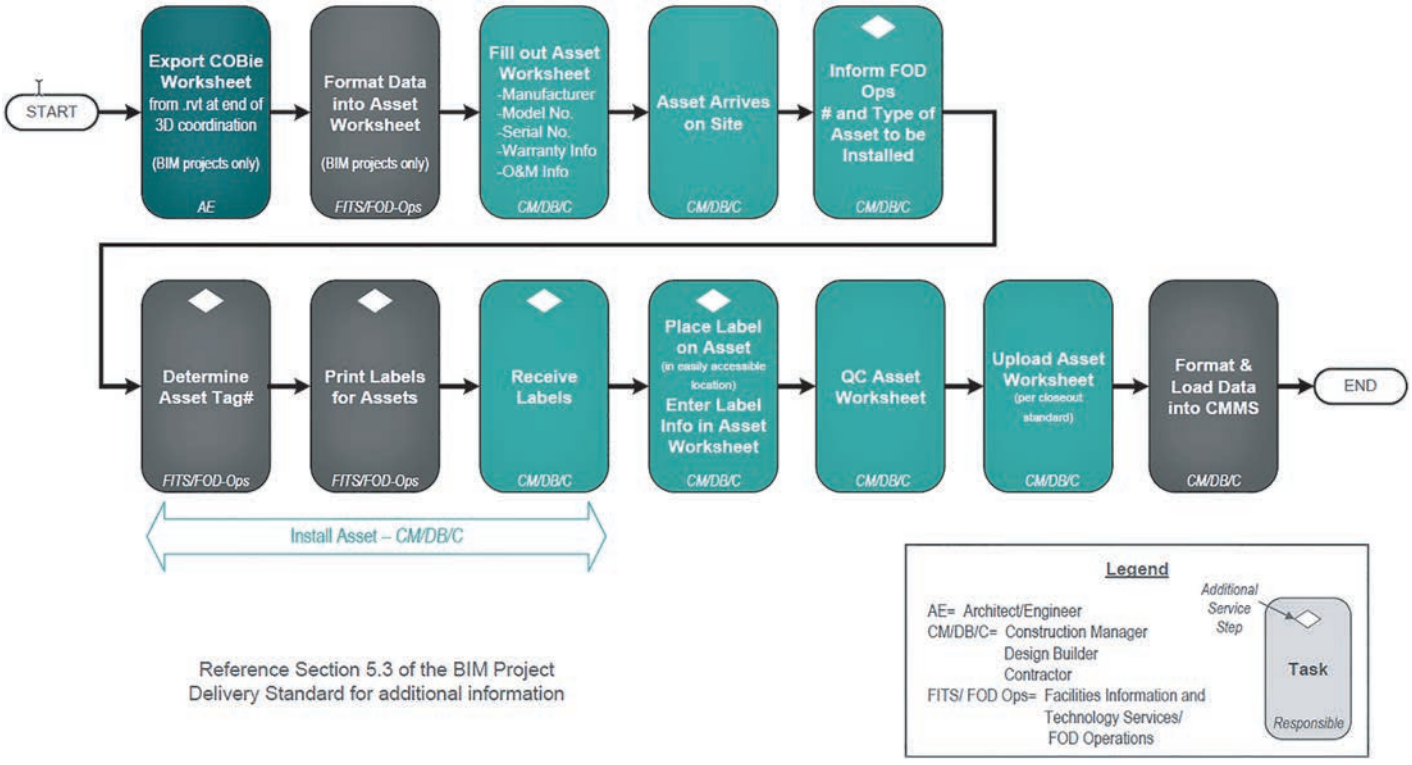
Attributes for each asset were captured in the COBie worksheet (detailed asset worksheet) as shown in Figure 14-8.

IFC data were developed for integration with other building systems (i.e., the BAS and energy analysis programs).

The initial effort used OSU architectural students to model the 53 buildings in the Wexner Medical Center and took a year to complete. Total student, consultant, and staff labor costs totaled $152,000. The measured rate of model development was approximately 900 sf/hour.

Creating BIM for existing buildings (see Figure 14-9) focused on creating "representational models" versus "buildable models." The base models (see Figure 14-10 for samples) included

- Exterior (detail)
- Walls/cubicles
- Doors
- Windows
- Roof (basic)
- Stairs (basic)



Source: OSU

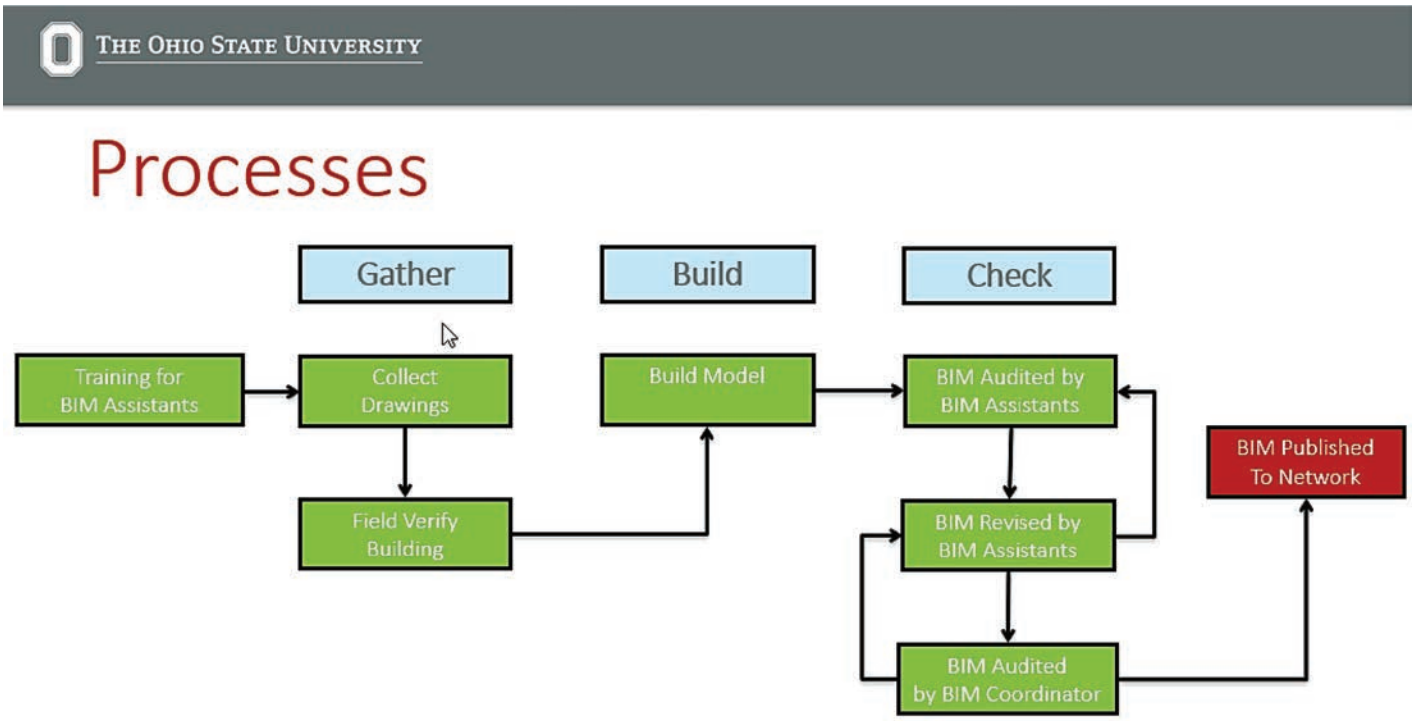
Figure 14-7. COBie and asset workflow.

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Parameter	Project Stakeholder Document Review	COBie Worksheet	Asset Worksheet
Building			
Building Number*	X	X	X
OSU Project Number*	X	X	X
Floor			
Floor Number*	X	X	X
Room			
Room Number*	X	X	X
Area*	X	X	X
Asset			
Revit Identifier*		X	X
Asset Tag			X
Asset Description*		X	X
Building Number*		X	X
Floor Number*		X	X
Room Number*		X	X
Serial Number			X
Manufacturer			X
Model Number			X
Warranty Start Date			X
Warranty Duration in Years			X
O & M Asset Link to BuckeyeBox Folder			X

Source: OSU

Figure 14-8. Asset worksheet.



Source: OSU

Figure 14-9. Existing building audit and BIM creation process.

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Source: OSU

Figure 14-10. Rendering of OSU facilities.

- Plumbing fixtures (basic)
- Columns/column grids
- Floors
- Ceiling (basic)
- Conference and classroom furniture

Additional selected modeling details included the following:

- Mechanical equipment (non-building systems)
- Fume hoods
- Catwalks
- Casework
- Furniture
- Plumbing fixtures (detail)

Detailed system models for the routing of HVAC, electrical, and plumbing were not modeled as part of the BIM development for existing buildings, although these would be available for facilities delivered through new construction under the building design standards.

BIM Uses

BIM uses include the following:

- Renovation planning
- Space planning

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- Congestion analysis
- Furniture layouts
- Energy analysis
- Asset management
- Disaster planning
- Wayfinding
- Faculty recruitment
- Donor recognition—sign and recognition designs
- Automated model checking

Organizational Readiness

OSU used internal staff and students to develop the BIM for its existing facility infrastructure, rather than outsourcing this effort to consultants. The use of internal staff and students not only reduced the cost, it also developed the internal BIM capabilities of the organization. The knowledge gained in developing the BIM has given the organization the depth of experience to take BIM to the next step: integration of BIM into work processes around the university.

OSU invested in training for staff and organized AEC town halls to further BIM training initiatives.

Business Case—ROI

OSU sees BIM as a measure to provide cost avoidance, rather than purely a measure for cost savings. Cost avoidance measures include the following:

- Improved quality and speed of decision making.
- Fewer owner-driven change orders based on the ability of staff to use new and renovated space to better visualize proposed designs.
- Reduced number of trips to perform maintenance.
- Reduced asset data handover time.
- Reduced long-term energy costs from energy usage analysis.

OSU used the benchmarks from a report prepared for the National Institute of Standards and Technology to project the ROI from its BIM facility development (Gallaher et al., 2004). In Figure 14-11, this calculation is shown for three of OSU's buildings: the Chiller Plant, McCampbell Hall, and Kennedy Commons.

Key Challenges

Key challenges were the following:

- Auditing as-built plans for accuracy before modeling was a challenge. Initially, field measurements were performed to verify data using handheld laser rangefinders. Audits could take 2 to 3 days per facility. Automated tools have reduced this time to less than 1 day.
- The main campus is a large campus composed of many separate facilities and organizations. Many departments have separate IT and maintenance groups (e.g., OSU Hospital).
- There has been no single linked BIM developed yet for the overall campus. Separate facility BIM, or small groups of linked facilities, must be managed and maintained.

Lessons Learned

Lessons learned include the following:

- Invest in up-front time getting the BIM standards and processes correct. As BIM is developed, changes become costlier and more difficult to implement.

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TOTAL COST AVOIDANCE
If a BIM Standard is fully adopted.

	Chiller Plant		McCampbell Hall		Kennedy Commons		Total
Project Statistics	Size: Cost: Const: Design:	95,737 SF $77,302,597 $54,784,978 $6,458,966	Size: Cost: Const: Design:	104,000 SF $17,171,045 $13,882,483 $1,363,045	Size: Cost: Const: Design:	30,000 SF $12,539,306 $7,931,390 $1,004,258	Total Size: 229,737 SF Total Cost: $107,012,948
2004 NIST Report **Planning Costs**	A/E: CM: Owner: Total:	$86,163 $151,264 $158,923 $396,350	A/E: CM: Owner: Total:	$93,600 $164,320 $172,680 $430,560	A/E: CM: Owner: Total:	$27,000 $47,400 $49,800 $124,200	Total $951,110
2007 NIST Report **Design Costs**	Total:	$322,948	Total:	$68,173	Total:	$50,212	Total $441,333
2007 NIST Report **Trade Package Creation**	Total:	$547,849	Total:	$138,824	Total:	$79,313	Total $765,986
2007 NIST Report **Coordination**	Total:	$3,865,129	Total:	$858,552	Total:	$626,965	Total $5,350,646
2004 NIST Report **Recapture of Data**	A/E: CM: Owner: Total:	$1,915 $20,105 $16,275 $38,295	A/E: CM: Owner: Total:	$2,080 $21,840 $17,680 $41,600	A/E: CM: Owner: Total:	$600 $6,300 $5,100 $12,000	Total $91,895
Total Project Cost Avoidance		$5,170,571		$1,537,709		$892,690	**Total** **$7,600,970**
% of Total Project Cost		6.6 %		8.9 %		7.1 %	Avg. = 7.1 %
Cost Per Square Foot		$54		$15		$30	Avg. = $33

Source: OSU

Figure 14-11. BIM ROI examples.

- Invest in training, and partner with BIM organizations to leverage their experience and knowledge.
- Pilot programs can help develop workable standards and create success stories that can be used to gain organizational acceptance.
- Do not include assets in the facility BIM that you do not have the resources to maintain.
- Leverage consultants where needed, but focus on developing internal capabilities.

14.7 Western Michigan University

Western Michigan University (WMU) serves over 25,000 students and has more than 400 employees. The total campus size is 1,200 acres, and there are more than 8 million sf of managed facility space. WMU has created a BIM for all facilities on its campuses, and BIM is

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available to all skilled trades via wireless tablets. WMU was awarded the APPA's "Effective and Innovative Practices Award" in 2015 for the processes it developed for "Building Information Modeling for Skilled Trades." WMU was an early adopter of BIM and one of the first universities to model an entire university for facility management purposes.

Goals

Goals included the following:

- Improved asset inventory and condition assessment
- Support of a continuous condition assessment process
- Asset life cycle cost analysis
- Consolidated asset management that supports system-wide strategic and capital planning
- Building management system (BMS) BIM integration

BIM Process

BIM for the WMU facilities was developed from new construction deliverables and from internal BIM development for existing buildings using student resources. Existing facility conditions were verified and updated by students from the College of Engineering and Applied Sciences, who performed laser scanning of those facilities. This not only lowered the cost but also provided valuable experience for the students.

WMU identified that the lack of a single source of easily accessible facility information contributed greatly to the time and labor required to complete maintenance work orders.

Developing BIM for WMU's existing facilities and making it available via field tablets greatly reduced the time to research work orders (see Figure 14-12) and eliminated most site initial visits required to collect data before work order completion. This not only saved time and cost but also contributed to improved customer satisfaction.

The WMU AIRs are defined in the WMU BIM execution plan in "Section N, Building Information Model Requirements—Life Cycle BIM." Each section of this BIM execution plan lists asset classification types and the asset data required for each. A few examples are listed below:

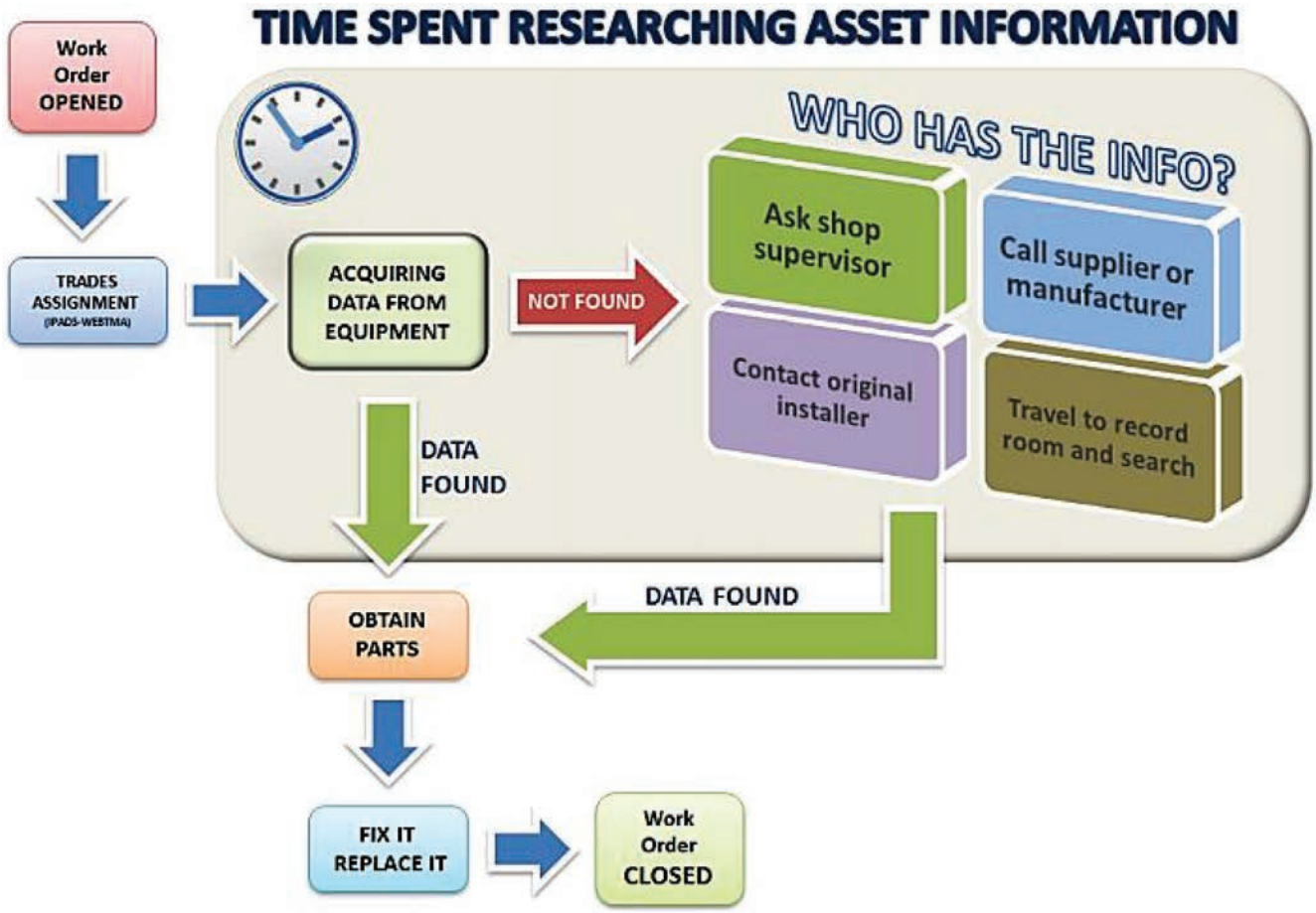
- Air distribution systems:
  - Supply fans—make, model, serial number, motor HP, belt size and quantity, fan curve
  - Exhaust fans—make, model, serial number, HP, frame
- Plumbing systems:
  - Meters—make, model, serial number, sequence of operations, O&M
  - Pressure relief valves—make, model, serial number, O&M
- Architectural:
  - Roof system—manufacturer, type, and material information, green roof details
  - Exterior doors—make, model, glass details, revolving/motorized details, sealant

BIM Uses

BIM uses are the following:

- Maintenance planning
- Asset management
- Asset inventory
- Space planning
- BIM field accessibility

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Source: WMU

Figure 14-12. Asset information research time before BIM.

Business Case—ROI

Use of BIM has resulted in

- Reduction in time spent to complete work orders of 15%.
- Reduction in preventive maintenance time of 6% to 9%.
- Improvement in labor efficiency of 7%.

Organizational Readiness

From its inception, the WMU BIM initiative has experienced executive-level support from the facilities management (FM) department. Staff from WMU's commissioning, IT, and database management departments have also been directly involved in BIM development.

The university also has been able to leverage its student workforce to develop and maintain BIM resources.

Key Challenges

Key challenges include the following:

- Developing initial BIM for a large campus.
- Educating skilled trades on how BIM could benefit their work process.

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Lessons Learned

Lessons learned include the following:

- BIM can have an immediate impact on the cost of asset maintenance and can improve maintenance labor efficiency.
- Mobile access to BIM for skilled trades is a key to maximizing the benefits from BIM.

14.8 Public Housing Work Order and Condition Assessment Analysis

A pilot study performed by CCI and Intellis was developed to demonstrate how BIM could be used to supplement the capital decision-making process and provide support for federal funding requests for facility repairs and improvements.

Goals

The goals included the following:

- Use BIM to perform root cause analysis of work orders and condition deficiencies.
- Use BIM to support capital planning and identify priorities.
- Use BIM to reduce TCO.
- Demonstrate BIM–CMMS integration benefits.

BIM Process

A housing authority had a database of 9 million work orders issued over 3 years and infrared imagery of the public housing facility rooftops that indicated the location of water infiltration and damage. The work order data identified each work order location by floor, apartment, and room number, but there was no linkage to floor plan layouts that could provide a coordinate-based location that could be used for spatial or root cause analysis.

A "SlimBIM" was rapidly developed from the existing as-built plans (see Figure 14-13) for a public housing facility where work order data were available. This model only included basic architectural structures for purposes of mapping the location of maintenance issues and roof conditions.

The work orders were exported from the IBM Maximo CMMS, and a custom script was developed to translate the floor-apartment-room location hierarchy into a coordinate-based location using the BIM as the master reference. A BIM project family was created that was



Figure 14-13. Work order integrated BIM.

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Figure 14-14. Building heat map.

a point-based object that could be placed in the BIM without clashing with any actual BIM architectural objects.

The infrared imagery was overlaid on the rooftop of the building (see Figure 14-14), and the color sampling script translated the infrared photo imagery into a mosaic of tiles (each with a color value representing the average color value over the sampled area). Spatial analysis was then performed between the areas of greatest water infiltration on the rooftop and water-related work orders in the building.

The spatial analysis highlighted a strong correlation between these defects and vertical spaces located below the areas of roof damage (see Figure 14-15).

The spatial analysis can be used to illustrate the linkage between the work order data and the associated cost. This analysis can also be used as input into the capital spending budgets to set priorities and to quantify potential future cost avoidance to justify federal grant dollars for repairs.

To demonstrate another potential benefit, asset condition data were color-coded into selected portions of the BIM to demonstrate how the model could be used as a planning tool to visually assess the condition of asset types within each public housing facility (see Figure 14-16). BIM could also assist in the prioritization of repairs that would provide the greatest benefit to the tenants.

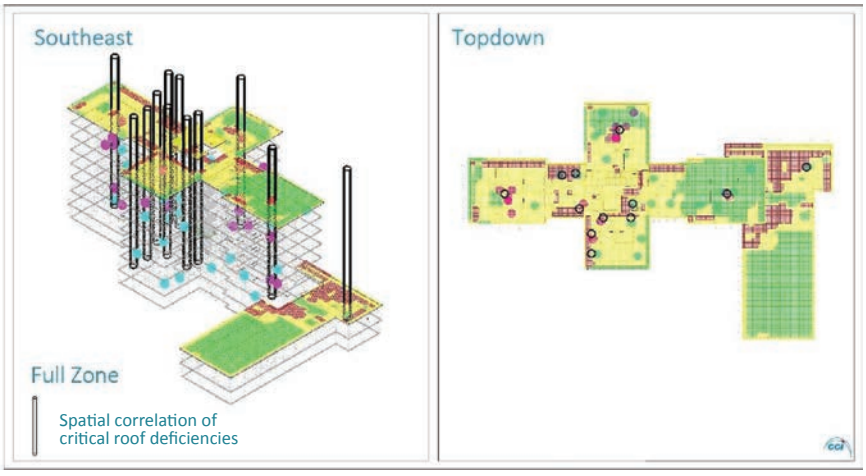


Figure 14-15. Work order spatial correlation.

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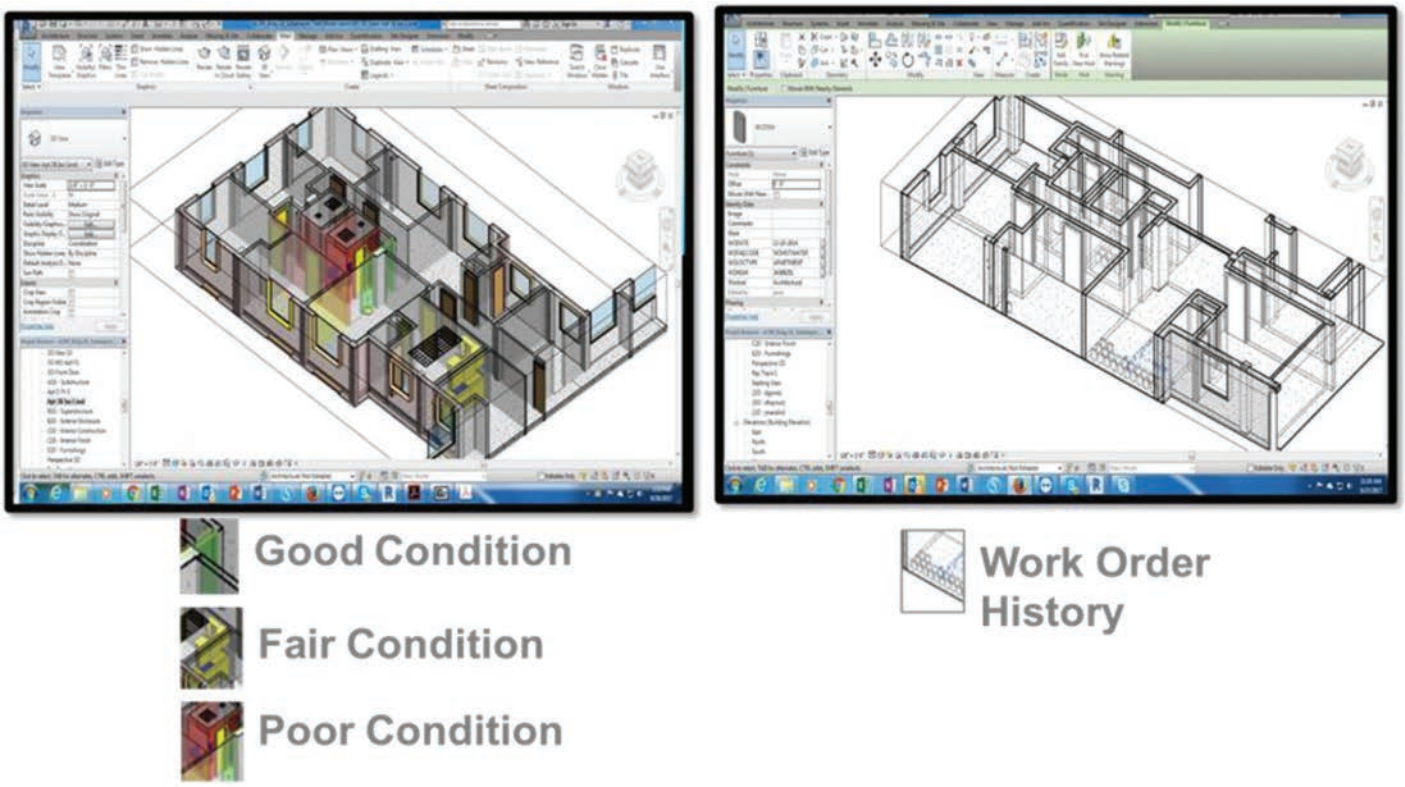


Figure 14-16. Condition assessment mapping.

Future potential applications of this approach include the following:

- 4D analysis of work orders over time to identify key indicators of trending failures based on past historical maintenance events. These indicators can be used to provide early warning of potential high-cost and imminent asset failures.
- With the inclusion of a statistically significant data pool, a data sciences AI-driven interface could be developed to greatly increase the accuracy of predicting future asset failures, as well as improve the response to existing failures.

BIM Uses

BMI uses include the following:

- Capital planning
- BIM authoring (SlimBIM)
- Maintenance planning
- Condition assessment
- Automated model checking

Business Case—ROI

No detailed ROI analysis was performed.

Financial benefits included improved utilization of capital dollars to minimize long-term facility TCO and to maximize the quality of life of the tenants.

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BIM Beyond Design Guidebook

Organizational Readiness

The owner did not have internal BIM skills and capabilities and relied on consultants and contractors to deliver actionable data generated from BIM.

Key Challenges

Key challenges included the following:

- A very large portfolio of facilities managed
- Verifying as-built data
- Collecting asset condition data

Lessons Learned

Lessons learned included the following:

- A low-cost SlimBIM approach for rapid BIM facility creation can be used to achieve financial benefits from BIM with little up-front investment.
- Automating CMMS–BIM integration can greatly reduce the level of effort in using BIM data.

14.9 New York Presbyterian Hospital

New York Presbyterian Hospital (NYP) is composed of seven separate medical campuses including the Columbia University Medical Center, Weill Cornell Medical Center, Allen Hospital, Lawrence Hospital, Lower Manhattan Hospital, Morgan Stanley Children's Hospital, and Westchester Division. The hospital network serves more than 2 million patients annually. NYP manages more than 10 million sf of facility space and over 100,000 infrastructure assets.

BIM was implemented as part of an asset management improvement effort that was designed to consolidate the asset inventory and management system to allow NYP to align and optimize asset utilization for operations, condition assessment, capital, and strategic planning.

Goals

Goals include the following:

- Improved asset inventory and condition assessment
- Support of a continuous condition assessment process
- Asset life cycle cost analysis
- Consolidated asset management that supports system-wide strategic and capital planning
- BMS BIM Integration

BIM Process

NYP developed "FM-AM-05 BIM/CAD Guidelines" that document its FM and asset management (AM) BIM requirements throughout the facility life cycle.

All assets are barcoded through design-construction and renovation, and barcode data are included in the BIM and included in any 2D plans generated from the BIM.

NYP maintains full copyright and license for all BIMs delivered through new design and construction projects and provides BIM to contractors on new projects. The existing condition

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BIM, however, is shared on the basis of the contractor assuming "sole risk" for its accuracy. These contractors are instructed not to rely upon the BIM for accurate coordinates.

NYP maintains a master BIM of the entire facility. On completion of new construction and renovation projects, a conformance BIM must be delivered that is used to update the master BIM. NYP's internal BIM support staff handles this integration task, including connecting any MEP elements in the project BIM to the NYP master BIM. NYP also maintains a BIM library of approved BIM asset models.

The asset data classification schema is UniFormat 2010, and NYP provides an asset inventory matrix that defines the delivery standards. Each asset type in the asset matrix indicates the barcoding requirements, whether the asset is inventoried, the modeling precision, and to which system the asset belongs.

NYP developed its BIM-authoring plug-in for designers to use when authoring BIM for NYP facilities. The plug-in facilitates the importing of barcode data into the BIM, manages the room and space number inventory, organizes rooms by NYP system type, and assists in naming ductwork systems within the correct parent equipment name.

BIM Uses

BIM uses include the following:

- Capital planning
- Space planning
- Asset management
- Condition assessment
- Asset inventory
- Maintenance planning
- BMS integration

Business Case—ROI

No detailed ROI analysis was performed. Financial benefits include the following:

- Improved capital planning, asset inventory, and condition assessment tracking
- Improved energy utilization and sustainability for new facilities and facility renovations

Organizational Readiness

NYP has invested in internal BIM staff to maintain the master BIM and integrate new BIM submittals from contractors.

Key Challenges

A key challenge has been consolidating facility asset management systems from seven separate facilities.

Lessons Learned

Lessons learned include the following:

- Developing a well-defined asset information matrix and BIM libraries improves BIM deliverables for facility asset management BIM uses.
- Linking assets in BIM with assets in the CMMS using barcode data.

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14.10 Perth Children's Hospital

Perth Children's Hospital is a $1.2 billion hospital designed and constructed as part of a public-private partnership between Western Australia and a private contractor. BIM for FM was a key deliverable. The public-private partnership nature of the contract provided incentive for the contractor to design the facility to minimize O&M costs.

Goals

Goals are the following:

- Improved life cycle asset management
- Improved space management
- Faster information exchanges and management
- Higher process automation
- Improved communication

BIM Process

The design and construction process required that a "whole of life" report be generated that detailed expected life of materials, replacement schedules, estimated costs, replacement strategies and expected disruptions, energy costs over 30 years, and estimated maintenance costs over 20 years.

AM-related benefits include asset management labor utilization, improved cost accounting, improved data and information management, and improved programming/scheduling. Asset information was delivered as COBie data with UniFormat classification.

The following were metrics suggested to measure the benefits from BIM on the project:

- Milestone BIM-related deliverables (number of drawings delivered on time)
- Number of work hours
- Number of resources needed by tasks
- Program compliance
- Smart clash reports
- Asset management element audit
- Asset management/smart handheld device interface audit
- Data accuracy
- Energy use (watts per square meter) versus modeled energy use and savings across life cycle
- Amount of rework
- Number of drawings made in a period of time versus traditional projects
- Costs associated with registering, validating, and responding to RFIs
- Number of RFIs
- Number and time required for resolution of issues register
- Saved time in preparing documents for handover and commissioning
- Surveys and other qualitative assessments
- Error rate (number of site surveys)
- Cost of design versus the traditional process
- Using elements of BIM in furniture, fixtures, and equipment through document management system; volumes, locations, equipment time, re-siting and placing, equipment and space management

Avatar walk-throughs were used to model patient and doctor traffic through the facility and to improve equipment and furniture layouts to optimize movements of nurses and patients in hospital rooms, labs, and testing facilities.

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The model was used to export data into a solar, wind, and thermal analysis software tool to determine best options for shading, facades, and building orientation. In this case, baseline metrics were available to demonstrate that this resulted in the effort taking several hours versus several days in previous projects.

BIM Uses

BIM uses include the following:

- ALCM (whole life report)
- Building management system integration
- Maintenance planning and work order management
- Design for sustainability—solar, wind, acoustic, and thermal analysis
- Design for operations—virtual walk-throughs

Business Case—ROI

No formal ROI analysis was performed for life cycle FM benefits.

Post-construction benefits included improved asset depreciation calculations, more precise asset management budgeting, and improved forecast LCC for individual assets.

Organizational Readiness

Ongoing BIM training will be required to receive full benefits from BIM for FM purposes.

Key Challenges

Key challenges included the following:

- Lack of BIM-experienced subcontractors involved in the project
- Value of BIM limited due to lack of trust in the BIM data
- Higher contractor costs due to BIM inexperience
- Hardware and software infrastructure
- For information exchanges, AIRs had not been defined
- Lack of standards

Lessons Learned

Lessons learned include the following:

- Owners need to engage the design team early in the process to ensure that facility management and operability issues are fully addressed.
- Asset information standards must be developed that define the data structure, asset classifications, standard libraries, life cycle phased requirements, and interoperability standards.

14.11 New Royal Adelaide Hospital: BIM O&M

The New Royal Adelaide Hospital (nRAH) is the largest BIM project completed in Australia, with a total cost of $1.85 billion. It was opened in September of 2017. The hospital is 260,000 sf and provides 800 patient beds. The project had a tight schedule that included 2 years of design and 2 years of construction. Prefabrication was optimized with 4.5-meter corridor modules being built and tested offsite to support accelerated construction goals.

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A Single Point of Truth nRAH (SPOTNIC) platform was developed to provide a collaboration tool that links the 3D model with QA and document management systems to provide a common facility data platform that can be accessed in real time by all team members. SPOTNIC can be accessed in the field using tablets.

The project developed facility management models to support O&M functions after project delivery. The facility had a design goal of 30% for improved efficiency over the existing Royal Adelaide Hospital.

Innovative aspects of the facility design that have been optimized for operations and maintainability include the following:

- The operational design was optimized to group critical care areas to reduce response time. The emergency department is located below the intensive care unit, trauma centers, and pathology centers with lifts connecting them for rapid transport.
- Automated guided vehicles deliver medications, meals, linens, and other supplies along designated corridors with embedded sensors.
- The automated pharmacy utilizes robots to manage inventory by expiration date and manage delivery to 74 dispensing stations via pneumatic tubes.
- Electronic asset tagging tracks all assets down to wheelchairs and blankets to optimize asset utilization.

Goals

Goals include the following:

- Design for operations and maintainability
- Design for sustainability
- Accelerated fast-track facility delivery
- Optimized prefabrication
- Increased collaboration across 200+ contractors/subcontractors
- Automated facility data handover
- Facility management models to support O&M

BIM Process

The project utilized the Australian BIM NATSPEC. Employing a public-private partnership project delivery method, the team was able to utilize BIM to maximize design for operations and maintainability considerations.

BIM Uses

BIM uses include the following:

- Design and construction
- Design for operations
- Design for maintainability
- Energy usage analysis
- Facility management model development
- CMMS integration

Business Case—ROI

No formal business case or ROI study was performed for this facility.

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Organizational Readiness

A facility management firm, Spotless, was involved early in the process. The project is a public-private partnership that will manage the project for the next 30 years. The public-private partnership project approach optimizes the ability of new capital projects to incorporate ALCM considerations early in the process.

Key Challenges

A key challenge was developing subcontractor BIM capabilities for such a large project, since the pool of experienced BIM contractors was limited.

Lessons Learned

Lessons learned include the following:

- Assess the regional BIM capabilities of contractors and subcontractors on large projects and build sufficient training and education time into the schedule to develop subcontractor skills.
- Leverage public-private partnership, or similar project delivery contract models, to maximize the ability to integrate architects, engineers, contractors, and owners to design for operations and maintainability.

14.12 Sydney Opera House

The Sydney Opera House was completed in 1973 after 14 years of construction. The roof consists of over 1 million roof tiles that define the building's uniquely recognizable design. BIM was implemented as part of the ongoing facility renovation efforts and included a life cycle BIM implementation (BIM4FM).

Goals

Goals include the following:

- Improved life cycle asset management
- Improved communication and collaboration across large and diverse stakeholder groups
- Common data environment to improve facility life cycle cost

BIM Process

Digital facility modeling at the Sydney Opera House was implemented using BIM as a national case for the use of BIM in FM in Australia. An open standards approach, using IFC data formats, was utilized to ensure maximum interoperability with existing and future systems. The case study goals were to examine the following:

- The reusability of BIM for FM purposes
- BIM as an integrated information model for facility management
- Extensibility of BIM to cope with specific business needs
- Commercial facility management software using standardized BIM
- The ability to add intelligence to the models

Assets were classified as anything worth more than $5,000. This included approximately 7,500 elements classified into 14 functional zones. The zones determined the level of detail for the BIM model elements based on the frequency of use. Annual asset audits are required.

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The building is composed of 7 theaters, 37 plant rooms, 12 lifts, and over 1,000 rooms. The building was designed with a lifespan of 250 years.

CAD plans were digitized in the 1980s, but the sequence of minor and major projects over the last three decades has made them less usable. Also, the design of the opera house is difficult to represent in 2D CAD.

A benchmarking system (the Building Presentation Index) was created, which can be automatically derived from the BIM.

Laser scanning was utilized to digitally capture the as-built condition of the opera house. A total of 800 scans and 56,000 digital photos were used to create the model.

Vehicle Access and Passenger Safety was the first full BIM project. It delivered BIM FM work products with an asset database tied directly to BIM.

The "BIM4FM" process was developed to create a web-based FM interface to link BIM to engineering, maintenance, and building control systems. This is a 7-year effort to align all building management systems with BIM.

A review of the project identified lessons learned that were categorized as defined expectations, in-house standards, hardware and software requirements, education, and disruptive culture:

- Defined expectations lessons learned included prioritizing needs, conducting workshops, and analyzing results.
- In-house standards lessons learned included creating BIM execution plans, model management plans, O&M guidelines, and contract language.
- Hardware and software lessons learned included properly scaling the system capabilities, capacity, and interoperability.
- Education lessons learned included providing ongoing training, management support, and proper expectations for the skill sets required to support the BIM4FM program.
- The disruptive change element included breaking down organizational silos and developing a collaborative environment, providing sufficient time for organizational adoption, and focusing on the cultural changes required.

BIM Uses

BIM uses include the following:

- Asset management
- Asset inventory
- Annual condition assessment
- Building management system integration

Business Case—ROI

No formal ROI analysis was performed for life cycle FM benefits.

The annual asset inventory and condition assessment was performed with a reduced cost.

Organizational Readiness

The organizational structure was not changed, but BIM improved the coordination and communication between departments to achieve long-term goals.

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Key Challenges

Key challenges included the following:

- Complexity of the facility
- Accuracy of as-built facility plans

Lessons Learned

Lessons learned include the following:

- Set manageable expectations and goals for the BIM FM program.
- Take a long-term view for developing BIM and its capabilities over time.

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Abbreviations

AEC	Architecture engineering construction
AGC	Associated General Contractors
AI	Artificial intelligence
AIA	American Institute of Architects
AIM	Asset information model
AIR	Asset information requirement
ALCM	Asset life cycle management
ANSI	American National Standards Institute
API	Application programming interface
ARFF	Aircraft Rescue and Firefighting
AVM	Aviation management
BAMie	Building Automation Modeling Information Exchange
BAS	Building automation system
BEP	BIM execution plan (only in Figure 3-3)
BIM	Building information modeling or building information model
BIMCAT	BIM Capabilities Assessment Tool
BMS	Building management system
BSI	British Standards Institution
CAD	Computer-aided design
CAFM	Computer-aided facility management
CIC	Computer Integrated Construction
CIM	Civil Integrated Management
CityGML	City Geography Markup Language
CMM	Capability maturity model
CMMS	Computerized maintenance management system
COBie	Construction Operations Building Information Exchange
COBIM	Common BIM requirements
CSI	Construction Specifications Institute
D&C	Design and construction
DEN	Denver International Airport
DFD	Data flow diagram
DFW	Dallas–Fort Worth International Airport
DOT	Department of Transportation
D4M	Design for maintenance (Fig 3-4)
EAM	Enterprise asset management
FBO	Fixed-base operator
FM	Facilities management
FMS	Facility management system (Fig 3-4)

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BIM Beyond Design Guidebook

gbXML	Green Building Extensible Markup Language
GC	General contractor
GFMAM	Global Forum on Maintenance & Asset Management
GIS	Geographic information system
GSA	U.S. Government Services Administration
HVAC	Heating, ventilation, and air conditioning
ICT	Information and communications technology
IFC	Industry Foundation Class
IFCXML	Industry Foundation Class Extensible Markup Language
IFMA	International Facilities Management Association
IoT	Internet of things
ISO	International Standards Organization
IT	Information technology
IWMS	Integrated workplace management system
JFK	John F. Kennedy International Airport
KPI	Key performance indicator
LAX	Los Angeles International Airport
LCC	Life cycle costing
LEED	Leadership in Energy and Environmental Design
LOD	Level of development
Massport	Massachusetts Port Authority
MEP	Mechanical, electrical, and plumbing
MPS	Model progression specification (Fig 3-4)
MVD	Model view definition
NATSPEC	National Specification (Australia)
NBIMS	National Building Information Modeling Standard
NCS	National CAD Standard
NFMT	National Facilities Management & Technologies
NIBS	National Institute of Building Sciences
NIBS-US	National BIM Standard-United States
NIST	National Institute of Standards and Technology
NPV	Net present value
NYP	New York Presbyterian Hospital
O&M	Operations and maintenance
OSU	Ohio State University
PANYNJ	Port Authority of New York & New Jersey
PAS	Publicly available specification
PDB	Progressive design build
QA/QC	Quality assurance/quality control
RFI	Request for information
RFP	Request for proposal
RFQ	Request for qualifications
ROA	Return on assets
ROE	Return on equity
ROI	Return on investment
SEA	Seattle-Tacoma International Airport
sf	Square foot
SFO	San Francisco International Airport
sUAS	Small unmanned aerial survey
TCO	Total cost of ownership
UK	United Kingdom

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WebGL	Web Graphics Language
WMU	Western Michigan University
W3C	World Wide Web Consortium
XML	Extensible Markup Language
YTV	Yleiset Tietomalli Vaatimukset
2D	Two-dimensional
3D	Three-dimensional
4D	Four-dimensional
5D	Five-dimensional

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APPENDIX A

Denver International Airport Contract Language

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BUILDING INFORMATION MODELING REQUIREMENTS

As used below, the term "Contractor" will mean and include Bidder/Contractor.

During the performance of this Agreement, the Contractor, for itself, its assignees, and successors in interest (hereinafter referred to as the "Contractor") agrees to comply with the <AGENCY> Building Information Modeling (BIM) requirements; including but not limited to:

- The Contractor recognizes and agrees that it shall be required to use the Autodesk BIM 360 Field (Field) and Autodesk BIM 360 Glue (Glue) platforms for this Project. Contractor recognizes and agrees that it shall make exclusive use of the <AGENCY> enterprise deployment of Field and Glue as provided by the <AGENCY>;

- The Contractor recognizes and agrees that it shall be required to execute a BIM Project Execution Plan in cooperation with the <AGENCY> representatives in compliance with the BIM Design Standards Manual and to adhere to the terms of that plan;

- Contractor shall produce a construction model and perform clash detection using Glue to the standards outlined in the BIM Design Standards Manual and to deliver the coordinated models to the <AGENCY> representatives;

- Contractor shall utilize Field to record required asset data for all <AGENCY> assets in compliance with the BIM Design Standards Manual within five working days of the installation of each identified asset;

- The Contractor recognizes and agrees that it shall be required to conform to all requirements of the BIM Design Standards Manual;

- The Contractor recognizes and agrees that it shall use the Autodesk BIM 360 Field (Field) platform for tracking issues discovered during construction.

- The Contractor recognizes and agrees that it shall use the Autodesk BIM 360 Field (Field) platform for tracking of warranty issues as defined by General Contract Conditions Title 18-warranties, guarantees, and corrective work.

Sanctions for Noncompliance. In the event of a Contractor's noncompliance with the provisions of this Agreement, the sponsor will impose such Contract sanctions as it may determine to be appropriate, including, but not limited to:

a. Withholding of payments to the Contractor under this Agreement until the Contractor complies, and/or;

b. Cancelling, terminating, or suspending this Agreement, in whole or in part.

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EXHIBIT R

BUILDING INFORMATION MODELING EXHIBIT

(CM At-Risk Projects)

I. GENERAL

A. Purpose of Exhibit.

This Building Information Modeling Exhibit ("Exhibit") establishes the Level of Development ("LOD") for the Building Information Model ("Model") for the Project. It is intended that such protocols will be memorialized in the Project's Building Information Modeling Project Execution Plan ("BIMPxP"), which will be developed soon after execution of the Agreement to which this Exhibit is appended. The BIMPxP will be used to create and modify the Model throughout the life of the Project. The BIMPxP will be updated throughout the Project and is hereby incorporated into the agreement by reference.

B. Application of Exhibit.

This Exhibit is a Contract Document and is intended to supplement both the professional services agreement ("Design Agreement") between the <AGENCY> ("Owner") and the Owner's Prime Design Consultant ("Consultant"), which this Exhibit is attached to. It is also intended that this Exhibit will apply to all project participants who receive Digital Data throughout the Project's development. Project participants may be required to verify that they have incorporated this Exhibit into their respective agreements and have likewise agreed to the most recent version of the Project's BIMPxP.

C. Necessity of the Model.

The Consultant recognizes that the Model is an indispensable deliverable under the Design Agreement and will be used by the <AGENCY> for future construction and maintenance projects. Failure to meet the LODs and/or deliver the model in accordance with the BIMPxP may result in a for cause termination of the Design Agreement.

II. RESPONSIBILITY FOR MODEL DEVELOPMENT

A. General.

The Consultant is responsible for the Models content and coordination among the Contractor, subcontractors, subconsultants and any other project participants and is assigned the responsibility for detecting and correcting errors.

B. Subsidiary Models.

It is anticipated that the Contractor and its Subcontractors may use Digital Data from the Model to develop several subsidiary construction models or such other uses as described in the BIMPxP. Any subsidiary models developed by the Contractor or its team shall be for the Contractor's own benefit and none of the subsidiary models shall be considered part of Contract Documents.

C. Responsibility for Model Content & Coordination among the Project Team.

1. Consultant's Responsibility: The Consultant retains ultimate responsibility for the Model's content as well as its development throughout the Project's entire design and construction. These responsibilities are non-delegable. The Consultant shall further be responsible for incorporating all

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appropriate Digital Data relating to the design's development into the Model. The Consultant will update the Model with Construction Information that it receives from the Contractor at intervals prescribed in the BIMPxP. (For purpose of this obligation, the term "Construction Information" includes, but is not limited to the identification, location and description of element properties of the Project's assets.) The Consultant will decide all issues involving design intent and will be responsible for coordinating the design efforts of its own subconsultants and integrating *design* comments from Owner.

2. Owner's Responsibility: Owner shall be responsible for and decide all issues involving the LOD for the various Model elements. The Owner will work with the Contractor to ensure that the format of the Digital Data it provides to the Consultant complies with the transfer protocols contained in the BIMPxP and Building Information Model Design Standards Manuel ("BIM DSM").

D. Responsibility for Detecting & Correcting Errors and Revising the Model.

1. Design issues: The Consultant shall be responsible for detecting and remedying interferences within the Model including any designs developed by consultants hired separately by the Owner. Once a solution is developed, the Consultant shall promptly revise the Model as required by the BIM DSM.

2. Construction issues: When requested by the Owner, the Consultant will be responsible for working with the Contractor to detect and remedy any interferences within the Model including any construction coordination issues with the Owner. Once a solution is developed, the Consultant shall promptly coordinate any revisions to the Model with the Owner.

3. Design/Construction Reconciliation Issues: The Consultant shall be responsible for detecting and remedying differences within the Model regarding the location and disposition of Owner designated facility assets as laid out in BIM DSM. Once a solution is developed, the Design shall promptly revise the Model as required by the BIM DSM.

E. Deadlines for Model Delivery.

The BIMPxP will contain a schedule for the Model's delivery (the "Model Delivery Schedule"). The Model Delivery Schedule identifies the intermediate and final deadline(s) for delivering each phase of the Model to the Owner for review and approval. It is anticipated that the Consultant will furnish Owner with a copy of the Model at the end of the phases as set forth in the Design Agreement. It is likewise acknowledged that the date for each deadline (as well as the LOD designations themselves) may be subsequently amended or modified in the BIMPxP. However, the Consultant needs a baseline schedule in order to price its efforts. Thus to the extent the BIMPxP subsequently modifies the original Model Delivery Schedule and such modifications impact the Consultant's initial pricing assumptions, the Consultant may be entitled to additional compensation. Failure by the consultant to meet the timelines or model phases, may result in the Owner holding pay applications until the Model phase is achieved and approved by the owner.

F. Owner's Review and Input.

Based upon the Consultant's submittals as described above, Owner shall review and provide comments to the Consultant and/or Contractor (depending upon the nature of Owner's comments) in a timely manner. The Consultant and/or Contractor shall take all necessary and reasonable steps to address

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such comments. Based upon the input of all concerned, and with Owner's approval, the Consultant shall revise the Model as soon as reasonably possible.

G. Standard of Care.

The Consultant shall perform its professional design services in accordance with the Standards of Performance described in the Agreement. With respect to the performance of the Consultant's Model Development services, the Consultant shall, at a minimum, conform to the requirements in this BIM Exhibit, the BIM DSM as well as the high standards of care and practice as outlined by the <AGENCY> State Board of Licensure for Architects, Professional Engineers and Professional Land Surveyors and the relevant statues and rule of the laws of the State of <AGENCY>.

III. BIM PROJECT EXECUTION PLAN ("BIMPxP")

A. BIMPxP Contract Status.

The BIMPxP will be modified as the Project develops. Nevertheless, the Parties are expected to comply with the requirements BIMPxP throughout the Term of the Agreement.

B. Process for Development.

Soon after being retained, the Consultant and the Owner will jointly develop the initial draft of the Project's BIMPxP using the Owner's BIMPxP Template as a starting point. If the Consultant believes that the protocols created in a subsequently developed BIMPxP represent a change in the scope of its services and believes that such change warrants an adjustment in compensation, contract sum, schedule or contract time, the Consultant is required to notify the Owner in writing in accordance with the terms of the Agreement. A failure to provide the required notice shall result in a waiver of any claim(s) for adjustments in compensation, contract sum, schedule or contract time as a result of the subsequently developed protocols.

C. Elements of the BIMPxP.

The Modeling protocols to be addressed in the BIMPxP shall:

1. Identify the Discipline Model Coordinator, Prime BIM Manager and when appropriate, the Construction BIM Manager and confirm their respective responsibilities per the roles described in the BIM Guidelines;
2. Define the various Authorized Uses and LOD for each Model and/or Model Element at prescribed Project milestones;
3. Identify the Project's construction classification systems per Owner's BIMGuidelines;
4. Define the process (and intervals) by which Project Participants will exchange and share the Model(s) and/or Model Element(s);
5. Define the process by which the Project Participants will identify, coordinate and resolve changes and updates to the Model(s) and/or Model Element(s); and
6. Include any other topics as may be required by the Owner.
7. Identify Technical Requirements, computer platform, software, etc.
8. Transmission protocols and procedures (CDs, e-mail, etc.)
9. Electronic signatures & stamps
10. Back-up, Format for exchange of As-Built information

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IV. RESPONSIBILITY FOR MODEL OWNERSHIP

A. Model Ownership.

The Parties agree that the Owner owns and hereby retains all legal rights of ownership and title to the Model and/or Model Elements and all other ancillary Project materials (electronic or otherwise) developed or prepared specifically for the Project by the Consultant, their subconsultants and all other project participants. Nothing contained in this Exhibit shall alter, diminish or be construed as a waiver by the Owner of such ownership rights.

The mere act of transmitting Digital Data or Confidential Digital Data does not convey any ownership right or legal interest in such data or in the software used to generate such data. Unless otherwise granted in a separate license, the Party receiving Digital Data or Confidential Digital Data (other than the Owner), may only use such data to design, construct, maintain, alter and/or add to the Project consistent with the terms of this BIM Exhibit, and nothing contained herein conveys any other right to use such data.

C. Owner's License to Project Team.

The Owner, as owner of all intellectual property rights associated with the Model(s) and/or Model Element(s) both at common law and by statute, hereby grants to the Consultant and Contractor, a revocable, non-exclusive, non-assignable, limited license to use the Model(s) and/or Model Element(s) and other relevant ancillary Project materials (electronic or otherwise) solely and exclusively to perform services for, or construction of the Project in accordance with the terms and conditions of the Design Agreement.

D. Authorized Use.

The BIMPxP lists the Authorized BIM Uses and LOD of the Model(s) and/or Model Element(s) at defined Project milestones.

E. Unauthorized Use.

If a Party uses a Model or Model Element in a way that is inconsistent with the Authorized Uses identified in the BIMPxP or not expressly authorized herein, such use shall be considered an Unauthorized Use and shall be at that Party's sole risk and without liability to any other Project Participant or Third-Party.

V. RESPONSIBILITY FOR MAINTENANCE OF DIGITAL DATA

A. Warranty of Authority to Transmit Digital Data.

The transmission of Digital Data constitutes an express warranty by the Party transmitting such data that it has legal permission to possess and transmit the Digital Data in accordance with the Authorized Uses described herein and in accordance with the BIMPxP.

B. Confidential Digital Data.

1. Confidential Digital Data: Confidential Digital Data is digital data expressly defined by the Owner as confidential. The transmission of Confidential Digital Data constitutes an express warranty by the Party transmitting such data that it is authorized and legally permitted to transmit the Confidential

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Digital Data in accordance with the Authorized Uses described herein and in Owner's BIMPxP. The Party receiving Confidential Digital Data shall keep such data confidential and shall not disclose it to any other person or entity except as provided below.

The Party receiving Confidential Digital Data may disclose such data only as required by law or court order, including a subpoena or other form of compulsory legal process issued by a court or governmental entity. Such Party may also disclose the Confidential Digital Data to its employees, consultants or contractors in order to perform services or work solely and exclusively for the Project, provided that such employees, consultants and contractors are likewise subject to the confidentiality restrictions set forth herein. In the event a Party in possession of Confidential Digital Data receives a subpoena seeking the production of such data, it shall immediately notify the Owner of the existence of such subpoena and give the Owner an opportunity to respond to the subpoena before releasing any such data.

2. <AGENCY> Open Records Act: Consultant acknowledges that the <AGENCY> is subject to the provisions of the <AGENCY> Open Records Act, and Consultant agrees that it will fully cooperate with the <AGENCY> in the event of a request or lawsuit arising under such act for the disclosure of any materials or information which Consultant asserts is confidential and exempt from disclosure. Any other provision of this Agreement notwithstanding, including exhibits, attachments and other documents incorporated into this Agreement by reference, all materials, records and information provided by Consultant to the <AGENCY> shall be considered confidential by the <AGENCY> only to the extent provided in the Open Records Act, and Consultant agrees that any disclosure of information by the <AGENCY> consistent with the provisions of the Open Records Act shall result in no liability of the <AGENCY>.

3. Sensitive Security Information: Sensitive Security Information ("SSI") is defined by 49 USC §1520. The management and handling of SSI is addressed in the BIMPxP, BIM DSM, and underlying Design Agreement and Construction Agreement (as appropriate).

IV. RESPONSIBILITY FOR MODEL MANAGEMENT

A. Archiving.

Subsequent to Final Completion of the Project, the Consultant shall be responsible for archiving all Digital Data in accordance with the obligations of its profession and the underlying Agreement. To the extent such obligations do not specifically pertain to the archiving of Digital Data, the protocols for retaining paper records shall apply.

The Consultant's obligations herein likewise include the duty to maintain the Project's Digital Data in an updated, accessible and readable format for a period of five (5) years from Final Completion. This obligation also includes the duty to convert the Project's Digital Data into a format compatible with any subsequently developed technology that might render the Project's existing Digital Data obsolete. The obligations herein survive the completion, termination or expiration of the underlying Agreement.

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B. Record Model.

Upon Substantial Completion of the Project, the Contractor is to provide the Consultant with As-Built information in accordance with the requirements of the BIMPxP. The Consultant is required to verify the information in accordance with the requirements and standards in the underlying Consultant Agreement.

C. Software Upgrades.

Upon receiving authorization to commence their respective scopes of work, the Consultant and the Contractor shall each represent to the Owner in writing that the software platform upon which the Model is to be developed is based upon the most recent version available. To the extent the Model's software platform is upgraded during the course of the Project's development, the Owner shall have the right (but not the obligation) to order all Project Participants to upgrade their respective platforms (at no cost to the Owner) in order to comply with the most recent version available. Responsibility to coordinate any such upgrades shall be on the Consultant and the Contractor for their respective teams (as applicable).

D. Governmental Approval of Model Drawings.

The Consultant shall produce printed paper and .pdf format drawings from the Model in order for the Contractor to obtain any necessary permits, approvals or government authorizations. Upon request, the Consultant shall also prepare an authentication (in a form prescribed by the Owner) verifying that the drawings are a true and accurate two-dimensional representation of the Model.

E. Signing and Sealing Model.

As required by the BIMPxP.

F. Standard of Care.

The Consultant shall perform its professional design services in accordance with the Standards of Performance described in the Agreement. With respect to the performance of the Consultant's Model Development services, the Consultant shall, at a minimum, conform to the requirements in this BIM Exhibit, the BIM DSM as well as the high standards of care and practice as outlined by the <AGENCY> State Board of Licensure for Architects, Professional Engineers and Professional Land Surveyors and the relevant statues and rule of the laws of the State of <AGENCY>.

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APPENDIX B

Penn State CIC Business Case Template

Organization Name

Business Case for Organization BIM Integration

Version 1.02
July 2, 2012

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Executive Summary:

[Abstract]

The executive summary of the Business Case provides a concise overview of the proposed BIM implementation and answers the question of why it should be supported. The executive summary allows reader to be quickly become acquainted with the contents of the business plan. It is intended to aid the decision makers within the organization and may be the most important part of a business plan. An executive summary of the Business Case for BIM Integration includes:

- the background of the BIM Planning process,
- the vision and objectives of BIM Implementation with organization,
- the proposed Uses of BIM with the organization,
- a cost/analysis summary of the BIM implementation, and
- outline recommendations.

The Executive Summary is written using short and concise sentences and paragraphs. It is no more than two pages in length. It is written in the same order as the business case and provides conclusions for the reader.

This template is a tool that is provided to assist in the development of an Organizational BIM Assessment Document. The template Document was created from the buildingSMART alliance™ (bSa) Project "BIM Execution Planning for Owners" as developed by The Computer Integrated Construction (CIC) Research Group of The Pennsylvania State University. The bSa project is sponsored by The Charles Pankow Foundation, US DoD Military Health System, Kaiser Permanente, US Department of Veterans Affairs, Penn State Office of Physical Plant (OPP), and The Partnership for Achieving Construction Excellence (PACE). The BIM Execution Planning Guide for Owners can be downloaded at http://bim.psu.edu.

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BIM Business Case

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1 Introduction and Background:

The introduction gives the background of BIM within the organization. Additionally, it includes a background on the organization itself. It discusses the mission and vision of the organization and its implementation of BIM. If the organization has used BIM in the past, even at a pilot level, it is summarized here.

2 Problem Definition, Goals & Objectives

(This section defines the problem that is being overcome through the use of BIM. It also documents the goals of the organization and includes the BIM objectives.)

2.1 Problem Definition

(What is the problem that is being solved through the use of BIM?)

2.2 Organizational Mission and BIM Vision

(Include a summary of the Organizational Mission)

(Include a summary of the BIM Vision)

2.3 Organizational Goals and BIM Objectives

(Include a summary of the Organizational Goals and BIM Objectives)

Table 2-1: Organizational Goals

PRIORITY	GOAL DESCRIPTION	BIM OBJECTIVE
Choose an item.		

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2.4 Planning Team Members:

(Document the personnel that help to create the Business Case.)

NAME	ROLE	DEPARTMENT	E-MAIL	PHONE
	BIM Champion			
	Discipline Lead			

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3 Proposed BIM Uses

(The proposed uses of BIM are documented in the business case. This section include methods in which BIM will be used to accomplish the objective and goals of the organization)

Table 3-1: Services Provided

X	CURRENT PROCESS	BIM USE	DESCRIPTION	CURRENT MATURITY	DESIRED MATURITY

Maturity Level	Description
(0) Non-Existent	At this maturity level, a process has not yet been incorporated into current business processes and does not yet have established goals and objectives.
(1) Initial	At this maturity level, a process produces results in which the specific goals are satisfied, however, they are usually ad hoc and chaotic. There no stable environment to support processes with the inability to repeat such and possible abandonment in time of crisis.
(2) Managed	At this maturity level, a process is planned and executed in accordance with policy; employs skilled people having adequate resources to produce controlled outputs; involves relevant stakeholders; is monitored, controlled, and reviewed; and is evaluated for adherence to its process description.
(3) Defined	At this Maturity level, a process is tailored to the organization's standard processes according to the organization's guidelines; has a maintained process description; and contributes process related experiences to the organizational process assets
(4) Quantitatively Managed	A this maturity level, a process is managed using statistical and other quantitative techniques to build an understanding of the performance or predicted performance of processes in comparison to the project's or work group's quality and process performance objectives, and identifying corrective action that may need to be taken.
(5) Optimizing	At this maturity level, a process is continually improved through incremental and innovative process and technological improvements based on a quantitative understanding of its business objectives and performance needs and tied to the overall organizational performance.

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4 Cost – Benefit Analysis

The implementation of BIM should be considered an investment into an organization's future. Just like any investment, a cost benefit analysis should be performed to ensure that it is a sound investment. A cost benefit analysis sets the financial and non-financial cost of implementing BIM along with the anticipated benefits. The cost-benefit analysis shows the anticipated return of implementing BIM.

The analysis answers the following questions:

- What is the benefit / income from BIM
- What are costs of implementing BIM? This includes both initial and long-term costs. It should also include a possible funding source for the investment.
- What are the major risks associated with BIM implementation?
- What are the major assumptions of this analysis?

4.1 Estimated Benefits and Metrics

A benefits analysis specifies the expected financial and non-financial returns from a given project. It compares 'with' and 'without' situations. The results of this analysis can be used to evaluate alternative options. It can strongly support a bid for management endorsement and resource allocation.

Some potential benefits that should be calculated based on the organization's planned implementation are:

- Improved project outcomes such as lower cost and shorter duration
- Streamlining of processes / reduced process time
- Improved quality of information
- Improved interoperability of data
- Reduced human error
- Reduced data entry time
- Centralization of information

4.2 Cost Estimates

The budget should include all identifiable costs to the Organization, including staffing, on-costs, software, legal, media, travel, physical resources, etc. The source of the funds should also be considered – is it an existing available fund or are new and additional funds required?

Some items to consider include

Cost for Planning:

BIM Champion(s) (Percentage of time allocated * salary for allocated timeframe)

Planning Team Costs (Percentage of time allocated * Number of Personnel * salary for allocated time frame)

Personnel:

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New / reallocated personnel ($/year including taxes and benefits)

Personnel Education and Training cost ($/course necessary)

Misc. Expenses (Travel Budget, etc)

Infrastructure

Software

Software Purchase($ / license)

Software Maintenance Fees ($ / license / year)

Hardware

Workstations ($/ workstation including accessories

Hardware infrastructure ($ / infrastructure item

Infrastructure maintenance costs ($ per year

Process change costs

Inefficiency Expenses (if applicable)

Learning Curve

Category	Item	Salary	Time	Factors	Cost	
Planning Cost	BIM Champion	Salary	Percentage of time allocated			
	Planning Team Members	Average Salary	Average Time All			

4.3 Risk Assessment

Performing a risk assessment is critical when developing a business case for BIM. Like other changes in work process, the integration of BIM within an organization has risks. Table 4-1 show an example of how the risk assessment summary may look. The steps of creating a BIM risk assessment include:

1. Risk identification
2. Risk evaluation including likelihood and impact
3. Risk mitigation
4. Risk summarization and recommendation
5. Risk assessment review and update

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Table 4-1: Risk Assessment Table

Risk	Risk Likelihood	Risk Impact	Risk Mitigation Measures	Recommendation
	Low	Low		Acceptable
	Medium	Med		Not Acceptable
	High			

4.4 Assumptions

Currently there is little data on the cost and benefit of implementing BIM within an owner organization, therefore assumptions of the cost and benefit analysis are documented. Each item is listed in this section.

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5 Implementation Timeline

The implementation timeline is an overview of the transition plan to building information modeling. It should include milestones and major objectives if the organization moves forward with BIM implementations

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6 Final Recommendations

The final recommendations include the conclusion that can be drawn about the business case for the implementation of BIM within the organization. The recommendations should include the key factors that need to be considered why determining the validity of the business case. It can also include highlights from the other sections of the document to support the business case for BIM

7 Appendices

Appendices include information that supports the business case for BIM with the organization. This information is often too detailed for the body of the business case, however is necessary for the analysis. It also helps to show the level of effort that when into creating the business case. The appendices could include items such as: the strategic plan, the organizational execution plan, a project execution plan, procurement documents, detailed financial analysis, definitions of terms, and other documents to support the business case.

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APPENDIX C

FAA Minimum BIM Requirements

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Minimum Building Information Modeling (BIM) Requirements

Introduction:

The following Building Information Modeling (BIM) requirements have been developed to define the BIM process within the FAA and establish requirements for the utilization of BIM in the design and construction of its new facilities. These requirements are based on the United States Army Corp of Engineers (USACE) BIM requirements.

Facility data requirements are based on the Department of Defense – Military Health System (DoD-MHS) requirements. Samples of typical facility data exports under the heading, "Medical Clinic Example," can be downloaded at http://www.buildingsmartalliance.org/index.php/projects/activeprojects/140#clinic .

Scope:

The Architectural/Engineering (A/E) firm will develop the Model with Facility Data for the purposes of coordination between various disciplines (also known as Clash Detection) and developing 2-dimensional extractions generated from the Model to produce accurate construction contract drawings required in the A/E Scope of Work (SOW) and A/E Project Manual. BIM associated submittals shall conform to the standards described in this document. Exceptions must be approved by FAA.

Section 1 - General:

1.1 Definitions. See Section 7 for definitions of terms used in this document.

1.2 Design Deliverables. Develop all designs using BIM. Design submittal drawings shall be D-size, suitable for half-size (11"x17") scaled reproduction.

1.3 Approved BIM authoring tools/Software. The FAA requires the use of object-oriented BIM software applications that comply with current industry interoperability standards for use in a collaborative environment. The models and analysis shall be used in support of the decision making process for high performance building design and other lifecycle facility management (FM) functions. All software platforms used for FAA projects must be compliant with:

a) Most current version of Industry Foundation Classes (IFC) file format
b) Commercially available collaboration software that provides interoperability between different software applications (e.g. NavisWorks or equal)

Approved BIM Software for FAA Projects:

Purpose	Tools/Software
Authoring-Design, Architectural, Structural	Revit Architecture, Bentley Architecture(Triforma), Tekla
Authoring-MEP, Design and Construction	Revit MEP, AutoCAD MEP, Bentley BIM, CAD-Duct, CAD-Pipe, AutoSprink, PipeDesigner 3D
Authoring-Civil	Bentley Inroads and Geopak, Autodesk Civil 3D
Coordination (spatial conflict, clash detection)	NavisWorks Manage or Bentley Navigator
4D Scheduling	Vico, NavisWorks Simulate, Primavera, MS Project, Bentley Navigator
Cost Estimation	Innovaya, Vico
Energy Analysis	Green Building Studio, IES, Ecotect, Hevacomp, TAS
Specifications	E-Specs
Model Checking/ Validation	Solibri
Water Management	Bentley WaterGem

Software other than those listed above may be used subject to the above compliance requirements and approval by the FAA CAEG Program's BIM Administrator.

A/E Scope of work
TUS-ACT
February 2012

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Section 2 – Design Requirements

2.1 Drawings. The use of BIM does not negate the need for delivery of CAD files used for the creation of the construction contract drawings. Specification of a CAD file format for these drawings submitted shall not be used to limit which BIM application(s) or software(s) may be used for project development and execution.

Deliver CAD files and Models used for the creation of the construction contact drawings per requirements in the A/E SOW, A/E Project Manual, and as noted herein. CAD drawings extracted from the Models shall comply with the latest version of FAA-STD-002 standards. Any deviations must be approved by the FAA. Any areas of conflict between this document, the A/E Project Manual, and FAA-STD-002 must be brought to the attention of the FAA.

Traditional 2D documentation shall be prepared with approved IFC Compliant BIM Authoring Software and, as such, the expectation shall be that plans, elevations, sections, schedules, and details are fully coordinated with the concurrent building model.

2.2 BIM Model and Facility Data. The A/E shall select the BIM application(s) and software(s) to develop the project design and use the 3D graphic model(s) (the "Model") and associated intelligent attribute data ("Facility Data")-created by this software to produce accurate construction documents. All submitted Models and associated Facility Data shall be fully compatible with Autodesk Revit 2009 or higher, Bentley BIM v8 or higher file format. Facility data contained in the Model shall be consistent with the structure of IFC 2x3 coordination view or better as defined in section 2.2.1

2.2.1 IFC Coordination View. The A/E's selected BIM application(s) and software(s) must be certified in the IFC Coordination View (2x3 or better. See www.iai-na.org). The coordination view shall contain the basic Model, as shared between the major disciplines within the design phase and shall include the following:

- Coordination between design disciplines (architecture, building services, structural) and with the client
- Building spatial structure of the Model
- Elements of the Model with their semantic information
- Logical structure of elements (within spatial structure, aggregation and decomposition, features, etc.)
- Spaces of the Model with their semantic information
- Logical structure of spaces (within spatial structure, elements included, etc., leading to a spacebook)
- Basic clash detection (3D geometry based)
- Basic communication about the design (2D/3D geometry based)
- Basic visual design intentions (2D/3D presentation information - color, hatching, rendering)
- The exchange of property information

Submit any deviations from or additions to the IFC property sets for any new spaces, systems, and equipment for FAA approval.

2.2.2 Submittal Requirements. BIM submittals shall conform to the requirements of Section 3, Section 4, and the A/E Project Manual.

2.2.3 Project Execution Plan (PxP): The A/E shall utilize the latest version of the USACE BIM PROJECT EXECUTION PLAN (USACE PxP) Template to develop an acceptable PxP. The template can be downloaded from the CAD/BIM Technology Center website, currently https://cadbim.usace.army.mil

2.2.3.1 The Project Planning Document (PPD) described in the A/E Project Manual shall include a PxP documenting viability of the BIM design and analysis technologies selected for the Model from concept development through As-Builts as a design, production, coordination, construction, and documentation tool and the collaborative process by which it shall be implemented.

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2.2.3.2 The PxP shall describe uses of BIM during design and construction phases to include value management, interference management, and design-change tracking, or such other uses as the A/E proposes.

2.2.3.3 The PxP shall identify how the Model data shall be managed and interoperate (data storage, sharing, viewing, quality control parameters in Section 2.3 Quality Control, and updating, as necessary) among all A/E team members.

2.2.3.4 After the PPD review phase, conduct a PxP demonstration at the 35% Design Review Meeting to review the PxP for clarification and to verify the functionality of Model technology workflow and processes. The FAA shall confirm acceptability of the PxP or advise as to additional processes or activities necessary to be incorporated into the Plan. If modifications are required, the A/E shall execute the modifications and resubmit the PxP for FAA approval. There will be no payment for design until the PXP is acceptable to the FAA. The FAA may also withhold payment for design for unacceptable performance in executing the PxP.

2.2.4 BIM Requirements.

2.2.4.1 Facility Data. Develop the Facility Data, consisting of a set of intelligent elements for the Model (e.g., doors, air handlers, electrical panels, ducts, beams etc.). This Facility Data shall be consistent with the structure of IFC 2x3 Coordination View and include all material definitions, qualities, attributes that are necessary for the facility design, construction, and support of operations and maintenance. The required parameters for the Model's project, room, area/zone, objects are as follows:

2.2.4.1.1 Minimum Project Parameters - The Model shall as a minimum have the project level parameters in accordance with Table 1.

2.2.4.1.2 Minimum Room Parameters - The Model shall have the room project level parameters in accordance with Table 2. The data shall be entered for each parameter which is applicable to the given room. The contractor shall maintain an exportable table from the Model that provides the parameters listed in Table 2 and Table 3.

Room Data to include: architectural, mechanical, electrical, plumbing, lighting and acoustical design criteria requirements.

2.2.4.1.3 Minimum Real Property Installed Equipment (RPIE) Parameters - The Model shall have the RPIE object level parameters in accordance with Table 4. The data shall be entered for each parameter which is applicable to the given RPIE item. The contractor shall maintain an exportable table from the Model that provides the parameters listed in Table 3, Table 4, and Table 5.

Data shall be exportable to COBie (version 2.3) for the FAA (file shall be named COBIE-FAA). All deliverables shall include a copy of the COBIE-FAA export for the RPIE. Samples can be obtained from the DoD-MHS BIM website, currently http://www.buildingsmartalliance.org/index.php/projects/activeprojects/162

2.2.4.1.4 Minimum Equipment & Furniture Parameters - The Model shall as a minimum have the Project Room Content Report (PRCR) object level parameters in accordance with Table 6. The data shall be entered for each parameter which is applicable to the given PRC item. The contractor shall maintain an exportable table from the Model that provides the parameters listed in Table 6, Table 7, and Table 8.

Data shall be exportable to COBie (version 2.3) for the FAA (file shall be named COBIE-FAA). All deliverables shall include a copy of the COBIE-FAA export for the equipment and furniture.

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Samples can be obtained from the DoD-MHS BIM website, currently http://www.buildingsmartalliance.org/index.php/projects/activeprojects/162

2.2.4.2 Model Content. The Model shall include, at a minimum, the requirements of Section 4. The FAA must agree with any proposed modifications to minimum requirements before incorporation into the Model.

2.2.4.3 Model Granularity. Models may vary in level of detail for individual elements within a model, but at a minimum must include all features that would be included on a quarter inch (1/4" = 1'0") scaled drawing (e.g. at least 1/16th, 1/8th and 1/4th) and all the elements of fire protection, or appropriately scaled civil drawings.

2.2.4.4 Output. All submitted drawings (e.g., plans, elevations, sections, schedules, details, etc.) shall be derived (commonly known as extractions, views or sheets) from the Model and Facility Data. Drawings derived from the Model shall remain connected to the Model for the life of the Project and documented in the PxP. FAA approval must be provided for drawings not derived from the Model. Any A/E request for approval shall also be documented in the PxP.

2.2.4.5 COBIE Compliance. The Model and Facility Data for the Project shall fulfill Construction Operations Building Information Exchange (COBIE) requirements, including all requirements for the indexing and submission of Portable Document Format (PDF) and other appropriate file formats that would otherwise be printed and submitted in compliance with Project operations and maintenance handover requirements.

2.2.4.5.1 Electronic Exchange. The National Building Information Model Standard (NBIMS) COBIE format shall be used for electronic exchange on this Project. Compile a COBIE index on the Microsoft Excel spreadsheet provided by NBIMS at www.nbims.org. Unless otherwise noted, also provide information identified in the COBIE Pilot Implementation Standard worksheets.

2.2.5 Baseline Model. The Contractor will not be provided a baseline multi-discipline BIM Model.

2.2.6 USACE BIM Workspace. If Contractor selects Bentley Systems BIM as the BIM platform of choice, the latest version of the USACE Bentley BIM Workspace must be used and can be downloaded from the CAD/BIM Technology Center website, currently https://cadbim.usace.army.mil If Contractor selects Autodesk Revit as the BIM platform of choice, a USACE Revit Workspace will not be provided; Contractor can select which Revit Workspace to use.

2.3 Quality Control. The A/E shall provide Coordination View Information Exchange (CVIE), deliverables to partially demonstrate their fulfillment of the requirements to fully coordinate design disciplines into a single cohesive design, as part of their approved Quality Control plan. The CVIE is based upon the International Alliance for Interoperability Industry Foundation Class (IFC) 2x3 Coordination View format as described in Section 2.2.1. The CVIE deliverables are comprised of: (1) the IFC Coordination View and (2) BIM model collision detection reports.

The A/E is responsible for implementing quality control (QC) parameters for the Model, including:

2.3.1 CAD Standards Checks. QC checking performed to ensure that the fonts, dimensions, line styles, levels and other contract document formatting issues are followed per the FAA-STD-002 CAD Standard (latest revision). Identify and report non-compliant content and submit a corrective action plan.

2.3.2 Model Integrity Checks. QC validation used to ensure that the Facility Data has no undefined, incorrectly defined or duplicated elements. Identify and report non-compliant elements and submit a corrective action plan. Provide the FAA with detailed justification and request FAA acceptance for any non-compliant element that the A/E proposes to be allowed to remain in the Model.

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2.3.3 Other Parameters. Develop such other QC parameters as A/E deems appropriate for the Project and provide to the FAA for concurrence.

2.4 Design and Coordination Reviews. Perform design and coordination reviews at each submittal stage under Section 3 to test the Model, including:

2.4.1 Visual Checks. Checking to ensure the design intent has been followed and that there are no unintended elements in the Model.

2.4.2 Interference Management Checks. Locating conflicting spatial data in the Model where two elements are occupying the same physical space. Log hard interferences (e.g., mechanical vs. structural or mechanical vs. mechanical overlaps in the same location) and soft interferences (conflicts regarding service access, fireproofing, insulation) in a written report and resolve.

2.4.3 IFC Coordination View. Provide an IFC Coordination View in IFC Express format for all deliverables. Provide exported property set data for all IFC supported named building elements.

2.4.4 Program for Design (PFD) Validation Check. Provide report of comparison of contracted PFD to actual design PFD. The comparison shall either be done within the model platform itself or an external project review program approved by the FAA. Actual NSF for the design shall be automatically generated within the model and not manually entered. The PFD validation check shall have a minimum of the following data points listed: PFD Room Code, Design Room Number, Department, Sub Department, Space Type, Room Name, PFD Target NSF, Design Actual NSF, Calculated Delta between Target and Actual NSF for room, calculated exceeds critical delta (yes or no), and “Project_Room_ID”. The project team shall establish a target “critical” delta or allowable variance for rooms at the beginning of the project (e.g. 2%). The PFD validation report will indicate rooms that fall outside of these established criteria.

2.4.5 Project Room Contents (PRC) Validation Check. Provide report of comparison of approved PRC list by room to actual design PRC. The comparison shall either be done within the model platform itself or an external project review program approved by the FAA. The report shall provide a list of rooms where the design PRC does not match the approved PRC and the specific items that do not match. The non-matching items list shall include at a minimum the PRC equipment item approved and expected, and the designed PRC item not matching.

2.4.6 Gross Area Tabulation Calculation. The contractor shall calculate the departmental gross square feet (GSF) and the building GSF using the model’s automatic calculation attributes in accordance with standard FAA Real Estate gross square footage calculation guidance.

2.4.6.1 Departmental Gross Area Tabulation report shall as a minimum identify the total departmental gross area factor.

2.4.6.2 Total building Gross Area Tabulation report shall as a minimum identify total mechanical gross, circulation gross, electrical gross and overall building gross area factor.

2.4.7 Over-the-Shoulder Reviews. Periodic quality control meetings shall include quality control reviews on the implementation and use of the Model, including interference management and design change tracking information.

2.4.8 Other Parameters. Develop such other Review parameters as the A/E deems appropriate for the Project and provide to the FAA for concurrence.

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Section 3 – Submittal Requirements

3.1 General Submittal Requirements. - Provide submittals in compliance with the A/E SOW deliverables and as described hereinafter.

3.1.1 Provide an A/E-certified written report with each design submittal, confirming that the Quality Control checks have been completed for the design submittal. This report shall be discussed as part of each design review meeting and shall address cross-discipline interferences, if any.

3.1.2 With each design submittal, provide the FAA a 3-D interactive visualization from the Model in Navisworks, ProjectWise Navigator, Adobe 3D PDF 7.0 (or later), Google Earth KMZ or equivalent format. The FAA may request other formats if needed to address Project-specific requirements.

3.2 PxP Review. At the 35% Design Review Meeting, demonstrate preliminary development of Model components and Facility Data identified in Section 2.2.4. Review the Model with the FAA for conformity to program, massing, circulation, fire protection, security, and sustainability `requirements consistent with the PxP.

3.3 Design Submittals.

3.3.1 BIM and CAD Data. The submitted Models shall include architectural, structural, electrical, mechanical, plumbing, fire protection, security systems, and Facility Data, to satisfy the review cycles prescribed in the A/E Project Manual and the A/E SOW. Submit rendering files, the Model, Facility Data, Workspace, CVIE dataset, and CAD Drawing files in native and interoperable formats, on DVD/CD-ROM.

The electronic submittals shall be organized and structured supportive of archival and retrieval. The electronic submittals shall have a “dash-board” type feature to assist viewers to navigate through the digital media and associated files. Files not using names which readily identify their content shall have appropriate Meta data attached to include searchable short descriptions of the file’s content or relevance.

3.3.2 Submittal packages. The A/E shall submit IFC Model View files and collision detection reports with each deliverable.

3.3.3 IFC CVIE deliverables. CVIE deliverables shall be included with all required design submissions that require review. In accordance with the A/Es’ Quality Control plan, the A/E shall use the CVIE deliverable to demonstrate that they have evaluated each collision identified by the BIM and or Model Checking software and documented the collision. The A/E shall provide their report with each CVIE Deliverable.

The A/E shall produce and submit CVIE reports directly from BIM or Model Checking software. The CVIE report shall be provided in PDF or XML Format. If provided in XML, a Cascading Style Sheet allowing review of the XML data in a web browser shall also be provided.

The collision detection report shall identify the GUID (Globally Unique Identifier) and human-readable name of each object involved in each collision. The collision detection report shall provide a graphic image for each collision highlighting those objects involved in the error. Images may be provided within the PDF report, or as separate .jpg files referenced in the XML report.

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Section 4 – BIM Model Minimum Requirements and Output

4.1 General Provisions. The Model shall be developed to include the systems described below as they would be built. The Model shall be developed to include as many of the systems described below as are necessary and appropriate for each design review cycle at that design stage. Existing conditions shall be exempt from the modeling requirements except at areas where the new systems interface with existing.

4.2 Architectural/Interior Design. The Architectural systems Model may vary in level of detail for individual elements, but at a minimum must include all features that would be included on a quarter inch (1/4”=1’0”) scaled drawing. Additional minimum Model requirements include:

4.2.1 Spaces. The Model shall include spaces defining accurate net square footage and net volume, and holding data for the room finish schedule for including room names and numbers. Include Program Information provided by the FAA to verify design space against programmed space, using this information to validate area quantities.

4.2.2. Walls and Curtain Walls. Each wall shall be depicted to the exact height, length, width and ratings (thermal, acoustic, fire) to properly reflect wall types. The Model shall include all walls, both interior and exterior, and the necessary intelligence to produce accurate plans, sections and elevations depicting these design elements.

4.2.3 Doors, Windows and Louvers. Doors, windows and louvers shall be depicted to represent their actual size, type and location. Doors and windows shall be modeled with the necessary intelligence to produce accurate window and door schedules.

4.2.4 Roof. The Model shall include the roof configuration, drainage system, major penetrations, specialties, and the necessary intelligence to produce accurate plans, building sections and generic wall sections where roof design elements are depicted.

4.2.5 Floors. The floor slab shall be developed in the Structural Model and then referenced by the Architectural Model for each floor of the Project building.

4.2.6 Ceilings. All heights and other dimensions of ceilings, including soffits, ceiling materials, or other special conditions shall be depicted in the Model with the necessary intelligence to produce accurate plans, building sections and generic wall sections where ceiling design elements are depicted.

4.2.7 Vertical Circulation. All continuous vertical components (i.e., non-structural shafts, architectural stairs, handrails and guardrails) shall be accurately depicted and shall include the necessary intelligence to produce accurate plans, elevations and sections in which such design elements are referenced.

4.2.8 Architectural Specialties. All architectural specialties (i.e., toilet room accessories, toilet partitions, grab bars, lockers, and display cases) and millwork (i.e., cabinetry and counters) shall be accurately depicted with the necessary intelligence to produce accurate plans, elevations and sections in which such design elements are referenced.

4.2.9 Signage. The Model shall include all signage and the necessary intelligence to produce accurate plans and schedules.

4.2.10 Schedules. Provide door, window, hardware, sets using Builders Hardware Manufacturers Association (BHMA) designations, flooring, and wall finish, and signage schedules from the Model, indicating the type, materials and finishes used in the design.

4.3 Furniture/Fixtures/Equipment (FFE). 3D representation of FFE elements is required. For projects with an extensive systems furniture layout that may impact BIM system performance the Contractor will

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contact the FAA for consideration of 2D representation. The FFE systems Model may vary in level of detail for individual elements, but at a minimum must include all features that would be included on a quarter inch (1/4”=1’0”) scaled drawing. Additional minimum Model requirements include:

4.3.1 Furniture. The furniture systems Model shall include all relevant office equipment and furniture system layouts, with necessary intelligence to produce accurate plans, sections, perspectives and elevations necessary to completely depict furniture systems locations and sizes.

4.3.2 Fixtures and Equipment. Fixtures and equipment shall be depicted to meet layout requirements with the necessary intelligence to produce accurate plans, elevations, sections and schedules depicting their configuration. Examples of equipment include but are not limited to copiers, printers, refrigerators, ice machines, microwaves, audio/visual equipment, FAA air traffic control (ATC) equipment racks and ATC displays/consoles.

4.3.2.1 Electronic Equipment – ATC Equipment provided in the PRC shall as a minimum be shown with accurate volumetric representations and loaded with appropriate FAA data set. The graphical representation (minimum volumetric representation) and parametric data shall be updated as the plan develops accurately representing the current planning details.

4.3.3 System Coordination. FFE that makes use of electrical, data, plumbing or other features shall include the necessary intelligence to produce coordinated documents and data.

4.3.4 Schedules. Provide FFE schedules from the model indicating the materials, finishes, mechanical, and electrical requirements.

4.4 Structural. The structural systems Model may vary in level of detail for individual elements, but at a minimum must include all features that would be included on a quarter inch (1/4”=1’0”) scaled drawing. Additional minimum Model requirements include:

4.4.1 Foundations. All necessary foundation and/or footing elements, with necessary intelligence to produce accurate plans and elevations.

4.4.2 Floor Slabs. Structural floor slabs shall be depicted, including all necessary recesses, curbs, pads, closure pours, and major penetrations accurately depicted.

4.4.3 Structural Steel. All steel columns, primary and secondary framing members, and steel bracing for the roof and floor systems (including decks), including all necessary intelligence to produce accurate structural steel framing plans and related building/wall sections.

4.4.4 Cast-in-Place Concrete. All walls, columns, and beams, including necessary intelligence to produce accurate plans and building/wall sections depicting cast-in-place concrete elements.

4.4.5 Expansion/Contraction Joints. Joints shall be accurately depicted.

4.4.6 Stairs. The structural Model shall include all necessary openings and framing members for stair systems, including necessary intelligence to produce accurate plans and building/wall sections depicting stair design elements.

4.4.7 Shafts and Pits. The structural Model shall include all necessary shafts, pits, and openings, including necessary intelligence to produce accurate plans and building/wall sections depicting these design elements.

4.5.8. Openings and Penetrations. All major openings and penetrations that would be included on a quarter inch (1/4”=1’0”) scaled drawing.

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4.5 Mechanical. The mechanical systems Model may vary in level of detail for individual elements, but at a minimum must include all features that would be included on a quarter inch (1/4"=1'0") scaled drawing. Small diameter (less than 1-1/2" NPS) field-routed piping is not required to be depicted in the Model. Additional minimum Model requirements include:

4.5.1 HVAC. All necessary heating, ventilating, air-conditioning and specialty equipment, including air distribution ducts for supply, return, and ventilation and exhaust ducts, including control system, registers, diffusers, grills and hydronic baseboards with necessary intelligence to produce accurate plans, elevations, building/wall sections and schedules.

4.5.1.1 Mechanical Piping. All necessary piping and fixture layouts, and related equipment, including necessary intelligence to produce accurate plans, elevations, building/wall sections, and schedules.

4.5.2 Plumbing. All necessary plumbing piping and fixture layouts, floor and area drains, and related equipment, including necessary intelligence to produce accurate plans, elevations, building/wall sections, riser diagrams, and schedules.

4.5.3 Equipment Clearances. All HVAC and Plumbing equipment clearances shall be modeled for use in interference management and maintenance access requirements.

4.5.4 Elevator Equipment. The Model shall include the necessary equipment and control system, including necessary intelligence to produce accurate plans, sections and elevations depicting these design elements.

4.6 Electrical/Telecommunications. The electrical systems Model may vary in level of detail for individual elements, but at a minimum must include all features that would be included on a quarter inch (1/4"=1'0") scaled drawing. Small diameter (less than 1-1/2"Ø) field-routed conduit is not required to be depicted in the Model. Additional minimum Model requirements include:

4.6.1 Interior Electrical Power and Lighting. All necessary interior electrical components (i.e., lighting, receptacles, special and general purpose power receptacles, lighting fixtures, panelboards and control systems), including necessary intelligence to produce accurate plans, details and schedules. Cable tray routing shall be modeled without detail of cable contents. Lighting and power built into furniture/equipment shall be modeled.

4.6.2 Special Electrical Systems. All necessary special electrical components (i.e., security, Mass Notification, Public Address, and other special occupancies, and control systems), including necessary intelligence to produce accurate plans, details and schedules.

4.6.3 Grounding Systems. All necessary grounding components (i.e., lightning protection systems, static grounding systems, communications grounding systems, bonding), including necessary intelligence to produce accurate plans, details and schedules.

4.6.4 Communications and Data. All existing and new communications and data service controls and connections, both above ground and underground with general purpose comm/data receptacles and the necessary intelligence to produce accurate plans, details and schedules. Cable tray routing shall be modeled without detail of cable contents.

4.6.5 Exterior Building Lighting. All necessary exterior lighting with necessary intelligence to produce accurate plans, elevations and schedules. The exterior building lighting Model shall include all necessary lighting, relevant existing and proposed support utility lines and equipment required with necessary intelligence to produce accurate plans, details and schedules.

4.6.6 Equipment Clearances. All lighting and communications equipment clearances and no-fly zones shall be modeled for use in interference management and maintenance access requirements.

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4.7 Fire Protection. The fire protection system Model may vary in level of detail for individual elements, but at a minimum must include all features that would be included on a quarter inch (1/4"=1'0") scaled drawing. Additional minimum Model requirements include:

4.7.1 Fire Protection System. All relevant fire protection components (i.e., branch piping, sprinkler heads, fittings, drains, pumps, tanks, sensors, control panels) with necessary intelligence to produce accurate plans, elevations, building/wall sections, riser diagrams, and schedules. All fire protection piping shall be modeled.

4.7.2 Fire Alarms. Fire alarm/mass notification devices and detection system shall be indicated with necessary intelligence to produce accurate plans depicting them.

4.8 Civil. The civil Model may vary in level of detail for individual elements, but at a minimum must include all features that would be included on a one inch (1"=100') scaled drawing. Additional minimum Model requirements include:

4.8.1 Terrain (DTM). All relevant site conditions and proposed grading, including necessary intelligence to produce accurate Project site topographical plans and cross sections.

4.8.2 Drainage. All existing and new drainage piping, including upgrades thereto, including necessary intelligence to produce accurate plans and profiles for the Project site.

4.8.3 Storm Water and Sanitary Sewers. All existing and new sewer structures and piping, including upgrades thereto, on the Project site with necessary connections to mains or other distribution points as appropriate, including necessary intelligence to produce accurate plans and profiles for the Project site.

4.8.4 Utilities. All necessary new utilities connections from the Project building(s) to the existing or newly-created utilities, and all existing above ground and underground utility conduits, including necessary intelligence to produce accurate plans and site-sections.

4.8.5 Roads and Parking. All necessary roadways and parking lots or parking structures, including necessary intelligence to produce accurate plans, profiles and cross-sections.

4.9 CVIE and BIM. The A/E shall utilize the native objects provided in the BIM software to create the models upon which the CVIE is based. Geometric-only representations of required Coordination View entities shall not be allowed. The A/E shall maintain a Globally Unique Identifier (GUID) for each BIM objects, physical room, and functional spaces. The A/E shall not change GUID's for objects and spaces that are submitted in subsequent deliverables. Multiple BIM files may be used to create the CVIE. If this is the case, then the A/E shall provide each of these individual files with each CVIE deliverable. The A/E shall utilize a common registration point for all design disciplines included in the CVIE even if this model is provided in multiple files. The A/E shall configure collision detection software, either embedded in their BIM software or separate software, to identify all physical overlaps of BIM objects.

Section 5 - Ownership and Rights in Data

5.1 Ownership. The FAA has ownership of and rights to all CAD files, the Models, and Facility Data developed for the Project in accordance with FAR Part 27. The FAA may make use of these files following any design phase deliverable.

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Section 6 – A/E Electives

6.1 Applicable Criteria. The A/E may elect to submit one or more of the following features as a separate cost line item in its contract proposal. The following criteria are requirements, as applicable to those elective feature(s).

6.2 Project Scheduling using the Model. In the PxP, provide an overview of the use of BIM in the development and support of the project construction schedule.

6.2.1 Submittal Requirements. During the Construction Submittal stages, the A/E shall associate the General Contractor's construction schedule with information derived from the Model.

6.2.1.2 Construction Submittals – Over-The-Shoulder Progress Reviews. Periodic quality control meetings or construction progress review meetings shall include quality control reviews on the implementation and use of the Model for construction project scheduling.

6.3 Cost Estimating. In the PxP, provide an overview of the use of BIM in the development and support of cost estimating requirements, or other applications such as cost analysis and estimate validation.

6.3.1 Submittal Requirements. During the Submittal stages, the A/E shall deliver cost estimating information derived from the Model.

6.3.2 Project completion. At project completion, the A/E shall provide an MII (Micro Computer Aided Cost Estimating System Generation II) Cost Estimate which follows the USACE Cost Engineering Military Work Breakdown System (WBS), a modified uniformat, to at least the sub-systems level and uses quantity information supplied directly from BIM output to the maximum extent possible, though other "Gap" quantity information will be included as necessary for a complete and accurate cost estimate.

6.3.2.1 Sub system level extracted quantities from the Model for use within the estimate shall be provided according to how detailed line items or tasks should be installed/built so that accurate costs can be developed and/or reflected. Therefore, when developing a BIM, the A/E shall be cognizant of what tasks need to be separated appropriately at the beginning stages of model development, such as tasks done on the first floor versus the same task on higher floors that will be more labor intensive and therefore need to have a separate quantity and be priced differently. Tasks and their extracted quantities from the Model shall be broken done by their location (proximity in the structure) as well as the complexity of its installation.

6.3.2.2 At all design stages it shall be understood that BIM output as described in this document will not generate all quantities that are necessary in order to develop a complete and accurate cost estimate of the project based on the design. An example of this would be plumbing that is less than 1.5" diameter and therefore not expected to be modeled due to granularity; this information is commonly referred to as The Gap. Quantities from The Gap and their associated costs shall be included in the final project actual cost estimates as well.

Section 7 - Definitions:

The following definitions apply specifically in the context of this BIM requirement document only:

"Model": An electronic, three-dimensional representation of facility elements with associated intelligent attribute data ("Facility Data").

"Facility Data": The non-graphical information attached to objects in the Model that defines various characteristics of the object. Facility Data can include properties such as parametric values that drive physical sizes, material definitions and characteristics (e.g. wood, metal), manufacturer data, industry standards (e.g. AISC steel properties), and project identification numbers. Facility Data can also define

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supplementary physical entities that are not shown graphically in the Model, such as insulation around a duct, or hardware on a door.

"Workspace": A collection of content libraries and supporting files that define and embody a BIM standard. A workspace includes BIM libraries such as wall types, standard steel shapes, furniture, HVAC fittings, and sprinkler heads. It also contains sheet libraries such as print/plot configurations, font and text style libraries, and sheet borders and title blocks. The USACE has developed Workspaces specific to USACE BIM standards; these workspaces are dependent on specific versions of the BIM applications they serve. All USACE BIM Workspaces can be downloaded from the CAD/BIM Technology Center (https://cadbim.usace.army.mil). In some cases, there is a specific Workspace for a given CoS Facility Standard Design.

"IFC": Industry Foundation Class, a standard and file format used for the exchange of BIM data; see www.iai-tech.org. Note: In the context of this BIM requirement document, IFC does not mean "Issued For Construction."

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NOTE: The table parameter names may change in the final, but that does not change the approval of this document. These tables accurately reflect data intent to achieve the FLCM goals of various users in the FAA FLCM process. The table parameter names are in final coordination with BuildingSmart, IFC (COBIE v2.30), other federal agencies (DoD MHS, VA, GSA, etc.). The number of variables is not expected to change.

Tables

The following tables reflect data element requirements, naming convention to be used for the various data elements, format restrictions, responsibility and data element responsible authority.

The note references apply to the following tables.

- Note 1 – To be determined in the Project Execution Plan (PxP). The A/E will at a minimum create the parameter in the model.
- Note 2 – Shall be automatically calculated by the BIM platform utilized.
- Note 3 – Derived from the SEPS BIM Export Files. The A/E shall coordinate with the FAA to adjust non-applicable DoD-MHS naming conventions in SEPS, DMLSS, and DMIS to FAA specific use.
- Note 4 – A/E
- Note 5 – Mechanical, electrical, administrative, circulation, inpatient bed room, etc.
- Note 6 – Shall be coordinated with the FAA to match intended signage.
- Note 7 – Reference Table 1 for parameter details and data should be drawn from the respective parameters.
- Note 8 – Reference Table 20-2 for parameter details and data should be drawn from the respective parameters.
- Note 9 – Shall be coordinated with the FAA ensure correct naming/classification structure/label is utilized. The A/E shall coordinate with the FAA to adjust non-applicable DoD-MHS naming conventions in SEPS, DMLSS, and DMIS to FAA specific use.
- Characters: 0.00 represents a number to two decimal places; if the type is “text” the number entered represents the maximum number of text characters allowed.
- Use established IFC parameters where relevant.

Table 1. Project Level Parameters

Parameter Name	Description	Type	Characters	Responsible Authority
CD_Const_A	Calendar days in contract for construction at award	Integer		Note 1
CD_Const_F	Calendar days in contract for construction at completion	Integer		Note 1
CD_Design_A	Calendar days in original design at award	Integer		Note 4

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Table 1. Project Level Parameters				
Parameter Name	**Description**	**Type**	**Characters**	**Responsible Authority**
CD_Design_F	Calendar days total for design at completion of design	Integer		Note 4
Const_Comp	Date construction complete actual	Date	DD/MM/YYYY	Note 1
Const_ContOff	Design & Construction Agent and office responsible for design award (e.g. USACE – NAD)	Text	50	Note 1
Cost_Arch	Construction Contract cost, architectural components for the project - actual	Number	0.00	Note 1
Cost_Control	Construction Contract cost, controls components for the project - actual	Number	0.00	Note 1
Cost_Cx	Construction Contract cost, commissioning (Cx) components for the project – actual	Number	0.00	Note 1
Cost_Elec	Construction Contract cost, electrical components for the project - actual	Number	0.00	Note 1
Cost_IT	Construction Contract cost, information technology components for the project - actual	Number	0.00	Note 1
Cost_LEED	LEED premium cost for construction	Number	0.00	Note 1
Cost_LOGCATA	Total cost of LOGCAT A items included in the project	Number	0.00	Note 1
Cost_Mech	Construction Contract cost, mechanical components for the project - actual	Number	0.00	Note 1
Cost_Plumb	Construction Contract cost, plumbing components for the project - actual	Number	0.00	Note 1
Cost_Struct	Construction Contract cost, structural components for the project - actual	Number	0.00	Note 1

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Table 1. Project Level Parameters				
Parameter Name	**Description**	**Type**	**Characters**	**Responsible Authority**
Cost_Total_A	Total construction cost for the project at contract award	Number	0.00	Note 1
Cost_Total_F	Total construction cost for the project at project completion	Number	0.00	Note 1
Currency	Currency used on project	Text	25	Note 1
Currency_Rate	Currency rate to USD for project	Number	0.00	Note 1
Design_Comp	Date that the design was completed, actual	Date	DD/MM/YYYY	Note 4
Design_ContOff	Design & Construction Agent and office responsible for design award (e.g. USACE – NAD)	Text	50	Note 4
Installation	Project location – installation, base, station name	Text	100	Note 1
Location_Country	Project location, country	Text	75	Note 1
Location_State	Project location, state	Text	75	Note 1
Location_Town	Project location, town	Text	75	Note 1
Location_Zip	Project location, postal code	Text	00000-0000	Note 1
NTP_Const	Date of notice to proceed was given for construction	Date	DD/MM/YYYY	Note 1
NTP_Design	Date of notice to proceed was given for design	Date	DD/MM/YYYY	Note 4
Organization	Organizational name (e.g. Tripler Army Medical Center)	Text	100	Note 1, 2
PFD_GSF_A	Gross square feet in the PFD at the award of the contract	Integer		Note 4
PFD_NSF_A	Net square feet in the PFD at the award of the contract	Integer		Note 4
PN	Project Number	Text	25	Note 1, 5
PName	Project name	Text	100	Note 1, 5
RPUID	Real property unique identification number	Text	25	Note 1
Total Items 36				

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Table 2. Room Level Parameters				
Parameter Name	**Description**	**Type**	**Characters**	**Responsible Authority**
Fire_Zone	Fire Safety Zone - to allow color diagram development	Text	10	Note 1
DMIS	DMIS ID (as associated with room's department)	Text	25	Note 3, 4
Gross_Count	Default YES. Yes indicates it counts in the space gross calculation for the building according to the UFC	Yes/No		Note 1
Hazardous_Materials	Room contains hazardous materials	Yes/No		Note 1
Master_RoomSpecs_Formal_Room_Name	Master room name from SEPS export	Text	150	Note 1, 3
NSF	Net square feet for the space	Integer		Note 2
NSF_Programmed	NSF programmed from the SEPS export	Integer	5	Note 1, 3
Omniclass_Code	Associated omni-class code	Text	17	Note 1
Project_Dept_AgencyName	Department owning space	Text	150	Note 1, 3
Project_FA_AgencyName	Department functional area classification	Text	150	Note 1, 3
Project_Room_ID	Project room ID	Integer		Note 1, 3
Project_Room_Name	Space name	Text	100	Note 1, 3
Room	Room Number	Text	15	Note 1, 4, 6
Room_Code	Room code from SEPS export	Text	5	Note 1, 3
Room_POC	Local point(s) of contact for the space	Text	100	Note 1
Security_Requirement	Security requirements for space	Text	45	Note 1
Shielded_Room	Space is shield (e.g. lead shielding)	Yes/No		Note 1
Total Items 17				

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Table 3. Associated Schedule Data from Project & Room Level Parameters				
Parameter Name	**Description**	**Type**	**Characters**	**Responsible Authority**
DMIS				Note 7
Floor_Number	Floor number	Text	15	Note 1, 5
Installation				Note 7
RPUID				Note 7
Total Items 4				

Table 4. Real Property Installed Equipment (RPIE) Object Parameters				
Parameter Name	**Description**	**Type**	**Characters**	**Responsible Authority**
Acquisition_Cost	Cost of the RPIE item	Number	0.00	Note 1
Areas_Supported	Areas of the building supported by this RPIE item	Text	40	Note 1
Assembly_Category	DMLSS RPIE Hierarchy	Text	60	Note 1, 9
Barcode	Unique barcode number assigned – no duplicates	Text	20	Note 1, 9
Capacity_Unit	Capacity of the piece of RPIE (e.g. BTUs, Tons, KVA, etc.) Must be a valid capacity from the “DMLSS Capacity Unit” table	Text	20	Note 1
Capacity_Value	The value of the capacity based on the “Capacity_Unit” associated	Text	11	Note 1
Catalog_No	RPIE item catalog number by manufacturer	Text	20	Note 1
Equipment_Hazard	Hazard associated with RPIE item. Must be a valid hazard in the “DMLSS Hazard” table.	Text	20	Note 1
Facility_SubSystem	DMLSS RPIE Hierarchy subsystem	Text	60	Note 1, 9
Facility_System	DMLSS RPIE Hierarchy system	Text	60	Note 1, 9
Index_Number	Unique identifier assigned by the facility – no duplicates	Text	25	Note 1, 9
Installation_Date	Date of installation	Date	DD/MM/YY YY	Note 1
Life_Expectancy	Expected operational life (in years) of the RPIE item	Integer		Note 1

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Table 4. Real Property Installed Equipment (RPIE) Object Parameters				
Parameter Name	**Description**	**Type**	**Characters**	**Responsible Authority**
Manufacturer	RPIE item manufacturer	Text	40	Note 1
Model	RPIE model number	Text	20	Note 1
Nomenclature	Generic name used for managed equipment in the DMLSS RPIE Hierarchy	Text	40	Note 1, 9
PM_Parent	Preventive maintenance schedule required on this item, but must be done with maintenance or a parent item	Text	1	Note 1
PM_Related	If “PM-Parent” is Yes, then list the parent RPIE item’s “Index_No” here	Text	25	Note 1
PM_Sched	Preventive maintenance schedule required Y – yes, or N – No	Text	1	Note 1
QA_Method	R – random, A – automatic; associated with government maintenance over life of RPIE item (Quality Assurance)	Text	1	Note 1, 9
QC_Method	R – random, A – automatic; associated with government maintenance over life of RPIE item (Quality Control)	Text	1	Note 1, 9
Risk_Assignment	Risk value assigned to the piece of equipment relating to the criticality of the specific piece of RPIE. Must be a valid DMLSS Risk Assignment value.	Text	25	Note 1, 9
Serial_No	RPIE serial number	Text	20	Note 1
Specification_Unit	Unit of measure for the RPIE item. (e.g. amperes, voltage, hertz, etc.)	Text	20	Note 1
Specification_Value	The value of the specification based on the “Specification_Unit” associated	Text	11	Note 1
Total Items 29				
Warranty_Labor_Expiration	Date on which the terms of the warranty for labor end	Date	DD/MM/YYYY	Note 1
Warranty_Parts_Expiration	Date on which the terms of the warranty for parts end	Date	DD/MM/YYYY	Note 1

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Table 4. Real Property Installed Equipment (RPIE) Object Parameters				
Parameter Name	**Description**	**Type**	**Characters**	**Responsible Authority**
Warranty_Start	Date the warranty starts for the RPIE item	Date	DD/MM/YYYY	Note 1
Warranty_Vendor	Vendor organization that holds the warranty responsibility	Text	40	Note 1
Total Items 30				

Table 5. Associated Schedule Data from Room Level Parameters				
Parameter Name	**Description**	**Type**	**Characters**	**Responsible Authority**
Room				Note 8
Total Items 1				

Table 6. Equipment and Furniture Object Parameters				
Parameter Name	**Description**	**Type**	**Characters**	**Responsible Authority**
Cost	Current estimated or actual cost of the item	Numeric	0.00	Note 1
Currency	Currency used on project	Text	25	Note 1
Currency_Rate	Currency rate to USD for project	Number	0.00	Note 1
JSN	Joint Service Number	Text	5	Note 1, 3
Logistics_Category	Actual logistics category used for this item in the project	Text	1	Note 1
Project_Equipment_ID	Project room contents ID	Integer		Note 1, 3
Project_Room_ID	Link in the SEPS data set to the unique room ID	Integer		Note 1, 3
QTY	Quantity, typically = 1	Numeric	0.00	Note 1
Total Items 8				

Table 7. Equipment and Furniture Object Parameters (as applicable to the specific piece of equipment)				
Parameter Name	**Description**	**Type**	**Characters**	**Responsible Authority**
Amps1	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Amps2	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Army_Cat	As defined in the MIL-STD 1691	Text	1	Note 1

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Table 7. Equipment and Furniture Object Parameters (as applicable to the specific piece of equipment)				
Parameter Name	**Description**	**Type**	**Characters**	**Responsible Authority**
BTUPerHour	As defined in the MIL-STD 1691	Text	6	Note 1
Date_Appr	As defined in the MIL-STD 1691	DATETIME	0.00	Note 1
Depth_CEN	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Depth_IN	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Desc	As defined in the MIL-STD 1691	Text	3000	Note 1
FS_Army	As defined in the MIL-STD 1691	Text	1	Note 1
FS_Navy	As defined in the MIL-STD 1691	Text	1	Note 1
FS_USAF	As defined in the MIL-STD 1691	Text	1	Note 1
FS_VA	As defined in the MIL-STD 1691	Text	1	Note 1
Height_CEN	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Height_IN	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Hertz	As defined in the MIL-STD 1691	Text	5	Note 1
Hertz_dep	As defined in the MIL-STD 1691	Text	1	Note 1
Hertz_swit	As defined in the MIL-STD 1691	Text	1	Note 1
JSN_Name	As defined in the MIL-STD 1691	Text	50	Note 1
Mean_Price	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Navy_Cat	As defined in the MIL-STD 1691	Text	1	Note 1
NSN	As defined in the MIL-STD 1691	Text	13	Note 1
Phase1	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Phase2	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Price_Date	As defined in the MIL-STD 1691	DATETIME	0.00	Note 1
Project_JSN_Comment	As defined in the MIL-STD 1691	Text	100	Note 1
Unit_Issue	As defined in the MIL-STD 1691	Text	2	Note 1
USAF_Cat	As defined in the MIL-STD 1691	Text	1	Note 1
Utility_1	As defined in the MIL-STD 1691	Text	1	Note 1
Utility_2	As defined in the MIL-STD 1691	Text	1	Note 1
Utility_3	As defined in the MIL-STD 1691	Text	1	Note 1
Utility_4	As defined in the MIL-STD 1691	Text	1	Note 1
Utility_5	As defined in the MIL-STD 1691	Text	1	Note 1
Utility_6	As defined in the MIL-STD 1691	Text	1	Note 1
VA_Cat	As defined in the MIL-STD 1691	Text	1	Note 1
Volts1	As defined in the MIL-STD 1691	Numeric	0.00	Note 1

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Table 7. Equipment and Furniture Object Parameters (as applicable to the specific piece of equipment)				
Parameter Name	**Description**	**Type**	**Characters**	**Responsible Authority**
Volts2	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Watts1	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Watts2	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Weight_AV	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Weight_MET	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Width_CEN	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Width_IN	As defined in the MIL-STD 1691	Numeric	0.00	Note 1
Total Items 42				

Table 8. Associated Schedule Data from Project and Room Level Parameters				
Parameter Name	**Description**	**Type**	**Characters**	**Responsible Authority**
PN	Project Number	Text	25	Note 1, 5
PName	Project name	Text	100	Note 1, 5
Room	Room number in which the equipment is located	Text		
TotalCostPerItem	Quantity x Cost	Numeric	0.00	Note 1
Total Items 4				

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Matrix of Submittals & Schedules

The following matrix represents a consolidated reference to submittal requirements defined in this document. The submittals are in concert with and do not replace the A/E Scope of Work (SOW).

Deliverable / Schedule	Description	Cross Reference to Section
Model Tests	The contractor shall conduct model tests to address clash detection / inference management validating discipline and trade coordination of submittal deliverables; PFD validation; PRCR validation; gross (total building and departmental) validation; other checks as proposed by the contractor.	*2.4 Associated with Section 3*
Schedule: Equipment & Furniture Data	Equipment & Furniture (associated with the PRCR) object data parameters to be displayed as a schedule and included as part of the project submittals.	2.2.4.1.4
Schedule: Project Data	Project level data parameters that shall be displayed as a schedule and included as part of the project submittals.	2.2.4.1.1
Schedule: Room Data	Room level data parameters that shall be displayed as a schedule and included as part of the project submittals.	2.2.4.1.2
Schedule: RPIE Data	Real Property Installed Equipment object data parameters to be displayed as a schedule and included as part of the project submittals.	2.2.4.1.3
Submittal: PxP	To be submitted to the FAA as defined in the referenced section.	2.2.3
Submittal: COBie-FAA Data for Equipment & Furniture	The contractor shall submit an extract of the data for the Equipment & Furniture in the COBie format. Submittal with interim deliverables, are intended to be used as “draft” submittal prior to the final to ensure the final is in compliance with the COBie-FAA structure and accepted by the FAA.	2.2.4.1.4
Submittal: COBie-FAA Data for RPIE	The contractor shall submit an extract of the data for the RPIE in the COBie format. Submittal with interim deliverables, are intended to be used as “draft” submittal prior to the final to ensure the final is in compliance with the COBie structure and accepted by the FAA.	2.2.4.1.3
Submittal: Design	Concurrent with the project submittals as defined in the project contract and defined in the accepted PxP for the project.	3.3
Submittal: Over-The-Shoulder	Over-the-shoulder progress reviews are used primarily as visual in process coordination reviews and as such typically have no electronic or hardcopy deliverable other than associated meeting minutes unless defined otherwise in the PxP or project contract.	2.4.7

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Acronyms List

The following is a list of acronyms relevant to this BIM requirement document:

Acronym	Represents
BHMA	Builders Hardware Manufacturers Association (http://www.buildershardware.com/)
BIM	Building Information Model or Building Information Modeling
CAEG	Computer Aided Engineering Graphics
CAFM	Computer Aided Facility Management
CMMS	Computerized Maintenance Management System
DMIS	Defense Medical Information System for DoD-MHS
DMLSS	Defense Medical Logistics Standard Support for DoD-MHS
DoD	Department of Defense
DTM	Digital Terrain Model
FAA	Federal Aviation Administration
FLCM	Facility Life Cycle Management
FM	Facility Management or Facility Manager
IFC	Industry Foundation Class
MEP	Mechanical, Electrical and Plumbing
MHS	Military Health System http://www.bim4healthcare.org/ http://www.tricare.mil/ocfo/ppmd/guideplates.cfm#Tools-SEPS
NAVFAC	Naval Facilities Engineering Command
NPS	Nominal Pipe Size
OmniClass	OmniClass Construction Classification System – Open Source (http://www.omniclass.org/)
PFD	Program for Design
PRC	Project Room Contents
PRCR	Project Room Contents Report
PxP	Project Execution Plan
RPIE	Real Property Installed Equipment
SEPS	Space and Equipment Planning System for DoD-MHS
SOW	Scope of Work or Statement of Work
SRM	Sustainment Restoration and Modernization
USACE	United States Army Corps of Engineers https://cadbim.usace.army.mil/BIM_Contract_Language

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APPENDIX D

Sample BIM Roadmaps: Korea Rail BIM 2030 Roadmap and Massport BIM Roadmap

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Executive Summary

This Rail BIM 2030 Roadmap explains the phases of the BIM adoption and diffusion strategies from 2018 to 2030 under "level of information utilization in BIM" for the advancement of railway facilities in the 4th Industrial Age. The five classification levels discussed below are based on the gradual maturity of the BIM capabilities of the participants taking part in the project.

Rail BIM 2030 Roadmap

Level 1: Target 2018 (BIM 1.0) 2D-to-3D Conversion BIM

Very few project participants will carry out BIM during the 2D-to-3D conversion BIM (level 1) phase in 2018. 2D drawings will remain the primary communication media. Participants can benefit from deploying BIM in their projects, as this will allow them to review designs and detect design errors while converting 2D drawings to BIM models. Once BIM models of the project become available, the models can also be utilized for public hearings, communicating with clients, constructability reviews, and so forth.

Level 2: Target 2020 (BIM 2.0) Two-Track BIM (Parallel BIM)

During the two-track BIM (level 2) phase, targeted for 2020, BIM will be used for the parts of a project that can significantly benefit from deploying BIM, such as in areas where multiple trades interface with one another, areas with complex geometry, and areas that require heavy engineering. The other parts of the projects will be carried out using the traditional 2D-based method.

Railway BIM2030 Roadmap

People / Process / Technology

BIM 1.0 **2D-3D Conversion BIM** → BIM 2.0 **Two-Track BIM** →

2018 — 2020

BIM Goals

Procurement & Planning	• Communication between managers and project participants using BIM visualization tools • Application of AR/VR for design reviews or public hearings
Design Expansion of BIM application	• Application for environmental assessment • Design review and coordination, budget review • Elimination of unnecessary social cost using visual information • Visualization/digitalization of construction plans and budgets

		2018	2020
PEOPLE Culture Organization Education	Company-wide level	Company-wide BIM team	Monitoring, support, and management of multiple BIM projects
	Individual project manager level	Ability to manage the 2D to 3D conversion process Ability to handle BIM models	Ability to lead two-track BIM Design review using BIM
PROCESS Policy Infrastructure Service & Reward Work Process	Coordination process	Design coordination between a limited number of team members	Increased importance of coordination between drawings and BIM models Limited use of BIM for specific issues by specific teams
	Tendering and management process	Application of existing tendering and project management methods	Little use of the lowest bid or the like Recommended use of services and CM at risk
TECHNOLOGY Software Hardware Information	Fundamental technology	BIM authoring tools BIM model checkers BIM viewers	BIM analysis tools Detailing tools Interoperability
	Management technology	A system that enables project participants to share BIM models and error reports acquired through design conversion (e.g., CDE, Big Room System) A system that tracks and manages design errors	Coordination technology between 2D and BIM processes (e.g., 3D scanning, model version management)

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At the end of each level, the lessons from each project should be integrated into updated rail BIM guides, ensuring efficient and effective use of BIM in the following stage. The rail BIM guides must be continuously updated. This report details rail BIM adoption and diffusion strategies in terms of the people, processes, and technologies necessary for achieving the five levels below.

Level 3: Target 2022 (BIM 3.0) Integrated BIM or Full BIM

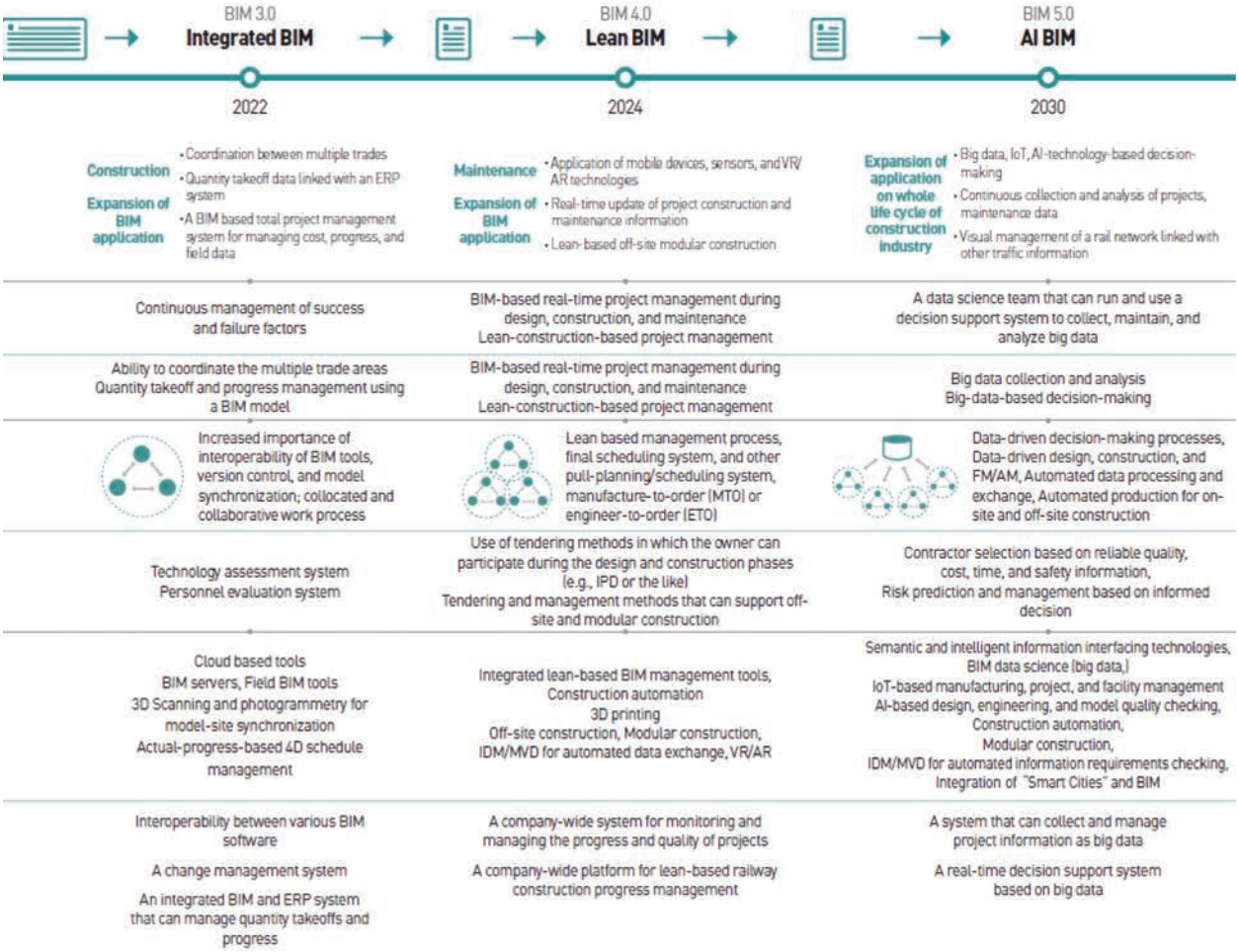
During the (fully) integrated BIM (level 3) phase, targeted for 2022, all the major project participants will generate and work with BIM models. This will enable the BIM-based, integrated project management of construction costs, scheduling, and processes, as well as issues concerning design and construction quality.

Level 4: Target 2024 (BIM 4.0) Lean BIM

During the lean BIM (level 4) phase, targeted for 2024, project management concepts influenced by the manufacturing industry, such as lean construction, offsite modular construction, construction automation, and integrated facility management, will be supported by BIM, generating synergy for improving the productivity and quality of projects.

Level 5: Target 2030 (BIM 5.0) Intelligent BIM (AI BIM)

During the intelligent BIM (level 5) phase, targeted for 2030, big data will be created by integrating BIM with sensors and multiple databases. This data will be used as a source for making informed decisions. The most optimal and effective solutions to many city-level problems, as well as individual facility-level problems, will be sought through the informed decision-making process.



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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TDC	Transit Development Corporation
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

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